Entropy-Based Learning

Principles, Methods, and Scientific Applications

Yatao Bian

RELEASED
August 2, 2026

LAST UPDATED
October 16, 2026



Blue Whale Lab

Contents

Preface . . . **1**
How complete is this draft? . . . 1
Citation . . . 2

1 Introduction . . . **3**
1.1 Why entropy? . . . 3
1.2 Roadmap . . . 4

Part I — Foundations . . . **5**

2 Entropy and the Maximum Entropy Principle . . . **6**
2.1 The many faces of entropy . . . 6
2.1.1 Information: Shannon entropy . . . 6
2.1.2 Heat: thermodynamic entropy . . . 6
2.1.3 Quantum states: von Neumann entropy . . . 7
2.1.4 One knob, a family: Rényi and Tsallis . . . 7
2.1.5 Minds: psychological entropy . . . 8
2.1.6 One form, many readings . . . 8
2.2 Why these formulas and no others: the axioms . . . 9
2.2.1 The load-bearing axiom . . . 9
2.2.2 Loosening one axiom at a time . . . 11
2.2.3 The other faces, axiomatized . . . 12
2.2.4 From entropy to diversity . . . 13
2.2.5 Similarity . . . 14
2.2.6 What axioms cannot settle . . . 18
2.3 The maximum entropy principle . . . 19
2.4 Minimum description length and minimax entropy . . . 19
2.5 Why it matters for learning . . . 20

3 Energy, Entropy, and Free Energy . . . **21**
3.1 Maximum entropy with an energy constraint . . . 21
3.2 Free energy . . . 21
3.3 What entropy buys you . . . 22
3.3.1 Isothermal expansion: a process with no energy change . . . 22
3.3.2 The rubber band: a force made of entropy . . . 22
3.3.3 Melting: where the competition changes hands . . . 23
3.4 From spins to networks . . . 23
3.5 How this reached AI . . . 23
3.6 The temperature dial . . . 24
3.7 Why it matters for learning . . . 25

4 Energy-Based Models . . . **26**
4.1 The energy view of inference . . . 26
4.2 Why learn an energy at all? . . . 27
4.3 Learning is shaping a landscape . . . 28

4.4 Strategy one: sample the pull-up . . . 29
4.5 Strategy two: match slopes, not heights . . . 30
4.6 Strategy three: learn by telling things apart . . . 31
4.7 One family, three estimators . . . 32
4.8 Latent variables, and the free energy again . . . 33
4.9 The modern landscape . . . 33
4.10 Where this sits in the book . . . 34
4.11 Takeaway . . . 35

Part II — Methods . . . 36

5 Maximum-Entropy Learning . . . 37
5.1 What this chapter means by maximum-entropy learning . . . 37
5.1.1 Representation: maximum-entropy models in two forms . . . 37
5.1.2 Inference: free-energy minimization over a tractable family . . . 38
5.1.3 Learning: fitting the energy through the inference . . . 39
5.1.4 Definition of maximum-entropy learning, and a running example . . . 39
5.2 Graphical models as an explanatory family . . . 41
5.3 A worked example: energy-based cooperative games . . . 43
5.3.1 Why this is useful . . . 43
5.4 Discrete maximum-entropy learning: set functions under the optimal subset oracle . . . 44
5.4.1 The optimal subset oracle . . . 44
5.4.2 Learning through mean-field inference: EquiVSet . . . 45
5.4.3 Which statistics may the energy depend on? INSET . . . 46
5.4.4 What the strand adds to the maximum-entropy programme . . . 47
5.5 Maximum-entropy reinforcement learning: energies over trajectories . . . 48
5.5.1 A Gibbs distribution over trajectories . . . 48
5.5.2 Soft Bellman equations as free energies . . . 49
5.5.3 Inverse reinforcement learning: the optimal subset oracle on trajectories . . . 49
5.6 What the family adds, and where it stops . . . 50
5.6.1 Maximum likelihood as a saddle-point problem . . . 50
5.6.2 Maximum entropy inside, minimum entropy outside: the bridge to minimax entropy 51
5.6.3 Where the family stops . . . 51
5.7 Takeaway . . . 52

6 Minimum-Entropy Learning . . . 54
6.1 Two regimes, and a warning about the name . . . 54
6.2 From word-level to meaning-level uncertainty . . . 55
6.3 Entropy minimisation as an objective: a short history . . . 55
6.4 EMPO: reward agreement, not correctness . . . 60
6.5 What "reward agreement" actually optimises . . . 61
6.6 The free-energy view, and why entropy collapses . . . 63
6.7 The information ceiling . . . 69
6.8 Schrödinger, read carefully . . . 72
6.9 Prigogine, and an analogy that does not hold . . . 73
6.10 So why call it minimum-entropy learning? . . . 74
6.11 When agreement means truth, and when it does not . . . 75
6.12 Takeaway . . . 75

7 Minimax-Entropy Learning . . . 76
7.1 The minimax entropy principle: what to measure . . . 76
7.1.1 Three equivalent views . . . 76

7.1.2 Solving it . . . 77
7.1.3 Implicit minimax entropy in energy-based models . . . 77
7.2 Minimax entropy on the joint space: modelling and predicting . . . 77
7.2.1 Two questions, one principle . . . 77
7.2.2 Classification is conditional minimax entropy . . . 78
7.2.3 A classifier is a joint maximum-entropy model . . . 79
7.3 When the label is missing . . . 81
7.3.1 A temperature on the label . . . 81
7.3.2 What the density term buys, and what it does not . . . 83
7.4 Boundaries, evidence, and a name clash . . . 85
7.5 Takeaway . . . 87

8 Energy-Based Guidance and Refinement . . . 89
8.1 The tilted reference . . . 89
8.2 Continuous guidance . . . 90
8.3 Discrete guidance: residual energies over a proposal . . . 91
8.4 Verifiers as energies, best-of-N, and sequential Monte Carlo . . . 92
8.5 Refinement and reasoning as energy minimization . . . 93
8.6 Guidance on the three levels . . . 94
8.7 Takeaway . . . 95

9 Maximum-Entropy Post-Training . . . 96
9.1 From the reference to its tilt . . . 96
9.2 Amortizing the tilt . . . 97
9.3 Learning the energy . . . 98
9.4 Self-referential energies . . . 99
9.5 The continuous case . . . 99
9.6 Post-training on the three levels, and where it stops . . . 100
9.7 Takeaway . . . 102

10 From Entropy-Based Learning to the Free-Energy Principle . . . 103
10.1 Foundations of the free-energy principle . . . 103
10.1.1 Origins . . . 103
10.1.2 Variational free energy and perception . . . 104
10.1.3 Predictive coding, precision, and attention . . . 105
10.1.4 Learning and model selection . . . 107
10.1.5 Action: active inference and the expected free energy . . . 107
10.1.6 Markov blankets, steady states, and Bayesian mechanics . . . 109
10.1.7 A map of the principle . . . 109
10.2 Recent developments, 2019–2026 . . . 110
10.2.1 Critiques and replies . . . 110
10.2.2 Planning and the theory of the expected free energy . . . 111
10.2.3 Deep and scalable active inference . . . 112
10.2.4 Experimental tests . . . 112
10.2.5 Generative AI and agents . . . 112
10.2.6 Timeline . . . 113
10.3 Connections to entropy-based learning . . . 114
10.3.1 One free energy . . . 114
10.3.2 Perception: two readings of the dial . . . 114
10.3.3 Learning: minimax entropy at $T = 1$ . . . 115
10.3.4 Model selection: Type I and description length . . . 116
10.3.5 Action: Type III, firm but fair . . . 117
10.3.6 Policies: Gibbs tilts, and precision as a temperature . . . 118

10.3.7 The closed loop and the information ceiling . . . 119
10.3.8 Self-referential tilts: learned preferences and habits . . . 119
10.3.9 Schrödinger, steady states, and the maximum-entropy duality . . . 120
10.3.10 The correspondence at a glance . . . 121
10.4 Takeaway . . . 122

Part III — Applications . . . 124

11 Entropy-Based Learning for Science . . . 125
11.1 Two synergistic directions . . . 125
11.2 Molecular foundation models . . . 125
11.3 Outlook . . . 125

12 Summary . . . 126

References . . . 127

Preface

Welcome to **Entropy-Based Learning**. This short book is an introductory, evolving companion to a research program on *modern entropy- and energy-based learning*, developed at the Blue Whale Lab led by Yatao Bian.

Entropy is one of the deepest bridges between physics, information theory, and machine learning — a single quantity reinvented, again and again, for heat engines, telephone lines, quantum ensembles, and even theories of human anxiety. Three complementary principles organize the material in this book:

- **The Maximum Entropy Principle** — among all models consistent with the observed constraints, prefer the one that is maximally noncommittal (highest entropy). This principle underlies statistical physics and Boltzmann machines, and it defines a family of learning methods that includes a principled class of game-theoretic valuations, neural set functions learned from their optimal subsets, and maximum-entropy and inverse reinforcement learning. With a pretrained model as the base measure, the same family contains energy-based guidance at inference time and post-training methods such as RLHF, GRPO, and DPO.
- **The Minimum Entropy Principle** — when a model's own predictions can serve as a training signal, reducing predictive uncertainty can elicit capabilities such as reasoning, *without* external supervision. It elicits but does not add capability, since the training loop never touches ground truth, and, left unchecked, it collapses the model onto one confident answer, right or wrong.
- **The Minimax Entropy Principle** — maximize entropy given the features a model includes, then choose the features that make that entropy as small as possible (Zhu et al. 1997; Carcamo et al. 2025). This couples the two principles above, and the same nesting answers two questions: *what to measure*, by choosing the features that give the shortest description of the data, and *what to commit to*, by minimizing the entropy of a prediction, which is what a softmax classifier with a linear head already does.

Binding them together is the **free energy** $F = U - TS$, in which temperature T sets the exchange rate between energy and entropy. It is the equation that carried statistical physics into machine learning, and it makes the three principles above three settings of one dial rather than three separate doctrines.

How complete is this draft?

> ⚠ Work in Progress
>
> This book is an ongoing project, released early so you can follow its development. Most chapters are incomplete. The table below shows the current status of each chapter. Please keep this in mind before quoting or citing any part of the text, or before judging drafts that are still being written.

Chapter	Status
1 — Introduction	early draft
2 — Entropy and the Maximum Entropy Principle	**mostly complete**
3 — Energy, Entropy, and Free Energy	**mostly complete**
4 — Energy-Based Models	**mostly complete**

Chapter	Status
5 — Maximum-Entropy Learning	**mostly complete**
6 — Minimum-Entropy Learning	**mostly complete**
7 — Minimax-Entropy Learning	first full draft
8 — Energy-Based Guidance and Refinement	first full draft
9 — Maximum-Entropy Post-Training	first full draft
10 — From Entropy-Based Learning to the Free-Energy Principle	first full draft
11 — Entropy-Based Learning for Science	early draft
12 — Summary	early draft

"Mostly complete" means the argument is in place and has been checked, though details may still move. "First full draft" means the chapter is written end to end but has not yet been checked with the same care. "Early draft" means little more than a scaffold. The table will be updated as chapters mature. Feedback and contributions are welcome at any stage.

A printable PDF of the current draft is available from the download button in the sidebar, or directly as Entropy-Based-Learning.pdf.

To learn more about the underlying research, visit https://yataobian.com/ and the Blue Whale Lab.

Citation

If you found this book useful, please cite it as:

```
@book{bian2026entropy,
  title     = {Entropy-Based Learning},
  author    = {Yatao Bian},
  year      = {2026},
  publisher = {Blue Whale Lab},
  url       = {https://yataobian.github.io/entropy-based-learning/}
}
```

Or, for attribution in text: Bian, Yatao. 2026. *Entropy-Based Learning*. Blue Whale Lab. https://yataobian.github.io/entropy-based-learning/.

1 Introduction

1.1 Why entropy?

Modern machine learning has long been nurtured by scientific disciplines. A canonical example is the **Boltzmann machine**, rooted in statistical physics and motivated by the *Maximum Entropy Principle*. This book takes entropy — and its close relative, energy — as the organizing thread for a family of learning methods that are both principled and practically effective.

That lineage is unusually direct. In 1982 John Hopfield borrowed the energy function of a magnetic spin system to build an associative memory (Hopfield 1982); within a year, simulated annealing had turned the competition between energy and entropy into an optimization schedule (Kirkpatrick et al. 1983); and by 1985 the Boltzmann machine placed Hopfield's network at a finite temperature and learned its couplings from data (Ackley et al. 1985). The same trade-off resurfaced as the evidence lower bound (Dayan et al. 1995; Neal and Hinton 1998), as the contrastive objectives of energy-based learning (LeCun et al. 2006), as the temperature parameter of every language-model decoder, and as the KL penalty with which language models are fine-tuned from human feedback (Ziegler et al. 2019; Ouyang et al. 2022). In 2024 the Nobel Prize in Physics was awarded to Hopfield and Hinton for this line of work (The Royal Swedish Academy of Sciences 2024). Chapter 3 develops the single equation, $F = U - TS$, that runs beneath all of it.

We study three directions that are connected:

1. **Maximizing entropy** to obtain the least-biased model consistent with given constraints. This yields exponential-family / Gibbs distributions and, as we will see, a principled framework for valuation problems in machine learning, in which the energy is given and nothing is learned (Bian et al. 2022). When the energy is itself learned rather than given, the same principle becomes a learning method on discrete domains — neural set functions trained from their optimal subsets alone (Ou et al. 2022; Xie, Bian, et al. 2024) — and, on trajectories, the inverse reinforcement learning that infers a reward from demonstrations (Ziebart et al. 2008). With a pretrained model as the base measure, the resulting Gibbs distribution is the common target of energy-based guidance at inference time and of post-training, which is how RLHF, GRPO, and DPO enter this book.
2. **Minimizing entropy** to sharpen a model's own predictions. When applied to large language models in a latent semantic space, this idea enables fully *unsupervised* elicitation of reasoning capabilities, as in EMPO (Zhang et al. 2025). The idea has two limits, and both are structural rather than defects of an implementation: it elicits capability but cannot add it, since its training loop never touches ground truth (Section 6.7), and, left unchecked, it collapses the model onto one confident answer through a first-order transition in a mean-field free energy (Section 6.6).
3. **Minimaxing entropy** to decide which features a model should include at all, and what it should commit to. The optimal features are those whose maximum-entropy model has the *minimum* entropy, i.e., gives the shortest description of the data (Zhu et al. 1997; Carcamo et al. 2025). The same nesting covers prediction: a softmax classifier with a linear head is a conditional maximum-entropy model, and the cross-entropy it minimizes is the largest conditional entropy its features allow (Section 7.2.2). Without labels, a temperature on the label moves the problem from maximum-entropy learning with missing data, at $T = 1$, to the classification likelihood, as $T \to 0$ (Section 7.3).

What connects the three is the free energy. Read at the level of algorithms, they share one inner problem, namely the minimization of a free energy over distributions, which Chapter 5 calls the representation and

inference levels, and they differ only in the outer criterion: i) fit the data, ii) sharpen the model's own output, or iii) choose what to measure and what to commit to. In the terms of Chapter 3, they are three settings of one temperature dial.

1.2 Roadmap

- **Part I — Foundations** surveys the thermodynamic, information-theoretic, quantum, and psychological notions of entropy, states the axioms that single out these formulas and loosens them one at a time, carries the same method over to the diversity of a set, and develops the maximum-entropy principle together with the minimum-description-length view that leads to minimax entropy (Chapter 2). It then derives the free-energy relationship $F = U - TS$ that connects entropy to energy (Chapter 3), and introduces energy-based models together with three classical ways of training them without computing the partition function (Chapter 4).
- **Part II — Methods** develops maximum-entropy learning, read through the three levels of representation, inference, and learning (Chapter 5), minimum-entropy learning (Chapter 6), and minimax-entropy learning (Chapter 7). It then applies the maximum-entropy machinery to pretrained models in two ways: energy-based guidance and refinement, which sample from a Gibbs tilt of a pretrained model at inference time, with the model itself frozen (Chapter 8), and maximum-entropy post-training, which reads RLHF, GRPO, DPO, and their relatives as the same tilt and pays for it once in training rather than at every query (Chapter 9).
- **Part III — Applications** connects these ideas to scientific foundation models and AI-for-science (Chapter 11).

Part I — Foundations

2 Entropy and the Maximum Entropy Principle

2.1 The many faces of entropy

Entropy was invented three separate times — for heat engines in the 1860s, for quantum ensembles in the 1920s, and for telephone lines in the 1940s — and has been rediscovered many times since. The names, the units, and the motivating problems all differ, but each version measures the same thing: given what we have specified about a system, how much remains unspecified. This section walks through those notions before the rest of the book settles on one of them.

2.1.1 Information: Shannon entropy

For a discrete random variable X with probability mass function $p(x)$, the **Shannon entropy** is

$$H(p) = -\sum_{x} p(x) \log p(x),$$

which quantifies the average uncertainty (in nats or bits) of X (Shannon 1948). Entropy is maximized by the uniform distribution, where it equals $\log |\mathcal{X}|$, and minimized (equal to zero) by a deterministic one.

This functional form is not an arbitrary choice. Shannon showed that three requirements — continuity in the probabilities, monotonicity in the number of equally likely outcomes, and consistency when outcomes are grouped — determine H uniquely up to the base of the logarithm; the requirements, their reformulation as the Shannon–Khinchin axioms, and the uniqueness theorem are stated in Section 2.2. It also carries a concrete operational meaning: $H(p)$ is the expected code length of an optimal lossless encoding of samples from p — exactly, in the limit of coding long blocks, and to within one bit if each symbol must be coded on its own. Uncertainty and description length are one quantity seen from two sides, a duality that returns in Chapter 7.

2.1.2 Heat: thermodynamic entropy

The original entropy is older and, on its face, unrelated to information. Studying the irreversibility of heat flow, Clausius defined a state function whose differential along a reversible path is

$$dS = \frac{\delta Q_{\text{rev}}}{T},$$

and named it from the Greek *τροπή*, transformation, deliberately shaping the word to rhyme with *energy* because the two quantities seemed to him so closely related (Clausius 1865). That 1865 paper closes with the two sentences that became the standard statement of the laws of thermodynamics: the energy of the universe is constant, and its entropy tends towards a maximum.

Boltzmann supplied the microscopic reading. A macrostate — a temperature, a pressure — is compatible with an enormous number W of microstates, and entropy simply counts them (Boltzmann 1877):

$$S = k_B \log W.$$

Gibbs then generalized the count to microstates that are not equally likely (Gibbs 1902):

$$S = -k_B \sum_i p_i \log p_i.$$

The Gibbs expression and the Shannon expression are the same formula, separated only by the constant k_B that carries the physical units. The coincidence is not accidental, and Jaynes turned it into the argument developed in Section 2.3: thermodynamic entropy is the Shannon entropy of our ignorance about the microstate, which makes equilibrium statistical mechanics a problem of inference rather than of mechanics (Jaynes 1957).

2.1.3 Quantum states: von Neumann entropy

Quantum mechanics replaces a probability distribution with a density matrix ρ, a positive semidefinite operator of unit trace. Von Neumann extended entropy to this setting in 1927, two decades before Shannon (Neumann 1927):

$$S(\rho) = -\operatorname{Tr}(\rho \log \rho).$$

Written in the eigenbasis of ρ, this reduces to the Shannon entropy of its eigenvalues, so the same functional form reappears with a different argument. It vanishes exactly for pure states, which are completely specified, and is maximal for the maximally mixed state.

The quantum case does break the analogy in one instructive place. Classically, conditional entropy is never negative: learning Y cannot leave us knowing more than everything about X. Quantum mechanically it can be — the joint state may be more sharply specified than either of its parts. A negative value therefore certifies entanglement, though the implication runs one way only: entangled states with non-negative conditional entropy exist.

2.1.4 One knob, a family: Rényi and Tsallis

Shannon's requirements can be relaxed, and each relaxation produces a one-parameter family. Rényi kept additivity over independent systems but dropped the requirement that entropy be a linear average of $-\log p$ (Rényi 1961):

$$H_\alpha(p) = \frac{1}{1-\alpha} \log \sum_x p(x)^\alpha, \qquad \alpha > 0,\ \alpha \neq 1.$$

The order α is a knob controlling how much weight rare outcomes receive. The limit $\alpha \to 1$ recovers Shannon entropy; $\alpha = 0$ gives the Hartley entropy $\log |\operatorname{supp} p|$, which counts possibilities and ignores their probabilities; $\alpha = 2$ gives the **collision entropy**

$$H_2(p) = -\log \sum_x p(x)^2,$$

whose argument $\sum_x p(x)^2$ is the probability that two independent draws coincide — a quantity that returns as a training objective in Section 6.5; and $\alpha \to \infty$ gives the **min-entropy**

$$H_\infty(p) = -\log \max_x p(x),$$

the worst-case measure used in cryptography, where what matters is not average uncertainty but the chance that an adversary guesses right on the first try.

A warning about that last name, since both readings appear later in this book. The min-entropy H_∞ is a *specific functional*, the most pessimistic member of the Rényi family. It should not be confused with the **minimum-entropy principle** of Chapter 6, which is not a functional at all but a class of learning *objectives* that reduce the entropy — usually the Shannon or the collision entropy — of a model's predictive distribution. The two are unrelated, and the shared prefix is an accident of terminology in two literatures that developed apart.

Tsallis relaxed the other requirement, replacing the chain rule by a q-deformed version (Section 2.2), so that additivity itself is lost (Tsallis 1988):

$$S_q(p) = \frac{1}{q-1}\Big(1 - \sum_x p(x)^q\Big).$$

Here $q \to 1$ again recovers the Boltzmann–Gibbs form, but for $q \neq 1$ two independent systems combine as $S_q(A,B) = S_q(A) + S_q(B) + (1-q)\,S_q(A)\,S_q(B)$. This *non-extensivity* was introduced to describe systems with long-range interactions and multifractal structure, where the usual assumption that entropy scales with system size fails.

Both families are useful in their own domains, and both make the same point by contrast: Shannon entropy is the unique member satisfying the full set of Shannon–Khinchin axioms, including the chain rule that makes conditional entropy and mutual information well behaved, as Section 2.2 spells out. That is why it is the form used throughout the remainder of this book.

2.1.5 Minds: psychological entropy

The most surprising reappearance is in psychology. Hirsh, Mar, and Peterson propose the **entropy model of uncertainty**, which applies Shannon's formula directly to the human information system (Hirsh et al. 2012). At any moment an organism faces a set of competing *affordances*: possible interpretations of the sensory input, and possible actions to take. The model treats these as a subjectively weighted probability distribution and defines **psychological entropy** as its Shannon entropy.

The reading is unusually literal. A familiar situation places nearly all the weight on one dominant affordance, giving low entropy and a fast habitual response. An unfamiliar or ambiguous one spreads the weight across many competing frames, giving high entropy and sustained neural competition — which, the authors argue, is what anxiety *is* subjectively, with anterior cingulate activity and noradrenaline release as its physiological correlates.

Two features of the account matter for what follows. First, goals act as constraints: adopting a goal reweights the distribution towards the actions that serve it, and thereby lowers entropy. This is structurally the same move as imposing moment constraints in Section 2.3, since a constraint is whatever narrows the space of live possibilities. Second, the model makes entropy reduction a *drive* rather than a calculation. That idea runs from Schrödinger's argument that organisms persist by exporting entropy to their environment (Schrödinger 1944) through to the free-energy principle (Friston 2010), and it is the biological precedent for the minimum-entropy learning objectives of Chapter 6.

2.1.6 One form, many readings

Notion	Probabilities over	Definition	Origin
Thermodynamic	no explicit distribution	$dS = \delta Q_{\text{rev}}/T$	Clausius, 1865
Statistical	microstates, equally likely	$k_B \log W$	Boltzmann, 1877
Gibbs	microstates, unequally likely	$-k_B \sum_i p_i \log p_i$	Gibbs, 1902

Notion	Probabilities over	Definition	Origin
Information	a random variable or message source	$-\sum_x p(x)\log p(x)$	Shannon, 1948
Quantum	eigenvalues of a density matrix	$-\operatorname{Tr}(\rho\log\rho)$	von Neumann, 1927
Rényi	any distribution, at order α	$\frac{1}{1-\alpha}\log\sum_x p(x)^\alpha$	Rényi, 1961
Tsallis	non-extensive systems, at order q	$\frac{1}{q-1}(1-\sum_x p(x)^q)$	Tsallis, 1988
Psychological	competing perceptions and actions	$-\sum_x p(x)\log p(x)$	Hirsh et al., 2012

Two observations follow. The first is mathematical: almost every row is the same functional $-\sum p\log p$, differing only in what the probabilities range over and what constant multiplies the result. Entropy is less a family of analogies than a single quantity applied to different objects.

The second is interpretive, and it is the one that does work in this book. The same number reads as dispersal of energy to a physicist, as expected code length to an engineer, and as felt uncertainty to a psychologist. Those readings are what make the two principles that follow feel necessary rather than arbitrary: maximizing entropy is being honest about what we have not specified, and minimizing it is committing to what we have.

2.2 Why these formulas and no others: the axioms

In two places so far, this chapter has referenced the uniqueness of Shannon entropy without stating the basis for that claim: first, in attributing the form of H to Shannon's trio of requirements (which specify H up to the logarithm's base), and again at the end of Section 2.1.4, which asserts that Shannon entropy alone satisfies the full set of Shannon–Khinchin axioms. In this section we state the axioms, give the uniqueness theorem in its sharpest known form, and then loosen one axiom at a time; H_α and S_q are as defined in Section 2.1.4.

The same method is then applied to a second quantity, which this book needs from Section 5.4 onward: the diversity of a set of objects, with or without a notion of similarity between them. Diversity has its own axiomatic literature, in ecology, economics and recently machine learning, and it is related to entropy by an exponential; we give the correspondence axiom by axiom and record what the axiomatic method cannot decide.

2.2.1 The load-bearing axiom

Let us start with a numerical example. Consider a distribution on five outcomes,

$$p = \left(\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{6}, \tfrac{1}{6}, \tfrac{1}{6}\right),$$

and think of drawing from it in two stages: first a coin toss $w = (\frac{1}{2}, \frac{1}{2})$ decides between a left block and a right block, then within the left block one draws from $p_1 = (\frac{1}{2}, \frac{1}{2})$ and within the right block from $p_2 = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. Written as a composite, $p = w \circ (p_1, p_2)$. Computed directly, the Shannon entropy of p is $\frac{1}{2}\log 4 + \frac{1}{2}\log 6 = \frac{3}{2}\log 2 + \frac{1}{2}\log 3$, about 2.29 bits. The two-stage computation gives $H(w) + \frac{1}{2}H(p_1) + \frac{1}{2}H(p_2) = \log 2 + \frac{1}{2}\log 2 + \frac{1}{2}\log 3$, which is the same number. The equality is an instance of the **chain rule**,

$$H\big(w \circ (p_1, \dots, p_n)\big) = H(w) + \sum_{i=1}^{n} w_i\, H(p_i),$$

valid for every $w \in \Delta_n$ (the simplex of distributions on n outcomes) and every choice of $p_i \in \Delta_{k_i}$. With X the block and Y the outcome within the block, it reads $H(X,Y) = H(X) + H(Y \mid X)$; for independent X and Y (all p_i equal to one p, a composite written $w \otimes p$) it reduces to additivity, $H(X,Y) = H(X) + H(Y)$, which is a strictly weaker statement.

Shannon's requirements. Shannon's own theorem (Shannon 1948, Theorem 2) assumes i) continuity in the p_i; ii) that $A(n) = H(1/n, \dots, 1/n)$ increases in n; iii) consistency under grouping, which is the two-stage computation above, and concludes that $H = -c \sum_i p_i \log p_i$ for a constant $c > 0$.

Khinchin's axioms. Most later work uses Khinchin's reformulation (Khinchin 1957) (the 1957 translation of his 1953 Uspekhi paper). Khinchin himself isolates three properties, namely maximality at the uniform distribution, the chain rule $H(AB) = H(A) + H_A(B)$ and expansibility, and proves that any H continuous in its arguments and satisfying them is $-\lambda \sum_i p_i \log p_i$ with $\lambda > 0$; later authors package continuity together with these three as the four **Shannon–Khinchin axioms**, with labels SK1–SK4. For a sequence of functions $H : \Delta_n \to \mathbb{R}$, $n \geq 1$, they are:

- SK1 (continuity): H is continuous in p;
- SK2 (maximality): $H(p) \leq H(u_n)$ for all $p \in \Delta_n$, where u_n is the uniform distribution;
- SK3 (expansibility): $H(p_1, \dots, p_n, 0) = H(p_1, \dots, p_n)$, i.e., adding an outcome of probability zero changes nothing;
- SK4 (strong additivity): $H(X,Y) = H(X) + \sum_x p(x)\, H(Y \mid X = x)$, i.e., the chain rule.

Any sequence satisfying SK1–SK4 is $-c \sum_i p_i \log p_i$ with $c \geq 0$, where $c = 0$ is the trivial $H \equiv 0$ that Khinchin's $\lambda > 0$ tacitly excludes (Khinchin 1957). Note that SK4 is strictly stronger than additivity for independent systems, which is the special case in which every conditional distribution $p(y \mid x)$ equals the marginal $p(y)$. SK4 is the load-bearing axiom in the following sense: every generalized entropy discussed below keeps SK1–SK3 and modifies SK4. This is what the phrase "unique member satisfying the full set of Shannon–Khinchin axioms" in Section 2.1.4 means.

Faddeev's theorem. Once SK4 is assumed, only continuity is needed from SK1–SK3, as Leinster's variant of Faddeev's theorem shows (Faddeev 1956; Leinster 2021, Theorem 2.5.1). Let $(I : \Delta_n \to \mathbb{R})_{n \geq 1}$ be a sequence of functions. Then the following are equivalent: i) the functions I are continuous and satisfy the chain rule $I(w \circ (p_1, \dots, p_n)) = I(w) + \sum_i w_i I(p_i)$ for all $n, k_1, \dots, k_n \geq 1$, $w \in \Delta_n$ and $p_i \in \Delta_{k_i}$; ii) $I = cH$ for some $c \in \mathbb{R}$.

Proof sketch. Firstly, the chain rule applied to $u_{mn} = u_m \circ (u_n, \dots, u_n)$ gives $I(u_{mn}) = I(u_m) + I(u_n)$; together with $I(u_{n+1}) - I(u_n) \to 0$, which follows from continuity, this gives $I(u_n) = c \log n$. Secondly, a distribution with non-zero rational entries $p = (k_1/N, \dots, k_n/N)$ satisfies $p \circ (u_{k_1}, \dots, u_{k_n}) = u_N$, so the chain rule gives $c \log N = I(p) + \sum_i p_i\, c \log k_i$, i.e., $I(p) = cH(p)$. Lastly, continuity extends the identity to all distributions (Leinster 2021, sec. 2.5).

We add some remarks on the hypotheses (Leinster 2021, Remarks 2.5.2; Faddeev 1956). Firstly, continuity is the only regularity condition; SK3 is not assumed, it follows from the conclusion $I = cH$ for every c, and SK2 follows as soon as the sign of c is fixed, e.g., by a normalization such as $I(u_2) = \log 2$. That Khinchin's maximality and expansibility axioms can be dispensed with is Faddeev's own observation: his system replaces the former by positivity of I at a single point and the latter, together with strong additivity, by a single recursion. Secondly, Faddeev's original 1956 theorem assumed in addition that I is symmetric, and assumed only the two-block special case $I(tw_1, (1-t)w_1, w_2, \dots, w_n) = I(w) + w_1 I(t, 1-t)$ (Faddeev split the last block, which is equivalent under symmetry), while requiring continuity only of the binary function $t \mapsto I(t, 1-t)$; under symmetry the two forms of the chain rule are equivalent by induction, so the general form allows symmetry to be removed (if symmetry is kept, continuity can instead be weakened to measurability, which is a 1964 theorem of Lee). Lastly, some regularity condition is necessary: for an additive but nonlinear $f : \mathbb{R} \to \mathbb{R}$ (which exists under the axiom of choice), $p \mapsto -\sum_{i \in \mathrm{supp}(p)} p_i f(\log p_i)$ satisfies the chain rule without being a multiple of H.

2.2.2 Loosening one axiom at a time

The five-outcome example makes the failure of SK4 concrete. For the collision entropy H_2 of Section 2.1.4, one computes directly $H_2(p) = -\log\left(2\cdot\frac{1}{16}+3\cdot\frac{1}{36}\right) = \log\frac{24}{5} \approx 1.569$ nats, whereas the chain-rule prediction is $H_2(w)+\frac{1}{2}H_2(p_1)+\frac{1}{2}H_2(p_2) = \frac{3}{2}\log 2+\frac{1}{2}\log 3 \approx 1.589$ nats. The two disagree, so H_2 violates SK4, although one can verify that it is still additive for independent X, Y. Now take the Tsallis entropy $S_2(p) = 1-\sum_i p_i^2 = \frac{19}{24}$. The chain-rule prediction fails again, but the modified rule

$$S_q\big(w\circ(p_1,\dots,p_n)\big) = S_q(w) + \sum_{i=1}^{n} w_i^q\, S_q(p_i)$$

gives $S_2(w)+\frac{1}{4}S_2(p_1)+\frac{1}{4}S_2(p_2) = \frac{1}{2}+\frac{1}{8}+\frac{1}{6} = \frac{19}{24}$, exactly. We now describe the two modifications one at a time.

Rényi: a different mean. Rényi kept additivity on independent systems (his Postulate 4) and replaced the arithmetic mean of $-\log p_i$ under p by a Kolmogorov–Nagumo quasi-arithmetic mean $g^{-1}\big(\sum_i p_i\, g(-\log p_i)\big)$ with g continuous and strictly monotone (Rényi 1961). Two choices of g are compatible with additivity: the linear functions, which return Shannon entropy, and the exponential functions $g(x) \propto 2^{(1-\alpha)x}$, which return H_α (Rényi's Theorem 2; his printed formula has the exponent $(\alpha-1)x$, but with his convention $H(\{p\}) = \log_2(1/p)$ the generator for H_α is $2^{(1-\alpha)x}$). Rényi himself called it an open question whether any other g is admissible for entropy, and proved the linear-or-exponential dichotomy only for the information gain $I(Q\,\|\,P)$ (his Theorem 3); for the entropy of incomplete distributions the dichotomy was proved by Aczél and Daróczy (Aczél and Daróczy 1963). Therefore the Rényi entropies satisfy SK1–SK3 and additivity on independent systems, and fail SK4 for $\alpha \neq 1$. Jizba and Arimitsu derive the family from SK-type axioms with a quasi-linear conditional entropy in place of SK4 (Jizba and Arimitsu 2004). Lazarev revisits Rényi's question without assuming additivity (Lazarev 2026) (a preprint at the time of writing): under the admissibility condition that a measure never has larger entropy than a reference measure dominating it, a continuous strictly monotone g is admissible exactly when $t \mapsto g(1/t)$ is strictly convex for increasing g, or strictly concave for decreasing g, and additivity on products then singles out the Rényi family.

Tsallis: a different chain rule. The functional S_q was introduced by Havrda and Charvát and by Daróczy in information theory, and independently by Tsallis in statistical physics (Havrda and Charvát 1967; Daróczy 1970; Tsallis 1988). Its defining property is the q-chain rule displayed above, in which the weights w_i of SK4 are replaced by w_i^q; for independent systems, where all p_i are equal, it reduces to the pseudo-additivity $S_q(A,B) = S_q(A) + S_q(B) + (1-q)\,S_q(A)\,S_q(B)$ stated in Section 2.1.4. The wording used there can accordingly be made more precise: rather than "giving up additivity itself", Tsallis replaced the chain rule by its q-deformation, and non-additivity on independent systems is the consequence. Uniqueness theorems with generalized SK axioms were given by Santos and by Abe (Santos 1997; Abe 2000). The shortest statement is Leinster's (Leinster 2021, Theorem 4.1.5): for real $q \neq 1$ and a sequence of functions $(I : \Delta_n \to \mathbb{R})_{n\geq 1}$, the following are equivalent: i) I is symmetric and satisfies

$$I(w\otimes p) = I(w) + \Big(\sum_{i\in\mathrm{supp}(w)} w_i^q\Big) I(p)$$

for all $n, k \geq 1$, $w \in \Delta_n$ and $p \in \Delta_k$; ii) $I = cS_q$ for some $c \in \mathbb{R}$. Notice that no regularity condition appears, in contrast to the Shannon case.

Hanel–Thurner: no SK4. One can also keep SK1–SK3 only and ask what freedom remains in a trace-form entropy $S = \sum_i g(p_i)$. Hanel and Thurner showed that the admissible g are classified by two asymptotic scaling exponents (c, d), with a representative

$$S_{c,d}(p) \propto \sum_i \Gamma\big(1+d,\, 1-c\log p_i\big)$$

for each class, up to constants, where $\Gamma(\cdot,\cdot)$ is the incomplete gamma function (Hanel and Thurner 2011). Shannon entropy is the point $(1,1)$, the Tsallis family is the line $(c,0)$, and the stretched-exponential entropies are the line $(1,d)$; the associated maximum-entropy distributions are Lambert-W exponentials. Tempesta's composability axiom requires, for statistically independent subsystems A and B, that $S(A\cup B)=\Phi(S(A),S(B))$ with Φ symmetric, associative and satisfying $\Phi(x,0)=x$, i.e., a commutative formal group law over the reals; the axiom replaces SK4 and generalizes the group entropies that Tempesta had introduced in 2011 (Tempesta 2016). Korbel's review surveys these frameworks (Shannon–Khinchin, Tempesta's group composability, Hanel–Thurner scaling, Shore–Johnson, Lieb–Yngvason) (Korbel 2026) (also a preprint). Once SK4 is dropped, choosing a trace-form entropy amounts, up to its asymptotic class, to choosing a point (c,d), i.e., a modelling decision, to which Chapter 7 returns.

The following table records which axiom each family keeps (Khinchin 1957; Faddeev 1956; Rényi 1961; Aczél and Daróczy 1963; Daróczy 1970; Abe 2000; Leinster 2021; Hanel and Thurner 2011).

axiom	Shannon H	Rényi H_α	Tsallis S_q	Hanel–Thurner $S_{c,d}$
SK1 continuity	yes	yes	yes	yes
SK2 maximal at uniform	yes	yes	yes	yes
SK3 expansibility	yes	yes	yes	yes
SK4 chain rule	yes	no	no (q-chain rule)	no
additive on independent systems	yes	yes	no (pseudo-additive)	no in general
arithmetic mean of $-\log p$	yes	no (quasi-arithmetic)	—	—
uniqueness theorem	Khinchin; Faddeev	Rényi, 1961 (given a Kolmogorov–Nagumo mean); Aczél–Daróczy, 1963	Daróczy, 1970; Abe, 2000; Leinster, Theorem 4.1.5	Hanel–Thurner, 2011 (classification into asymptotic (c,d)-classes; $S_{c,d}$ is a representative)

The rows SK1–SK3 hold for Rényi and Tsallis when $\alpha, q>0$; for negative orders the uniform distribution is not the maximizer (Leinster 2021, Remarks 4.4.4(ii)). Hanel and Thurner assume the trace form $S=\sum_i g(p_i)$ with g continuous and concave, $g(0)=0$, and $c\in(0,1]$.

2.2.3 The other faces, axiomatized

For three readings in the table closing the preceding section, the axiomatic method gives an independent derivation.

Entropy as information loss. Return to the five-outcome distribution and merge the two outcomes of probability $\frac{1}{4}$ into one. The result is $q=(\frac{1}{2},\frac{1}{6},\frac{1}{6},\frac{1}{6})$ with $H(q)=\log 2+\frac{1}{2}\log 3$, so the merge loses $H(p)-H(q)=\frac{1}{2}\log 2$, exactly the term $w_1H(p_1)$ that the chain rule attributes to the left block. Baez, Fritz and Leinster take this loss as the object of the axioms (Baez et al. 2011). On the category FinProb, whose objects are finite probability spaces (X,p) and whose morphisms $f:(X,p)\to(Y,q)$ are measure-preserving maps, an information loss assigns a number $F(f)\in[0,\infty)$ to each morphism and is i) functorial, $F(g\circ f)=F(g)+F(f)$; ii) convex-linear, $F(\lambda f\oplus(1-\lambda)g)=\lambda F(f)+(1-\lambda)F(g)$; iii) continuous. Their theorem states that every such F equals $c\,[H(p)-H(q)]$ for a constant $c\geq 0$; replacing the weights $\lambda,1-\lambda$ in convex-linearity by $\lambda^\beta,(1-\lambda)^\beta$, $\beta>0$, gives $c\,[S_\beta(p)-S_\beta(q)]$ with the Tsallis entropy S_β instead (Baez

et al. 2011, Theorem 7). Baez and Fritz characterize relative entropy in the same style on a category FinStat (Baez and Fritz 2014).

Quantum states. Ochs gave a list of postulates for a quantity measuring the intrinsic dispersion of a quantum state, including invariance under unitaries and partial isometries, additivity on product states, subadditivity, continuity, and some technical conditions, and proved that these single out $S(\rho) = -k \operatorname{Tr} \rho \log \rho$ up to the scale k (Ochs 1975). Section 2.2.5 applies this functional to a normalized similarity kernel.

Heat. Earlier in this chapter, the argument connecting the thermodynamic row to the rest was Jaynes's interpretive one (Jaynes 1957); Lieb and Yngvason give an axiomatic one, which assumes nothing about heat, temperature or microscopic constituents (Lieb and Yngvason 1999). The primitive is the relation of adiabatic accessibility $X \prec Y$ between equilibrium states, meaning that Y can be reached from X by a process whose only net effect on the surroundings is that a weight has been raised or lowered. The axioms are properties of this relation: reflexivity, transitivity, consistency under composition of systems, scaling invariance, splitting and recombination, and stability, together with a comparison hypothesis (any two states of the same system are comparable under $\prec$). From these they prove that there is a function S on states, unique up to affine transformations, with $X \prec Y$ if and only if $S(X) \le S(Y)$, additive under composition and extensive under scaling. Weilenmann, Krämer, Faist and Renner apply the framework, including its 2013 extension to non-equilibrium states, to finite-dimensional quantum systems ordered by majorization; the resulting entropy coincides with the von Neumann entropy on the equilibrium states, in this setting those with flat spectrum, and with the min- and max-entropies of one-shot information theory, $H_{\min}(\rho) = -\log \|\rho\|_\infty$ and $H_{\max}(\rho) = \log \operatorname{rank} \rho$, away from them (Weilenmann et al. 2016). The result gives an axiomatic identification of the Clausius and von Neumann rows of the table, where the chapter so far had only Jaynes's interpretive argument; the thermodynamic side returns in Chapter 3.

The inference rule. Shore and Johnson axiomatize the rule that takes a prior and a set of constraints on expected values to a posterior, rather than the entropy functional. They require of such a rule, realized by minimizing a functional $H(q, p)$, uniqueness, invariance under coordinate changes, system independence and subset independence, and prove that every rule satisfying these is equivalent to minimization of relative entropy (their "cross-entropy", $\int q \log(q/p)$), which reduces to the maximum-entropy principle of Section 2.3 for a uniform prior (Shore and Johnson 1980). Chapter 5 uses this result, which axiomatizes a procedure rather than H. Csiszár's survey (Csiszár 2008) treats the characterizations of functionals, of entropic vectors and of inference rules in one place.

2.2.4 From entropy to diversity

The following example is due to Jost (Jost 2009; Jost et al. 2010), in the form given by Leinster (Leinster 2021, Example 2.4.11); Jost's original has 20 islands with Barro Colorado abundance data, and the oil company, the lawyers and the figures below are Leinster's re-casting. An oil company plans work on 16 equally sized islands which will destroy all wildlife on 8 of them. The islands share no species and each has diversity 4, in the sense of the effective number defined below, so the group has diversity $16 \times 4 = 64$ before the work and 32 after, a loss of 50%. With Shannon entropy as the index, as ecologists have long used it, the company's lawyers compute $\log 64$ before and $\log 32$ after, hence $\log 32 / \log 64 = 5/6 \approx 83\%$ of the diversity preserved; the environmentalists' lawyers compute the entropy of the islands to be destroyed, $\log 32$ out of $\log 64$, and claim that 83% is lost. The two conclusions contradict each other, and both are wrong, since half of the diversity is lost; the reason is that H fails the replication principle stated below.

Effective numbers. The remedy, argued by Jost under the slogan that entropies are not diversities (Jost 2006), is to exponentiate. For an abundance distribution p on n species and the Rényi entropy H_q of Section 2.1.4, the **Hill number** of order q is

$$D_q(p) = \Big(\sum_{i=1}^{n} p_i^q\Big)^{1/(1-q)} = \exp H_q(p),$$

introduced into ecology by Hill and, for $q = 1$, by MacArthur (Hill 1973; MacArthur 1965). $D_q(p)$ is the **effective number** of species: a community with $D_q = m$ counts as diverse as one of m equally abundant, completely distinct species. Specifically, D_0 is species richness (the Hartley count of possibilities of Section 2.1.4), $D_1 = e^H$, $D_2 = 1/\sum_i p_i^2$ is the inverse of Simpson's concentration (Simpson 1949), and $D_\infty = 1/\max_i p_i$. Patil and Taillie read a diversity index as an average rarity $\sum_i p_i R(p_i)$, with R a rarity function (Patil and Taillie 1982); the choice $R(p) = (1 - p^{q-1})/(q-1)$ gives the q-logarithmic entropies S_q, whose numbers equivalents are the Hill numbers. D_q itself is an average rarity, namely the power mean $M_{1-q}(p, 1/p)$ of the rarities $1/p_i$ weighted by p (Leinster 2021, Definition 4.3.4), which is Hill's own reading of D_q as the reciprocal of a mean proportional abundance (Hill 1973). The property on which the oil-company example depends is the **replication principle**: if n communities of equal size, sharing no species and each of diversity D, are pooled, the pooled community has diversity nD. Formally, $D(u_n \otimes p) = n\,D(p)$, whereas $H(u_n \otimes p) = \log n + H(p)$.

Leinster's characterization. The Hill numbers are characterized by this property plus a few elementary ones (Leinster 2021, Theorem 7.4.3). One condition needs a definition (Leinster 2021, Definition 7.4.1): a sequence $D : \Delta_n \to (0, \infty)$ is *modular-monotone* if $D(p_i) \le D(\tilde{p}_i)$ for all i implies $D(w \circ (p_1, \dots, p_n)) \le D(w \circ (\tilde{p}_1, \dots, \tilde{p}_n))$, i.e., if no subcommunity's diversity decreases then neither does that of the whole. It implies modularity, i.e., dependence on the p_i only through the $D(p_i)$. The theorem states that for a sequence of functions $(D : \Delta_n \to (0, \infty))_{n \ge 1}$ the following are equivalent: i) D is symmetric, absence-invariant (deleting a species of abundance zero does not change D), continuous in the positive probabilities, normalized ($D(u_1) = 1$), modular-monotone, and satisfies the replication principle; ii) $D = D_q$ for some $q \in [-\infty, \infty]$. A refinement in the same section excludes negative q under an additional condition.

Compared with the SK axioms, the correspondence holds exactly for SK3 and for additivity, and up to a deliberate weakening for SK4. Expansibility (SK3) corresponds to absence-invariance. Additivity on independent systems corresponds to multiplicativity $D(w \otimes p) = D(w)\,D(p)$, of which the replication principle is the special case $w = u_n$; the theorem assumes only this special case and derives multiplicativity from it (Leinster 2021, Lemma 7.4.7). The chain rule (SK4), whose exact diversity form is $D(w \circ (p_1, \dots, p_n)) = D(w) \prod_i D(p_i)^{w_i}$ (Leinster 2021, Corollary 2.5.8), corresponds to the strictly weaker modular-monotonicity, i.e., the diversity of the whole depends monotonically only on w and the diversities of the parts; this weakening is the reason the theorem produces the whole family D_q instead of D_1 alone. Finally, the effective-number property $D(u_n) = n$, which is the exponential-scale counterpart of the normalization $H(u_n) = \log n$ rather than of the maximality inequality SK2, is not assumed at all, because it follows from normalization and replication: $D(u_n) = D(u_n \otimes u_1) = n\,D(u_1) = n$ (Leinster 2021, Lemma 7.4.4). Maximality at the uniform distribution itself is not a consequence of the six hypotheses: the theorem admits Hill numbers of negative order, which are not maximized at u_n (Leinster 2021, Remarks 4.4.4(ii)); it holds for the refinement $q \ge 0$. Since $H_q = \log D_q$, the theorem is also a new characterization of the Rényi entropies, without Rényi's quasi-arithmetic mean among the hypotheses (Leinster 2021, Remark 7.4.15).

2.2.5 Similarity

Hill numbers treat every pair of distinct species as equally distinct. A similarity matrix removes this restriction, and the two constructions below use it differently. Let two objects have similarity $s \in [0, 1]$ and equal weight, so that $Z = K = \begin{pmatrix} 1 & s \\ s & 1 \end{pmatrix}$. The eigenvalues of $K/2$ are $(1 \pm s)/2$, and the exponential of their Shannon entropy is $\exp H\big(\frac{1+s}{2}, \frac{1-s}{2}\big)$, about 1.75 for $s = 0.5$. Alternatively, the expected similarity of a random individual to either object is $(Zp)_i = (1 + s)/2$, and $1/(Zp)_i = 2/(1 + s)$, which is 1.33 for $s = 0.5$. The two functionals coincide when K is block-diagonal with all-ones blocks, in which case the non-zero eigenvalues of K/n are

the block masses (as noted in Section 6.5), and they differ in general.

Similarity-sensitive diversity. Rao's quadratic entropy $\sum_{i,j} p_i p_j d_{ij}$, the expected dissimilarity between two randomly drawn individuals, is the best-known early index that admits inter-species distances (the same form appears as Nei and Li's 1979 nucleotide diversity) (Rao 1982); Stirling decomposes diversity into variety, balance and disparity and proposes the heuristic $\sum_{i\neq j} d_{ij}^{\alpha}(p_i p_j)^{\beta}$, whose $\alpha = \beta = 1$ case is Rao's index (Stirling 2007). The family that contains the Hill numbers is Leinster and Cobbold's **similarity-sensitive diversity** of order q (Leinster and Cobbold 2012): for a similarity matrix Z with $Z_{ii} = 1$ and $0 \le Z_{ij} \le 1$, and $p \in \Delta_n$,

$$D_q^Z(p) = \Big(\sum_{i\,:\,p_i>0} p_i \big((Zp)_i\big)^{q-1} \Big)^{1/(1-q)}, \qquad q \neq 1,$$

where $(Zp)_i = \sum_j Z_{ij} p_j$ is the expected similarity between species i and an individual chosen at random, and $q = 1, \infty$ are taken as limits. $Z = I$ recovers D_q. The properties they establish are of three kinds: i) elementary ones (symmetry, absent species, identical species, effective number); ii) the effect of similarity (monotonicity in Z, the naive model $Z = I$ as an upper bound, and the range $1 \le D_q^Z(p) \le n$); iii) partitioning (modularity and replication for mutually dissimilar subcommunities).

Maximum diversity. For a symmetric Z, i) there exists a distribution that maximizes D_q^Z for all $q \in [0, \infty]$ simultaneously, and ii) the maximum value $\sup_p D_q^Z(p)$ is independent of q (Leinster and Meckes 2016; Leinster 2021, Theorem 6.3.2). The maximum equals the magnitude $|Z_B|$ of some principal submatrix Z_B of Z (for invertible Z_B, the sum of all entries of Z_B^{-1}) (Leinster 2021, Theorem 6.3.13), and in important classes of cases the magnitude of Z itself. The same quantity had appeared in 1994, when Solow and Polasky proposed $V(S) = \mathbf{1}^{\mathsf{T}} F^{-1} \mathbf{1}$ with $F_{ij} = f(d_{ij})$ for a positive-definite, decreasing f with $f(0) = 1$ and $f(\infty) = 0$, their leading example being $f(d) = \exp(-\theta d)$, and called it the effective number of species (Solow and Polasky 1994); for the exponential choice it is exactly the magnitude of the scaled metric space, a coincidence that Leinster records as a historical surprise (Leinster 2021). Leinster's earlier note on the result is titled "A maximum entropy theorem with applications to the measurement of biodiversity" (Leinster 2009). The title is meant in a broader sense than the principle of Section 2.3: there are no moment constraints, the maximized quantity is the similarity-sensitive entropy $\log D_q^Z$, and the maximizer does not depend on q.

Diversity of a set: the economics tradition. Economists measure the diversity of a plain set, without abundances. Weitzman defines a diversity function recursively from pairwise distances, $V(S) = \max_{i\in S} \big[V(S \setminus \{i\}) + d(i, S \setminus \{i\})\big]$ with $d(i, Q) = \min_{j\in Q} d_{ij}$, and shows that it has monotonicity in species, the twin property, and continuity and monotonicity in distances (Weitzman 1992). Solow and Polasky set out three natural requirements: monotonicity in species (adding a new distinct species does not decrease diversity), twinning (adding an exact copy of a present species does not increase it; the term is Weitzman's), and monotonicity in distance; their $V(S)$ satisfies the first two, but they note that it is not monotone in distance in general, and conjecture that it is for exponential f when the distances satisfy the triangle inequality (Solow and Polasky 1994; as read by Emmerich et al. 2026b). Nehring and Puppe model diversity as the total weight of the attributes realized by some member of a set, $v(S) = \sum_{A\,:\,A\cap S\neq\emptyset} \lambda_A$ with $\lambda_A \ge 0$, and show that these are exactly the set functions with non-negative *conjugate* Möbius inverse, equivalently the monotone, totally submodular set functions, which are the duals of the totally monotone belief functions (Nehring and Puppe 2002). In particular such functions are submodular, which places set diversity among the set functions learned in Section 5.4.

Kernels in machine learning. The Vendi Score of Friedman and Dieng is the eigenvalue-based construction of the opening example (Friedman and Dieng 2023). Given a positive semidefinite kernel k with $k(x, x) = 1$ and samples $x_1, \dots, x_n$, let $K_{ij} = k(x_i, x_j)$ and $\rho = K/n$, a positive semidefinite matrix of unit trace, i.e., a density matrix. The **Vendi Score** is

$$\mathrm{VS}_k(x_1, \dots, x_n) = \exp\Big(-\sum_i \lambda_i \log \lambda_i \Big) = \exp\big(-\operatorname{Tr} \rho \log \rho \big),$$

with λ_i the eigenvalues of ρ, i.e., the exponential of the von Neumann entropy of ρ, hence the name. Their Theorem 3.1 lists four properties: 1) effective number: if $k(x_i, x_j) = 0$ for all $i \neq j$ then VS is maximal and equals n, and if $k(x_i, x_j) = 1$ for all i, j it is minimal and equals 1; 2) identical elements: if $k(x_i, x_j) = 1$ for some $i \neq j$, then merging the weights, $p_i' = p_i + p_j$ and $p_j' = 0$ in the weighted form of the score, leaves VS unchanged; 3) partitioning: if $S_1, \dots, S_m$ are mutually dissimilar collections ($k = 0$ across groups) of relative sizes p_i, then $\mathrm{VS}(S_1, \dots, S_m) = \exp\big(H(p_1, \dots, p_m)\big) \prod_i \mathrm{VS}(S_i)^{p_i}$; 4) symmetry under permutation. No reference dataset or labels are required. Property 3) is the chain rule in exponential form. Pasarkar and Dieng replace the Shannon entropy of the spectrum by H_q, giving $\mathrm{VS}_q = \exp H_q(\lambda)$ (Pasarkar and Dieng 2024); this is the matrix-based Rényi entropy of Giraldo et al. (Sánchez Giraldo et al. 2015), as Section 6.5 notes. At $q = 2$ no eigendecomposition is needed, since $\operatorname{Tr} \rho^2 = n^{-2} \sum_{i,j} K_{ij}^2$; the entropy $-\log\big(n^{-2} \sum_{i,j} K_{ij}^2\big)$ is the RKE of Jalali, Li and Farnia, computable in $O(n^2)$ (Jalali et al. 2023). FKEA accelerates the general case with random features (Ospanov et al. 2024), and quality-weighted variants exist (Nguyen and Dieng 2024). A ninth row could be added to the table of readings in the preceding section: diversity, over the eigenvalues of a normalized similarity kernel, $\exp(-\operatorname{Tr} \rho \log \rho)$, Friedman and Dieng, 2023.

DCScore. A cheaper summarization of the same kernel reads diversity as a classification problem (Zhu et al. 2025). With embeddings h_i, kernel matrix K and temperature $\tau > 0$, form $P = \mathrm{softmax}(K/\tau)$ row-wise, $P_{ij} = \exp(K_{ij}/\tau) / \sum_{j'} \exp(K_{ij'}/\tau)$, and set $\mathrm{DCScore} = \operatorname{tr}(P) = \sum_i P_{ii}$, the total probability that each sample is classified back into its own class. Zhu et al. prove four properties, named after those of Leinster and Cobbold; stated precisely, they are the following. i) Effective number: for a kernel whose rows are maximal on the diagonal (cosine similarity of normalized embeddings, or an RBF kernel), $P_{ii} \geq 1/n$, so $1 \leq \mathrm{DCScore} < n$; the value is 1 exactly when all samples coincide, and the value n for distinct samples is reached only in the limit, which the proof states as "tending to n", i.e., as $\tau \to 0$. ii) Identical samples: merging a dataset with an identical copy of itself leaves the score unchanged. This is a dataset-level property, not Leinster and Cobbold's merging of two identical species, and it holds exactly for every kernel and every τ, since duplication halves each P_{ii} and doubles their number. iii) Symmetry under permutation. iv) Monotonicity, a comparative statement: two datasets of n mutually distinct samples, both with DCScore n, receive the same new sample, which is more similar to the samples of the second; then the first augmented dataset scores higher. Since $\mathrm{DCScore} = n$ holds only as $\tau \to 0$, the version at a fixed τ is the elementwise one: for symmetric K and $i \neq j$,

$$\frac{\partial\, \mathrm{DCScore}}{\partial K_{ij}} = -\frac{1}{\tau}\big(P_{ii}P_{ij} + P_{jj}P_{ji}\big) < 0,$$

where $K_{ij} = K_{ji}$ varies jointly, so the score strictly decreases whenever an off-diagonal similarity increases, and iv) holds whenever the two augmented kernel matrices differ only in the row and column of the new sample, with the entries of the second at least as large and one of them larger. Given K, the summarization costs $O(n^2)$ for DCScore against $O(n^3)$ for the eigendecomposition in the Vendi Score; with an inner-product kernel and $n \gg d$, the Vendi Score can use the $d \times d$ matrix $X^\top X$ instead, at total cost $O(d^2 n)$, against $O(n^2 d)$ for DCScore (Zhu et al. 2025, Table 2). The following table collects the properties of these measures and of the constructions of the next subsection, where the rows A1–A3 are defined.

property	Hill D_q	Leinster–Cobbold D_q^Z (equal weights, $q \in [0, \infty)$, $q \neq 1$)	Vendi / q-Vendi	DCScore	Mironov–Prokhorenkova constructions
symmetry	yes	yes	yes	yes	yes

property	Hill D_q	Leinster–Cobbold D_q^Z (equal weights, $q \in [0,\infty)$, $q \neq 1$)	Vendi / q-Vendi	DCScore	Mironov–Prokhorenkova constructions
absence-invariance / identical elements	yes	yes	yes (weighted form)	dataset duplication only	—
effective number	yes	yes	yes	1 for identical samples; n only as $\tau \to 0$	no (not normalized; inferred from the definitions)
modularity / partitioning	yes	yes	yes	no	—
replication	yes	yes	yes	no	—
strict monotonicity in distances (A1)	n/a	yes	no for $q = 1$; yes for RKE ($q = 2$, non-negative similarities)	yes, if $K_{ij} = f(d_{ij})$ with f strictly decreasing	yes
uniqueness (A2)	n/a	no	no	no	yes
continuity (A3)	yes	yes	yes	yes	yes
cost	$O(n)$	$O(n^2)$	$O(n^3)$ given K, $O(d^2 n)$ with a linear kernel on d-dimensional embeddings; $O(n^2)$ for RKE	$O(n^2)$ given K, $O(n^2 d)$ with a linear kernel	NP-hard

Several cells carry qualifications. The A2 failure of D_q^Z is established in Mironov and Prokhorenkova (2025) numerically, on one three-point example for $q \in [0, 100]$, and not by a proof for all q; Leinster and Cobbold's own monotonicity in Z (Leinster 2021, Lemma 6.2.2) is non-strict, and the strict A1 is Mironov and Prokhorenkova's statement for the equal-weight case. "Not normalized" is not stated in Mironov and Prokhorenkova (2025) but follows from the definitions: MultiDimVolume equals $n - 1$ for n points at unit distance and 0 for coincident points, so it is not an effective number in $[1, n]$. The DCScore entries for A1–A3, partitioning and replication are not stated in Zhu et al. (2025) and follow from the definition. A1 holds whenever the similarity is a strictly decreasing function of the distance, e.g., cosine similarity of normalized embeddings, $1 - \|h_i - h_j\|^2/2$, by the derivative above; the argument is unaffected by duplicates and, unlike that for RKE, needs no sign condition on the similarities. A2 fails: on the three points of Mironov and Prokhorenkova (2025) used there against RKE (on a circle with cosine similarity, arcs 1.1 and 0.4 radians from x_1 to x_2 and from x_2 to x_3), making x_2 a copy of x_3 raises DCScore from 1.337 to 1.394 at $\tau = 1$. A smaller τ does not restore A2: by Proposition 6.1 of Mironov and Prokhorenkova (2025), a continuous measure satisfying A2 cannot depend on which element is duplicated, whereas for points at angles $0, 0.3, 1.0, 2.2$ and $\tau = 0.1$, adding a copy of x_2 or of x_3 gives 2.998 or 3.095. Partitioning and replication fail at every $\tau > 0$, since a similarity of 0 still contributes $e^0 = 1$ to the softmax denominator: two samples with $K_{11} = K_{22} = 1$ and $K_{12} = 0$ score $2e^{1/\tau}/(e^{1/\tau} + 1) < 2$, e.g., 1.462 at $\tau = 1$. The two limits are degenerate: as $\tau \to \infty$ every dataset scores 1, and as $\tau \to 0$ DCScore tends to the number of distinct samples, provided $K_{ij} < K_{ii}$ whenever $x_j \neq x_i$, i.e., to the measure Unique of Mironov and Prokhorenkova (2025), which satisfies A2 only.

2.2.6 What axioms cannot settle

Consider three points on a circle with cosine similarity, with arc 0.6 radians from x_1 to x_2 and 1.4 radians from x_2 to x_3. Move x_3 a further 0.1 radians away from the other two. No pairwise distance decreases, and the Vendi Score falls from 1.941 to 1.916. Now let the arcs be 0.2 and 0.3 radians and replace x_2 by a copy of x_1. The set now contains a duplicate, and the Vendi Score rises from 1.187 to 1.233. Both computations are from (Mironov and Prokhorenkova 2025, sec. 3 and Appendix B).

Three axioms for the diversity of a set. Mironov and Prokhorenkova formalize the intuitions violated here (Mironov and Prokhorenkova 2025). The setting is n objects, possibly repeated, with pairwise distances d_{ij} satisfying $d_{ij} \geq 0$, $d_{ii} = 0$, symmetry, and the consistency condition that $d_{ij} = 0$ implies $d_{ik} = d_{jk}$ for all k; the triangle inequality is not required. A diversity function is a permutation-invariant map from this space $\mathcal{D}_n$ to $\mathbb{R}$, with n fixed. The axioms are: A1 (monotonicity), the function is strictly increasing in every pairwise distance, and this must hold also when the set contains duplicates; A2 (uniqueness), replacing an element that coincides with some other element by any element that coincides with none strictly increases diversity; A3 (continuity) in the distances. Their Table 2 checks the existing measures. Average pairwise distance and its sum, RKE, and Species(q), the latter being Leinster–Cobbold diversity with equal weights, satisfy A1 and A3 and fail A2, at cost $O(n^2)$ (for RKE, which depends on the similarities only through s_{ij}^2, A1 presupposes $s_{ij} \geq 0$); Diameter, Bottleneck and the Energy measures fail A1 and A2; the count of unique elements satisfies A2 only; the Vendi Score and the DPP determinant fail A1 and A2 and satisfy A3, at cost $O(n^3)$; #Circles(t) fails all three and HamDiv fails A1 and A2, and both are NP-hard. Two new constructions, MultiDimVolume and IntegralMaxClique, satisfy all three axioms, and both are NP-hard to compute. The paper leaves open whether a polynomial-time measure satisfying A1–A3 exists. Velikonivtsev et al. state slightly weaker variants of A1 and A2 (they assume all objects pairwise distinct), and the two-object "dissimilarity" axiom of Xie et al., i.e., non-strict monotonicity of the diversity of a pair in its distance, is implied by A1 (Velikonivtsev et al. 2024; Xie et al. 2023, Axiom 4.3; cf. Mironov and Prokhorenkova 2025, sec. 4.4). Emmerich et al. show, in two preprints, that selecting a fixed-cardinality subset of maximum Solow–Polasky diversity is NP-hard in general metric spaces, and remains so for points in the Euclidean plane (Emmerich et al. 2026b, 2026a).

It is important to read the Vendi row correctly. That the Vendi Score fails A2 does not contradict its own identical-elements property. The latter is an invariance of the weighted representation (two identical samples may be merged into one of double weight without changing the score), whereas A2 asks for a strict increase when a duplicate is replaced by a new element; the two axiom sets are about different properties. Propositions 6.1 and 6.2 of Mironov and Prokhorenkova (2025) explain the difficulty. Proposition 6.1 states that under A2 and A3 the diversity of a set does not depend on which of its elements are duplicated. Proposition 6.2 states that no measure satisfying A1–A3 decomposes as $F(d_{12}, \ldots, d_{1n}) + G(\text{distances among } x_2, \ldots, x_n)$, so no additive split into one element's contribution plus the diversity of the rest is available.

Which similarity. The axiomatic method fixes a functional only after the similarity, or the embedding that induces it, has been chosen. Friedman and Dieng note the dependence themselves: a too sensitive similarity makes every set look diverse, and one that is not sensitive enough gives every set a low score (Friedman and Dieng 2023). Zhao et al., reviewing 135 image and text datasets presented as diverse, borrow the vocabulary of measurement theory (D. Zhao et al. 2024): conceptualization, operationalization, and evaluation of the indicators for reliability (inter-annotator agreement, test–retest) and validity (convergent and discriminant); scale, they note, is not diversity. In these terms the choice of k or Z is a construct-validity question rather than a theorem, and, since diversity metrics depend on the embedding space, they recommend benchmarking a dataset across several embedding spaces suited to the chosen definition of diversity.

> ⚠ Terminology
>
> Several words in this section have different meanings in different literatures. i) "Uniqueness" is a theorem in the axiomatics of entropy (the functional is unique given the axioms) and an axiom in Mironov and Prokhorenkova (a duplicate is worth less than a new element). ii) "Monotonicity" has at least four senses: Shannon's increasing $A(n)$; Solow and Polasky's monotonicity in species and in distance, both non-strict; Mironov and Prokhorenkova's strict monotonicity in every distance, duplicates included; and the comparative property proved for DCScore. iii) "Effective number", $D(u_n) = n$ for all n, is stronger than "normalized", $D(u_1) = 1$; Leinster's theorem assumes the latter and derives the former from replication. Compare the warning on min-entropy in Section 2.1.4.

In summary, this section supplies the following for learning. The chain rule is the reason the remainder of this book works with H by default; when a later chapter writes a Rényi or Tsallis quantity instead, one named axiom has been traded for a computational or modelling advantage, and Chapter 7 treats that trade as part of model selection. The kernel relaxation of collision entropy in Section 6.5 is the $q = 2$ member of the Vendi family, i.e., RKE up to a monotone transformation, and the table above records that it keeps A1 for non-negative similarities and gives up A2. Finally, the drop of output diversity after RLHF (Kirk et al. 2024) and the diversity collapse under RLVR (Yuan et al. 2026) (a preprint) are statements about the effective number of a model's outputs under some similarity, and Chapter 9 discusses them in these terms.

2.3 The maximum entropy principle

Given constraints — typically moment constraints of the form $\mathbb{E}_p[f_k(X)] = \mu_k$ — the **maximum entropy** distribution solves

$$\max_p \; H(p) \quad \text{s.t.} \quad \sum_x p(x) = 1, \;\; \mathbb{E}_p[f_k(X)] = \mu_k.$$

Its solution is the **Gibbs / Boltzmann** distribution

$$p(x) = \frac{1}{Z(\boldsymbol{\lambda})} \exp\Big(\textstyle\sum_k \lambda_k f_k(x) \Big),$$

where $\boldsymbol{\lambda}$ are Lagrange multipliers and Z is the partition function. This is the least-biased distribution consistent with what we know (Jaynes 1957).

The single most important instance of this framework is the one in which the constrained feature is the energy of a physical system. That case, worked out in Chapter 3, is what turns the multipliers above into a temperature and connects the whole construction to thermodynamics.

2.4 Minimum description length and minimax entropy

The maximum-entropy principle can be *derived* rather than postulated. Under the **minimum description length** (MDL) view (Grünwald 2007), a model P encodes each state with a code word of length $-\log P(x)$, and its description length is the expected code length across all datasets consistent with the measured features. Among models matching those features, description length is minimized by the maximum-entropy model (Feder 1986).

MDL also poses a question that maximum entropy alone cannot answer: which features should be measured in the first place? For maximum-entropy models the description length coincides with the model's own entropy,

$$L(P_F) = S(P_F),$$

so the best feature set F is the one whose maximum-entropy model has the *smallest* entropy. This is the **minimax entropy** principle, developed in Chapter 7.

2.5 Why it matters for learning

The maximum-entropy principle explains why exponential families are ubiquitous, grounds the Boltzmann machine in statistical physics, and — as developed in Bian et al. (2022) — provides a principled basis for defining valuations in machine learning. The next chapter supplies the missing half of the picture: what happens when entropy is traded off against energy, and what the exchange rate between them turns out to be.

3 Energy, Entropy, and Free Energy

The previous chapter maximized entropy subject to abstract moment constraints and obtained a Gibbs distribution written in terms of Lagrange multipliers. The next chapter writes probabilities as $p(x) \propto e^{-E(x)}$ and calls the exponent an energy. Something has to connect the two, and that something is one of the most useful equations in science:

$$F = U - TS.$$

At a fixed temperature, nature neither minimizes energy nor maximizes entropy. It minimizes a combination of the two, and the temperature is the exchange rate between them. This chapter derives that statement, illustrates it with three classical examples, and follows it into artificial intelligence.

3.1 Maximum entropy with an energy constraint

The maximum-entropy problem of Section 2.3 was stated for arbitrary features f_k. Historically the original problem had exactly one constraint: the average energy of the system is known to be U. Written out,

$$\max_p \; H(p) \quad \text{s.t.} \quad \sum_x p(x) = 1, \;\; \sum_x p(x)\,E(x) = U,$$

and solved with Lagrange multipliers, it yields the **Boltzmann distribution**

$$p(x) = \frac{1}{Z}\, e^{-\beta E(x)}, \qquad Z(\beta) = \sum_x e^{-\beta E(x)},$$

where the multiplier enforcing the energy constraint is the inverse temperature, $\beta = 1/k_B T$.

That identification is worth pausing on, because it is what gives temperature a meaning outside physics. A Lagrange multiplier is a *price*: it measures how much of the objective must be surrendered to buy one unit of the constrained quantity. Here it measures how much entropy must be given up to raise the expected energy by one unit. Temperature is that price. A system at high temperature buys energy cheaply and stays spread out; a system at low temperature pays dearly for it and concentrates on low-energy states.

This also settles a notational discrepancy the attentive reader will have noticed. In Section 2.3 the solution appeared as $p(x) \propto \exp\big(\sum_k \lambda_k f_k(x)\big)$, with a plus sign; here it appears as $\exp(-\beta E(x))$, with a minus. The two are the same object under the correspondence $E = -\sum_k \lambda_k f_k$. Energy is negative log-probability up to a constant, and the sign convention comes from physics, where stable configurations should sit at *low* energy. Chapter 4 follows the physics convention and absorbs β into E by setting $k_B T = 1$.

3.2 Free energy

Given the Boltzmann distribution, the two quantities of interest follow at once: the average energy $U = \langle E \rangle$ and the entropy $S = -k_B \sum_x p(x) \log p(x)$. The combination that governs equilibrium is the **Helmholtz free energy** $F = U - TS$, and it has a compact closed form (Callen 1985):

$$F = -k_B T \log Z.$$

Free energy is the log partition function. What Chapter 4 will present as the central computational obstacle of energy-based modelling is, read from the other side, the central thermodynamic potential of the system.

The relationship has a third face, and it is the one that matters most for learning. Let q be any distribution over the same states, and define its **variational free energy**

$$F(q) = \langle E \rangle_q - T\,S(q).$$

A short calculation relates this to the equilibrium value:

$$F(q) = F + k_B T \cdot \mathrm{KL}\big(q \,\|\, p\big),$$

where p is the Boltzmann distribution. Because the Kullback–Leibler divergence is non-negative, $F(q) \geq F$ with equality exactly when $q = p$. Equilibrium is therefore not merely *described* by the Boltzmann distribution, it is *characterized* by it, as the unique minimizer of a variational objective. Nearly everything in later chapters that goes by the name of a variational bound is this one inequality in disguise.

3.3 What entropy buys you

The trade-off is easiest to believe after seeing cases in which entropy alone decides the outcome.

3.3.1 Isothermal expansion: a process with no energy change

An ideal gas of N particles expands from volume V_1 to V_2 while held at temperature T. The internal energy of an ideal gas depends only on temperature, so $\Delta U = 0$: not one joule of stored energy is released. The gas expands regardless, because

$$\Delta S = N k_B \log \frac{V_2}{V_1} > 0 \quad \Longrightarrow \quad \Delta F = -T\Delta S < 0.$$

The work extracted, $W = N k_B T \log(V_2/V_1) = T\Delta S$, is drawn entirely from the heat bath. An account of this process in terms of energy alone cannot even say which direction it runs.

3.3.2 The rubber band: a force made of entropy

Stretch a rubber band and it pulls back. The restoring force is not stored in stretched chemical bonds. A polymer chain is a random walk of N segments of length a, and there are vastly more coiled configurations than extended ones, so extending the chain to end-to-end length L costs entropy. The force is the gradient of that cost:

$$f = -T\frac{\partial S}{\partial L} \approx \frac{3 k_B T}{N a^2}\, L.$$

The energy landscape here is essentially flat, and yet there is a real force, one measurable with a spring scale. The formula also makes two predictions that run against intuition, and both hold. The force is *proportional to temperature*, so heating a stretched rubber band makes it contract harder — the opposite of a metal rod, which expands. And stretching a band quickly warms it, an effect readily confirmed by holding one against your lip.

3.3.3 Melting: where the competition changes hands

Ice has lower energy than liquid water, since melting must pay $\Delta H > 0$ to break hydrogen bonds; water has the higher entropy. At the melting point the two contributions balance exactly, $\Delta H - T_m \Delta S = 0$, so

$$T_m = \frac{\Delta H}{\Delta S}.$$

Below T_m the energy term dominates and ice is the stable phase; above it the entropy term dominates and water is. A phase transition is precisely the temperature at which the exchange rate tips the competition from one side to the other.

3.4 From spins to networks

The simplest system with an energy is a single two-state variable $s = \pm 1$ with energy $\mp\varepsilon$. Its partition function and average value are

$$Z = 2\cosh(\beta\varepsilon), \qquad \langle s \rangle = \tanh(\beta\varepsilon),$$

and the temperature limits are exactly what the trade-off predicts: as $T \to 0$ the variable freezes into its low-energy state, and as $T \to \infty$ it becomes a fair coin with entropy $k_B \log 2$.

Now couple many such variables together. The **Ising model** assigns to a configuration $s = (s_1, \ldots, s_N)$ the energy

$$E(s) = -\sum_{i<j} J_{ij}\, s_i s_j - \sum_i h_i s_i,$$

with J_{ij} the interaction between units i and j and h_i a local bias. When the couplings are disordered, the landscape acquires a large number of local minima and the system is called a *spin glass*.

That equation is the hinge of this chapter. It is a model of magnetism, and it is also, term for term, the energy function of a Hopfield network and of a Boltzmann machine. What follows is what happened when physicists noticed.

3.5 How this reached AI

Year	Contribution	Role of the entropy–energy trade-off
1982	Hopfield network (Hopfield 1982)	The Ising energy becomes an associative memory: stored patterns are local minima and recall is descent. The $T = 0$ limit — energy only
1983	Simulated annealing (Kirkpatrick et al. 1983)	Turns the trade-off into a *schedule*: start hot so entropy drives exploration, then cool so energy drives commitment
1985	Boltzmann machine (Ackley et al. 1985)	Places the Hopfield network at finite temperature, $p(s) \propto e^{-E(s)/T}$; learning compares correlations in a clamped and a free phase

Year	Contribution	Role of the entropy–energy trade-off
1995–98	Helmholtz machine (Dayan et al. 1995); free-energy view of EM (Neal and Hinton 1998)	The evidence lower bound is identified as a negative variational free energy
2002	Contrastive divergence (Hinton 2002)	Makes the intractable $\log Z$ gradient practical and revives deep generative models
2006	Energy-based learning (LeCun et al. 2006)	Names the log-partition term "the Helmholtz free energy of the ensemble" and unifies probabilistic and non-probabilistic learning
2024	Nobel Prize in Physics (The Royal Swedish Academy of Sciences 2024)	Awarded to Hopfield and Hinton; the citation rests on the correspondence between neural networks and spin models in statistical physics

Two threads in that table deserve to be stated plainly, because the rest of the book leans on them.

The first is that **the evidence lower bound is a free energy**. Take a latent variable z, set $E(z) = -\log p(x, z)$ and $T = 1$, and the variational inequality of the previous section reads

$$-\log p(x) \;\le\; \big\langle -\log p(x, z) \big\rangle_q - H(q) \;=\; F(q),$$

with the gap equal to $\mathrm{KL}\big(q \,\|\, p(\cdot \mid x)\big)$. Variational autoencoders, variational inference, and the training objectives of diffusion models are all instances of minimizing $U - TS$ at unit temperature. The physics is not an analogy here; it is the same algebra.

The second is that **annealing is the trade-off used as a control signal**. Simulated annealing, the noise schedules of Langevin and diffusion samplers, and the practice of decoding a language model at a temperature that falls during generation are one idea: begin where entropy dominates and the search is broad, end where energy dominates and the answer is definite.

3.6 The temperature dial

The modern instance is worth making explicit, because it is the version most readers use daily. A language model produces logits $\ell(x)$ and samples

$$p(x) \propto \exp\big(\ell(x)/T\big),$$

which is the Boltzmann distribution with energy $E = -\ell$. Greedy decoding is $T \to 0$; sampling uniformly from the vocabulary is $T \to \infty$. The `temperature` field in every inference API is the same T that appears in $F = U - TS$.

Read this way, the three principles that organize this book are three settings of a single dial.

Principle	Where the dial sits	What it buys
Maximum entropy (Chapter 5)	High: entropy dominates	Honesty about what the constraints leave undetermined
Minimum entropy (Chapter 6)	Low: energy dominates	Commitment — sharpened predictions and elicited reasoning

Principle	Where the dial sits	What it buys
Minimax entropy (Chapter 7)	Both, nested	Maximize on the inside to model the world, minimize on the outside to choose what to model

The principles are not competing claims about the right amount of uncertainty. They are claims about *when* to be uncertain, and the free energy is what makes that question well posed. Read at the level of algorithms rather than slogans, the three also share one *inner* problem — the minimization of a free energy over distributions, which Chapter 5 calls the representation and inference levels — and differ only in the *outer* criterion: fit the data (maximum entropy), sharpen the model's own output (minimum entropy), or choose the features whose maximum-entropy model has the least entropy (minimax entropy).

3.7 Why it matters for learning

The equation $F = U - TS$ supplies three things the rest of this book uses. It explains where the Gibbs form of Chapter 4 comes from and why its exponent carries a minus sign. It identifies the partition function — the central computational difficulty of energy-based modelling — as a free energy, which is why variational bounds are the natural way around it. And it gives *temperature* a precise meaning, so that annealing schedules, decoding parameters, and the coalition temperature τ of Chapter 5 can all be recognized as one knob rather than three coincidences.

The next chapter takes the Gibbs distribution as its starting point and asks what happens when the energy function is learned from data.

4 Energy-Based Models

The previous chapter left us with an open question: what changes when the Gibbs distribution's energy function is no longer prescribed by physics, but instead *learned from data*? This chapter answers that question.

Two things change the moment the energy becomes learnable. First, the energy stops being handed to us by physics — the Ising couplings, the hydrogen bonds of Chapter 3 — and becomes a neural network $E_\theta(x)$ whose parameters we must fit. Second, the partition function, which for a physicist is a bookkeeping device, becomes for us the central computational villain of the story: nearly every algorithm in this chapter is best understood as a different strategy for *not computing it*.

An **energy-based model (EBM)** associates to each configuration x a scalar energy $E_\theta(x)$ and defines the Gibbs distribution

$$p_\theta(x) = \frac{e^{-E_\theta(x)}}{Z(\theta)}, \qquad Z(\theta) = \int e^{-E_\theta(x)}\, dx, \tag{4.1}$$

with a sum in place of the integral when x is discrete. This is the Boltzmann distribution of Chapter 3 at unit temperature, with a learnable energy in place of a physical one. Low energy means high probability; the sign convention and the absorbed temperature were settled at the end of that chapter.

The material here draws chiefly on two tutorials, written fifteen years apart and complementary in spirit. The first (LeCun et al. 2006) treats the energy as a *decision surface*: inference is energy minimization, learning is shaping the landscape, and probabilities are an optional purchase. The second (Song and Kingma 2021) treats the energy as a *density*: the model is Equation 4.1 taken literally, and the question is how to estimate it when $Z(\theta)$ cannot be computed. Reading them together — decision surface first, density second — is the plan of this chapter. A regularly updated companion list of papers, tutorials, and software is maintained at *Awesome-EBM* (Bian 2020).

4.1 The energy view of inference

Start with the decision surface. In the energy-based view, a model is a function $E_\theta(x, y)$ that measures the *compatibility* between an observed input x and a candidate answer y: small energy means "these two belong together," large energy means "they do not." Inference is then not a forward pass but an *optimization*:

$$y^* = \arg\min_{y \in \mathcal{Y}} E_\theta(x, y). \tag{4.2}$$

When $\mathcal{Y}$ is a handful of class labels, the arg min is a table lookup and Equation 4.2 is just classification wearing unfamiliar clothes. The formulation earns its keep when $\mathcal{Y}$ is large or structured: all consistent segmentations of an image, all sentences of English, all conformations of a protein. In those cases no model can afford to enumerate answers; what it can afford is to *score* any proposed answer, and to search — by dynamic programming, by gradient descent on a continuous y, by annealing — for a good one. The energy function is precisely the object that makes such a search well posed.

This view immediately clarifies what a model owes us, and it is less than one might think. LeCun et al. distinguish the questions a model may be asked (LeCun et al. 2006):

1. **Decision.** "Which y is best for this x?" Only the *ordering* of energies matters. Any strictly increasing transform of E_θ gives the same answers.
2. **Ranking.** "Is y_1 better than y_2?" Again only comparisons matter.
3. **Detection.** "Is this y good enough?" Now energies are compared to a threshold, so their *scale* begins to matter.
4. **Probability.** "How confident are we in each y?" Only here must energies be converted into calibrated numbers that sum to one.

Only the fourth question requires the Gibbs conversion

$$p_\theta(y \mid x) = \frac{e^{-\beta E_\theta(x,y)}}{\int_{y'} e^{-\beta E_\theta(x,y')}\, dy'}, \tag{4.3}$$

(a sum over y' when $\mathcal{Y}$ is discrete), restoring the inverse temperature β of Chapter 3 as an explicit dial. The rest of the chapter works at $\beta = 1$ unless a limit in β is at issue, matching the unit-temperature convention of Equation 4.1. The denominator is the partition function — the free-energy bill of Chapter 3. This is the first strategic lesson of energy-based learning: *probabilities are expensive, and you should buy them only when the application actually needs them* (LeCun et al. 2006). A robot that must steer left or right needs the arg min, not a posterior. Much of this chapter is about what becomes possible when we decline to pay, and what it costs to pay when we must.

4.2 Why learn an energy at all?

If unnormalized models create such trouble, why prefer them? Three reasons, in increasing order of depth.

Freedom of form. A normalized density must integrate to one, and maintaining that constraint dictates the architecture: autoregressive models must factor into conditionals, normalizing flows must be invertible with tractable Jacobians, variational autoencoders must route everything through a directed latent-variable structure. An energy function must merely be a scalar-valued function, bounded below well enough for Equation 4.1 to make sense. Any regression architecture qualifies — a convolutional network for images, a graph neural network for molecules, a transformer for text. In the words of Song and Kingma, density estimation is thereby "reduced to a nonlinear regression problem" (Song and Kingma 2021). The modeling question stops being "what distributions can my architecture normalize?" and becomes "what shape should the landscape have?"

Compatibility is often what we actually know. Domain knowledge usually arrives as statements about what configurations are good — physical constraints, grammatical constraints, consistency conditions — not as recipes for sampling. An energy function absorbs such knowledge directly, as terms that raise the energy of violating configurations.

Energies add. If E_1 scores grammaticality and E_2 scores factual accuracy, then $E_1 + E_2$ scores both at once, and at the level of distributions the sum of energies is the *product* of the corresponding Gibbs factors — a product of experts (Hinton 2002), in which each expert can veto a candidate by assigning it high energy. Normalized models do not compose this way: the product of two normalized densities is not normalized, and renormalizing it is exactly the partition-function problem again. This additivity is what makes energies the natural currency for *inference-time* composition — steering a generator with extra constraints — a theme that returns in force in Chapter 8.

The price of this freedom is stated by Equation 4.1 itself: $Z(\theta)$ is an integral over every configuration the model can represent, and for any interesting architecture it is hopelessly intractable. Neither exact likelihoods nor exact samples are available. Everything that follows is about living well anyway.

4.3 Learning is shaping a landscape

What should a learned energy landscape look like? Configurations resembling the training data should sit in valleys; everything else should sit high. The subtlety — and it is *the* central subtlety of EBM training — is that a learning rule which only digs valleys will destroy the landscape.

Consider the most innocent possible objective, the **energy loss**: for each training pair (x^i, y^i), minimize $E_\theta(x^i, y^i)$. Push down on the data; done. For some architectures this works — in least-squares regression, $E_\theta(x, y) = \|y - G_\theta(x)\|^2$, pushing the correct answer's energy toward zero automatically raises every other answer's, because the architecture allows only one minimum per input. But give the model more freedom and the innocent objective finds the cheat: make the energy *low everywhere*. A constant landscape assigns the data the lowest possible energy and is perfectly useless — it has **collapsed** (LeCun et al. 2006). Flat landscapes answer every question with "everything is equally fine."

The cure is as old as the perceptron: every push down on the correct answer must be paired with a **pull up** on incorrect ones. Losses that implement this pairing are called *contrastive*, and the zoo of them is organized by one question — *which* incorrect answers get pulled up, and how hard?

The most offending incorrect answer. The sharpest choice is to pull up only on the incorrect answer the model currently likes best, $\bar{y}^i = \arg\min_{y \neq y^i} E_\theta(x^i, y)$ — the *most offending incorrect answer* (LeCun et al. 2006). The **perceptron loss** $E_\theta(x^i, y^i) - \min_y E_\theta(x^i, y)$ does this but demands no gap, and so can still be satisfied by a nearly flat landscape. Margin losses fix that. The **hinge loss** $\max\big(0,\, m + E_\theta(x^i, y^i) - E_\theta(x^i, \bar{y}^i)\big)$ insists the correct answer sit at least m below its closest rival — the loss of support vector machines, re-read as landscape shaping. The **log loss** $\log\big(1 + e^{E_\theta(x^i, y^i) - E_\theta(x^i, \bar{y}^i)}\big)$ is its soft, infinite-margin sibling. The **square–square loss** $E_\theta(x^i, y^i)^2 + \max(0, m - E_\theta(x^i, \bar{y}^i))^2$ pins correct answers near zero and rivals above m, which is natural when the energy is a distance and hence bounded below.

All incorrect answers at once. The other extreme is the **negative log-likelihood (NLL) loss**. Write the Gibbs conversion Equation 4.3, take $-\log$, and per sample:

$$\mathcal{L}_{\text{nll}} = E_\theta(x^i, y^i) \;+\; \frac{1}{\beta} \log \int_y e^{-\beta E_\theta(x^i, y)}\, dy. \tag{4.4}$$

(Replace the integral by a sum when $\mathcal{Y}$ is finite.) The second term is *minus* the Helmholtz free energy $F = -(1/\beta) \log \int e^{-\beta E}\, dy$ of Chapter 3 — at $\beta = 1$ it is $\log Z$, not F itself. The first term pushes down on the data; the $\log Z$ term pulls up on *every* answer. How hard? Differentiate (at $\beta = 1$):

$$\frac{\partial \mathcal{L}_{\text{nll}}}{\partial \theta} = \frac{\partial E_\theta(x^i, y^i)}{\partial \theta} - \int_y p_\theta(y \mid x^i)\, \frac{\partial E_\theta(x^i, y)}{\partial \theta}\, dy. \tag{4.5}$$

Each answer is pulled up with force proportional to *the probability the model currently assigns it*. The landscape's own valleys attract the correction: wherever the model is confidently wrong, the pull is strongest. This is a beautiful rule and an expensive one — the integral in Equation 4.5 is an expectation over the model distribution, which is the partition-function problem in gradient form. Two limits knit the zoo together (LeCun et al. 2006). As $\beta \to \infty$, $(1/\beta) \log \int e^{-\beta E}\, dy \to -\min_y E_\theta(x^i, y)$, so Equation 4.4 becomes the perceptron loss (only the minimum-energy rival matters). And when $\mathcal{Y}$ has just two elements $\{y^i, \bar{y}^i\}$, the $\beta = 1$ case collapses algebraically onto the log loss: $E(y^i) + \log\big(e^{-E(y^i)} + e^{-E(\bar{y}^i)}\big) = \log\big(1 + e^{E(y^i) - E(\bar{y}^i)}\big)$.

> **i** Collapse is a recurring character
>
> Hold on to the failure mode, because this book meets it twice more. In Section 6.6, a language model trained on its own agreement signal can collapse to confident nonsense — entropy minimization with the pull-up term missing. And in modern self-supervised representation learning, joint-embedding archi-

> tectures collapse to constant embeddings unless a contrastive or regularization term props the landscape up (LeCun 2022; Dawid and LeCun 2024). The lesson is always the same: *an objective that only rewards fitting what you see will be gamed by a model that makes everything look equally good.* The interesting design space is in how — and how cheaply — you pull up.

The remainder of the chapter follows the density view (Song and Kingma 2021): three families of estimators, distinguished by how they approximate or evade the pull-up integral of Equation 4.5. In caricature: **sample it** (maximum likelihood with MCMC), **differentiate it away** (score matching), or **turn it into a classification problem** (noise contrastive estimation).

4.4 Strategy one: sample the pull-up

Take maximum likelihood seriously and estimate the intractable term by Monte Carlo. For the unconditional model Equation 4.1, the log-likelihood gradient splits into the same two forces as Equation 4.5:

$$\nabla_\theta \log p_\theta(x) = -\nabla_\theta E_\theta(x) \; + \; \mathbb{E}_{\tilde{x}\sim p_\theta}\big[\nabla_\theta E_\theta(\tilde{x})\big]. \tag{4.6}$$

The derivation is worth doing once by hand, because the punchline is neat. Start from $\nabla_\theta \log Z(\theta)$, use the chain rule, and watch the model distribution assemble itself:

$$\nabla_\theta \log Z = \frac{1}{Z}\nabla_\theta \int e^{-E_\theta(x)}dx = \int \frac{e^{-E_\theta(x)}}{Z}\big(-\nabla_\theta E_\theta(x)\big)\,dx = -\,\mathbb{E}_{x\sim p_\theta}\big[\nabla_\theta E_\theta(x)\big]. \tag{4.7}$$

The intractable normalizer has become an *expectation under the model* — not computable in closed form, but estimable from a single model sample, without bias. Substituting into $\nabla_\theta \log p_\theta = -\nabla_\theta E_\theta - \nabla_\theta \log Z$ gives Equation 4.6.

Read the two terms as the two phases of learning. The **positive phase** lowers the energy of real data. The **negative phase** raises the energy of whatever the model itself likes to produce — its *fantasies*, in the old Boltzmann-machine vocabulary, where the phases were called *clamped* and *free* (Ackley et al. 1985) (the 1985 row of the table in Chapter 3). Learning stops exactly when the fantasies become statistically indistinguishable from the data, at which point the two forces cancel in expectation. Confabulate, compare, correct.

The entire difficulty hides in four words: *sample from the model*. The standard tool is **Langevin dynamics**, which needs only the gradient of the log-density in x — and here EBMs catch a genuine break, because the normalizer does not depend on x:

$$\nabla_x \log p_\theta(x) = -\nabla_x E_\theta(x) - \underbrace{\nabla_x \log Z(\theta)}_{=0} = -\nabla_x E_\theta(x). \tag{4.8}$$

This quantity — the gradient of the log-density with respect to the *input*, not the parameters — is called the **score**, and Equation 4.8 says an EBM hands it to us free of charge. Langevin MCMC then iterates

$$x_{k+1} = x_k - \frac{\epsilon^2}{2}\nabla_x E_\theta(x_k) + \epsilon\, z_k, \qquad z_k \sim \mathcal{N}(0, I)\ : \tag{4.9}$$

roll downhill, but shake while doing so. The reader will recognize finite-temperature descent — energy pulling toward minima, injected noise buying the entropy that keeps the walker exploring (Chapter 3). The *ratio* of noise amplitude to drift sets the temperature; Equation 4.9 is the unit-temperature case already built into Equation 4.1. The step size ϵ is a discretization parameter, not a temperature: as $\epsilon \to 0$ and the number of steps grows, the iterates distribute as p_θ (Song and Kingma 2021).

"Long enough" is the problem: fresh chains per gradient step are ruinously slow, so practice truncates. **Contrastive divergence (CD)** starts chains at data points and runs a handful of steps (Hinton 2002); **persistent CD** never restarts, carrying chains across parameter updates (Tieleman 2008); replay buffers reinitialize chains from a reservoir of past samples (Du and Mordatch 2019); short-run MCMC accepts non-convergence and reinterprets the truncated sampler itself as the generative model (Nijkamp et al. 2019). These approximations power most large-scale EBM image models, but the bias of truncated chains is real, and taming it remains a research area of its own (Song and Kingma 2021).

4.5 Strategy two: match slopes, not heights

Here is a different idea, and it begins with an observation about Equation 4.8. The score $\nabla_x \log p_\theta(x)$ does not contain $Z(\theta)$ at all. Two normalized densities with the same score everywhere on a connected domain are the same density — if their log-densities have equal gradients, they differ by a constant, and normalization forces that constant to zero. (When the domain falls into well-separated pieces the argument fails, and that failure is the mode-weight blind spot below.) So instead of matching *heights* of the log-density (which requires knowing sea level, i.e. Z), match *slopes*. Picture two hikers comparing terrain maps: they need not agree on altitude above sea level to verify they hold maps of the same mountain — agreeing on the steepness and direction of the slope at every point is enough.

Score matching (Hyvärinen 2005) makes this precise by minimizing the **Fisher divergence** between data and model:

$$D_F(p_{\text{data}} \,\|\, p_\theta) = \mathbb{E}_{p_{\text{data}}}\left[\frac{1}{2}\left\|\nabla_x \log p_{\text{data}}(x) - \nabla_x \log p_\theta(x)\right\|^2\right]. \tag{4.10}$$

An objection leaps out: Equation 4.10 contains the score of the *data distribution*, which nobody knows. The resolution is a small miracle of integration by parts, and it is worth seeing in one dimension. Expand the square in Equation 4.10; the term $\frac{1}{2}\mathbb{E}[(\partial_x \log p_{\text{data}})^2]$ is a constant in θ and can be dropped, and the term $\frac{1}{2}\mathbb{E}[(\partial_x \log p_\theta)^2]$ is computable. The troublesome cross-term is where the trick lands:

$$-\mathbb{E}_{p_{\text{data}}}\big[\partial_x \log p_{\text{data}} \cdot \partial_x \log p_\theta\big] = -\int p_{\text{data}}(x)\,\frac{\partial_x p_{\text{data}}(x)}{p_{\text{data}}(x)}\,\partial_x \log p_\theta(x)\,dx = -\int \partial_x p_{\text{data}}(x)\cdot \partial_x \log p_\theta(x)\,dx.$$

The unknown density has been reduced to a plain factor, and one integration by parts (boundary terms vanishing for well-behaved densities) moves the derivative off it entirely:

$$= \int p_{\text{data}}(x)\,\partial_x^2 \log p_\theta(x)\,dx = \mathbb{E}_{p_{\text{data}}}\big[\partial_x^2 \log p_\theta(x)\big].$$

The unknown score has become a second derivative of the *model* — which for an EBM is a second derivative of the energy — averaged over data samples we have. In d dimensions (Hyvärinen 2005):

$$D_F = \mathbb{E}_{p_{\text{data}}}\left[\sum_{i=1}^{d}\frac{1}{2}\left(\frac{\partial E_\theta(x)}{\partial x_i}\right)^2 - \frac{\partial^2 E_\theta(x)}{\partial x_i^2}\right] + \text{const.} \tag{4.11}$$

Signs deserve a moment of care here, because they are easy to slip on. Since $\log p_\theta = -E_\theta - \log Z$, second derivatives of the log-density are *negatives* of second derivatives of the energy, which is where the minus sign in Equation 4.11 comes from; the squared first-derivative term is immune because squaring erases the sign. A one-line sanity check: fit a Gaussian $E_\theta(x) = x^2/2\sigma^2$ to standard-normal data. Equation 4.11 gives $\frac{1}{2\sigma^4} - \frac{1}{\sigma^2}$, minimized at $\sigma^2 = 1$ as it must be; with the sign flipped, the objective would be minimized by

sending $\sigma^2 \to \infty$ — a model dissolving into uniformity. (For *location* families the second-derivative term is constant in the parameter, so both signs give the same fit; that is exactly why the slip is easy to miss.)

No sampling, no partition function, and a consistent estimator. The catch is the Hessian trace: for deep networks and high-dimensional x, exact second derivatives cost roughly d backpropagations. Two descendants remove the bottleneck, one by adding noise and one by throwing dice.

Denoising score matching (DSM). Perturb each data point with Gaussian noise, $\tilde{x} = x + \sigma z$, and match the model's score to the score of the *noisy* data distribution. Vincent's identity (Vincent 2011) shows this equals (up to a constant) a beautifully simple objective:

$$\mathbb{E}_{x\sim p_{\text{data}},\, z\sim\mathcal{N}(0,I)}\left[\frac{1}{2}\left\|\nabla_{\tilde{x}}\log p_\theta(\tilde{x}) + \frac{z}{\sigma}\right\|^2\right], \qquad \tilde{x} = x + \sigma z. \tag{4.12}$$

Intuitively: given a noisy point, the score of the noised distribution points back toward the clean data that produced it, so *learning the score is learning to denoise*. Second derivatives are gone. The costs are a bias — the optimum matches the noisy distribution, not the data — and, if one shrinks σ to reduce that bias, a variance in the z/σ term that blows up as $\sigma \to 0$ (Song and Kingma 2021). Noise is a knob with no free setting.

Sliced score matching (SSM). Keep the clean data, and check the score match only along random directions: project both scores onto a random vector v and match the resulting scalars (Song et al. 2019). The Hessian-trace term of Equation 4.11 is replaced by the directional second derivative $v^\top(\nabla_x^2 E_\theta)\,v$ — the Skilling–Hutchinson estimator — *keeping the minus sign*. Both the squared directional derivative and that Hessian–vector product are computable with two backpropagations at any dimension. SSM is consistent, like exact score matching, at the price of extra estimator variance from the random projections.

The blind spot, and how it launched diffusion. Score matching has one structural weakness worth internalizing. Suppose the data has two well-separated modes — a mixture $\pi\, p_0 + (1-\pi)\, p_1$ with essentially disjoint supports. *Inside* each mode's region, the mixture's score is just that component's own score: the weight π enters the density as a constant factor, and gradients of log-densities kill constant factors. The relative weight of the modes lives precisely in the no-man's-land *between* them, where there is no data to enforce it. A score-matched model can therefore reproduce each island perfectly and still apportion probability between the islands arbitrarily (Song and Ermon 2019; Song and Kingma 2021).

The fix is to flood the landscape: perturb the data with noise large enough to bridge the modes — making the weights visible in the score — and, since heavy noise distorts fine detail, do it at *many scales*, learning a noise-conditional score for each. Sampling then anneals from coarse to fine (Song and Ermon 2019). The reader of Chapter 3 will recognize the annealing schedule — start where entropy dominates, finish where energy does — and the reader of the wider literature will recognize the result: this multi-scale, score-based recipe, together with its diffusion-process formulation (Sohl-Dickstein et al. 2015; Ho et al. 2020; Song, Sohl-Dickstein, et al. 2021), became the generative-modeling workhorse of the 2020s. Diffusion models are the energy-based lineage's most successful child: what they inherit is the score Equation 4.8, the Langevin walk Equation 4.9, and the annealed temperature dial of Chapter 3.

4.6 Strategy three: learn by telling things apart

The third strategy is the slyest. If estimating a density is hard, compare it to one you know.

Noise contrastive estimation (NCE) (Gutmann and Hyvärinen 2010) introduces a noise distribution p_n with tractable density and easy sampling — a Gaussian, say — and mixes noise samples with data samples. Now play a game: given a point, was it data or noise? By Bayes' rule, the ideal referee computes

$$P(\text{data} \mid x) = \frac{\nu\, p_{\text{data}}(x)}{\nu\, p_{\text{data}}(x) + p_n(x)}, \tag{4.13}$$

with ν the data-to-noise mixing ratio. NCE builds a classifier of exactly this functional form, but with the model density in the data slot,

$$P_\theta(\text{data} \mid x) = \frac{\nu\, p_\theta(x)}{\nu\, p_\theta(x) + p_n(x)}, \tag{4.14}$$

and trains it by ordinary logistic regression on the mixed samples. If the classifier family is rich enough, training drives Equation 4.14 to the true posterior Equation 4.13 for every x, and matching posteriors at fixed p_n and ν forces $p_\theta = p_{\text{data}}$. Density estimation has been smuggled inside a classification problem — the one problem deep learning solves in its sleep.

Two features distinguish NCE in the EBM toolkit (Song and Kingma 2021). First, the normalizer Z_θ can be treated as an explicit learnable scalar — the classification game is only won when the model's *absolute* density matches the data's, so the game itself calibrates Z. Alternatively, fix $Z = 1$ and obtain a *self-normalized* model, a trick beloved of language modeling, where NCE-style objectives trained word embeddings and early neural language models at vocabulary scales where softmax normalization was unaffordable (Mnih and Teh 2012). Second, the choice of p_n is the whole game in practice: the classification task must be neither trivial nor impossible, which means the noise should resemble the data — a requirement that becomes brutal in high dimension, and has spawned a cottage industry of learned, adaptive noise distributions (Song and Kingma 2021).

One more connection ties the room together. Choose the noise to be the data itself, shifted by a small vector v, and the NCE objective Taylor-expands — as $\|v\| \to 0$ — into the sliced score matching objective with projection direction v (Song and Kingma 2021). Telling the data apart from a nudged copy of itself means, in the limit, matching directional slopes.

4.7 One family, three estimators

The three strategies look like three separate inventions. They are better seen as one family, distinguished by which divergence they minimize and what they demand of us.

Table 4.1: Three roads around the partition function. Each row answers: how does this method avoid computing $Z(\theta)$, and what does it pay instead?

	MCMC-based MLE (Section 4.4)	Score matching (Section 4.5)	NCE (Section 4.6)
Minimizes	KL divergence	Fisher divergence	KL of classification posteriors
Pull-up term handled by	sampling it	differentiating it away	canceling it against a reference
Requires	model samples (MCMC)	input derivatives of E_θ	a tractable noise density
Access to Z?	no	no (energy determined up to a constant)	yes (Z learned, or fixed to 1)
Characteristic failure	biased truncated chains	invisible mode weights	noise too far from data

And the roads interconnect. Contrastive divergence run with a single infinitesimal Langevin step *is* score matching, up to rescaling (Hyvärinen 2007). NCE against infinitesimally shifted data is sliced score matching, as we just saw. Underneath both coincidences sits a single identity — a relative form of de Bruijn's identity

— relating the two divergences. Perturb data and model with the *same* Gaussian noise of variance t, write q_t and $p_{\theta,t}$ for the smoothed densities, and

$$\frac{d}{dt} D_{\mathrm{KL}}(q_t \,\|\, p_{\theta,t}) = -\tfrac{1}{2} D_F(q_t \,\|\, p_{\theta,t}) \,: \tag{4.15}$$

the Fisher divergence is the *rate* at which the KL divergence dissolves under noise (Song and Kingma 2021). (The classical de Bruijn identity is the special case that tracks entropy rather than KL.) KL is the integral view, Fisher the differential view, of one and the same discrepancy — which is why an MCMC shortcut and a derivative trick keep converging on the same estimator.

There are further roads — minimizing Stein discrepancies, adversarially training a sampler network against the energy, differences of KL divergences — for which Song and Kingma's survey (Song and Kingma 2021) and the running list at *Awesome-EBM* (Bian 2020) are the entry points. The taxonomy above is enough to read that literature: whenever a new EBM training method appears, ask first *how it implements the pull-up*.

4.8 Latent variables, and the free energy again

Often the energy needs helper variables that are observed never: the pose of a face, the segmentation of a sentence, the rotation of a molecule. Give the energy an extra argument $E_\theta(x, y, z)$ and let inference minimize over z as well. But minimizing over z is a $T = 0$ policy — it credits y with only its single best explanation. The finite-temperature policy sums over explanations:

$$F_\beta(x, y) = -\frac{1}{\beta} \log \int_z e^{-\beta E_\theta(x,y,z)} \, dz, \tag{4.16}$$

which the reader will recognize as a free energy — Equation 4.16 *is* $F = -\frac{1}{\beta} \log Z$ with the latent variable playing the role of microstate, and it interpolates between marginalization ($\beta = 1$: an answer is good if its explanations are *collectively* probable) and minimization ($\beta \to \infty$: an answer is as good as its best explanation) (LeCun et al. 2006). Latent-variable EBMs thereby fold the machinery of Chapter 3 inside the energy function itself, and the variational bound of that chapter — minimize $F(q) = \langle E \rangle_q - T\,S(q)$ over tractable q, with $T = 1/\beta$ — becomes the standard training route when the integral is intractable. The evidence lower bound of variational autoencoders is this construction at $T = 1$, as Chapter 3 already made explicit.

This is also where the energy-based view connects to the present frontier of self-supervised learning. LeCun's proposal for world-model architectures — JEPA, the joint-embedding predictive architecture — is at bottom a latent-variable EBM over *representation* space: energy measures prediction error between embeddings, latent variables absorb what cannot be predicted, and the central design problem is once again preventing collapse, either by contrastive pull-up or by regularizers that keep the embedding distribution spread out (LeCun 2022; Dawid and LeCun 2024). Decades of this subject can be compressed into one sentence: *dig valleys where the data is, and find an affordable way to keep the rest of the landscape high.*

4.9 The modern landscape

A brief tour of where energies appear in current practice, with pointers rather than treatments — the companion list (Bian 2020) tracks the full literature.

Your classifier was an EBM all along. A softmax classifier computes logits $f_\theta(x)[y]$ and normalizes over labels only. Reuse those logits as a joint energy, $E_\theta(x, y) = -f_\theta(x)[y]$, and the *marginal* energy of an input is $E_\theta(x) = -\log \sum_y e^{f_\theta(x)[y]}$ — the free energy of the label ensemble. Training this hidden generative model alongside the classifier improves calibration and robustness (Grathwohl et al. 2020). Even without

retraining, the same marginal energy is a useful *score* for out-of-distribution detection, with higher energy read as more unfamiliar (Liu et al. 2020). (A classifier that was never trained as a generative model does not automatically place OOD inputs high; the energy score is still often a better OOD detector than the raw softmax confidence.) This is the "energy dial" of Chapter 3 read in reverse — the logits every deep classifier already emits are energies in thin disguise.

Generation and its guidance. Diffusion and score-based models (Section 4.5) carry the EBM inheritance in generation, and the additivity of energies (Section 4.2) becomes *composability at inference time*: adding a classifier's gradient to a sampler's score steers generation toward a class (Dhariwal and Nichol 2021); adding hand-built constraint energies composes concepts (Du et al. 2020); in text, a lightweight energy can re-rank or steer a powerful autoregressive proposal — residual EBMs (Deng et al. 2020), constraint-guided decoding (Qin et al. 2022). Discrete spaces lack Langevin's gradient walk, and a lively literature builds gradient-informed proposals for them (Grathwohl et al. 2021). This inference-time story is the subject of Chapter 8; its training-time counterpart, in which the tilt is absorbed into the weights of the pretrained model, is the subject of Chapter 9.

Science. Molecules are the native habitat of energy functions — the word itself came from physics, and here it goes home. Denoising-style score matching over molecular geometries drives conformation generation (Shi et al. 2021); SE(3)-equivariant energies score protein structures end-to-end (Wu et al. 2021). The learned energy (or its score) sits in the same seat as the physical force field it approximates or replaces — Chapter 11 takes up this thread.

4.10 Where this sits in the book

This chapter closes Part I, and it is the load-bearing wall for Part II. Backward: Chapter 2 supplied entropy and the maximum-entropy principle; Chapter 3 derived the Gibbs distribution and identified $-\log Z$ as a free energy; this chapter made the energy learnable and catalogued the ways around the normalizer. Forward, each methods chapter picks up a specific thread:

- **Maximum-entropy learning** (Chapter 5) is the family in which Gibbs distributions are *used* as models and the entropy term is explicit in the inference, whereas this chapter supplied *estimators* for a Gibbs model, in which the entropy is implicit in $\log Z$ (maximum likelihood, contrastive divergence) or replaced by a generalized entropy (score matching, NCE; see Section 5.6.3). The family is defined through the three levels of representation, inference, and learning (Section 5.1), and it contains classical MaxEnt, probabilistic graphical models as an explanatory family (Section 5.2), maximum-entropy and inverse reinforcement learning (Section 5.5), energy-based guidance, and two strands of our own — a Gibbs distribution over coalitions in a *given* cooperative game, from which game-theoretic valuations are derived (Bian et al. 2022) (Section 5.3), and a Gibbs distribution over subsets whose energy is a neural set function *learned* from optimal subsets alone (Ou et al. 2022), a fourth road around the partition function that Table 4.1 does not list (Section 5.4).
- **Minimum-entropy learning** (Chapter 6) trains a model on its own agreement signal; its collapse mode (Section 6.6) is the energy loss's flat-landscape failure in a new costume, and the free-energy decomposition used to analyze it is Equation 4.4 seen from the other side.
- **Minimax-entropy learning** (Chapter 7) asks which *features* an exponential-family model should include — which is precisely the architecture-selection problem for the EBMs of this chapter, posed variationally.
- **Energy-based guidance and refinement** (Chapter 8) keep a pretrained generator frozen and spend inference-time compute tilting it by an energy: guidance is gradient descent on a composed energy, refinement is repeated local descent, and both inherit the annealing wisdom of Chapter 3. The energy may itself have been trained — by the NCE of this chapter, for instance — but it enters only through the sampler.

- **Maximum-entropy post-training** (Chapter 9) performs the same tilt at the learning level: the pre-trained model is fitted to absorb the tilt, or the energy is learned through the model's log-ratio to its reference, which is how RLHF, GRPO, and DPO enter this book.

4.11 Takeaway

An energy-based model is a learnable landscape: inference rolls downhill on it, probability — when needed — is bought from it at the price of a partition function. Training is landscape-shaping, and its non-negotiable rule is that digging valleys under the data must be paired with pulling up elsewhere, lest the landscape collapse into a useless plain. The three classical training strategies are three ways to afford the pull-up: sample it (MCMC-based maximum likelihood), sidestep it by matching slopes (score matching, the road that led to diffusion models), or fold it into a classification game against known noise (NCE). One identity — a relative form of de Bruijn's — reveals them as views of a single discrepancy, and one design question organizes the whole field, from Boltzmann machines to JEPA: *how do you keep the rest of the world high?*

Part II — Methods

5 Maximum-Entropy Learning

5.1 What this chapter means by maximum-entropy learning

Throughout this book the term *maximum-entropy learning* is used in a broad sense: it refers to a family of methods rather than to one algorithm. The family contains methods in which nothing is learned at all, e.g., the cooperative games of Section 5.3, as well as methods in which an energy function is fitted from data, e.g., the set functions of Section 5.4. We therefore need a vocabulary that keeps the two apart. We borrow it from probabilistic graphical models, whose standard textbooks are organized into three parts, *representation*, *inference*, and *learning* (Koller and Friedman 2009), and we read every method through these three levels in turn. In this section we make each level precise, state the definition of the family, and fix the notation used in the rest of the chapter.

Notation. We write $\mathrm{T}(x) = (f_1(x), \dots, f_K(x))$ for the vector of features whose expectations are constrained, i.e., the *sufficient statistics* of the resulting exponential family. The upright sans-serif T is used on purpose: the italic T is reserved for temperature throughout the book, and the two must not be confused. $h(x)$ denotes a *base measure*, i.e., a reference distribution, taken to be uniform unless stated otherwise; $\boldsymbol{\lambda}$ denotes the vector of Lagrange multipliers (the natural parameters), Z the partition function, and $\mathrm{KL}(q \,\|\, p)$ the Kullback–Leibler divergence.

5.1.1 Representation: maximum-entropy models in two forms

The first level asks which model class a method uses. In maximum-entropy learning the answer is the Gibbs distribution, equivalently the exponential family, and it can be written in two forms.

The **constrained form** is the one of Section 2.3, stated here with a base measure (Jaynes 1957; Kullback 1959; Csiszár 1975):

$$\begin{aligned} p^\star &= \arg\min_q \ \mathrm{KL}(q \,\|\, h) \quad \text{s.t.} \quad \mathbb{E}_q[\mathrm{T}(x)] = \boldsymbol{\mu}, \\ p^\star(x) &= \frac{h(x)\, \exp\big(\boldsymbol{\lambda}^\top \mathrm{T}(x)\big)}{Z(\boldsymbol{\lambda})}. \end{aligned} \tag{5.1}$$

When h is uniform, minimizing $\mathrm{KL}(q \,\|\, h)$ is the same as maximizing the entropy $H(q)$, and Equation 5.1 is exactly the Jaynes problem of Section 2.3. For a general h the principle is that of *minimum relative entropy*, known as minimum discrimination information in Kullback's work (Kullback 1959) and as minimum cross-entropy in the axiomatic treatment of Shore and Johnson (1980); the solution $p^\star$ is the *I-projection* of h onto the constraint set (Csiszár 1975).

The **free-energy form** is the one introduced in Chapter 3. Given an energy $E(x)$ and a temperature $T > 0$, define the variational free energy

$$\begin{aligned} \mathcal{F}_T(q; E) &= \mathbb{E}_q[E] \ + \ T\, \mathrm{KL}(q \,\|\, h), \\ p_{E,T} &= \arg\min_q \ \mathcal{F}_T(q; E) \ = \ \frac{h(x)\, e^{-E(x)/T}}{Z}, \qquad \min_q \mathcal{F}_T(q; E) \ = \ -T \log Z. \end{aligned} \tag{5.2}$$

For uniform h this is $\mathbb{E}_q[E] - T\,H(q)$ up to a constant, i.e., the $U - TS$ of Chapter 3, and the identity $\min_q \mathcal{F}_T = -T \log Z$ is the variational representation of the log-partition function (Wainwright and Jordan 2008, sec. 3.6), known in statistical physics as the Gibbs variational principle.

The two forms are equivalent by Lagrangian duality. Setting $E = -\boldsymbol{\lambda}^\top \mathrm{T}$ and $T = 1$ in Equation 5.2 recovers Equation 5.1, and the constraint values and the multipliers are in one-to-one correspondence through $\boldsymbol{\mu} = \nabla_\lambda \log Z(\boldsymbol{\lambda})$, provided that $\boldsymbol{\mu}$ lies in the interior of the marginal polytope (the regularity condition discussed in Section 5.2). One can also read the equivalence geometrically. For the empirical distribution $\hat{p}$ and the maximum-likelihood member $p^\star$ of the exponential family, Csiszár's Pythagorean identity $\mathrm{KL}(\hat{p} \,\|\, h) = \mathrm{KL}(\hat{p} \,\|\, p^\star) + \mathrm{KL}(p^\star \,\|\, h)$ holds (Csiszár 1975), so the maximum-entropy solution and the maximum-likelihood solution are one point approached from two directions (Della Pietra et al. 1997).

Both forms will be used in this chapter, and the choice is a matter of emphasis. We state a model in the constrained form when the point is *what is known*, i.e., the form of T declared by the modeler: clique indicators in a Markov random field, n-gram features in a maximum-entropy classifier (Berger et al. 1996), the invariant statistics of INSET (Section 5.4.3), reward features together with the dynamics in reinforcement learning (Section 5.5). We state it in the free-energy form when the point is *what is computed*: an ELBO, a mean-field iteration, a soft Bellman equation, an annealing schedule. It is worth noting where the knowledge sits. It does not sit in the value of $\boldsymbol{\mu}$, which the data supply, but in the form of T: the content of the theorem behind Equation 5.1 is that the energy is *confined to the span of the declared features*, $E = -\boldsymbol{\lambda}^\top \mathrm{T}$, not that it has an exponential shape, since any strictly positive p can be written as e^{-E} with $E := -\log p$.

A related point is that the two forms constrain different variables. The constrained form restricts the distribution q; an architecture for E_θ restricts the energy, i.e., the dual variable, and the two descriptions coincide exactly only when the admissible energies form a linear space (equality constraints, with free multipliers) or a convex cone (inequality constraints, with nonnegative multipliers). For a neural energy the correspondence holds slice by slice through the last linear layer, and at every stationary point through the likelihood equations of Section 5.1.3, but not for the family as a whole, which is a curved rather than an exponential family (Efron 1975). Symmetry is the exception that lives on both sides at once: invariance of the distribution under a group G is a set of homogeneous linear constraints $\mathbb{E}_q[f \circ g - f] = 0$, the maximum-entropy solution inherits it (Jaynes 1968), and it is equivalent to restricting the features to functions of the maximal invariant of the group (Bloem-Reddy and Teh 2020), which is the route taken in Section 5.4.3.

5.1.2 Inference: free-energy minimization over a tractable family

The second level asks how the distribution is computed or approximated. In maximum-entropy learning, inference is the minimization of the free energy of Equation 5.2 over a tractable family $\mathcal{Q}$ of distributions,

$$q^\star_{\mathcal{Q}}(E, T) = \operatorname*{arg\,min}_{q \in \mathcal{Q}} \mathcal{F}_T(q; E). \tag{5.3}$$

The choice of $\mathcal{Q}$ is what distinguishes the algorithms. If $\mathcal{Q}$ is the set of all distributions, the inference is exact and returns $p_{E,T}$. If $\mathcal{Q}$ is the set of product distributions, Equation 5.3 is mean-field variational inference (Jordan et al. 1999; Wainwright and Jordan 2008), the workhorse of Section 5.3 and Section 5.4. If $\mathcal{Q}$ is the set of trajectory distributions induced by a policy under the true dynamics, it is maximum-entropy reinforcement learning (Section 5.5). Markov chain Monte Carlo replaces the minimization by sampling, and a path in T is an annealing schedule (Chapter 8). The two limits are instructive: as $T \to 0$, Equation 5.3 becomes energy minimization, i.e., the inference of energy-based models in the sense of Section 4.1; as $T \to \infty$, it returns the base measure.

A remark on terminology is in order. We deliberately avoid the phrase "maximum-entropy inference". In Jaynes' writing, *inference* means choosing a distribution from constraints, which is our representation level; in the graphical-model literature it means computing marginals or posteriors, which is the present level. For

the latter we say *variational inference* or *free-energy minimization*.

5.1.3 Learning: fitting the energy through the inference

The third level is active when the energy itself carries parameters, E_θ, which are fitted from data $\mathcal{D}$ through the inference of the second level:

$$\min_\theta \; \mathcal{L}\big(\mathcal{D};\; q_Q^\star(E_\theta, T)\big). \tag{5.4}$$

The supervision in $\mathcal{D}$ can take several forms: i.i.d. samples, as in maximum likelihood; prescribed moments, as in the classical setting; observed *optimal solutions*, as in the optimal subset oracle of Section 5.4; demonstrated trajectories, as in inverse reinforcement learning (Section 5.5); or preferences. In every case the division of labour is the same. The learner chooses the form of T, i.e., an architecture or a symmetry; the data supply the constraint content; and the maximum-entropy principle fixes the distribution. The inner problem Equation 5.3 is strictly convex in q, whereas the outer problem Equation 5.4 is in general non-convex in θ.

Maximum likelihood deserves a separate comment, because it is where the two levels meet. For the Gibbs model $p_\theta \propto h\, e^{-E_\theta}$ (we set $T = 1$ and absorb the scale into E_θ), the gradient of the negative log-likelihood is $\mathbb{E}_{\hat{p}}[\nabla_\theta E_\theta] - \mathbb{E}_{p_\theta}[\nabla_\theta E_\theta]$ (Equation 4.5), so at any stationary point

$$\mathbb{E}_{\hat{p}}\big[\nabla_\theta E_\theta\big] \;=\; \mathbb{E}_{p_\theta}\big[\nabla_\theta E_\theta\big]. \tag{5.5}$$

These are the classical *likelihood equations*, and they say that the model matches the data on the moments of the features $\nabla_\theta E_\theta$. For a linear energy $E = -\boldsymbol{\lambda}^\top \mathrm{T}$ they are exactly the constraints of Equation 5.1 with $\boldsymbol{\mu}$ the empirical moments — this is the maximum-entropy / maximum-likelihood duality — and the Boltzmann machine learning rule, clamped correlations minus free correlations (Ackley et al. 1985), is their best-known early instance in machine learning. For a neural energy the features move with θ, so the constrained form holds only at each stationary point; the general case is better written in the free-energy form, as a saddle-point problem (Section 5.6).

5.1.4 Definition of maximum-entropy learning, and a running example

We can now state the definition used in this chapter.

> **i** Definition: maximum-entropy learning
>
> A method belongs to **maximum-entropy learning** if i) its model class is the Gibbs distribution of Equation 5.1 or Equation 5.2, determined by a declared form of constraint; ii) its inference is a free-energy minimization Equation 5.3 in which the entropy term is explicit and load-bearing, i.e., $T > 0$; and, when the energy is fitted from data, iii) the fitting is carried out through that inference, as in Equation 5.4. A method satisfying i) and ii) is a *maximum-entropy model* with variational inference; a method satisfying i)–iii) is termed a **maximum-entropy learner**. When the Kullback–Leibler divergence in i)–ii) is replaced by the Bregman divergence of another proper scoring rule, the same structure defines a *generalized* entropy (Grünwald and Dawid 2004); we then speak of maximum-entropy learning *in the generalized sense*, and reserve the unqualified term for the Shannon case.

Two consequences of the definition are worth spelling out. First, the energy-based models of Chapter 4 are all maximum-entropy models in the sense of i), and their training criteria differ in how they treat the inner problem that carries the entropy. Maximum likelihood and its approximations — sampled, as in per-

sistent contrastive divergence (Tieleman 2008); truncated, as in contrastive divergence (Hinton 2002) and short-run MCMC (Nijkamp et al. 2019); or amortized, as when a generator is trained to maximize its own entropy (Kumar et al. 2019) — are maximum-entropy learners in the strict sense of iii). Score matching and noise-contrastive estimation replace the log score by another proper scoring rule; each such rule carries its own generalized entropy and Bregman divergence (Grünwald and Dawid 2004), and we call these methods maximum-entropy learners in the generalized sense, naming the entropy in each case (Section 5.6.3). The zero-temperature losses of energy-based learning, e.g., the perceptron and hinge losses, lie on the boundary of the family. Second, methods in which the energy is *given* — a cooperative game (Section 5.3), a reward function in the soft actor–critic (Section 5.5), a composed energy used for guidance (Chapter 8) — are maximum-entropy models with variational inference and no learner. Nothing is learned in them, and yet they belong to the family, because the entropy term does the work. Table 5.1 places the strands of this chapter, and two neighbouring ones, on the three levels.

Table 5.1: The strands of maximum-entropy learning read through the three levels. A dash means that the level is not active: the energy is given rather than learned.

Strand	Representation	Inference	Learning
Classical MaxEnt, log-linear models, CRFs (Section 5.2)	Gibbs with declared features	exact, or message passing	multipliers by maximum likelihood
Undirected graphical models (Section 5.2)	Gibbs over cliques	variational or sampling	parameters by maximum likelihood
Energy-based models by maximum likelihood or CD (Chapter 4)	Gibbs with learned features	MCMC, sampled or truncated	energy by maximum likelihood
Energy-based models by score matching or NCE (Chapter 4)	Gibbs with learned features	none (local divergence) or a fixed noise proposal	energy by a generalized (Bregman) divergence; see Section 5.6.3
Energy-based cooperative games (Section 5.3)	Gibbs over coalitions, one constraint	mean field	— (game given)
Optimal subset oracle (Section 5.4)	Gibbs over 2^V, neural set function	mean field, unrolled	energy from optimal subsets
Maximum-entropy RL with a given reward (Section 5.5)	Gibbs over trajectories	soft Bellman recursion	— (reward given)
Inverse RL (Section 5.5)	Gibbs over trajectories	soft Bellman recursion	reward from demonstrations
Guidance and refinement (Chapter 8)	Gibbs tilt of a pretrained generator	guided or annealed sampling, reranking	— (energy given to the sampler; if trained, by a separate learner)
Maximum-entropy post-training (Chapter 9)	Gibbs tilt of a pretrained model	amortized into the weights	reward from preferences, or through the closed form (DPO)

The status of the temperature follows the same pattern. When the energy is given and no constraint level is prescribed, T is a genuine hyperparameter: the coalition temperature τ of Section 5.3, the entropy coefficient α of the soft actor–critic, the KL coefficient β of policy optimization (Equation 6.2). When the energy is learned and its family is closed under rescaling, T is absorbed into the scale of E_θ and is not identifiable; this is the case of Section 5.4. When a constraint level is prescribed, T is the reciprocal of a multiplier and is determined by that level, as in the canonical ensemble of Chapter 3.

A running example. The simplest maximum-entropy learner is the maximum-entropy classifier of classical natural language processing (Berger et al. 1996), i.e., multinomial logistic regression. The features are

indicator functions $f_k(x, y)$, e.g., "word k occurs in document x and the label y is *sports*". The constrained form asks for the conditional distribution closest to uniform that matches the empirical expectations of all f_k; the solution is $p(y \mid x) \propto \exp\left(\lambda^\top \mathrm{T}(x, y)\right)$, and fitting λ by maximum likelihood is the dual problem, so the likelihood equations Equation 5.5 are precisely the moment constraints. In the free-energy form, the negative log-likelihood of one example is $-\lambda^\top \mathrm{T}(x, y) + \log \sum_{y'} \exp\left(\lambda^\top \mathrm{T}(x, y')\right)$, i.e., the energy of the observed label plus the free energy of the label ensemble. Inference is exact because the label set is small, and the temperature is absorbed into the scale of λ. Levels i)–iii) are all present and all three are trivial; the rest of this chapter is about what happens when the space is the power set 2^V or the set of trajectories, and the inference is no longer exact.

5.2 Graphical models as an explanatory family

Probabilistic graphical models (PGMs) organize a joint $p(x)$ by a graph that *encodes* a set of conditional independencies (Pearl 1988; Koller and Friedman 2009). The encoding is not the same in the two standard realizations. On an undirected graph, a missing edge is pairwise independence given all other variables (and, when p is strictly positive, the pairwise, local, and global Markov properties coincide). On a directed acyclic graph the criterion is *d-separation*. A missing arrow still carries an independence — the local Markov property says each variable is independent of its non-descendants given its parents — but not the *same* independence as a missing undirected edge: conditioning on all the remaining variables can *open* a path through a collider, so the MRF's pairwise reading has no directed analogue (Pearl 1988). The field is older than the modern energy-based revival of Chapter 4, larger than this chapter, and still the working language for structured uncertainty in statistics and in much of classical machine learning — vision, speech, computational biology, and expert systems among them.

An **undirected** model, or Markov random field (MRF), scores *compatibilities*. On a finite graph G, a strictly positive distribution is Markov with respect to G if and only if it is a Gibbs distribution that factors over the cliques of G (Hammersley–Clifford; the published proof usually cited is Besag (1974)). In energy language that is already the definition of Chapter 4, with the extra discipline that E is a sum of local terms:

$$p(x) \;=\; \frac{1}{Z}\,\exp\Bigl(-\sum_{C \in \mathrm{cliques}(G)} E_C(x_C)\Bigr).$$

(The sum may equally be taken over maximal cliques: smaller clique terms can be absorbed.) The Gibbs $\Rightarrow$ Markov direction is elementary and needs no positivity; it is Markov $\Rightarrow$ Gibbs that does. Without it the implication fails: Moussouris' four-cycle example is a distribution that is Markov with respect to G but admits no Gibbs factorization over the cliques of G (Moussouris 1974).

A **directed** model, or Bayesian network, tells a *generative* story. If pa_i are the parents of coordinate i in a directed acyclic graph,

$$p(x) \;=\; \prod_i p(x_i \mid \mathrm{pa}_i) \tag{5.6}$$

(Pearl 1988). The arrows are often read as explanation or cause; the undirected edges of an MRF are not. Pearl's own contrast is that a Bayes net is a knowledge base of directed influences, while a Markov network is a system of symmetric associations. That reading is a stretch beyond what the joint distribution alone licenses: from observational data a DAG is identifiable only up to its **Markov equivalence class** — every DAG with the same skeleton and the same v-structures — so "the graph is the explanation" should be heard as a claim about which independencies the graph asserts, not about which arrow points which way.

This book treats the pair as an **explanatory family**, in a local sense that is not a standard name in the literature. The graph is the explanation: it says which independencies the joint is required to satisfy, and the factorization

is what makes the model intelligible rather than a mere scoring rule. That reading is closest to Pearl's directed case. It is more of a stretch for MRFs, whose native language is compatibility rather than cause. Both, however, explain the joint by a sparse independence hypothesis, and that is the sense in which they belong together here.

How much of this is classical maximum entropy? Enough to be worth saying, not enough to be a slogan. The maximum-entropy distribution of Section 2.3 that matches prescribed expectations of clique-supported features is an exponential-family MRF: the Lagrange multipliers are the feature weights, and the solution is the Gibbs form above (with $E_C = -\sum_k \lambda_k f_k$ over the features supported on C; the sign is the usual energy convention) (Jaynes 1957; Della Pietra et al. 1997; Wainwright and Jordan 2008). On a finite discrete state space every strictly positive MRF arises this way, because the clique-configuration indicators are a sufficient statistic. The modern synthesis of Wainwright and Jordan writes undirected graphical models, exponential families, and entropy duality as one subject. Feature induction for random fields (Della Pietra et al. 1997) and the maximum-entropy models of classical NLP (Berger et al. 1996) are the same *exponential-family* construction; they are MRFs only when the features have local graphical support — Stephen Della Pietra, Vincent Della Pietra, and John Lafferty are explicit that their fields may be non-Markovian. Conditional random fields (Lafferty et al. 2001) take the construction over to $p(y \mid x)$ and became, for a decade, the default of structured prediction. None of that is controversial.

The duality itself has a real regularity condition, not just a name: the program $\max_p H(p)$ subject to $\mathbb{E}_p[f] = \mu$ is dual to maximum likelihood in the exponential family exactly when μ sits in the *interior* of the marginal polytope; on the boundary the MLE fails to exist, and the maximum-entropy optimum is attained only in the closure of the exponential family (Wainwright and Jordan 2008, sec. 3.4).

What is *not* settled — and should not be written as if it were — is the claim that "PGM *is* classical maximum entropy." Three qualifications matter.

1. The PGM literature organizes itself around *representation, inference, and learning* — the triad we borrowed in Section 5.1 — not around Jaynes. Standard treatments (Pearl 1988; Koller and Friedman 2009) may discuss log-linear Markov networks, whose maximum-likelihood dual *is* a maximum-entropy problem, without taking that dual as the field's founding slogan. The maxent identification is a precise reading of the exponential-family parameterization, not the community's self-description.
2. The tight theorem is undirected, finite, and strictly positive. Directed models are a different organizing principle: Equation 5.6 is not a Jaynes problem. A Bayes net can be *moralized* into an MRF on the graph obtained by marrying co-parents and dropping directions; the resulting undirected model is an **I-map** (every independence it asserts by missing edges does hold in p), but it need not be a *perfect* map — moralization can lose independencies that the DAG had (v-structures are the usual example, since marrying the parents removes exactly the edge whose absence encoded that independence). The rewriting $E(x) = -\sum_i \log p(x_i \mid \mathrm{pa}_i)$ is an energy in the sense of Chapter 4, and in this case $Z = 1$ already, so the network sits inside the energy-based tent as a *normalized* model. That does not make it a maximum-entropy model, and the precise reason is the dimensional cousin of the loss of independencies above: a Jaynes problem with *linear* constraints has a solution confined to a *linear* exponential family. Fully observed discrete **undirected** graphical models are exactly such linear exponential families; fully observed discrete **directed** (DAG) models are in general only a *curved* exponential family (Geiger et al. 2001), with equality precisely when the DAG has no immoralities (v-structures) — that is, when its Markov equivalence class contains a decomposable (chordal) undirected model. Away from finite discrete variables the identification loses its grip for a further, dimensional reason: the clique potentials now range over an infinite-dimensional function space, so the MRFs on a fixed graph no longer form a *finite-dimensional* exponential family. (A single Gibbs distribution is always trivially exponential-family — the claim has content only for the model class.) Gaussian MRFs remain a finite-dimensional exponential family; potentials given by mixtures do not.
3. Even on the undirected side, maxent fixes the *distribution given the constraints*. Choosing the graph, or which clique features to include, is a *feature-selection* question with more than one answer — the

greedy KL-gain feature induction of Della Pietra, Della Pietra, and Lafferty (Della Pietra et al. 1997), and the minimax entropy of Zhu, Wu, and Mumford (Zhu et al. 1997), the subject of Chapter 7 — and neither is part of Jaynes' original programme.

So the accurate statement, and the one this chapter will use, is narrower than a slogan and stronger than a metaphor: **a strictly positive exponential-family MRF on a finite graph is a classical maximum-entropy model with structured constraints**; Bayesian networks are an explanatory sibling that can be rewritten as (already normalized) energies but should not be renamed maxent; the PGM field as a whole is larger than either identification, and still doing work that this book will not reproduce. A future revision should include full derivations of both the Hammersley–Clifford theorem and the moralization construction. For now, this chapter simply points to them as essential references.

5.3 A worked example: energy-based cooperative games

The variational machinery of the previous section is not only for graphs. The same mean-field relaxation that Wainwright and Jordan use to approximate a graphical model's log-partition function applies verbatim to a Gibbs distribution over coalitions, and what it produces there turns out to be a familiar object.

As one concrete instance of the programme above, consider a **valuation problem** in machine learning: how much does a feature, data point, or player contribute to an outcome? The classical answer is the **Shapley value** from cooperative game theory, but it is one particular choice among many.

In *Energy-Based Learning for Cooperative Games* (Bian et al. 2022), the valuation problem is recast through an energy / maximum-entropy lens. A cooperative game with characteristic function $v(S)$ over coalitions $S \subseteq N$ is turned into a distribution over coalitions by a single Jaynes constraint: match a prescribed expected payoff and nothing else. The solution is the Gibbs distribution

$$p(S) \propto \exp\big(v(S)/\tau\big),$$

with energy $-v(S)$. Here τ is a temperature in exactly the sense of Chapter 3: as $\tau \to 0$ the mass concentrates on the highest-value coalitions, while as $\tau \to \infty$ it flattens towards uniform. In the strict Jaynes reading $\tau = 1/\lambda$ is not a free knob — it is pinned by whichever expected payoff the constraint matches — so sweeping τ below is the same move as sweeping that matched level.

Player valuations are then read from a mean-field approximation to p. The payoff term of that mean-field ELBO is exactly Owen's **multilinear extension** of the game (Owen 1972, 1975). What the variational reading adds is the entropy term — which turns the multilinear extension from an interpolation device into an ELBO with a temperature — and the trajectory beyond one step: further mean-field iterations define a family of variational valuations whose fixed point — the minimizer of the mean-field KL divergence, i.e. the member with the best conceivable decoupling error — is the **Variational Index**. Shapley is therefore not a special case of the maximum-entropy *distribution*, but a one-step mean-field reading of it.

5.3.1 Why this is useful

- **Principled interpretation.** Valuations become variational parameters of a mean-field approximation to a maximum-entropy distribution over coalitions, rather than isolated axiomatic scores.
- **A family, not a point.** The temperature τ and the number of mean-field steps trace a spectrum of valuations, with Shapley and Banzhaf as one-step readings rather than the only options.
- **Bridges to physics.** The same Gibbs machinery is the one that carried the Ising energy into neural networks in the first place, by way of the Hopfield network (Hopfield 1982) and the Boltzmann machine (Ackley et al. 1985); see Chapter 3.

5.4 Discrete maximum-entropy learning: set functions under the optimal subset oracle

Everything in Section 5.3 presupposed the game. The characteristic function $v(S)$ was given, the Gibbs distribution over coalitions was *derived* from it by one Jaynes constraint, and mean-field inference on that distribution was an *interpretation* device — a way to read valuations off a distribution nobody had to fit. Nothing was learned. This section drops that assumption. The set function is unknown, the only supervision is which subset turned out to be optimal, and the same Gibbs-plus-mean-field machinery becomes a *training loop* on the power set 2^V — an exponentially large discrete space on which none of the Langevin machinery of Chapter 4 applies. This section most concretely embodies the concept of maximum-entropy *learning* as defined in Section 5.1: here, the energy function is not given a priori but is instead learned directly on a discrete set space, so the strand is a maximum-entropy *learner* in the sense of Section 5.1.4. The published line runs through EquiVSet (Ou et al. 2022) and INSET (Xie, Bian, et al. 2024), and has since been taken up by others (Xie, Wang, et al. 2024; Özcan et al. 2025); a further step of our own is still a preprint and is noted only at the end. We present the line as a paradigm of **discrete maximum-entropy learning**, and in Section 5.4.4 say what it adds to the programme and what it leaves open.

5.4.1 The optimal subset oracle

A **set function** $F(S; V)$ assigns a utility to every subset S of a ground set V: the products a customer would want among those displayed, the anomalous images in a batch, the compounds worth advancing from a screening library. The object of interest is the maximizer,

$$S^* \;=\; \arg\max_{S \subseteq V} F(S; V), \tag{5.7}$$

and the question is how to learn F_θ from data. The classical setting is the **function-value (FV) oracle**: pairs $(S_i, F(S_i))$ on a fixed ground set, from which F_θ is fit by regression. It is expensive — one utility label per subset, on a space of $2^{|V|}$ subsets — and the learnability theory built on it carries strong inapproximability results even for submodular functions (Balcan and Harvey 2018). The **optimal subset (OS) oracle** is weaker and far more common: pairs (V_i, S_i^*), a ground set together with the subset that was actually chosen, and no utility values at all. The two are not comparable in general. Max-cut has a trivial FV oracle and an NP-hard OS oracle; a recommender log has a natural OS oracle — what the customer selected from what was shown — and no FV oracle, because nobody knows the customer's utility.

Learning under the OS oracle was posed by Ou et al. (2022) as maximum likelihood over a distribution on subsets whose mass grows with the utility:

$$\max_\theta \; \mathbb{E}_{(V,S^*)}\big[\log p_\theta(S^* \mid V)\big], \qquad p_\theta(S \mid V) \;=\; \frac{\exp\big(F_\theta(S; V)\big)}{\sum_{S' \subseteq V} \exp\big(F_\theta(S'; V)\big)}. \tag{5.8}$$

This is a conditional energy-based model in the sense of Section 4.1 — input $x = V$, answer $y = S$, answer space $\mathcal{Y} = 2^V$, energy $E_\theta(S; V) = -F_\theta(S; V)$ — and the Gibbs form is not a convenience but the maximum-entropy choice. Among all distributions on 2^V that match a prescribed expected utility $\mathbb{E}_p[F(S)] = \mu$, the entropy maximizer is unique and is $p(S) \propto \exp(\lambda F(S))$ (Ou et al. 2022, Thm. 1); the argument is Jaynes' own (Jaynes 1957). It is the single-constraint problem of Chapter 3 and exactly the coalition distribution of Section 5.3 with v replaced by F_θ; the multiplier $\lambda = 1/\tau$ is absorbed into the scale of F_θ, as β was absorbed into E in Chapter 4. The paper calls the property *minimum prior*: the model assumes nothing about the subset beyond what its utility says. The contrast is with the probabilistic greedy model (Tschiatschek et al. 2018), which builds $p(S)$ by summing an order-dependent, autoregressive greedy-sampling distribution over all orderings of S — importing an inductive bias the data never asked for, and at factorial cost.

It is worth seeing what the likelihood asks for. In the normalization of Equation 4.4, with an explicit inverse temperature, the negative log-likelihood of one pair is $-F_\theta(S^*;V)+\frac{1}{\beta}\log\sum_S e^{\beta F_\theta(S;V)}$. As $\beta\to\infty$ the second term tends to $\max_S F_\theta(S;V)$, and the loss to $\max_S F_\theta(S;V)-F_\theta(S^*;V)$: the perceptron loss of Section 4.3 on the answer space 2^V, which vanishes exactly when S^* is a maximizer — Equation 5.7 read as a loss. Maximum likelihood under the OS oracle is therefore the finite-temperature relaxation of "make the observed subset the arg max." It buys, at the price of a partition function, a smooth objective whose pull-up (Equation 4.5) is spread over every rival subset in proportion to the probability the model currently assigns it. In the taxonomy of Section 4.1 this is the *decision* question answered through the *probability* question, and the detour is deliberate.

The same objective can be read in the constrained form of Section 5.1.1. What the OS oracle tells us about the unknown set function is a collection of *optimality constraints*, $F(S^*;V)\geq F(S;V)$ for all $S\subseteq V$, i.e., one polyhedral cone per instance, and the set functions satisfying them are exactly those under which the observed subset is optimal. Inferring an objective from observed optimal solutions is the classical problem of *inverse optimization* (Burton and Toint 1992; Ahuja and Orlin 2001; Heuberger 2004), and the zero-temperature loss above is the structured perceptron of structured prediction (Collins 2002; Taskar et al. 2003; Tsochantaridis et al. 2005). Seen this way, the setting is not new in kind, and it is worth being explicit about its nearest relatives. Max-margin learning of submodular set functions from reference summaries (Lin and Bilmes 2012; Sipos et al. 2012) is the $T\to 0$ version on a fixed function class; structured prediction energy networks (SPENs) place a deep energy on a structured output and train it through unrolled gradient-based inference (Belanger and McCallum 2016; Belanger et al. 2017). What EquiVSet adds is the combination: a permutation-invariant neural set function on a variable ground set, the maximum-entropy justification of the set mass function, amortized mean-field inference on the power set, and a copula that restores the correlations which mean field discards. The finite temperature is what turns the optimality constraints into a likelihood, and the likelihood into an objective that a gradient method can train.

Two further requirements shape the architecture. The set mass function must be **permutation invariant**, and it must accept a **varying ground set**, since every instance arrives with its own V. Both are met by parameterizing F_θ as a DeepSets network, $F(S)=\rho\big(\sum_{s\in S}\kappa(s)\big)$, a form that is universal for permutation-invariant set functions under mild conditions (Zaheer et al. 2017). With minimum prior and **scalability** — training in polynomial rather than factorial time — these are the four desiderata of Ou et al. (2022), and scalability is where the difficulty lives.

5.4.2 Learning through mean-field inference: EquiVSet

The difficulty is the one this book keeps meeting. The pull-up term of Equation 5.8 is an expectation under $p_\theta(\cdot\mid V)$ over $2^{|V|}$ subsets; there is no continuous S on which to run Langevin dynamics, and running a discrete sampler (Grathwohl et al. 2021) or contrastive divergence inside the training loop is costly and, on large ground sets, unstable (Ou et al. 2022). EquiVSet takes a road absent from the table of Section 4.7: *approximate the model by a tractable family, and train through the approximation.*

The tractable family is mean field — the product of independent Bernoullis $q(S;\psi)=\prod_{i\in S}\psi_i\prod_{i\notin S}(1-\psi_i)$ with $\psi\in[0,1]^{|V|}$, ψ_i the probability that element i belongs to the optimal subset. Minimizing $\mathrm{KL}(q_\psi\,\|\,p_\theta)$ is maximizing the evidence lower bound,

$$\mathrm{ELBO}(\psi)\;=\;f_{\mathrm{mt}}^{F_\theta}(\psi)+H(q_\psi),\qquad f_{\mathrm{mt}}^{F_\theta}(\psi)\;:=\;\sum_{S\subseteq V}F_\theta(S;V)\prod_{i\in S}\psi_i\prod_{i\notin S}(1-\psi_i),\tag{5.9}$$

and the payoff term is again the **multilinear extension** (Owen 1972; Calinescu et al. 2007) — the object that carried the valuations of Section 5.3, now built from a learned F_θ rather than a given v. Read through Chapter 3, Equation 5.9 is minus a variational free energy at unit temperature, $\mathcal{F}(\psi)=\langle E_\theta\rangle_{q_\psi}-H(q_\psi)$, with the multilinear extension of the energy as the expected energy. The stationarity condition of the bound is the

mean-field fixed point $\psi_i = \sigma\big(\partial_i f_{\mathrm{mt}}(\psi)\big)$, and the partial derivative has a form that makes it estimable:

$$\partial_i f_{\mathrm{mt}}^{F_\theta}(\psi) \;=\; \mathbb{E}_{S\sim q(\cdot;\psi_{-i})}\big[F_\theta(S\cup\{i\};V) - F_\theta(S;V)\big], \tag{5.10}$$

the expected *marginal gain* of i over a random subset of the other elements, drawn from q with ψ_i set to zero. That is the quantity read off as a player's value in Section 5.3; here it is estimated by Monte Carlo and pushed through a sigmoid, and K such updates, run in parallel over coordinates, define a differentiable map $\psi^* = \mathrm{MFVI}(\psi^{(0)}, V, K)$.

Learning then never touches $\log Z$. The parameters are trained by the cross-entropy of the observed optimal subset under the marginals the inference algorithm produces,

$$\mathcal{L}(\theta) \;=\; -\sum_{j\in S^*} \log \psi_j^*(\theta) \;-\; \sum_{j\in V\setminus S^*} \log\big(1-\psi_j^*(\theta)\big), \tag{5.11}$$

with gradients back-propagated through the K fixed-point steps. This is Domke's *marginal-based loss* (Domke 2013) — the training rule behind differentiable mean field for dense random fields and CRF-as-RNN (Krähenbühl and Koltun 2013; Zheng et al. 2015) — carried over to a deep energy on a variable ground set. In the terminology of the broader literature it is end-to-end training through *unrolled inference*, also termed deep unfolding: the iterations of an inference algorithm are treated as the layers of a network, and the loss is back-propagated through them (Gregor and LeCun 2010; Stoyanov et al. 2011; Domke 2012; Hershey et al. 2014; Belanger et al. 2017). Two properties should be stated plainly. The loss is not a bound on the log-likelihood of Equation 5.8: what is fit is the output of an approximate inference algorithm, which is a virtue (inference error is accounted for at training time) and a limitation (the link to the maximum-entropy distribution runs only through the fixed point). And the mean-field family cannot represent interactions between elements — the decoupling error of Section 5.3 returns as a modelling limit.

EquiVSet addresses cost and decoupling together. A permutation-*equivariant* recognition network $\psi^{(0)} = \mathrm{EquiNet}(V;\phi)$ amortizes inference by proposing the initial marginals; it is trained to maximize Equation 5.9 (through a Gumbel–softmax relaxation of the discrete samples), optionally with a Gaussian copula that correlates the Bernoullis. A single mean-field step from that initialization then carries the gradient to θ, and the two networks are trained cooperatively (Xie et al. 2018). At test time the same pipeline — proposal, one step, top-k rounding — returns a subset. A telling ablation: taking *more* mean-field steps from the copula initialization hurts, because the iteration converges to the plain mean-field fixed point and washes out the correlations the copula supplied.

On product recommendation (Amazon baby-registry categories), set anomaly detection (Double MNIST, CelebA), and compound selection from PDBBind and BindingDB, EquiVSet outperformed the probabilistic greedy model by large margins and, more tellingly, outperformed a DeepSets network trained directly on membership labels — the comparison that isolates the value of learning an explicit set function. The last application is the one that matters for Part III. A virtual-screening pipeline filters by bioactivity, diversity, and ADME in stages whose intermediate labels are costly or proprietary, but the final selected library is observed. That is an OS oracle and nothing else (Chapter 11).

5.4.3 Which statistics may the energy depend on? INSET

EquiVSet's set function looked only at S. But a subset's utility depends on the ground set it was drawn from: an image is anomalous *relative to* its batch, a bundle is chosen *from* what was displayed. INSET (Xie, Bian, et al. 2024) asks what an energy $F_\theta(S,V)$ is permitted to depend on, and derives the answer from symmetry. The conditional law the network must represent — that of the membership probabilities $Y \in [0,1]^{|V|}$ given (S,V) — should be invariant under permutations of S and under the nested permutations of V viewed as a set of disjoint subsets (a wreath-product group). The theorem, proved through the noise-outsourcing lemma

of probabilistic symmetry (Bloem-Reddy and Teh 2020), is that $P(Y \mid S, V)$ has this invariance if and only if $Y \overset{\text{a.s.}}{=} f\big(\xi, M(S,V)\big)$ for independent noise ξ and a *maximal invariant* M — a statistic that is constant on the orbits of the group and separates them. Such an M is then an **adequate statistic**: conditioning on it loses nothing about Y. The invariant sufficient representations of S and of V may be built separately and combined, and each is approximated by a DeepSets sum, which gives the architecture

$$F_\theta(S,V) \;=\; \sigma\Big(\theta_1 \sum_{x\in S} \phi(x) \;+\; \theta_2 \sum_{x\in V} \phi(x)\Big). \tag{5.12}$$

The move is the one Section 5.2 attributed to Jaynes, with invariance in place of moment constraints: the constraints fix *which statistics* the distribution may depend on — there, clique indicators; here, the maximal invariant of the symmetry group — and leave the function of those statistics free for the data to determine. In the vocabulary of Section 5.1.1, invariance is the one structural constraint that can be stated on either side of the duality, as linear equality constraints on the distribution or as the requirement that the features factor through the maximal invariant, and INSET takes the second route. The two-branch design improved on EquiVSet across every benchmark (on Double MNIST the mean Jaccard index rose from 0.575 to 0.707), converged in fewer epochs, and an ablation shows the gain is not bought with parameter count.

5.4.4 What the strand adds to the maximum-entropy programme

Five points, and then what is open.

1. **The feature is learned, not the multipliers.** Classical maximum entropy fixes the distribution given the features; Section 5.3 read valuations off that distribution for a given game. Here the confinement of Section 5.1.1 collapses to a single feature — the utility itself — and that feature is a neural network trained from data. In the vocabulary of Section 5.1.4 this is a maximum-entropy *learner* on a discrete domain, with the coalition temperature τ of Section 5.3 reappearing as the learned scale of F_θ.
2. **Learning from observed optimal solutions.** The OS oracle is neither function values nor samples from the model; it is arg-max points of the energy, i.e., the supervision of inverse optimization. Maximum likelihood is the finite-temperature relaxation of the perceptron condition "the observed subset is a maximizer," and the pull-up over 2^V is the price of smoothing it. The data constrain F only through its maximizers, so what is learned is a landscape with the right valleys, not a calibrated utility: with noiseless optimal subsets the likelihood alone would send the scale of F_θ to infinity, as in separable logistic regression, and regularization and early stopping are what pin it.
3. **A fourth road around the partition function.** Table 4.1 listed three ways to afford the pull-up: sample it, differentiate it away, cancel it against noise. EquiVSet's road — relax the discrete distribution to a tractable family, unroll the inference, and train the marginals it outputs — is a fourth, with its own characteristic failure: the decoupling error of the family, and a loss that is not a likelihood bound. The bridge between 2^V and $[0,1]^{|V|}$ is always the multilinear extension, which is why continuous DR-submodular maximization (Bian et al. 2019; Hassani et al. 2017) is the right theory for the guarantees. In Section 5.3 mean-field inference was an interpretation device; here it is the training loop.
4. **Constraints from symmetry.** INSET shows that the statistics an energy may depend on can be derived from the invariances of the conditional distribution — sufficiency and adequacy in the classical sense, reached by the same route as Jaynes' clique features in Section 5.2. What the data are left to learn is the function of those statistics.
5. **The free energy.** The ELBO that drives these methods is $-\mathcal{F}(\psi)$ at $T = 1$ (Chapter 3), so variational inference on the learned energy is the minimization of $U - TS$ that runs beneath the whole book; and inference in the trained model — iterate toward the fixed point, then round — is energy-based reasoning on a discrete landscape in the sense of Chapter 8.

What is open is at least as interesting. A discrete EBM trainer under OS supervision whose objective *bounds*

the likelihood does not yet exist; EquiVSet says so in as many words, and the candidates are gradient-informed discrete samplers for the pull-up (Grathwohl et al. 2021) and contrastive objectives against structured negatives (Section 4.6). Beyond mean field, structured variational families — the copula is a patch — would let the multi-step trajectory of Section 5.3 be used rather than avoided; that more steps *hurt* EquiVSet is a symptom worth diagnosing. Priors on the set function — diminishing returns (Bilmes and Bai 2017), bounded submodularity ratio (Bian et al. 2017) — would let the guarantees of continuous DR-submodular maximization apply to the learned F_θ rather than be assumed of it. And the setting generalizes: multisets and integer lattices already have provable mean-field inference (Sahin et al. 2020); sequences, permutations, and graphs are the natural next domains for the same recipe — a discrete energy learned from its own optimizers. That recipe is what we mean by discrete maximum-entropy learning, and compound selection is only its first scientific case (Chapter 11).

The most immediate bottleneck is the Monte Carlo estimate in Equation 5.10, repeated at every unrolled step. A preprint of ours (Shi et al. 2026) removes it by learning the continuous relaxation directly — a neural set-function extension (Karalias et al. 2022) climbed by gradient ascent in place of the sampled fixed point — with a constant-factor guarantee under weak DR-submodularity and a reading of the ELBO as a free energy whose temperature is learned rather than fixed at one. It is not yet published, and we record it here as the direction the strand is taking rather than as a result.

5.5 Maximum-entropy reinforcement learning: energies over trajectories

Reinforcement learning (RL) supplies the largest instance of the family outside our own work, and the one that most closely resembles Section 5.4. We do not survey the field here. The aim of this section is to place maximum-entropy RL on the three levels of Section 5.1, and to show that inverse RL is the optimal subset oracle with trajectories in place of subsets.

5.5.1 A Gibbs distribution over trajectories

Consider a Markov decision process with states s, actions a, transition kernel $p(s' \mid s, a)$, and reward $r(s, a)$, and let $\tau = (s_1, a_1, \dots, s_H, a_H)$ denote a trajectory. The representation level is a Gibbs distribution over trajectories,

$$\begin{aligned} p(\tau) &= \frac{1}{Z}\, h(\tau)\, \exp\Big(\frac{1}{\alpha}\sum_{t=1}^{H} r(s_t, a_t)\Big), \\ h(\tau) &= p(s_1)\prod_{t=1}^{H-1} p(s_{t+1} \mid s_t, a_t)\prod_{t=1}^{H} u(a_t), \end{aligned} \tag{5.13}$$

with energy $E(\tau) = -\sum_t r(s_t, a_t)$, temperature α, and a base measure that is not uniform: h is the law of the trajectories generated by the dynamics under a uniform action distribution u, i.e., the *passive dynamics* of the system in the terminology of KL control (Todorov 2006, 2009; Kappen 2005). The base measure is what keeps the model from placing mass on physically impossible transitions. For deterministic dynamics h is uniform over the feasible trajectories, and Equation 5.13 is exactly the trajectory distribution of maximum-entropy inverse RL (Ziebart et al. 2008); for stochastic dynamics the same paper carries an explicit transition factor together with an approximation, and the principled repair is the *maximum causal entropy* discussed next (Ziebart et al. 2010).

5.5.2 Soft Bellman equations as free energies

The inference level restricts $\mathcal{Q}$ to the distributions an agent can actually realize, namely those induced by a policy under the true dynamics, $q_\pi(\tau) = p(s_1)\prod_t p(s_{t+1} \mid s_t, a_t)\,\pi(a_t \mid s_t)$. This restriction has a pleasant consequence. In $\mathrm{KL}(q_\pi \,\|\, h)$ the transition factors cancel, and what remains is $\mathbb{E}_{q_\pi}\big[\sum_t \log \pi(a_t \mid s_t) - \log u(a_t)\big]$, i.e., minus the *causal entropy* of the policy (Ziebart et al. 2010) up to a constant. Minimizing the free energy Equation 5.2 over q_π is therefore

$$\max_\pi \; \mathbb{E}_{q_\pi}\Big[\sum_t r(s_t, a_t) \;+\; \alpha H\big(\pi(\cdot \mid s_t)\big)\Big], \tag{5.14}$$

which is the objective of maximum-entropy RL (Ziebart et al. 2010; Fox et al. 2016; Haarnoja et al. 2017). Its optimality conditions are the *soft* Bellman equations,

$$\begin{aligned} Q(s,a) &= r(s,a) + \mathbb{E}_{s' \sim p(\cdot \mid s,a)}\big[V(s')\big], \\ V(s) &= \alpha \log \sum_a \exp\big(Q(s,a)/\alpha\big), \\ \pi^\star(a \mid s) &= \exp\big((Q(s,a) - V(s))/\alpha\big), \end{aligned} \tag{5.15}$$

and the soft value $V(s)$ is a log-partition function, i.e., a free energy at temperature α, one per state. As $\alpha \to 0$ one recovers the hard Bellman equations and the greedy policy, exactly as $T \to 0$ recovers energy minimization in Section 5.1.2. The reading of RL as inference in a graphical model, in which the policy-induced family plays the role of $\mathcal{Q}$, goes back to Todorov, Toussaint, and Kappen (Todorov 2009; Toussaint 2009; Kappen et al. 2012) and is reviewed by Levine (2018); the theory of entropy-regularized Markov decision processes states the same facts in the language of convex regularization (Neu et al. 2017; Geist et al. 2019). When the reward is given, as in soft Q-learning and the soft actor–critic (Haarnoja et al. 2017, 2018), this is the whole story: a maximum-entropy model with variational inference and no learner, i.e., the same class as the cooperative games of Section 5.3, with α a genuine dial. The KL-regularized policy optimization of Equation 6.1 is the same construction on a single decision, with a reference policy as the base measure and the KL coefficient as the temperature; imposing the KL as a constraint instead, as trust-region methods do (Schulman et al. 2015), is the constrained form of the same problem. Chapter 9 develops this reading for the post-training of pretrained models.

5.5.3 Inverse reinforcement learning: the optimal subset oracle on trajectories

The learning level becomes active when the reward is unknown and one observes expert trajectories τ^* instead, which is the problem of inverse RL (Ng and Russell 2000). Maximum-entropy inverse RL fits a parametric reward r_θ by maximum likelihood under Equation 5.13. For a linear reward $r_\theta(s,a) = \theta^\top \phi(s,a)$, the likelihood equations Equation 5.5 read

$$\mathbb{E}_{\text{expert}}\Big[\sum_t \phi(s_t, a_t)\Big] \;=\; \mathbb{E}_{p_\theta}\Big[\sum_t \phi(s_t, a_t)\Big], \tag{5.16}$$

i.e., the learned reward makes the model reproduce the expert's expected feature counts. This is the *feature-expectation matching* that apprenticeship learning had imposed directly (Abbeel and Ng 2004), and the model expectation is computed by the soft Bellman recursion of Equation 5.15, i.e., the inference of the second level runs inside the training loop. At zero temperature the same objective becomes a max-margin structured prediction problem over policies, termed maximum margin planning (Ratliff et al. 2006).

The parallel with Section 5.4 is now line by line, and Table 5.2 records it. In both cases the supervision consists of *observed optimal solutions* of an unknown objective: a chosen subset in one case, a demonstrated

trajectory in the other. In both cases the finite temperature turns the optimality constraints into a likelihood, the zero-temperature limit is a perceptron or max-margin loss, the inner problem is a free-energy minimization over a tractable family (mean field on the multilinear extension; soft value iteration over policies), and the learning signal is the difference between an expert statistic and a model statistic. The difference lies in the base measure and in the family: the power set has no dynamics and h is uniform, whereas trajectories are constrained by the transition kernel, which is why the causal entropy replaces the Bernoulli entropies of Equation 5.9. We regard this parallel as the strongest external evidence that "learning an energy from its own optimizers" is a definition and not the description of one paper.

Table 5.2: The optimal subset oracle and maximum-entropy inverse RL, read through the same three levels.

	Optimal subset oracle (Section 5.4)	Maximum-entropy inverse RL
Observed optimal solution	subset $S^* \subseteq V$	trajectory τ^*
Energy to be learned	$-F_\theta(S; V)$	$-\sum_t r_\theta(s_t, a_t)$
Base measure h	uniform on 2^V	passive dynamics
Constrained reading	optimality constraints (inverse optimization)	feature-expectation matching (Abbeel and Ng 2004)
Zero-temperature limit	structured perceptron; max-margin submodular learning (Lin and Bilmes 2012; Sipos et al. 2012)	maximum margin planning (Ratliff et al. 2006)
Inner problem	mean field on the multilinear extension	soft value iteration
Entropy that does the work	Bernoulli entropies of q_ψ	causal entropy of π
Learning signal	marginal-based loss through unrolled steps	expert minus model feature counts

5.6 What the family adds, and where it stops

Having placed the strands, we close the chapter with three general facts that the definition of Section 5.1 makes available, and with a statement of what the definition excludes.

5.6.1 Maximum likelihood as a saddle-point problem

For the Gibbs model $p_\theta = h\, e^{-E_\theta/T}/Z(\theta)$, the variational representation $-T \log Z(\theta) = \min_q \mathcal{F}_T(q; E_\theta)$ of Equation 5.2 turns the negative log-likelihood into

$$\begin{aligned} T \cdot \mathrm{NLL}(\theta) &= \mathbb{E}_{\hat{p}}[E_\theta] + T \log Z(\theta) \\ &= \max_q \Big\{ \mathbb{E}_{\hat{p}}[E_\theta] - \mathbb{E}_q[E_\theta] - T\, \mathrm{KL}(q \,\|\, h) \Big\} + \text{const}, \end{aligned} \tag{5.17}$$

so maximum likelihood is $\min_\theta \max_q$: a saddle-point, or primal–dual, problem in which the inner player is the inference of Equation 5.3 and the outer player shapes the energy (Kim and Bengio 2016; Dai et al. 2019). This is the form in which the pull-up of Equation 4.5 appears as a variational player, and it is also the right place to locate the estimators of Table 4.1. Maximum likelihood with MCMC samples from the inner maximizer; contrastive divergence truncates the chain that would reach it; the fourth road of Section 5.4 solves the inner problem over a tractable family; score matching and noise-contrastive estimation do something different in kind — they replace the Kullback–Leibler divergence behind Equation 5.17 by another divergence — and we return to them in Section 5.6.3. We call Equation 5.17 the *saddle-point form* and we do not call it minimax, since that word is reserved for the different nesting of Chapter 7.

5.6.2 Maximum entropy inside, minimum entropy outside: the bridge to minimax entropy

Let the energy be linear in its last layer, $E_\theta(x) = -w^\top \phi_{\theta'}(x)$, as is the case for every network whose output is a linear read-out of its penultimate features, e.g., the last layer of ρ in a DeepSets set function. For fixed θ', the maximum-likelihood w solves the Jaynes problem Equation 5.1 with $\mathrm{T} = \phi_{\theta'}$ and $\boldsymbol{\mu}$ the empirical moments, and the likelihood equations in w state that $\mathbb{E}_{\hat{p}}[\phi_{\theta'}] = \mathbb{E}_{p_\theta}[\phi_{\theta'}]$. At such a point, and for uniform h,

$$\begin{aligned} \mathbb{E}_{\hat{p}}\big[-\log p_\theta\big] &= -w^\top \mathbb{E}_{\hat{p}}[\phi_{\theta'}] + \log Z \\ &= -w^\top \mathbb{E}_{p_\theta}[\phi_{\theta'}] + \log Z \;=\; H(p_\theta), \end{aligned} \tag{5.18}$$

i.e., the cross-entropy of the data under the model equals the entropy of the model itself. Consequently, minimizing the negative log-likelihood over the feature parameters θ' is minimizing the entropy of the maximum-entropy model built on the features $\phi_{\theta'}$:

$$\min_{\theta'} \ \mathrm{NLL}\big(\theta', w^\star(\theta')\big) \;=\; \min_{\theta'} \ H\big(p_{\theta', w^\star(\theta')}\big).$$

This is the minimax entropy principle of Chapter 7, with a continuous family of features and with gradient descent in place of greedy selection. The incremental version is classical: in feature induction for random fields, the gain of adding a feature is the reduction of the model's entropy, which equals the reduction of the KL divergence to the data (Della Pietra et al. 1997), and in the FRAME model of texture the filters are selected by the largest entropy drop (Zhu et al. 1998). Replacing "select a filter from a bank" by "take a gradient step on the filter parameters" is what a deep energy model does (Ngiam et al. 2011). The vision lineage of energy-based models describes itself in exactly these terms: the generative ConvNet is an exponential tilting of a Gaussian reference distribution by a ConvNet, and its authors present it as a hierarchical FRAME model, i.e., a maximum-entropy model whose features are learned filters (Xie et al. 2016). In this precise sense, every energy-based model trained by maximum likelihood is a minimax-entropy learner. The hypotheses should be kept in mind — a linear last layer (or a family closed under rescaling), an exact fit of w, and a uniform base measure — and Chapter 7 studies the explicit, budgeted, combinatorial version of the outer problem, which is where the open questions lie.

5.6.3 Where the family stops

The definition also says where the family stops, and which of its neighbours sit on which side of the line. We go through them in turn, starting with the estimators of Chapter 4, since the answer there is not the same for every estimator.

- Contrastive divergence (Hinton 2002), its persistent variant (Tieleman 2008), and short-run MCMC (Nijkamp et al. 2019) are maximum-entropy learners in the strict sense. They approximate the inner maximization of Equation 5.17 by sampling from, or truncating, a Markov chain whose target is the maximum-entropy distribution, so the entropy term is carried by that target. The truncated versions share the structure and the caveat of Section 5.4.2: what is fit is the output of a truncated inference procedure, the update is biased and in general not the gradient of any objective (Carreira-Perpiñán and Hinton 2005; Bengio and Delalleau 2009; Sutskever and Tieleman 2010), and the learned energy need not be a valid density (Nijkamp et al. 2020). In this respect the optimal subset oracle and CD-k stand or fall together, and we keep both inside the family.
- Score matching (Hyvärinen 2005) and noise-contrastive estimation (Gutmann and Hyvärinen 2010) fit the same Gibbs model by minimizing a different divergence, and we call them maximum-entropy learners *in the generalized sense*. The construction behind the word is that of proper scoring rules. Every proper scoring rule S defines a generalized entropy $H_S(p) = \mathbb{E}_p[S(x, p)]$ and a Bregman divergence

$d_S(p, q) = \mathbb{E}_p[S(x, q)] - H_S(p)$, and maximizing H_S under constraints is dual to minimizing the worst-case expected score, exactly as for the Shannon entropy and the log score (Grünwald and Dawid 2004; Dawid 2007). Score matching minimizes the Fisher divergence, which is d_S for the Hyvärinen score — the basic second-order local proper scoring rule, computable without the normalizing constant (Parry et al. 2012) — and the corresponding generalized entropy is $-\frac{1}{2}\,\mathbb{E}_p\|\nabla_x \log p\|^2$, i.e., minus one half of the Fisher information. Score matching is therefore a *minimum-Fisher-information* learner, a principle with its own history in physics (Frieden 1990). Its bridges to the strict case are quantitative: it is the infinitesimal-step limit of Langevin contrastive divergence (Hyvärinen 2007), the derivative of the KL divergence under Gaussian smoothing (Lyu 2009), and, with likelihood weighting in diffusion models, an upper bound on the negative log-likelihood (Song, Durkan, et al. 2021). Noise-contrastive estimation is the maximum-entropy classifier of Section 5.1.4 with the energy as its single learned feature; it is consistent, and for normalized models it becomes asymptotically as efficient as maximum likelihood as the noise-to-data ratio grows (Gutmann and Hyvärinen 2012). The constant it estimates in place of $\log Z$ plays the role of a free-energy difference between the model and the noise distribution, and the estimator is, up to the roles of data and noise samples, Bennett's acceptance-ratio method of statistical physics, which is itself a logistic-regression maximum-likelihood estimator (Bennett 1976; Shirts et al. 2003). Both estimators, together with ratio matching, belong to one Bregman-divergence framework for unnormalized models (Gutmann and Hirayama 2011), and the Kullback–Leibler divergence of maximum likelihood is itself a Bregman divergence. Their derivations remain in Chapter 4.

- The zero-temperature losses — perceptron, hinge, maximum margin planning — are the $T \to 0$ limits of the family and lie on its boundary: the entropy term has been switched off.
- Directed graphical models fitted by maximum likelihood or EM are learners at levels ii)–iii): the E-step is the I-projection of Equation 5.1, and the free-energy view of EM is that of Section 4.8. But their model class is not a linear exponential family (Section 5.2), so they are maximum-entropy learners without being maximum-entropy models.
- Minimum-entropy learning (Chapter 6) uses the same inner Gibbs tilt (Equation 6.2) but reverses the outer criterion; it is excluded by

  ii) by design, and the two chapters are the two directions of the dial of Chapter 3.

- Entropy bonuses added to an arbitrary objective as a heuristic, without a Gibbs model behind them, are in the penumbra: they become maximum-entropy learning exactly when the objective can be written as Equation 5.2.

Two remarks connect the definition to finite samples and to decision theory. When the moments $\boldsymbol{\mu}$ are estimated from data, relaxing the constraints of Equation 5.1 to a box $|\mathbb{E}_q[\mathrm{T}] - \hat{\boldsymbol{\mu}}| \le \boldsymbol{\beta}$ is dual to ℓ_1-regularized maximum likelihood, so constraint slack and regularization strength are one quantity (Dudík et al. 2007; Altun and Smola 2006). And the maximum-entropy distribution is the minimax strategy of a zero-sum game against nature under logarithmic loss (Topsøe 1979; Grünwald and Dawid 2004), which gives the principle a decision-theoretic rationale that is independent of Jaynes' informational one.

5.7 Takeaway

Maximizing entropy reframes choosing a distribution over coalitions as the process of specifying a constraint; mean-field inference on this distribution transforms the question from "which valuation should I adopt?" to "how far along the mean-field path should I iterate?"—placing Shapley, Banzhaf, and related values on a continuous spectrum rather than as disconnected axioms. This echoes the main probabilistic graphical models insight from earlier in the chapter: a strictly positive MRF corresponds to a maximum-entropy model with structured constraints, and it is the mean-field relaxation that unifies different valuation principles as points along a single trajectory.

If we no longer assume the game is given, the same framework becomes a learning algorithm: we model a Gibbs distribution over subsets, where the energy is a neural set function, and learn this energy directly from observations of optimal subsets—by differentiating through mean-field inference on the multilinear extension. This is maximum-entropy learning as defined in Section 5.1, but now on a discrete space with an energy function discovered (not specified) from data. Here, the ELBO becomes the free energy to optimize, and the resulting energy landscape is itself shaped by the training data—a direct example of the chapter's central idea.

Read through the three levels of Section 5.1, the chapter has one shape. Every strand shares the representation and inference levels—a Gibbs model and a free-energy minimization—and the strands differ in whether the learning level is active and in what supervises it. The cooperative games and maximum-entropy RL with a given reward stop at inference; the optimal subset oracle and inverse RL learn the energy from observed optimal solutions, and the two are parallel line by line (Table 5.2). Finally, by the identity Equation 5.18, maximum likelihood over the features of any such model is already a minimax-entropy problem, which is related to Chapter 7.

6 Minimum-Entropy Learning

A model that answers the same question five different ways is telling us something. This chapter is about turning that signal — a model's disagreement with itself, measurable without any ground truth answer — into a training objective, and about what such an objective can and cannot deliver.

The true result is more limited than people often say, and it is important to say this clearly from the beginning. Minimising the entropy of a model's own answers is a strikingly effective way to *elicit* capability the model already has: on mathematics it matches verifier-based reinforcement learning while using no labels at all. It cannot add capability, and that is not a defect of present implementations but a consequence of a training loop that never touches ground truth (Section 6.7). Its characteristic failure is equally structural. Left unchecked, the objective drives the model to a single confident answer, right or wrong; Section 6.6 shows that this is not an implementation nuisance to be tuned away but a first-order transition in a mean-field free energy — an abrupt jump rather than a gradual drift — with a computable critical point and, whenever three or more distinct answers compete, hysteresis: a memory effect that makes the jump one-way. That is why the methods that survive their own objective all hold the entropy above a floor — by reset, by fairness term, or by filter — and why no choice of regularisation strength can be made to serve instead. In the implementation this chapter examines most closely, the floor currently sits at zero — a configuration its own ablations arrived at, and one whose published runs completed stably. Section 6.6 argues there is no contradiction in that, once the floor's role is named correctly: it is a stopping rule that decides how much of the model's diversity sharpening spends, not a stability device, and in the published runs a cruder stopping rule — the finite training budget — was doing its work.

Three cautions run through what follows. The name is unfortunate: minimum-entropy learning is a family of *objectives* and has nothing to do with the Rényi min-entropy H_∞ of Chapter 2. The physics is routinely oversold: Schrödinger's "order from order" turns out to be a genuine and clarifying reading of what these methods do, whereas Prigogine's minimum-entropy-production principle does not survive contact with the details, and Section 6.9 says why in more detail. And the history is longer than the modern literature usually recounts — the objective and the term that balances it appear side by side as early as 1991, and the pairing has since been arrived at independently in several communities (Section 6.3).

6.1 Two regimes, and a warning about the name

The maximum-entropy principle of Section 2.3 answers a question about *honesty*: given that we have measured a few things and nothing else, which distribution commits us to the least? The minimum-entropy principle answers a question about *commitment*, and it applies in a regime the first one never contemplates. Suppose the distribution in front of us is not a summary of our ignorance but the output of a model that has already absorbed an enormous amount of information. Its dispersion is then no longer a virtue to be preserved; it is a symptom. A model that answers the same question five different ways is telling us something, and what it is telling us can be turned into a training signal.

The two principles are therefore not rivals. They act on different objects at different moments: maximum entropy shapes the distribution we *assume*, minimum entropy sharpens the distribution a trained model *produces*. A system that only maximises entropy generates candidates forever and never decides; one that only minimises it becomes rigid and confidently wrong. This chapter mainly discusses the second failure, as it is the main problem for the methods covered here.

Before going further, a terminological issue has to be cleared. Chapter 2 introduced the **min-entropy** $H_\infty(p) = -\log \max_x p(x)$, the $\alpha \to \infty$ member of the Rényi family, used in cryptography to bound an adversary's first-guess probability. That quantity has nothing to do with the subject of this chapter. Here, "minimum-entropy learning" names a class of *objectives* — procedures that reduce the entropy of a model's predictive distribution — not a particular entropy functional. The collision is unfortunate and it is now standard in both literatures, so it is best met head on. As it happens, the connection is not quite empty: Section 6.5 shows that the objective actually implemented by the method at the centre of this chapter is the Rényi entropy at $\alpha = 2$, a different member of the same family.

6.2 From word-level to meaning-level uncertainty

Turning "the model contradicts itself" into a number requires deciding what counts as a contradiction, and the obvious choice fails immediately. Ask a language model for the capital of France five times and it may return *Paris*, *It is Paris*, *The capital of France is Paris*, *Paris, France*, and *Paris — it has been since 987*. These are five distinct token sequences. If we calculate entropy at the token level, based on the distribution over next words, it might suggest high uncertainty—even when, in reality, the model is certain. For example, if you ask which paper first documented an obscure phenomenon, the model might return five distinct and seemingly plausible citations, each actually incompatible with the others. In such cases, the token-level entropy could be the same as, or even lower than, situations where the correct answer is always given. This happens because the model generates convincing but fabricated responses with confidence on a word-by-word basis.

The solution is to group together outputs with the same meaning before making any measurement. Let $\sim$ be an equivalence relation on responses that identifies those with the same meaning, and let $\mathcal{C}$ be the resulting set of clusters. Sampling G responses $o_1, \dots, o_G$ from the model for a prompt q and writing $\hat{p}_c = |c|/G$ for the empirical frequency of cluster c, the **semantic entropy** is

$$H_{\text{sem}}(q) = -\sum_{c \in \mathcal{C}} \hat{p}_c \log \hat{p}_c.$$

This is Shannon's formula applied to a coarser σ-algebra, which is the whole of the idea. In the Paris case all five responses fall into one cluster and $H_{\text{sem}} = 0$. The measurement now tracks what we care about, and the repetition exploit disappears, because repeating oneself does not create a new cluster.

Semantic entropy was introduced for uncertainty quantification (Kuhn et al. 2023) and reached wide attention as a hallucination detector (Farquhar et al. 2024): resample an answer ten times, cluster by meaning, and flag the high-entropy cases before a user sees them. Its appeal is that the signal is available with no answer key, no reference database, and no change to the model — which makes it deployable exactly where verification is hardest. But it is only a detector. It says when not to trust an answer; it does not produce a better one. The rest of this chapter concerns the obvious next question: if this uncertainty can be measured, can it be optimised?

6.3 Entropy minimisation as an objective: a short history

The idea of adding an entropy term to a loss and descending it is not new, and its history is worth tracing, because the same pair of ideas has surfaced independently in several communities, under different names and with different emphases — a pattern that is common wherever one piece of mathematics serves problems with different constraints. Everything the next sections struggle with has been met before.

Both terms are already present in 1991. Bridle et al. (1992) asked how to train a classifier on unlabelled data alone, and proposed maximising the mutual information between the input and the predicted class. Decomposed, that objective reads

$$I(c\,;x) \;=\; \underbrace{H(\bar{y})}_{\text{fair}} \;-\; \underbrace{\overline{H(y)}}_{\text{firm}},$$

where y is the predictive distribution for one input and $\bar{y}$ its average over the dataset. The second term is entropy minimisation exactly as it is practised today: make each individual prediction decisive. The first is the thing that stops the second from being satisfied by fraud, since a classifier that answers "class 3" to everything is perfectly decisive and carries no information at all. Bridle and his co-authors state the point in so many words, observing that it is trivial to achieve either desideratum alone, and their summary of what a working objective must be — *firm but fair* — remains the clearest available description of what is at stake.

The two terms then travelled separately. Grandvalet and Bengio (2004) brought entropy minimisation to semi-supervised classification: augment the likelihood on labelled data with $-\lambda \sum_{x \in \mathcal{U}} H\big(p_\theta(y \mid x)\big)$ over the unlabelled set, pushing the classifier towards confident predictions on data it has never been told the answer for. The justification is the cluster assumption — that decision boundaries should fall in low-density regions — and under it, confident predictions on unlabelled points are evidence of a boundary in the right place. It is the firm term on its own, and the setting explains why that is a reasonable formulation there: the labelled likelihood anchors the classifier to all K classes, so the firm term can stand alone in a way it cannot in the fully unsupervised case, and a fairness term has no obvious work to do. This anchored formulation became the standard reference point for the modern literature, and pseudo-labelling (Lee 2013) — whose author explicitly identifies it with entropy regularisation — operates in the same anchored regime. The fairness term returned to wide use in 2020, when Berthelot et al. (2020) introduced it under the name *distribution alignment* and traced the idea back to Bridle's mutual-information objective, observing that it had been proposed more than twenty-five years earlier and had, to their knowledge, not been in use in contemporary semi-supervised learning. Read as a whole, the record is less a story of anything being lost than of the two terms mattering in different regimes: as long as a supervised anchor was present, the firm term alone was enough; the fair term becomes essential exactly when the anchor is removed — which is the situation of this chapter.

What the field eventually converged on is not one fix but a three-term structure. Drawing on what has proven effective, any objective for unlabelled data must include all of the following components, with each omission leading to its own characteristic failure:

term	what it buys	what its absence costs
conditional entropy down — *firmness*	decisive predictions	predictions stay diffuse and carry no usable signal
marginal entropy up, or an equipartition constraint — *fairness*	use of the whole label space	**everything collapses onto one class**
a capacity, smoothness or invariance constraint	classes that mean something	the partition fragments, or becomes balanced and arbitrary

That two of the three do not suffice is not a conjecture. Gomes et al. (2010) demonstrate both failures side by side on data constructed to expose them: dropping fairness lets decision boundaries vanish and clusters disappear, dropping the capacity term shatters the data into a large number of categories. The equipartition methods make the third failure vivid — a constant encoder paired with a uniform code satisfies a balance constraint *exactly*, which is why Asano et al. (2020) and Caron et al. (2020) must couple it to an augmentation-invariance objective. And Caron et al. (2021) separate the two collapse attractors quantitatively: without centring the output entropy falls to zero, without sharpening it rises to $\log K$, and only the two together hold a representation in between.

One mechanism, six notations. The most striking feature of this literature is how many communities arrived at the fairness term independently — as a bonus in the loss, as a reweighting of the pseudo-label, as an edit to the logits, and as a constraint on the assignment matrix. These are objects of four different kinds, and it is

worth saying precisely in what sense they are the same thing, because the ways in which they are *not* turn out to matter.

Fix the notation for the rest of this section. A network maps an input x to logits $z_c(x)$ over K classes, giving predictions $p_c(x) = \mathrm{softmax}(z(x))_c$, and over a batch or dataset of N inputs the **class marginal** is the average prediction

$$\bar{p}_c \;=\; \frac{1}{N}\sum_{n=1}^{N} p_c(x_n), \qquad \textstyle\sum_c \bar{p}_c = 1.$$

π_c denotes the **target class distribution** for class c—that is, the desired proportion of examples assigned to each class. In most typical cases, we want all classes to be equally represented, so π_c is set to the uniform distribution: $\pi_c = 1/K$, where K is the total number of classes. In some scenarios, prior knowledge may justify setting a non-uniform target distribution.

Collapse is the statement that $\bar{p}$ concentrates on one class; fairness is any mechanism that resists it. The unifying observation is that all six act on the predictions in the same way:

> **i** Each replaces the logits $z_c(x)$ by $z_c(x) - \gamma_c$ for a **per-class offset** γ_c increasing in how over-represented class c currently is. What distinguishes them is how γ_c is estimated, and how far the resulting marginal is driven towards its target.

mechanism	offset γ_c	enforcement	acts on
marginal-entropy bonus $+\lambda H(\bar{p})$	$\propto p_c(x)\big(\log \bar{p}_c - \mathbb{E}_{p(x)} \log \bar{p}\big)$	soft penalty, one gradient step	the loss
$-\lambda D_{\mathrm{KL}}(\bar{p} \,\|\, \mathrm{Unif})$	as above	soft penalty, one gradient step	the loss
distribution alignment	$\log \bar{p}_c - \log \pi_c$	one Sinkhorn step	the pseudo-label
logit debiasing	$\lambda \log \bar{p}_c$	one Sinkhorn step ($\lambda = 1$)	the logits
centring	$\overline{z_c}$, a moving average	one step, on a lagged estimate	the teacher's logits
Sinkhorn equipartition	the γ making $\bar{p}$ exactly uniform	hard constraint, at convergence	the assignments

Reading in order: the marginal-entropy bonus of Bridle et al. (1992), Hu et al. (2017) and Liang et al. (2020); the same quantity written as a divergence from uniform; the distribution alignment of Berthelot et al. (2020); logit debiasing (Menon et al. 2021; Wang et al. 2022); the centring operation of Caron et al. (2021); and the Sinkhorn equipartition constraint of Asano et al. (2020) and Caron et al. (2020).

Rows three, four and six are the same operation at different stopping times. With a uniform target $\pi_c = 1/K$, distribution alignment reweights the prediction to $\tilde{q}_c \propto p_c(x)/\bar{p}_c$, which in logits is $z_c(x) - \log \bar{p}_c$ up to a per-example constant — logit debiasing at $\lambda = 1$. And if the predictions are arranged into the matrix $Q_{cn} = p_c(x_n)/N$, whose columns already sum to $1/N$, its row sums are exactly $\bar{p}_c$, so the first row-normalisation of the Sinkhorn iteration towards $U\big(\frac{1}{K}\mathbf{1}, \frac{1}{N}\mathbf{1}\big)$ rescales row c by $1/(K\bar{p}_c)$: the same offset once more. **Logit debiasing is one Sinkhorn step; Sinkhorn is logit debiasing run to convergence.**

This distinction is important, and it is a common source of confusion. One debiasing step does not make the marginal uniform. It makes the *row* sums uniform, but the per-example softmax that follows is a column renormalisation, and that undoes part of the correction — which is precisely why Sinkhorn alternates rather than stopping. The residual shrinks by a roughly constant factor of about three per pass: with $K = 10$ classes and $N = 256$ examples drawn from a moderately confident network, the total-variation distance from uniform runs $0.060 \to 0.020 \to 0.0064 \to 0.0021$, falling to about 10^{-7} after a dozen passes and to machine precision

after about thirty. A single step therefore buys most of the correction and none of the guarantee, which is the operative difference between a method that *discourages* collapse and one that *forbids* it. Row two, by contrast, is identical to row one: $D_{\mathrm{KL}}(\bar{p} \,\|\, \mathrm{Unif}) = \log K - H(\bar{p})$, so the two objectives differ by a constant and have the same gradient everywhere.

The remaining two relations are approximations, and both gaps have a definite sign. The gradient of $H(\bar{p})$ is *not* the flat offset $\lambda \log \bar{p}_c$: it carries a factor $p_c(x)$, so it acts on each example only through the classes that example is already predicted to belong to, and it is centred, so within an example it merely redistributes. The hand-designed offset applies the full correction to every example unconditionally. They agree on sign and on monotonicity in $\bar{p}_c$, and on little else — which is also why the two are calibrated differently: $\lambda = 1$ is a canonical, tuning-free setting for the logit correction, being exactly the Bayes adjustment that maps a model with implicit prior $\bar{p}$ onto a uniform one (Menon et al. 2021), whereas the entropy bonus has no distinguished coefficient and must be tuned.

Centring differs along another axis entirely. Subtracting the mean *logit* $\overline{z_c}$ is, because $\log p_c(x) = z_c(x) - \log Z(x)$ and the log-partition is common to all classes, the same as subtracting $\overline{\log p_c}$ — the logarithm of the **geometric** mean of the predictions, where debiasing subtracts the logarithm of the **arithmetic** mean:

$$\begin{aligned} \gamma_c^{\text{debias}} &= \log\Big(\tfrac{1}{N}\textstyle\sum_n p_c(x_n)\Big) \\ &\geq \tfrac{1}{N}\textstyle\sum_n \log p_c(x_n) \;=\; \gamma_c^{\text{centring}} \;-\; \tfrac{1}{N}\textstyle\sum_n \log Z(x_n), \end{aligned}$$

by Jensen, with equality only if p_c is constant across the dataset; the trailing term is the same for every class and so invisible to the softmax. The gap widens with the variance of p_c over examples, so centring discounts classes that dominate a few inputs strongly and penalises only classes predicted uniformly often — a weaker and smoother intervention, which is consistent with Caron et al. (2021) needing to pair it with sharpening rather than with an explicit capacity term.

i Gradient of the marginal-entropy bonus

Differentiating $H(\bar{p}) = -\sum_c \bar{p}_c \log \bar{p}_c$ through $\bar{p}_c = \frac{1}{N}\sum_n p_c(x_n)$ and the softmax Jacobian $\partial p_c / \partial z_j = p_c(\delta_{cj} - p_j)$ gives, for one example x,

$$\begin{aligned} \frac{\partial H(\bar{p})}{\partial z_j(x)} &= \frac{1}{N}\sum_c \big(-\log \bar{p}_c - 1\big)\, p_c\big(\delta_{cj} - p_j\big) \\ &= -\frac{p_j(x)}{N}\Big(\log \bar{p}_j - \sum_c p_c(x) \log \bar{p}_c\Big), \end{aligned}$$

the -1 terms cancelling because $\sum_c p_c = 1$. Ascending H therefore *subtracts* this quantity from the logits, which is the offset in the first row of the table. Two features are worth noting. The bracket has p-weighted mean zero, so the update is a pure redistribution within each example; and the prefactor $p_j(x)$ vanishes for classes the example is confidently not in, so an example already committed elsewhere contributes almost nothing to correcting an over-represented class. Neither property is shared by $\gamma_j = \lambda \log \bar{p}_j$.

Six communities, six notations, one mechanism — with the caveat that the two notations most often treated as interchangeable, the entropy bonus and the logit correction, are the two that are genuinely distinct.

A second and orthogonal question — what the penalty does *at* the boundary of the simplex — matters for what follows, and appears not to be remarked on anywhere. Tanaka et al. (2018) penalise $D_{\mathrm{KL}}(\pi \,\|\, \bar{p})$, a fixed prior π against the marginal prediction $\bar{p}$ — with the prior in the *first* slot, so the term *diverges* as any class's marginal mass approaches zero: an infinitely high barrier that no finite reward can buy through. The marginal entropy $H(\bar{p})$ is bounded on the simplex, so collapsing costs at most $\log K$, and a sufficiently strong firmness

term will simply pay it. One is a hard floor and the other a soft one, and the distinction decides whether collapse is impossible or merely expensive.

Test-time adaptation is the closest analogue to the unsupervised setting. When a deployed network adapts to a shifted input distribution by minimising the entropy of its own predictions on the incoming stream (Wang et al. 2021), the labelled anchor is gone and collapse stops being hypothetical. Niu et al. (2023) devote a subsection to it, under a heading that says all that needs saying — online entropy minimisation tends to predict all samples to the same class. Their measurements are the strongest empirical statement available that this objective can be actively harmful when nothing holds it — accuracy on ImageNet-C at the highest corruption severity, averaged over all fifteen corruptions, with a batch-normalised ResNet-50:

stream	no adaptation	Tent	EATA
imbalanced label order, batch 64	18.0	2.1	0.9
balanced labels, batch size 1	18.0	0.1	0.1

Adaptation by entropy minimisation takes a network from 18% to a fraction of one per cent — far worse than leaving it alone. Both of these are deliberately adverse streams rather than the standard benchmark, which is the point: the objective is fine until the conditions that were implicitly propping it up are withdrawn. Note also that the weight regularisation of Niu et al. (2022) does not rescue it here, though it does help on backbones whose normalisation is not batch-dependent, which is itself evidence that anchoring is a weaker remedy than a structural one.

Two details of the fix Niu et al. (2023) propose are worth carrying forward. The first is that they monitor a moving average of the entropy loss and *reset the model* when it falls below a threshold, set to 0.2 — an entropy floor placed strictly inside the simplex, implemented as a discrete intervention rather than a filter. The same object returns in Section 6.4 with its role transformed: here it is a stability device rescuing a network from destruction; there, where the degenerate endpoint turns out to be quiet rather than catastrophic, a floor would serve as a stopping rule — and Section 6.6 argues that distinction is the key to reading its current setting of zero correctly. The second is that in their diagnostic experiment, on a single corruption with a group-normalised backbone, the model collapses at severity level five while at level three the gradients stay in a stable range throughout: the behaviour changes sharply as a control parameter is turned. That is the closest thing in the empirical literature to the critical value derived in Section 6.6, though it should not be read as a general safe threshold — under mixed corruptions the same backbone collapses at level three as well. Running the other way, Lee et al. (2024) argue that entropy is not a trustworthy reliability signal in the first place, since under spurious correlation confidently wrong samples are exactly the harmful ones — a result that weakens the entire filtering family and, by elimination, strengthens the case for structural remedies.

The language-model literature has adopted two of the three families and not the third. Reinforcement learning from internal feedback has taken up filtering — the two-sided entropy band described in Section 6.4, the substitution of self-certainty for entropy in Zhao et al. (2025) — and anchoring, in the form of covariance-targeted clipping and KL penalties (Cui et al. 2025). A fairness term is absent: no work in this line maximises a dataset-level marginal entropy. Part of the reason is a genuine type mismatch. The answer space of a language model is prompt-dependent, so "cluster 1" for one question has nothing to do with "cluster 1" for another and the marginal $\bar{p}$ has no well-defined referent. Candidate substitutes exist — a prompt-independent output attribute such as length, format or refusal rate; a population-level coverage or pass@k target; or a floor on cluster mass within each prompt, which is what δ_{low} would approximate if it were set anywhere above zero — but none has been tried. That the gap is real is confirmed by what happens without them: in the plain entropy-minimisation baseline that Zhao et al. (2025) run against their own method, the model "converges to producing the same character regardless of the prompt" after a few updates. That is the vision literature's collapse, arriving in the language literature under a different name and from the same cause.

What is new in the method described next is precisely the removal of the supervised anchor. There is no

labelled term at all. This is what makes the approach interesting, and it is also what returns the degeneracy in full force — a point developed in Section 6.6, where it turns out to be sharper than a mere risk of a bad optimum.

6.4 EMPO: reward agreement, not correctness

EMPO — entropy-minimised policy optimisation — applies the objective to the elicitation of reasoning in large language models, in a fully unsupervised setting (Zhang et al. 2025). The context is the reinforcement-learning recipe behind recent reasoning models (Shao et al. 2024; DeepSeek-AI 2025): sample several attempts at a question, score each against a ground-truth answer with an automatic verifier, and reinforce the winners,

$$r_i = \mathbb{1}\big[\,\mathrm{verify}(o_i, a) = \mathrm{true}\,\big].$$

The recipe works remarkably well, but the bottleneck sits entirely in the symbol a. Every training question needs a gold answer and a cheap, reliable checker. Mathematics has one — compare the final expression — and code has a test suite. A differential diagnosis does not; neither does a proposed mechanism, a study design, or a molecule nobody has synthesised. The algorithm is not the scarce resource. The verifier is.

EMPO removes a from the loop. Given an unlabelled question q, it samples G responses, partitions them by meaning as in Section 6.2, and pays each response the share of the group that agrees with it:

$$r_i = \hat{p}_{c(i)} = \frac{|c(i)|}{G},$$

where $c(i)$ is the cluster containing o_i. Responses in the majority cluster are reinforced and outliers are suppressed, with no reference to whether the majority is right. In the released implementation, the equivalence relation is a pipeline of answer extraction and symbolic equality for mathematics, and a small dedicated verifier model for open-ended reasoning — cheap in both cases, and cheap in a way that does not require knowing the answer, though for the learned verifier that last clause turns out to need a careful audit, taken up in Section 6.7.

Two guards accompany the reward. The first is a band on the entropy: training proceeds only on questions with $\delta_{\text{low}} < H_{\text{sem}}(q) < \delta_{\text{high}}$. Too much dispersion and the majority cluster is noise; too little and there is nothing left to sharpen but overconfidence. The second is a set of anti-gaming penalties — an empty answer is penalised outright, and an answer that merely echoes the question earns nothing — which prevent the model from manufacturing agreement without saying anything.

It is worth examining how the first guard is realized in practice, because the implementation ended up stricter than the design intended—and understanding why is revealing. The results above were obtained with the band active. When the experiments were later rebuilt on a different training framework, an ablation found the thresholds made limited difference, and they were relaxed to keep any group whose entropy is neither zero nor close to its ceiling: $0 < H_{\text{sem}} < 0.6 \log G$, over a group of G samples. **The two endpoints (0 and $\log G$) are redundant.** A group in which all G responses agree has $H_{\text{sem}} = 0$ and pays every response the same reward, 1; a group in which no two agree has $H_{\text{sem}} = \log G$ and again pays every response the same reward, $1/G$. Since the advantage subtracts the group mean, both cases contribute an identically zero update whether they are filtered or not. A filter acting at the endpoints removes only groups the optimiser was already ignoring — consistent with the ablation's verdict that the original thresholds could be relaxed at little cost.

What remains is the upper bound alone, and it is a real constraint: at $G = 16$ it admits a group only if its meanings are no more dispersed than roughly five equally-sized clusters. But it guards against one failure only — a majority that is really noise. Nothing in the current configuration bounds the entropy from *below* at any value strictly inside the simplex. That asymmetry is not a detail, but reading it correctly takes some

care, and Section 6.6 supplies the reading: the floor turns out to be a *stopping rule* — the mechanism that decides where sharpening halts, per prompt — rather than a stability device, which is why its absence coexists peacefully with stable published runs governed by a short horizon, and why the ablation that retired it, run at exactly such a horizon, was not positioned to see what it does.

Empirically the method closes most of the gap to supervision without using any. On five mathematics benchmarks with a 7B base model, averaging over the suite:

Method	What it needs	Score
Base model	—	33.7
Supervised fine-tuning	$\{q, r, a\}$	39.0
Verifier-based RL (GRPO)	$\{q, a\}$	51.7
EMPO	$\{q\}$	51.6

Here q is the question, a the gold answer, and r a reasoning trace. Two things in this table deserve to be said plainly. The first is that supervised fine-tuning, which sees the most, does worst of the three — imitation of traces is a weaker signal than selection among the model's own attempts. The second is that the tie between EMPO and verifier-based RL holds on mathematics; on open-ended reasoning a real gap of several points remains. The honest summary is that removing labels costs little in a domain where the model is already strong, and more elsewhere.

6.5 What "reward agreement" actually optimises

It is worth computing what the reward above does in expectation, because the answer is both cleaner and different from the one usually stated.

Averaging r_i over the group, each response contributes the frequency of its own cluster, so a cluster of size $|c|$ contributes $|c|$ times $\hat{p}_c$:

$$\bar{r} \;=\; \frac{1}{G}\sum_{i=1}^{G} \hat{p}_{c(i)} \;=\; \sum_{c\in\mathcal{C}} \hat{p}_c^{\,2} \;=\; \exp\big(-H_2(\hat{p})\big),$$

where $H_2(p) = -\log \sum_c p_c^2$ is the **Rényi entropy of order two**, also called the collision entropy (Rényi 1961). The quantity being maximised is the *collision probability*: the chance that two independent samples from the model land on the same meaning.

This deserves a frank note, and then a correction to the way such paper-versus-code mismatches are usually described. The objective as written in the paper is the Shannon semantic entropy of Section 6.2; the objective the code implements is H_2. These are genuinely different functions: their gradients differ, and H_2 weights the dominant cluster more heavily than Shannon does. But they are not different *kinds* of object, and naming the family they belong to turns the discrepancy from an inconsistency into a choice that can be argued for.

A unified family, parametrized by order. Consider the Rényi entropy H_α from Section 2.1.4. Instead of rewarding responses based on their cluster's simple share, generalize by awarding a power of that share:

$$r_\alpha(c) \;=\; \frac{p_c^{\alpha-1}-1}{\alpha-1}, \qquad \mathbb{E}_{c\sim p}\big[r_\alpha\big] \;=\; \frac{\sum_c p_c^{\alpha}-1}{\alpha-1},$$

an expectation monotone in H_α for every $\alpha > 1$, so that maximising it minimises the entropy of that order. Two members are already in front of us. As $\alpha \to 1$ the reward tends to $\log p_c$ and its expectation to $-H_{\text{sem}}$, which is the paper's objective. At $\alpha = 2$ the reward is $p_c - 1$, which differs from the implemented reward p_c by a constant — and constants are invisible here, since the advantage subtracts the group mean before anything

is done with it. **The stated objective and the implemented one are the $\alpha \to 1$ and $\alpha = 2$ members of a single one-parameter family.**

A third member turns up in the filters. The surviving band is written in terms of H_{sem}, but the majority-share cutoffs that have at various points stood in for it are thresholds on $\max_c p_c$, which is e^{-H_∞}. Between the derivation, the reward and the gate, the algorithm therefore touches three orders at once: $\alpha \to 1$, $\alpha = 2$, $\alpha = \infty$. The min-entropy that Section 2.1.4 was at pains to separate from minimum-entropy learning turns out to appear here after all — not as the objective, but as the gatekeeper.

Why order two is the defensible choice. Three reasons, of which the first is decisive. A group holds only G samples — seven to sixteen in the released configurations — so p_c is never known, only estimated by $\hat{p}_c = |c|/G$. Shannon entropy admits *no unbiased estimator at any finite G*: the expectation of any estimator is a polynomial in p, and entropy is not a polynomial (Paninski 2003, Prop. 8). Collision probability is a polynomial, of degree two, and is therefore estimable without bias by $\sum_c |c|(|c|-1)/G(G-1)$. The gap is not academic: at $G = 8$ over five clusters the plug-in Shannon estimator is biased downward by 0.28 nats, a fifth of the quantity being measured, while the degree-two estimator is exact. Second, the reward stays bounded: a singleton cluster earns $1/G$, where the order-one reward $\log \hat{p}_c$ hands it $-\log G$, a spike that grows with the group size. Third —

$$\log K - H_\alpha \;=\; \frac{\alpha}{2K}\sum_c \varepsilon_c^2 \;+\; O(\varepsilon^3), \qquad p_c = \frac{1+\varepsilon_c}{K}, \quad \textstyle\sum_c \varepsilon_c = 0,$$

so near the uniform distribution every order is the *same* function of the distribution, differing only by the factor α. A Shannon derivation and a collision-entropy implementation give parallel gradients wherever the meanings are near-balanced, which — since the band admits only high-entropy questions — is where training begins. They part company as p concentrates, which is precisely the regime Section 6.6 is about.

They are not, for all that, the same function. Both are Schur-concave, so they agree on any pair ordered by majorization, and both are maximised by the uniform distribution and minimised by a point mass. But majorization is a partial order, and on incomparable pairs the two can disagree outright. A group placing half its mass on one meaning and scattering the rest over nine others, $p = (0.5,\ 0.056^{\times 9})$, has Shannon entropy 1.79 against 1.39 for a group split evenly four ways — while its collision entropy is 1.28 against 1.39. One says the first group is the more diverse; the other says it is the less. Which order is used is a modelling decision, not a formality.

A second and independent axis: from a partition to a kernel. Everything above holds the hard clustering fixed. Relaxing it is a separate generalisation, and the two compose. Replace the equivalence relation with a positive semi-definite similarity kernel over the sampled answers, normalise it to unit trace as $\rho = K/\operatorname{tr} K$, and read the entropy off the spectrum, $H_\alpha(\rho) = \frac{1}{1-\alpha}\log \operatorname{Tr}\rho^\alpha$, with $\alpha \to 1$ giving the von Neumann entropy of Chapter 2. The classical cases are recovered exactly rather than approximately: when K is block-diagonal with all-ones blocks, the non-zero eigenvalues of ρ *are* the cluster masses p_c (Pasarkar and Dieng 2024), so $\alpha \to 1$ returns H_{sem} and $\alpha = 2$ returns $-\log \sum_c p_c^2$. The resulting two-parameter family — an order, and a choice of hard or soft — is the order-q Vendi score (Friedman and Dieng 2023; Pasarkar and Dieng 2024), which coincides with the matrix-based Rényi entropy introduced independently by Sánchez Giraldo et al. (2015), and which is, under the name Hill numbers, the standard family of diversity indices in ecology (Hill 1973; Leinster and Cobbold 2012).

This is worth spelling out because of one claim it would be easy to make and wrong. The kernel relaxation is *not* available only for collision probability. Nikitin et al. (2024) build exactly the $\alpha \to 1$ version for language models, under the name kernel language entropy, and prove that for any semantic clustering there exists a kernel whose von Neumann entropy equals the semantic entropy — by the same block-diagonal construction. What distinguishes $\alpha = 2$ is cost, not existence: $\operatorname{Tr}\rho^2 = \|\rho\|_F^2$ is a sum of squares needing no eigendecomposition, while every other order needs the spectrum. Over a group of sixteen that is free either way; over a corpus it is not.

The gap worth recording is the empty cell. The kernel literature has developed the $\alpha \to 1$ corner and the clustering literature the $\alpha = 2$ one, and a soft, order-two semantic objective — the natural home for a reward that is already a collision probability — appears not to have been tried.

There is also a pleasing coincidence in the expression $\sum_c p_c^2$. It is exactly the chance-agreement term p_e in Cohen's κ, the standard correction applied when two human raters are scored for agreement (Cohen 1960). In the inter-rater setting one computes the agreement two independent raters would reach by luck alone, and subtracts it. Here the same expression is computed over one model's repeated answers and *maximised*. The statistical object that measures how much of an agreement is worthless is, read in the other direction, the training signal.

6.6 The free-energy view, and why entropy collapses

Here is the conclusion of this section, before the argument for it. Rewarding agreement is a rich-get-richer process, and whether it runs away is decided by a single number crossing one. On one side of that threshold, an accidental imbalance between meanings dies out and the model stays honest about what it does not know; on the other side, the imbalance feeds on itself and the model commits to one meaning — and nothing in the objective checks whether that meaning is correct. Two properties of the crossing make it worse than an ordinary tuning problem. It is abrupt: the entropy barely moves for most of the approach, then falls off a cliff. And when more than two meanings compete, it has memory: the collapsed state is already stable on the *safe* side of the threshold, so a single large fluctuation can tip the system over, and moving the knob back afterwards does not undo it. Setting the knob near its critical value is therefore not a defence. A floor on the state is — though what exactly it defends, and what stood in for it in the published runs, takes the rest of the section to state precisely.

Everything below is the derivation, and the machinery is that of Chapter 3.

The control parameter is a temperature, and it is a familiar one. Anyone who has sampled from a language model has already met the object doing the work here. Sampling at temperature τ replaces the model's distribution by $p(o)^{1/\tau}$ and renormalises: large τ flattens the distribution, small τ sharpens it, and $\tau \to 0$ is arg-max decoding. What follows is that same operation with two modifications. The exponent is a reward rather than the model's own log-probability, and — this is the entire source of the difficulty — the reward is itself the probability being adjusted. It is a softmax whose logits are its own outputs.

Start from the canonical form of the policy-optimisation problem, in which reward is traded against a Kullback–Leibler penalty that keeps the updated policy near a reference π_0:

$$\max_{\pi} \;\; \mathbb{E}_{\pi}[\, r \,] \;-\; \beta\, D_{\mathrm{KL}}\big(\pi \,\|\, \pi_0\big). \tag{6.1}$$

This is the minimisation of a free energy, with $-r$ in the role of energy and β in the role of temperature, and it can be solved in closed form. Add a Lagrange multiplier λ to enforce that the probabilities sum to one, and differentiate with respect to $\pi(o)$:

$$\begin{aligned} &\frac{\partial}{\partial \pi(o)} \left[\sum_{o'} \pi(o') r(o') - \beta \sum_{o'} \pi(o') \log \frac{\pi(o')}{\pi_0(o')} + \lambda \Big(1 - \sum_{o'} \pi(o') \Big) \right] \\ &\qquad = r(o) - \beta \log \frac{\pi(o)}{\pi_0(o)} - \beta - \lambda = 0. \end{aligned}$$

Solving for $\pi(o)$ and absorbing everything independent of o into a normaliser gives the **Gibbs tilt of the reference**,

$$\pi^\star(o) = \frac{\pi_0(o)\, e^{r(o)/\beta}}{Z}, \qquad Z = \sum_o \pi_0(o)\, e^{r(o)/\beta}, \tag{6.2}$$

which is the unique maximiser, since the KL term is strictly convex in π and the reward term is linear. Substituting back, the optimal value is $\beta \log Z$ — a log-partition function, exactly the free energy of Chapter 3. This is the same exponential family that Section 2.3 produces from the other direction: there, maximising entropy under a moment constraint; here, maximising reward under a divergence constraint.

The temperature reading is worth making concrete. Take three semantic clusters, a uniform reference $\pi_0 = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, and rewards $r = (1.0, 0.5, 0.0)$. Then Equation 6.2 gives

β	$\pi^\star$	$\mathbb{E}[r]$
∞	(0.333, 0.333, 0.333)	0.500
2	(0.419, 0.326, 0.254)	0.582
1	(0.506, 0.307, 0.186)	0.660
0.25	(0.867, 0.117, 0.016)	0.925
$\to 0$	(1, 0, 0)	1.000

At high temperature the update does nothing and the policy stays at its reference; at zero temperature it becomes greedy and places all mass on the best cluster. The KL coefficient is precisely the knob that decides how far towards the arg-max a single update is allowed to travel — the role played by τ in temperature sampling, now attached to a reward instead of a log-probability.

Nothing so far is specific to entropy minimisation. What *is* specific is that under EMPO the reward is not an external function of the response: by Section 6.4, r is the cluster frequency of the very policy being optimised. Writing p_c for the probability the policy assigns to cluster c and p_c^0 for the reference, substituting $r(o) = p_{c(o)}$ into Equation 6.2 turns an explicit formula into a **self-consistency condition**:

$$p_c \;=\; \frac{p_c^0\, e^{p_c/\beta}}{\sum_{c'} p_{c'}^0\, e^{p_{c'}/\beta}}. \tag{6.3}$$

This is a mean-field equation, of exactly the kind that describes a magnet whose spins feel the average field they themselves generate. The consequences are worth spelling out, because they turn a vague worry about instability into a quantitative statement. Take the KL anchor seriously for the moment and suppose π_0 is genuinely fixed; the paragraphs after next remove that supposition, which is what real implementations do.

A critical value. Take a reference that is uniform over K clusters, $p_c^0 = 1/K$, and nudge the balanced solution slightly: $p_c = 1/K + \varepsilon_c$ with $\sum_c \varepsilon_c = 0$. Substituting into Equation 6.3, the factor $e^{1/(K\beta)}$ is common to every cluster and cancels, leaving $p_c \propto e^{\varepsilon_c/\beta} \approx 1 + \varepsilon_c/\beta$ to first order, so that after normalisation

$$\varepsilon_c \;\longmapsto\; \frac{\varepsilon_c}{K\beta}.$$

An imbalance between meanings is thus multiplied by $1/(K\beta)$ at every step. This is the number promised at the opening, and the rest of the section is a matter of where it sits relative to one. The balanced state survives a perturbation if and only if $K\beta > 1$, so there is a critical value

$$\beta_c = \frac{1}{K},$$

below which the policy commits to a single cluster. The appearance of K is not an artefact: the more distinct meanings a model is entertaining, the smaller each one's share of the group, the weaker the positive feedback any one of them can exert, and so the less braking is needed to contain it. For $K = 2$ the self-consistency

condition can be solved exactly — writing $m = p_1 - p_2$, Equation 6.3 becomes

$$m = \tanh\Big(\frac{m}{2\beta}\Big),$$

the Curie–Weiss equation — the textbook description of a magnet that spontaneously magnetises when cooled below a critical temperature — with β in the place of temperature and $\beta_c = 1/2$ in the place of that critical temperature (the Curie point). One warning for readers arriving from physics, where β conventionally denotes *inverse* temperature: here it is the KL coefficient and it plays the role of T itself. Large β means strong regularisation, a policy held near its reference and high entropy — the hot, disordered phase; small β is the cold, ordered, collapsed one.

i Derivation of the $K = 2$ case

With two clusters and a uniform reference, the factor $\frac{1}{2}$ cancels from numerator and denominator of Equation 6.3, leaving a softmax whose logits are the cluster masses themselves. Since $p_1 + p_2 = 1$ there is only one degree of freedom; write $p_{1,2} = \bar{p} \pm \delta$ with $\bar{p} = \frac{1}{2}$ and $\delta = m/2$. Then

$$m = \frac{e^{p_1/\beta} - e^{p_2/\beta}}{e^{p_1/\beta} + e^{p_2/\beta}} = \frac{e^{\bar{p}/\beta}\big(e^{\delta/\beta} - e^{-\delta/\beta}\big)}{e^{\bar{p}/\beta}\big(e^{\delta/\beta} + e^{-\delta/\beta}\big)} = \tanh\Big(\frac{\delta}{\beta}\Big) = \tanh\Big(\frac{m}{2\beta}\Big),$$

by the definition of the hyperbolic tangent. The cancellation of $e^{\bar{p}/\beta}$ is worth pausing on, because it is the same one that occurs at general K in the perturbation argument above: a softmax is blind to a common shift of its logits, so only *differences* between clusters drive the dynamics. It is also why subtracting the group baseline $\bar{r}$ leaves the update direction untouched.

Whether a non-zero solution exists is then settled entirely by the slope at the origin,

$$\frac{d}{dm}\tanh\Big(\frac{m}{2\beta}\Big)\Big|_{m=0} = \frac{1}{2\beta},$$

which is the amplification factor $1/(K\beta)$ evaluated at $K = 2$, as it must be — linear stability analysis is exactly this derivative. Because tanh is concave for $m > 0$ and bounded by one, a slope below one lets the curve fall away from the diagonal immediately and never return, so $m = 0$ is the only solution; a slope above one carries the curve above the diagonal, and boundedness forces it back across, creating a pair of solutions $\pm m^\star$. The two regimes are separated by $1/(2\beta) = 1$.

Expanding $\tanh x = x - \frac{1}{3}x^3 + O(x^5)$ and dividing through by $m \neq 0$ gives $1 = 1/(2\beta) - m^2/(24\beta^3)$, hence

$$m^\star = 2\beta\sqrt{3\,(1 - 2\beta)} \simeq \sqrt{3\Big(1 - \frac{\beta}{\beta_c}\Big)}$$

near the critical point. The imbalance m — the *order parameter*, in physics language — grows continuously out of zero with exponent $1/2$, the standard mean-field value. That continuity is what makes the $K = 2$ transition *second order*: the state changes smoothly, with no jump. It is precisely this smoothness that fails for $K \geq 3$.

The two regimes, step by step. The algebra is easier to trust once one has watched it happen. Both runs start from the same barely-perturbed state $p = (0.36, 0.32, 0.32)$ with $K = 3$; only β changes.

Above the threshold ($\beta = 1.5\,\beta_c$), the perturbation is erased and the entropy returns to its ceiling $\log 3 = 1.0986$:

step	p	H
0	0.360, 0.320, 0.320	1.0970

step	p	H
2	0.345, 0.327, 0.327	1.0983
4	0.339, 0.331, 0.331	1.0985
6	0.336, 0.332, 0.332	1.0986
8	0.334, 0.333, 0.333	1.0986

Below the threshold ($\beta = 0.75\,\beta_c$), the same perturbation compounds and the distribution is most of the way to a point mass:

step	p	H
0	0.360, 0.320, 0.320	1.0970
2	0.384, 0.308, 0.308	1.0931
4	0.432, 0.284, 0.284	1.0775
6	0.539, 0.231, 0.231	1.0097
8	0.750, 0.125, 0.125	0.7358

The column that deserves attention is the entropy in the second table. Over the first four steps it falls by less than two per cent while the gap between the leading cluster and the others nearly quadruples, and only then does it drop. Entropy is a *lagging* indicator of a process that is already well advanced, which is exactly why collapse is reported as sudden: from the outside it looks like training proceeding normally and then failing without warning. The warning was in the gap between clusters, not in the entropy.

For more than two clusters the transition is discontinuous. At $K = 2$ the dominant cluster's mass grows continuously from $1/2$ as β falls through β_c. For $K \geq 3$ it does not, and the cleanest evidence is a decisive experiment: hold β fixed, start the iteration from two different places, and see whether they agree. The balanced start carries a 10^{-3} perturbation, because starting from perfect symmetry would test nothing: a perfectly balanced state stays balanced by symmetry at every β. For $K = 3$, reporting the converged $\max_c p_c$:

β	from balanced start	from concentrated start	
$1.200\,\beta_c$	0.333	0.333	balanced only
$1.080\,\beta_c$	0.333	0.674	**both stable**
$1.035\,\beta_c$	0.333	0.768	**both stable**
$1.020\,\beta_c$	0.333	0.788	**both stable**
$1.000\,\beta_c$	0.811	0.811	collapsed only

Throughout the window $\beta \in [1.000, 1.093]\,\beta_c$ the two states are *both* stable, and which one the system occupies depends on where it came from. That path dependence is called *hysteresis* — the system remembers its history — and it is the defining signature of a *first-order* transition, one where the state jumps rather than drifts. The window is empty at $K = 2$, widens to $[1.000, 1.403]\,\beta_c$ at $K = 5$ and $[1.000, 1.882]\,\beta_c$ at $K = 8$ — the more meanings in play, the larger the region of ambiguity. This matches a classical model of statistical physics, the mean-field Potts model, whose transition is likewise smooth at $q = 2$ states and discontinuous for $q \geq 3$ (Wu 1982) — so the behaviour is a structural property of self-referential dynamics of this type rather than a peculiarity of the present problem.

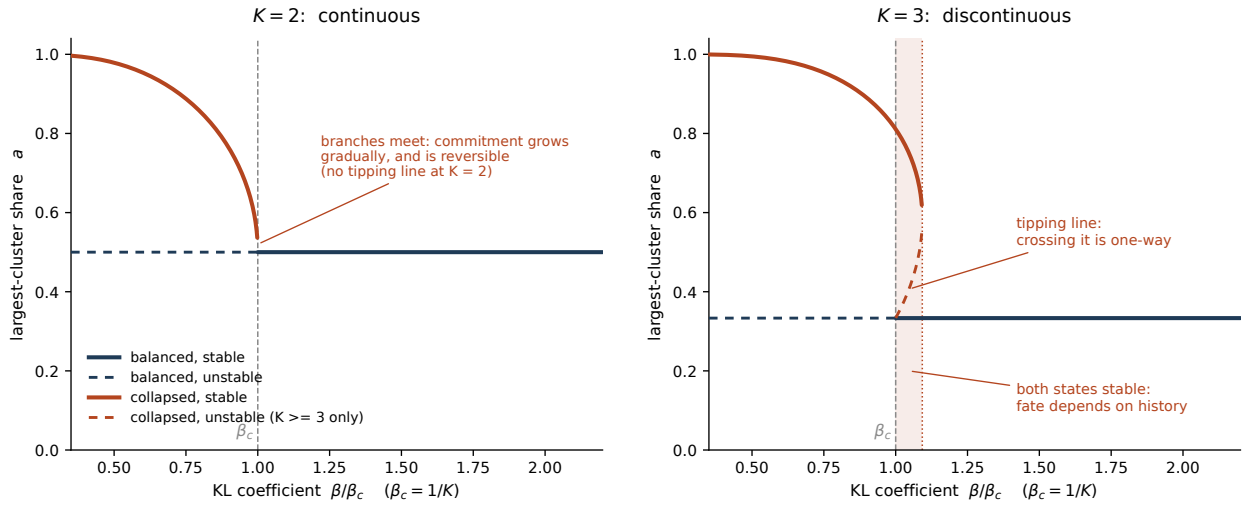


Figure 6.1: Where training can come to rest, and why the number of competing meanings changes everything. Each panel shows the resting states of the self-consistency condition Equation 6.3 as the anchor strength β varies: strong anchor on the right, weak on the left. Height is the share of the largest meaning cluster — $1/K$ means the model is keeping its options open (navy), 1 means it has committed to a single answer (rust). Solid curves are states the system can rest in; dashed curves are unstable and cannot be occupied. **Left ($K = 2$):** weaken the anchor past β_c and commitment grows *gradually* out of the balanced state; strengthen it again and the system walks back. No red dashed line appears because none exists at $K = 2$ — there is no tipping line to cross, so the transition is continuous and reversible. **Right ($K = 3$):** the committed state is already available and stable *before* the threshold is reached. In the shaded window both states are stable, separated by an unstable ridge (red dashed): which state the system occupies depends on where it came from, and a fluctuation across the ridge is one-way. This is the figure's practical message — with three or more meanings in play, setting β "just above the critical value" parks the system inside the window where a single large fluctuation commits it irreversibly. Generated by `code/min-entropy-phase-diagram.py`.

Figure 6.1 makes the practical consequence visible. The threshold $\beta_c = 1/K$ is not where the danger begins; it is only the last point of the safe regime — what physicists call a *spinodal*, the place where the balanced state finally stops surviving even small knocks. The collapsed state has already been available and stable for some distance above it. Tuning β to sit just above β_c is therefore not a safety margin — it places the system exactly in the region where both states coexist, and where a single fluctuation large enough to cross the dividing line is never recovered from. What is needed is not a carefully chosen distance from the critical point but a mechanism that acts on the state itself.

In practice there is no such reference, and collapse becomes unconditional. The stabilising mechanism above depends entirely on the KL term in Equation 6.1 anchoring the policy to a *fixed* π_0. It is worth checking whether real implementations have one, and the answer is that they do not. In the released EMPO code both available training stacks disable the penalty outright: the reference-model implementation sets its KL coefficient `beta: 0.0`, and the large-scale implementation switches the KL penalty off everywhere it can appear — both the penalty added to the reward and the auxiliary KL loss, along with both of their coefficients — in every one of its published run scripts. The entropy bonus is likewise set to zero. There is no term anywhere in the optimiser that opposes entropy reduction.

What remains is only the implicit restraint of taking small steps. There is a standard way to read each gradient step in the language of this section (formally, as *mirror descent*): every update applies Equation 6.2 with the *current* policy in place of π_0 and the step size in place of β — instead of tilting a fixed reference, each step tilts wherever the policy already stands, so the cluster probabilities evolve as $p_c \propto p_c\, e^{p_c/\beta}$. The same perturbation analysis now gives

$$\varepsilon_c \longmapsto \varepsilon_c\Big(1 + \frac{1}{K\beta}\Big),$$

an amplification factor strictly greater than one for *every* $\beta > 0$. No step size stabilises the balanced state. A trust region — whether imposed by a small learning rate or by the unusually tight clipping range the released configuration uses — bounds how fast the policy moves, not where it ends up. Within this idealised dynamics, collapse is not a risk that a step size can tune away; it is the asymptotic behaviour, on a clock the trust region can stretch but not stop.

Two features of the group-relative advantage sharpen this rather than soften it. Advantages are computed as $A_i = (r_i - \bar{r})/s_r$ within the group of G samples. The baseline $\bar{r}$ cancels exactly in Equation 6.2, since a constant shift in the exponent is absorbed by the normaliser, so subtracting the mean has no effect on the direction of the update. The division by the group standard deviation s_r does have an effect: it acts as an adaptive inverse temperature — rescaling the pull without redirecting it — and because $s_r \to 0$ as the group becomes homogeneous, the effective temperature *shrinks* precisely as collapse proceeds. The pull towards the dominant cluster does not weaken near the degenerate state; it is held back only by clipping.

This is the sharpest justification available for the entropy floor of Section 6.4, and stating it correctly also dissolves what looks, at first sight, like a contradiction. The chapter has argued that a floor strictly inside the simplex is structurally necessary; the released configuration sets $\delta_{\text{low}} = 0$; and yet the published runs completed stably and an ablation found the thresholds made little difference. If the floor were a *stability device* — a thing that keeps the optimiser from blowing up — those three facts could not sit together. They sit together because that is not what the floor is, and seeing what it is instead requires separating three questions that the word *collapse* runs together.

Does training diverge without a floor? No — and not because anything opposes the dynamics, but because the endpoint silences itself. A prompt whose group has fully agreed pays every response the same reward, the advantage is identically zero, and the prompt stops contributing gradient. The fully-agreed state is what dynamicists call *absorbing* — once a prompt reaches it, it never leaves — but it is also quiet: nothing visibly breaks, and reaching it looks like convergence, not failure. The catastrophic collapses of Section 6.3 — a classifier sending every input to one class, a policy emitting one character for every prompt — are *cross-input* degeneracies, and EMPO's structure blocks the direct route to them: per-prompt clusters leave no shared label space for a marginal to collapse onto, and the anti-gaming penalties tax the cheapest constant policies.

Where does each prompt end without a floor? With all of its probability on a single meaning — at a corner of the simplex. That is what the mean-field analysis establishes: once the anchor is gone the balanced state is unstable at every β, the transition is first order, and past the point of no return the commitment cannot be undone by a luckier sample later. Nothing in the optimiser selects a stopping point short of full commitment, and no setting of β can be made to select one — with an anchor it merely positions the system inside the window where both states coexist, and without one there is nothing to position.

Is full commitment what the method wants? Half of it is. Sharpening all the way to that corner is exactly the elicitation the method is for, on prompts where the majority is right. The other half is the cost, and it is not hypothetical, because the paper's own measurements price it: on prompts where the majority is wrong, the commitment is equally final — and the pass@k curves of Section 6.7, converging to the base model and crossing below it on hard benchmarks, are the coverage bill for sharpening run to the end.

On this reading the floor is a **stopping rule**, not a safety rail: it decides how much of the distribution's diversity sharpening is allowed to spend, per prompt, and it is the only instrument in the design that can make that decision at the level of the state. The ceiling cannot substitute — it rejects dispersion, and full commitment approaches from the other side. What operated in its place in the published runs was a second, cruder stopping rule: the finite training budget, a global and undirected version of what the floor would do per prompt. A fixed horizon stops the average prompt in time, the already-committed prompt too late, and the still-diffuse prompt too early; but under a horizon short enough, and clipping as tight as the released configuration's, most prompts never enter the region where a per-prompt floor would bind. The ablation's null verdict is then the expected signature rather than a refutation — it compared two configurations both governed, in effect, by the horizon — and the same reasoning says where the floor's value should become measurable: longer runs,

sparser prompts, weaker base models, and any deployment in which locking in the current majority early is expensive. The degeneracy is the one Section 6.3 traced through three decades of this idea, now out in the open because the supervised anchor that used to contain it has been removed. Read against that history, even the band as designed is a *filtering* remedy — the weakest of the three families Section 6.3 describes, since it acts on which examples are used rather than on the objective, and it has no counterpart to the fairness term that the semi-supervised and self-supervised literatures found necessary. The analysis also predicts the failure mode reported in follow-up work on this family of methods, in which performance improves for a while and then falls below the untrained model: that is what running past a first-order transition looks like from the outside — a horizon that outlasted its usefulness as a stopping rule.

Three caveats on the scope of this analysis. It is an idealised treatment of the *reward dynamics* in the space of cluster probabilities, not of the neural network; it treats each prompt independently, ignoring the fact that all prompts share one set of weights, so an update for one prompt shifts the policy for all others. It treats p_c as known rather than estimated from G samples, which for the group sizes actually used adds sampling noise of the same order as the effects being described. And it reads the optimiser as taking the idealised update of Equation 6.2 exactly, whereas the clipped surrogate actually used departs from that whenever the clip binds. What survives these caveats is the qualitative structure — self-consistency, a critical temperature, a discontinuous transition, and the absence of any stabilising term — not the precise numerical threshold.

A note on what here is new, since the ingredients are not. Reading a learning objective as a free energy and locating its critical temperature is the method of deterministic annealing, where Rose et al. (1990) computed the temperature at which a cluster splits in two — the same construction, run in the opposite direction, since there cooling *creates* structure out of a single-cluster symmetric state whereas here it *destroys* the balanced one. The information bottleneck has its own sharp transition, below which the trivial representation becomes the global optimum (Wu et al. 2019; Wu and Fischer 2020), and entropy-regularised learning dynamics are known to switch behaviour abruptly at a critical exploration rate (Kianercy and Galstyan 2012). Closest of all, GX-Chen et al. (2025) analyse precisely the object of Equation 6.2 and conclude, as this section does, that collapse is a property of the objective rather than of the optimiser, and that β is the parameter controlling it. What that work characterises is the optimum; what is added here is the self-consistency that arises when the reward is the policy's own cluster frequency, and with it a critical value, an order of transition, and a stability analysis. The claim worth defending is not that this is a phase transition — any physicist would frame it so — but the chain that follows from the transition being first order: collapse has memory, the threshold at which the balanced state loses stability is consequently the wrong safety criterion, and no choice of β can substitute for a floor on the state.

6.7 The information ceiling

There is a limit on what any of this can achieve, and it is worth stating formally because it is easy to lose sight of amid the empirical results.

Let π_0 be the base policy, π_T the policy after training, and $a^\star$ the true answer to a question q. Every update EMPO performs is computed from samples drawn from the current policy and from the equivalence relation $\sim$; the ground truth enters nowhere. Two facts about that loop do all the work: the randomness of sampling — seeds, batch order; call it R — carries no information about the answer, and $\sim$ is frozen before training begins — whatever it is, it does not change in response to anything the run produces. Everything the run ends up with is therefore determined by four ingredients: the base model, the question, the metric, and the dice. The trained policy is a deterministic function $\pi_T = f(\pi_0, q, \sim, R)$, and the data-processing inequality — the rule that no amount of computation can create information about a quantity its inputs do not already carry — gives

$$I\big(\pi_T;\, a^\star \mid \pi_0,\, q,\, \sim\big) = 0.$$

All the content sits in the words after the bar. The statement is not that the trained policy knows nothing about the truth — it plainly does — but that *given what the base model already was and what the equivalence relation already knew*, the training run added nothing. The same argument, run once more (the callout below has the steps), yields the inequality of this section's title, and it has two terms:

$$I(\pi_T;\, a^\star \mid q) \;\le\; \underbrace{I(\pi_0;\, a^\star \mid q)}_{\text{pre-training}} \;+\; \underbrace{I(\sim;\, a^\star \mid \pi_0,\, q)}_{\text{the metric}}.$$

Information about the truth can be rearranged and expressed more reliably, but the budget is fixed before the first update — by pre-training, plus whatever the equivalence relation itself carries about the answer. On mathematics the second term is not merely small but identically zero: the clustering there is answer extraction followed by symbolic equality, a hand-written rule with no trained parameters and no access to the question, so the ceiling is pre-training alone, exactly as the one-term reading suggests. On open-ended reasoning the second term cannot simply be declared zero, because the equivalence relation is itself a trained model — and that turns out to deserve a section of its own, taken up below. Either way, whatever the trained policy gained, it gained by redistributing probability mass over a support that the base model already had.

i Derivation, and a world small enough to check it

The assumption doing the quiet work is that the training randomness R — sampling seeds, batch order — is independent of $a^\star$ given $(\pi_0, q, \sim)$, and that $\sim$ is frozen before the run. Both hold in the released code, where rewards and filters are computed from cluster frequencies alone. Then, in three steps. *Determinism*: every update uses only the G samples and their cluster frequencies, so $\pi_T = f(\pi_0, q, \sim, R)$ with f deterministic. *Markov chain*: given $(\pi_0, q, \sim)$, π_T is a function of R alone and R is conditionally independent of $a^\star$, so $P(\pi_T \mid \pi_0, q, \sim, a^\star) = P(\pi_T \mid \pi_0, q, \sim)$ — the chain $a^\star \to (\pi_0, q, \sim) \to \pi_T$. *Data processing*: $0 \le I(\pi_T; a^\star \mid \pi_0, q, \sim) \le I(R; a^\star \mid \pi_0, q, \sim) = 0$. The two-term ceiling then takes two short steps: since π_T is a function of $(\pi_0, \sim, R)$ once q is fixed, data processing gives $I(\pi_T; a^\star \mid q) \le I(\pi_0, \sim, R;\, a^\star \mid q)$; and expanding the right side with the chain rule, the R term vanishes and the other two are the budget and the metric term. Supervision breaks the chain at the first step: the verifier reward $\mathbb{1}[\text{verify}(o_i, a)]$ makes π_T a function of $a^\star$ as well, and each verification deposits bits about the answer into the weights.

The theorem is easiest to trust in a world small enough to enumerate. Let $a^\star$ be uniform on $\{0, 1\}$; let pre-training set a parameter θ_0 equal to $a^\star$ with probability 0.8; let the base policy answer θ_0 with probability 0.7; and let training sample $G = 5$ answers and sharpen towards their majority, so that θ_T is a function of θ_0 and the sampling noise alone. Exact enumeration of the joint distribution gives $I(\theta_T; a^\star \mid \theta_0) = 0$ — machine-exact, not approximately — while $I(\theta_T; a^\star) = 0.121$ bits against a pre-training budget of $I(\theta_0; a^\star) = 0.278$ bits, and replacing the majority vote by a verifier-guided update lifts the conditional information to 0.358 bits. Two details repay attention. The inherited information is *less* than the budget: majority voting is a noisy compression of θ_0, and processing loses information more readily than it preserves it. And one loophole sits outside the theorem entirely — selecting a checkpoint on a labelled validation set is itself an update that touches ground truth, worth at most $\log_2$ of the number of candidates in bits; small, but not zero. The numbers are reproduced by `code/min-entropy-information-ceiling.py`.

The metric is not free. The second term of the ceiling deserves to be followed into the implementation, because what sits there is more interesting than a technicality. For open-ended reasoning, the released code clusters answers with a small generative verifier (Ma et al. 2025) — a mathematics-specialised 1.5B model fine-tuned on roughly 230,000 questions *with ground-truth answers*, whose equivalence verdicts were annotated by a much larger commercial model. The weights of $\sim$ are, in other words, a deterministic function of a labelled dataset. Three details of how it is called sharpen the point. The prompt contains the full question, not

just the two answers being compared. One of the two policy outputs is placed in a slot literally named *Ground Truth Answer* and the other in a slot named *Student Answer* — the model's native supervised-verification format, with a sampled answer standing in for the gold one. And the prompt instructs the model not to solve the question itself, which is an instruction, not an architectural constraint: nothing prevents the forward pass from forming its own belief about the answer on the way to a verdict.

So can the metric smuggle supervision into a "fully unsupervised" run? The answer has a clean structure: leakage requires two conditions *jointly*, and either one alone transmits nothing. First, **knowledge** — the verifier must actually carry information about this question's answer, $I(\sim; a^\star \mid \pi_0, q) > 0$, whether from partially solving q, from having seen related questions in its training set, or from the annotation model whose judgments it distils. Second, a **behavioural channel** — its same-or-not-the-same verdicts must *vary* with that knowledge: merging a pair more readily when both members agree with its own belief, or fragmenting clusters of answers it considers wrong. A verifier that knows the answer but judges equivalence without ever consulting that knowledge leaks nothing: if the verdicts do not vary with what it knows, nothing of what it knows can reach the training run. The toy world above extends to check this exactly: add a verifier belief V that equals $a^\star$ with probability 0.9, and a merging rule that consolidates clusters agreeing with V while fragmenting the rest.

clustering metric	$I(\theta_T; a^\star \mid \theta_0)$	$I(\theta_T; a^\star)$ vs. one-term ceiling 0.278
uninformed rule (regex-like)	0	0.044 — holds
informed, correctness-blind verdicts	0	0.121 — holds
informed, verdicts follow the belief	0.114	0.340 — **exceeded**

In the third row the training run genuinely adds information about the truth — the quantity the one-term reading says is zero is 0.114 bits, and the trained model ends up knowing more than pre-training supplied. No theorem is violated: conditioning on the verifier still gives exactly zero in every row, and the two-term ceiling (0.636 bits in this world) holds throughout. The bits came from the verifier's training labels, carried into the run through pairwise equivalence verdicts. The numbers are reproduced by `code/min-entropy-verifier-leak.py`.

Two things follow. The defence offered for such metrics — they answer only "are these two the same", never "which is better", and provide no gradient — is right in spirit, because it narrows the behavioural channel; the middle row of the table is exactly that defence working. But it does not close the channel while the question sits in the prompt, and how far the verdicts of this particular verifier follow its beliefs has not yet been measured — a natural piece of follow-up work, and a cheap one. The measurements are easy to specify: cluster with the question replaced by an empty string and compare training curves; measure how often the verdict flips when the two answers swap slots; on a labelled held-out set, compare the merge rate for pairs of equivalent *wrong* answers against pairs of equivalent correct ones. Until then, a careful accounting prices the two settings differently. The mathematics results — including the pass@k curves below — rest on a parameter-free symbolic rule and inherit the one-term ceiling exactly. The open-ended results are unsupervised *at the level of per-question labels*, with a metric distilled from labelled verification data; their ceiling is the two-term one, and the second term — plausibly small, since the verifier is a 1.5B model and the benchmarks in question sit well above its own solving ability — awaits measurement.

A last distinction keeps the accounting honest in the other direction. Even a metric carrying zero bits about $a^\star$ imports something real: a definition of *what counts as the same meaning*, learned from labelled entailment or equivalence data (Kuhn et al. 2023). That is supervision about how the space of answers is carved into meanings, not about which meaning is the true one — and it is load-bearing, since an earlier attempt with a weaker entailment model produced clusters poor enough that training collapsed into reward hacking. The theorem is silent about this kind of supervision, and should be: it is the same status as the tokeniser, fixed machinery that defines the problem rather than labels that answer it. The information ceiling concerns the

other kind, and for the metric actually deployed on open-ended tasks, the other kind is not ruled out — only unmeasured.

The ceiling is not a technicality, and it has a testable consequence: a method that adds no information cannot enlarge the set of problems the model is able to solve given enough attempts. Measuring pass@k — the probability that at least one of k sampled attempts is correct — the trained and untrained models should converge as k grows, and on some benchmarks the untrained model should win, since sharpening a distribution costs coverage. That cost is not a side effect — the objective works at it directly: a correct answer rare enough to appear once among G samples earns the group's lowest reward and a negative advantage, so the objective actively pushes down exactly the answers that only pass@k can see. Both predictions are observed — on mathematics benchmarks, where the metric is the parameter-free rule and the ceiling argument applies in its exact form. At $k = 1$ the trained model is far ahead; by $k = 256$ the curves meet, and on harder benchmarks the base model overtakes. Minimum-entropy learning changes which answer comes out first. It does not change which answers are reachable.

6.8 Schrödinger, read carefully

The preceding result has an old precedent, and getting the precedent right requires more care than the usual invocation supplies.

In *What Is Life?* (Schrödinger 1944), Schrödinger asks how an organism avoids the decay to thermodynamic equilibrium that the second law seems to demand. His answer, in the sections titled "It feeds on 'negative entropy' " and "The statistical meaning of entropy," is that it does not evade the law but pays for its exemption:

> "Thus a living organism continually increases its entropy — or, as you may say, produces positive entropy — and thus tends to approach the dangerous state of maximum entropy, which is death. It can only keep aloof from it, i.e. alive, by continually drawing from its environment negative entropy."

and, a few pages later, that the organism "attract[s], as it were, a stream of negative entropy upon itself, to compensate the entropy increase it produces by living and thus to maintain itself on a **stationary and fairly low entropy level**."

Two details in that sentence are usually dropped and both matter here. First, the claim is a *balance*, not a minimisation: a level is held constant because what flows out offsets what is produced inside. Nothing is being pushed to its lowest possible value. Second, the level is "fairly low," not zero. Schrödinger is describing a floor, and Section 6.6 gives the machine-learning analogue of why a floor rather than a minimum is the right target.

The part of the book that bears most directly on Section 6.7 is a different argument, the one Schrödinger frames as **order from order** as against order from disorder. Ordinary physical law, he observes, is statistical: the regularity of a gas law is order extracted from the disorder of enormous numbers. Heredity cannot work that way, because the structure carrying it is far too small for statistics to smooth anything. It must instead be order copied from pre-existing order — his "aperiodic crystal," a structure whose information resides in not repeating.

The mapping onto the pass@k result is exact enough to be worth stating as a point-by-point correspondence rather than a loose analogy. Pre-training is the aperiodic crystal: the reservoir of structure, acquired at enormous cost from an external source. Entropy minimisation is the copying mechanism: it makes that structure come out reliably, and it makes it come out first. It creates nothing. An organism does not synthesise its own order either; it imports it, and Schrödinger's whole point is that the import is what has to be accounted for.

Where the analogy must stop is also clear. Schrödinger's entropy is the physical entropy of an open system exchanging real heat with a real environment, carrying units of k_B and constrained by the second law. The

entropy of Section 6.2 is the Shannon entropy of a predictive distribution over meanings. Chapter 2 argued that these are the same functional applied to different objects, which licenses the shared vocabulary and the shared mathematics — but not the transfer of a thermodynamic theorem. Nothing in this chapter is derived from the second law, and nothing in it needs to be.

One further detail is worth recording, because it turns out to favour the treatment given here. In a note appended to the chapter in later editions, Schrödinger conceded to his critics that he would have framed the discussion in terms of **free energy** had he been writing for physicists alone, choosing entropy only because it connects more transparently to Boltzmann's order–disorder principle for a general reader. The concession is narrower than it is often reported to be — he does not retract the argument — but it points in a useful direction. Section 6.6 shows that the object doing the real work in minimum-entropy learning *is* a free energy, with the KL coefficient as its temperature. On this point Schrödinger's second thoughts and the mathematics agree.

6.9 Prigogine, and an analogy that does not hold

There is a second thermodynamic principle with "minimum entropy" in its name, formulated a year after Schrödinger's book by a scientist who worked on open systems and later won a Nobel Prize for the thermodynamics of self-organisation. The temptation to enlist it here is considerable. It should be resisted, and setting out why is more instructive than the endorsement would have been.

Prigogine's theorem (Prigogine 1945, 1947) concerns the **entropy production rate**. For an open system the second law is written as a balance, $dS/dt = d_eS/dt + d_iS/dt$, separating entropy exchanged across the boundary from entropy generated internally, with the internal term $\sigma \equiv d_iS/dt \geq 0$. Near equilibrium, each flow responds in proportion to the push that drives it — heat flow to a temperature difference, diffusion to a concentration difference: $J_i = \sum_j L_{ij}X_j$, and

$$\sigma = \sum_i J_iX_i = \sum_{i,j} L_{ij}X_iX_j.$$

The theorem states that if some pushes are held fixed by external constraints and the rest are free, minimising σ over the free ones is equivalent to the vanishing of the flows they drive — that is, to the steady state. In that regime σ only ever decreases as the system evolves (it is what dynamicists call a Lyapunov function), which is what makes the steady state stable.

Three things about this result rule it out as a foundation for anything in this chapter.

It is about a rate, not a level. The quantity minimised is d_iS/dt, with units of entropy per unit time, and it is not the system's entropy. Schrödinger's "fairly low entropy level" and Prigogine's minimum entropy production are different claims about different quantities, and they are logically independent: a system could hold Schrödinger's balance while producing entropy at any rate whatever (Gnaiger 2009). What Section 6.2 minimises is a level — the entropy of a distribution — with no rate anywhere in sight.

Its hypotheses have no counterpart here, and they are more restrictive than usually advertised. The theorem requires the flow–push relations to be linear, symmetric ($L_{ij} = L_{ji}$, Onsager reciprocity), with coefficients L_{ij} that are *constant* rather than state-dependent — and it requires the system's variables to look the same when the film is run backwards (Verschaffelt 1954; Maes and Netočný 2013). That last condition is the one most often omitted, and it is precisely the one Landauer (1975) exploited: a resistance in series with an inductance settles at the current $I = E/R$, which manifestly does not minimise $\sigma = RI^2/T$ — because a current reverses sign when the film runs backwards, and the theorem's hypotheses exclude such variables. Jaynes noted that the principle already fails for a network of resistors held at different temperatures (Jaynes 1980). Extending the statement from a point to a whole region, by minimising $\int_V \sigma\, dV$ to determine spatial profiles, produces field equations that contradict the conservation laws (Barbera 1999; Martyushev et al. 2007). Modern work has settled the status of the whole construction: the theorem is a small-driving approximation to an

exact variational principle, accurate only when the system is barely pushed (Maes and Netočný 2007). It is a narrow theorem, not a general law.

It does not apply to living systems, and Prigogine said so. In his Nobel lecture he was explicit that the result holds only in "a 'strictly' linear theory in which the deviations from equilibrium are so small that the phenomenological coefficients may be treated as constants," and that "far from equilibrium the thermodynamic behavior could be quite different, in fact, even opposite to that indicated by the theorem of minimum entropy production" (Prigogine 1978). Elsewhere he noted that the linear relations, adequate for heat conduction and diffusion, "are generally not valid for the conditions of chemical kinetics" — which is to say, for biochemistry. The modern verdict is blunter. Martyushev and Seleznev (2013) point out that if minimum entropy production governed development, the goal of an organism's life would be the soonest possible death, since σ attains its minimum value of zero at equilibrium.

The history here contains its own warning. Prigogine did apply the principle to biology, in a 1946 paper with Wiame proposing that living systems evolve towards states of least entropy production and offering, in the authors' words, "a physicochemical interpretation of Lamarckism" (Prigogine and Wiame 1946). That paper is a substantial part of why the principle took hold among biologists, and its destination is a fair indication of what happens when a theorem with narrow hypotheses is carried into a domain that satisfies none of them.

Two further cautions belong here, since they generalise beyond this case. The first is that minimum and maximum entropy production are not the contradiction they appear to be: one minimises and the other maximises the same function, under different constraints and with respect to different variables, and which applies depends on the choice of thermodynamic variables and their behaviour under time reversal (Martyushev and Seleznev 2006). The second is a piece of history that ought to make anyone cautious about arguing from the word "entropy" in a title. The founding paper of the maximum-entropy-production literature in climate science was published as a theory of *minimum* entropy exchange; the two descriptions are the same physics, differing only in whether one counts the flux at the boundary as leaving or entering (Paltridge 1975).

The conclusion for this chapter is that the shared phrase is a coincidence. Minimum-entropy learning is not an instance of the minimum entropy production principle, does not inherit its guarantees, and would gain nothing if it did, since those guarantees hold only in a regime with no analogue here. There is one genuine structural inversion worth noticing on the way out. In Prigogine's setting, low entropy production is where the dynamics *ends up on its own*; the principle is descriptive. In the setting of Section 6.6, low entropy is what we *pay* for, and having paid, we must then pay again to stop the system going lower. The two are not the same shape of statement at all.

6.10 So why call it minimum-entropy learning?

Given the preceding two sections, the name deserves a defence. It is defensible on two of three possible grounds, and it is worth being explicit about which.

As a literal description, it holds. The method minimises a named entropy functional over a well-defined space — the Shannon entropy of a distribution over semantic clusters as stated, the Rényi entropy at $\alpha = 2$ as implemented. Nothing metaphorical is happening; a specific entropy of a specific distribution goes down.

As a structural claim, it holds, and this is the more interesting case. The objects that appear when the method is analysed are not borrowed imagery. The KL-regularised optimum is a Gibbs distribution; the objective is a free energy; the coefficient β enters exactly where a temperature enters; the self-consistency condition is a mean-field equation with a genuine critical point and a genuine first-order transition. These are the same mathematics as statistical physics, not a picture of it — which is the same standard that justifies calling a Boltzmann machine a Boltzmann machine.

As a claim to physical law, it does not hold and is not needed. No thermodynamic theorem is being invoked, no second law is doing any work, and no result transfers from Prigogine's setting or from Schrödinger's. The

biological reading of Section 6.8 is an analogy about the *provenance of order* — a claim about where structure comes from, which the pass@k result independently establishes on its own terms — and not a derivation.

Keeping these three levels apart is the discipline this whole area most needs. The entropy vocabulary is powerful enough to describe learning objectives precisely, and seductive enough to make unearned claims sound rigorous. The first two grounds are earned. The third would not be.

6.11 When agreement means truth, and when it does not

The method rests on treating consensus as a proxy for correctness, and it is worth being precise about when that substitution is legitimate.

It works when the knowledge is already in the model but unstably expressed. The model can reach the answer; it simply does not do so reliably, and sharpening the distribution converts an occasional success into a dependable one. This is the regime Section 6.7 describes: the answer is already among the things the model can say, and training moves probability onto it. It also requires the question to have a stable meaning, since grouping by meaning is the only thing preventing "be consistent" from degenerating into "repeat yourself."

It fails in two ways. The model may never have seen the material, in which case there is nothing to sharpen — no amount of self-agreement conjures knowledge the model never had. More dangerously, the model may be confidently and consistently wrong, in which case low entropy certifies a falsehood and training reinforces it. The paper's own appendix contains a clean instance: on a probability problem the model silently substitutes a different goal, then twice announces that something looks wrong and reworks the solution — each rework landing on the arithmetic and never on the substituted goal. It returns 325 with high confidence; the answer is 265. Nothing in the reward can see the substitution: a model that re-derives the same misreading on resampling lands in the same cluster, the semantic entropy is low, and agreement is paid for being wrong. Fluent self-correction is not verification.

Three design rules follow, and they summarise most of what this chapter has established.

Threshold, do not minimise. Stop reducing entropy at a floor, leaving the model room to change its mind. Section 6.6 shows this is not a matter of taste: without a floor the fixed point is degenerate and the approach to it is discontinuous. Schrödinger's "fairly low," not zero.

Group meanings, not strings. Otherwise consistency collapses into repetition, which is free to produce and worth nothing.

Never evaluate on the training signal. Consistency drives the optimisation; held-out ground truth is what gets reported. The moment those become the same number, the method is measuring its own success by the criterion it was trained to satisfy, and the failure case above becomes invisible.

6.12 Takeaway

Minimum-entropy learning turns a model's disagreement with itself into a training signal, which buys reliable elicitation of existing capability without any labels at all — and buys nothing beyond that, by an information-theoretic argument that the pass@k curves on mathematics confirm in its exact form. Its central pathology, degeneration to a single confident answer, is not an implementation nuisance but a phase transition in a mean-field free energy, which is why the durable remedies are floors on the entropy of the state — stopping rules for how much diversity sharpening may spend — rather than settings of a knob.

Set against Chapter 5, the pair is now complete: maximum entropy supplies possibility and minimum entropy supplies commitment, and neither is a general recipe for intelligence on its own. The next chapter shows that the two are not merely complementary but composable — that choosing *which features to model* turns out to require maximising entropy on the inside and minimising it on the outside, simultaneously.

7 Minimax-Entropy Learning

Maximum entropy (Chapter 5) answers a precise question: given a set of measured features, which model is least biased? It is silent on a prior question — *which features should we measure at all?*

Every model is a lossy compression that trades complexity against uncertainty. Include too few features and the model stays vague; include too many and it ceases to be a compression. For a fixed budget of features, we want the description that leaves the least uncertainty.

Many learning problems raise a second question as well. A classifier, a test-time adapter, or a reasoning model has to *commit*: it must put its probability on one value of a designated variable Y, e.g., a label or an answer, and describing its inputs well is not enough. The same nesting answers this second question. The inner maximum-entropy model describes the data as honestly as the features allow, and the outer minimization acts on the entropy of the prediction.

The chapter takes the two questions in turn. We state the principle for the first in Section 7.1 and answer the second on the joint space of inputs and labels, with labels in Section 7.2 and without them in Section 7.3; Section 7.4 then marks the boundaries of the principle and weighs the evidence.

7.1 The minimax entropy principle: what to measure

Starting from the **minimum description length** (MDL) principle (Grünwald 2007), two results follow in sequence. First, among all models consistent with a feature set F, the shortest description is the maximum-entropy model (Feder 1986). Second, for such models the description length equals the model's own entropy, $L(P_F) = S(P_F)$. Choosing features to minimize description length therefore means choosing them to minimize that entropy.

The optimal features are the ones producing the maximum-entropy model with *minimum* entropy (Carcamo et al. 2025):

$$F^* = \arg\min_F \; S(P_F), \qquad P_F = \arg\max_P \left\{ S(P) \;:\; \mathbb{E}_P[f_\mu] = \langle f_\mu \rangle_{\text{exp}},\; f_\mu \in F \right\}.$$

The nesting gives the principle its name: maximize entropy on the inside, minimize it on the outside. The best features are those that maximally constrain our expectations while leaving the model maximally random about everything unobserved. The principle was first proposed for texture modeling in computer vision (Zhu et al. 1997). Throughout this chapter, *minimax entropy* is meant in this sense of Zhu, Wu, and Mumford; a method of the same name in domain adaptation is unrelated, as we note in Section 7.4.

7.1.1 Three equivalent views

Minimizing $S(P_F)$ optimizes three quantities at once. In what follows, P_{true} denotes the distribution that actually generated the data, and P_{ind} the featureless baseline model in which the variables are treated as independent.

- **Shortest description.** Since $L(P_F) = S(P_F)$, the optimum is the most compressed account of the data.
- **Closest to the truth.** The entropy gap to the true distribution is exactly a divergence, $S(P_F) - S(P_{\text{true}}) = D_{\text{KL}}(P_{\text{true}} \| P_F)$, so the optimum is also the most accurate model.

- **Most informative features.** The information carried by the selected features, $I_F = S(P_{\text{ind}}) - S(P_F)$, is maximized at the same point.

Compression, accuracy, and informativeness coincide — a reassuring sign that the criterion is the right one.

7.1.2 Solving it

The difficulty is combinatorial. Each candidate feature set requires solving a maximum-entropy problem for its parameters, and the number of candidate sets grows super-exponentially, so brute-force search is infeasible beyond small systems (Carcamo et al. 2025). The practical workhorse is a **greedy** algorithm that repeatedly adds whichever remaining feature reduces the entropy most.

Structure sometimes restores exactness. For trees of pairwise correlations the entropy reduction decomposes into a sum of mutual informations, the problem becomes a maximum-spanning-tree problem, and greedy is optimal. More generally, if entropy reduction is **submodular** — diminishing returns as features accumulate — greedy comes within $1 - 1/e$ of the optimum (Nemhauser et al. 1978). Whether the minimax entropy objective is submodular in general remains open; where it is not, guarantees can still be recovered in terms of how far the objective departs from submodularity (Bian et al. 2017).

7.1.3 Implicit minimax entropy in energy-based models

The principle is closer to everyday practice than its combinatorial statement suggests. Let an energy-based model (Chapter 4) be linear in its last layer, $E_\theta(x) = -\sum_\mu w_\mu \phi_\mu(x;\theta')$, so that for fixed θ' the functions $\phi_\mu(\cdot;\theta')$ play the role of the feature set F. For fixed θ' the maximum-likelihood w is the maximum-entropy model P_F of these features, and at that point the cross-entropy of the data equals $S(P_F)$; this is the identity Equation 5.18 of Section 5.6.2, whose hypotheses are a linear last layer, an exact fit of w, and a uniform base measure. Maximum likelihood over θ' is therefore $\min_F S(P_F)$ over a continuous family of feature sets: minimax entropy, with gradient descent in place of greedy selection. In this sense every neural energy-based model trained by maximum likelihood is a minimax-entropy learner, whether or not anyone says so. What this chapter adds, and what the implicit version lacks, is the *explicit* selection of a feature set under a description-length budget — a combinatorial outer problem whose solution can be inspected, bounded, and compared across budgets. It also adds a second kind of outer problem, which acts on the entropy of a prediction rather than on the entropy of the whole model (Section 7.2).

7.2 Minimax entropy on the joint space: modelling and predicting

So far the outer problem has chosen features, i.e., it has decided *what to measure*. We now turn to the second question raised at the beginning of the chapter, namely what a model should *commit to*. Let us consider the most common case, a classifier over K labels, with labelled data in this section and without labels in Section 7.3. As in Chapter 5 and Chapter 6, we write H for the entropy that Section 7.1 denotes by S, and Δ_K for the probability simplex over the labels.

7.2.1 Two questions, one principle

Let $f_\theta : \mathcal{X} \to \mathbb{R}^K$ be the logits of a classifier and h a base measure on $\mathcal{X}$. The joint energy-based model (JEM) of Grathwohl et al. reads the logits as a negative joint energy (Grathwohl et al. 2020), as in Section 4.9:

$$p_\theta(x,y) = \frac{h(x)\, e^{f_\theta(x)[y]}}{Z(\theta)}, \qquad p_\theta(x) = \frac{h(x)\, e^{-E_\theta(x)}}{Z(\theta)}, \tag{7.1}$$

where $E_\theta(x) = -\log \sum_y e^{f_\theta(x)[y]}$ is the marginal energy and $Z(\theta) = \int h(x) \sum_y e^{f_\theta(x)[y]}\, dx$. The normalizer cancels in the conditional, $p_\theta(y \mid x) = \operatorname{softmax}\big(f_\theta(x)\big)_y$, which is the usual softmax classifier. Note that adding the same input-dependent scalar to all the logits, a redundant degree of freedom for an ordinary classifier, now changes $p_\theta(x)$.

Let the last layer be linear, $f_\theta(x)[y] = w_y^\top \phi_{\theta'}(x) + b_y$, with $\theta = (\theta', W, b)$. For a fixed representation θ', the model Equation 7.1 is the exponential family with the sufficient statistics

$$\mathrm{T}_{\theta'}(x, y) = \big(e_y \otimes \phi_{\theta'}(x),\ e_y\big), \tag{7.2}$$

where e_y is the y-th unit vector of $\mathbb{R}^K$; it is the maximum-entropy distribution relative to h that matches the moments of these statistics (Equation 5.1). For a classifier read in this way, the phrase "the model takes maximum entropy" is therefore exact, at the representation level of Section 5.1. Over such a model the outer problem can minimize two different entropies, and we term them two types.

> **i** Definition: minimax-entropy learning
>
> A method belongs to **minimax-entropy learning** if i) *inner*: its model is the maximum-entropy (Gibbs) distribution over the modelled variables that matches the moments of a set of features, possibly relative to a base measure; ii) *outer*: the features, and the assignment of a designated variable Y where Y is unobserved, are chosen to minimize an entropy of that model, namely
>
> - **Type I** (*what to measure*): the entropy of the model itself, with the features ranging over a budgeted combinatorial family, as in the principle of Section 7.1, or over a continuous family, as in the implicit version of Section 7.1.3; or
> - **Type II** (*what to commit to*): the predictive entropy of Y, measured by the cross-entropy where Y is observed (Section 7.2.2), and through a posterior sharpened at a label temperature $T < 1$ where it is not (Section 7.3.1);
>
> and iii) *anchor*: the inner model is a maximum-entropy learner in the sense of Section 5.1.4, fitted to the same observed data on which the outer entropy is minimized; a frozen reference model does not suffice. When the inner model is fitted by score matching or noise-contrastive estimation instead of maximum likelihood, we speak, as in Section 5.1.4, of minimax-entropy learning *in the generalized sense*.

Type I is the principle of Section 7.1. Type II is what the slogan of this chapter, *minimize entropy to commit to what matters*, refers to, and the next two subsections show that, for classification, the slogan can be taken literally. Condition iii) is what separates the present chapter from Chapter 6, and we return to it in Section 7.4.

7.2.2 Classification is conditional minimax entropy

Let us start with labelled data $\{(x_i, y_i)\}_{i=1}^n$ with empirical distribution $\hat{p}$, and write $\hat{H}_q(Y \mid X) = \frac{1}{n} \sum_i H\big(q(\cdot \mid x_i)\big)$ for the conditional entropy of a conditional distribution q on the observed inputs.

The conditional bridge. Fix the representation θ', let $(W^\star, b^\star)$ be the maximum-likelihood softmax head, and write $\theta = (\theta', W^\star, b^\star)$. Let $\mathcal{Q}_{\theta'}$ be the set of conditional distributions q that match the data on the statistics of Equation 7.2 under the empirical input marginal, i.e., $\mathbb{E}_{\hat{p}(x)\, q(y|x)}\big[\mathrm{T}_{\theta'}\big] = \mathbb{E}_{\hat{p}(x,y)}\big[\mathrm{T}_{\theta'}\big]$. Then

$$\underbrace{\mathbb{E}_{\hat{p}}\big[-\log p_\theta(y \mid x)\big]}_{\text{cross-entropy}} = \hat{H}_{p_\theta}(Y \mid X) = \max_{q \in \mathcal{Q}_{\theta'}} \hat{H}_q(Y \mid X). \tag{7.3}$$

Minimizing the cross-entropy over the representation is therefore $\min_{\theta'} \max_{q \in \mathcal{Q}_{\theta'}} \hat{H}_q(Y \mid X)$: the outer prob-

lem minimizes, over the features, the largest conditional entropy that the features allow.

> **Derivation of the conditional bridge**
>
> Write $\lambda = (W, b)$, so that $\log p_\theta(y \mid x) = \lambda^\top \mathrm{T}_{\theta'}(x, y) - \log Z_\lambda(x)$ with $Z_\lambda(x) = \sum_y \exp\big(\lambda^\top \mathrm{T}_{\theta'}(x, y)\big)$. The likelihood equations in λ are the moment conditions
>
> $$\frac{1}{n}\sum_i \mathrm{T}_{\theta'}(x_i, y_i) \;=\; \frac{1}{n}\sum_i \mathbb{E}_{p_\theta(y \mid x_i)}\big[\mathrm{T}_{\theta'}(x_i, y)\big].$$
>
> Substituting them into the cross-entropy $\frac{1}{n}\sum_i \big[\log Z_\lambda(x_i) - \lambda^\top \mathrm{T}_{\theta'}(x_i, y_i)\big]$ replaces the data moments by the model moments, and gives $\frac{1}{n}\sum_i \mathbb{E}_{p_\theta(y \mid x_i)}\big[-\log p_\theta(y \mid x_i)\big] = \hat{H}_{p_\theta}(Y \mid X)$, which is the first equality. The second is the duality of Berger, Della Pietra, and Della Pietra (Berger et al. 1996): among the conditional distributions that satisfy these moment constraints under the empirical input marginal, the one with the largest conditional entropy is the conditional exponential family with the maximum-likelihood parameters.

It is worth noting that Equation 7.3 removes an apparent conflict of signs. Berger et al. (Berger et al. 1996) and Farnia and Tse (Farnia and Tse 2016) describe classification as *maximizing* a conditional entropy, and Farnia and Tse identify the maximum-conditional-entropy rule with the robust Bayes decision rule under the logarithmic loss, i.e., with the game against nature of Section 5.6. Training a classifier, on the other hand, is commonly described as *minimizing* a cross-entropy, which at the fitted head equals the conditional entropy of its predictions. Both descriptions are correct, at different levels of the nesting. The inner problem commits as little as the features allow, and the outer problem chooses the features such that this least committal prediction is still as sharp as possible. Zhou et al. use the name *minimax conditional entropy* for an analogous nesting in crowdsourcing (Zhou et al. 2015).

The hypotheses are the analogues of those of Equation 5.18: a linear head, an exact fit of the head, and no regularization; no base measure is needed, since the conditional does not involve one. A deep network is trained jointly rather than by nested optimization, so the identity holds at stationary points of the head. With weight decay it becomes an identity of regularized maximum entropy, in which the moment constraints are relaxed rather than exact (Section 5.6). If the features separate the labelled data, the maximum-likelihood head does not exist. The constraint set then contains only the conditional that puts all its mass on the observed labels, and both sides of Equation 7.3 equal zero, the cross-entropy as an infimum.

7.2.3 A classifier is a joint maximum-entropy model

Let us now fit the joint model Equation 7.1 instead of the conditional one.

The joint bridge. Fix θ', let $(W^\star, b^\star)$ be the maximum-likelihood head of the joint model, and let h be uniform. Then

$$\begin{aligned}\underbrace{\mathbb{E}_{\hat{p}}\big[-\log p_\theta(x)\big]}_{\text{generative}} + \underbrace{\mathbb{E}_{\hat{p}}\big[-\log p_\theta(y \mid x)\big]}_{\text{discriminative}} \\ = H\big(p_\theta(X, Y)\big) \;=\; \underbrace{H_\theta(X)}_{\text{modelling}} + \underbrace{H_\theta(Y \mid X)}_{\text{predicting}},\end{aligned} \tag{7.4}$$

where $H_\theta(Y \mid X) = \mathbb{E}_{p_\theta(x)} H\big(p_\theta(\cdot \mid x)\big)$ and the entropy of X is differential; for a non-uniform h, the entropies are taken relative to h. Training JEM with equal weights on its two terms is therefore the principle of Section 7.1 on the joint space (X, Y). The outer problem minimizes the joint entropy of the maximum-entropy model, and the chain rule splits this entropy into a part that models the inputs and a part that predicts the label. In the terms of the definition above, JEM is of both types at once.

Derivation of the joint bridge

With $\lambda = (W, b)$, the joint negative log-likelihood is $-\log p_\theta(x, y) = \log Z(\theta) - \lambda^\top \mathrm{T}_{\theta'}(x, y) - \log h(x)$, which, for uniform h, is affine in the statistics of Equation 7.2. The likelihood equations in λ equate the data and model moments of $\mathrm{T}_{\theta'}$, so the expected negative log-likelihood under $\hat{p}$ equals that under p_θ, i.e., $H\big(p_\theta(X, Y)\big)$; this is Equation 5.18 for the joint exponential family. Splitting $p_\theta(x, y) = p_\theta(x)\, p_\theta(y \mid x)$ on the left gives the two data terms, and the chain rule of entropy gives the right-hand side.

One should be careful about what is split. The identity Equation 7.4 contains two different splits of one total: a data-side split into a generative negative log-likelihood and a cross-entropy, and a model-side split into $H_\theta(X)$ and $H_\theta(Y \mid X)$. Only the totals are equal. The reason is that $-\log p_\theta(x) = E_\theta(x) + \log Z(\theta) + \text{const}$, and the marginal energy E_θ is a log-sum-exp of the statistics rather than a linear function of them, so the moment conditions do not equate its data and model expectations. One can verify on small, exactly solvable examples that the two splits indeed differ term by term. The precise statement behind "JEM minimizes the entropy of its predictions" is therefore that it minimizes the total $H_\theta(X) + H_\theta(Y \mid X)$, and the two parts can trade against each other.

Grathwohl et al. train JEM with the factorization $\log p_\theta(x, y) = \log p_\theta(x) + \log p_\theta(y \mid x)$: the second term is the exact cross-entropy, and the gradient of the first is estimated by stochastic gradient Langevin dynamics (SGLD) with persistent chains (Grathwohl et al. 2020). On CIFAR-10 the model reaches an accuracy of 92.9%, against 95.8% for the purely discriminative Wide ResNet, with an Inception Score of 8.76 and an FID of 38.4. On CIFAR-100 it reaches 72.2% against a baseline of 74.2%, and it is, in the authors' words, nearly perfectly calibrated, whereas the baseline is very poorly calibrated. The authors also note that their gradient estimators are quite unstable. Later work narrows the gaps in accuracy and in sample quality with sharpness-aware minimization and by excluding data augmentation from the maximum-likelihood estimate (Yang et al. 2023), improves accuracy, stability, and speed with a proximal variant of SGLD and an informative initialization (Yang and Ji 2021), and addresses the insufficient quality of the SGLD negative samples (Sustek et al. 2023).

One ablation of the original paper is instructive for our reading. The factorization above was chosen so that $\log p_\theta(y \mid x)$, which is computed exactly, receives no bias from the sampler. Prior work factorizes the objective instead as $\log p_\theta(x \mid y) + \log p_\theta(y)$, with a separate normalizer for every class, so that neither $p(y \mid x)$ nor $p(x)$ can be computed. Trained to maximize this objective, the model reaches only 30.1% accuracy (Grathwohl et al. 2020, Table 1). In the authors' terms, the two objectives are alternative factorings of the same likelihood. The principle fixes the criterion, the joint likelihood, and leaves the estimator open, and the estimator can matter as much as the criterion.

The observation that a discriminative network contains a generative model predates JEM, and it comes from the same line of work as the minimax entropy principle. Dai, Lu, and Wu construct a generative model of a ConvNet as an exponential tilting of a reference distribution (Dai et al. 2015). Xie et al. derive the generative ConvNet from the discriminative ConvNet, by assuming that one of the categories is a base category generated by a reference distribution, and present it as a hierarchical FRAME model (Xie et al. 2016); FRAME is the texture model for which the minimax entropy principle was proposed (Zhu et al. 1997, 1998). A second line learns generative models through discriminative classifiers (Tu 2007; Jin et al. 2017). What JEM adds is a single normalizer for $p_\theta(x)$, $p_\theta(x, y)$, and an exact $p_\theta(y \mid x)$, which makes the joint form trainable with modern architectures. Placing JEM in the present chapter is thus a return to the original line rather than a graft.

7.3 When the label is missing

Without labels, the data moments in Equation 7.3 and Equation 7.4 cannot be computed, and one has to decide how the missing label enters the objective. A natural device is a distribution $q(\cdot \mid x)$ over the label, with a temperature on it. We develop this device first, and then turn to the term that distinguishes a joint model from a purely discriminative one.

7.3.1 A temperature on the label

Free energy at a label temperature. For an unlabelled input x and a label temperature $T > 0$, let

$$\ell_T(\theta; x) \;=\; \min_{q \in \Delta_K} \Big\{ \mathbb{E}_q\big[-\log p_\theta(x, y) \big] - T\, H(q) \Big\}. \tag{7.5}$$

The minimizer is $q^\star \propto p_\theta(y \mid x)^{1/T}$, and

$$\begin{aligned} \ell_T(\theta; x) \;&=\; -T \log \sum_y p_\theta(x, y)^{1/T} \\ &=\; -\log p_\theta(x) + (1 - T)\, H_{1/T}\big(p_\theta(\cdot \mid x)\big), \end{aligned} \tag{7.6}$$

where H_α is the Rényi entropy of order α (Section 2.1.4). In particular, $\ell_1 = -\log p_\theta(x)$, and as $T \to 0$,

$$\ell_T(\theta; x) \;\to\; -\log \max_y p_\theta(x, y) \;=\; -\log p_\theta(x) + H_\infty\big(p_\theta(\cdot \mid x)\big),$$

with H_∞ the min-entropy.

> **Derivation of the label free energy**
>
> The objective Equation 7.5 is the free energy Equation 5.2 on the label, with the energy $-\log p_\theta(x, y)$ and the counting measure as the base measure. Its minimizer is the Gibbs distribution $q^\star \propto p_\theta(x, y)^{1/T} \propto p_\theta(y \mid x)^{1/T}$, and its minimum is $-T$ times the log-partition function, $-T \log \sum_y p_\theta(x, y)^{1/T}$. Writing $p_\theta(x, y) = p_\theta(x)\, p_\theta(y \mid x)$ and $\alpha = 1/T$, one has $\log \sum_y p_\theta(y \mid x)^\alpha = (1 - \alpha) H_\alpha$, hence $-T \log \sum_y p_\theta(y \mid x)^{1/T} = -T(1 - 1/T) H_{1/T} = (1 - T) H_{1/T}$. The value at $T = 1$ is immediate, and the limit $T \to 0$ follows from $-T \log \sum_y a_y^{1/T} \to -\log \max_y a_y$ for positive a_y.

We note that Equation 7.6 places maximum-entropy and minimum-entropy learning of the missing label on one dial, and that the entropy it produces is a Rényi entropy whose order is the reciprocal of the label temperature. At $T = 1$ the missing label is assigned by the posterior with its full entropy, and the objective reduces to the marginal likelihood. This is maximum-entropy learning with missing data, i.e., EM, or the semi-supervised JEM of Zhao et al. (Zhao et al. 2020), and it belongs to Chapter 5. For $T < 1$ the entropy enters with the positive weight $1 - T$, and the objective penalizes the uncertainty of the prediction. This is where the problem becomes minimax: the input side is still a maximum-entropy learner, while the label side is pushed towards low entropy. As $T \to 0$ the objective becomes the classification likelihood. Lowering T along a path is deterministic annealing (Rose 1998), here applied to the label, and Table 7.1 places known methods on the dial.

Table 7.1: The label-temperature dial on an unlabelled input, where T is the label temperature. The objective is $\ell_T(\theta; x) = -\log p_\theta(x) + (1 - T)\, H_{1/T}$ of Equation 7.6, and the third column gives its label-side term, with $H_{1/T}$ the Rényi entropy of order $1/T$ of $p_\theta(\cdot \mid x)$. The instances include joint and purely discriminative methods; dropping the term $-\log p_\theta(x)$ gives the purely discriminative version of each row.

T	entropy of the missing label	label-side term	known instances
$T > 1$	rewarded	$-(T-1)\, H_{1/T}$	discriminative analogue: the confidence penalty (Pereyra et al. 2017)
$T = 1$	full, from the posterior	0	EM; semi-supervised JEM (Zhao et al. 2020); TEA on test inputs (Yuan et al. 2024)
$0 < T < 1$	penalized, from a sharpened posterior	$(1 - T)\, H_{1/T}$	deterministic-annealing EM for entropy regularization (Grandvalet and Bengio 2006)
$T \to 0$	none, a hard label	H_∞	classification EM (Celeux and Govaert 1992); self-training with pseudo-labels (Lee 2013)

Relation to entropy regularization. Grandvalet and Bengio solve their entropy-regularized criterion with a deterministic-annealing EM (Grandvalet and Bengio 2006). Its E-step distributes an unlabelled input over the classes with the Gibbs weights $g_m \propto P(m \mid x)^{1/(1-\lambda)}$; $\lambda = 0$ recovers EM, $\lambda = 1$ gives the hard assignments of classification EM (Celeux and Govaert 1992), and the authors describe $T = 1 - \lambda$ as the analogue of a temperature. These weights are exactly the minimizer $q^\star$ of Equation 7.5 at $T = 1 - \lambda$. Two remarks are in order. Firstly, their criterion involves only $P(y \mid x)$ and favours low density separation, in their words, "without any modeling of the density of input features". The extra term $-\log p_\theta(x)$ in Equation 7.6 is precisely the maximum-entropy learner on the inputs that a joint model supplies, so Type II can be described as entropy regularization with a maximum-entropy model of the inputs. Secondly, their criterion penalizes the Shannon entropy, λH, and they present it as the subproblem that deterministic annealing solves. The iterations with the weights g_m, however, are block-coordinate descent on Equation 7.5, with $-\log P(y \mid x)$ in place of $-\log p_\theta(x, y)$, and the value at the optimal q is the Rényi penalty $\lambda H_{1/(1-\lambda)}$. The two penalties agree to first order in λ and differ beyond it. At $\lambda = 1$, which they identify with self-training, the iterations reach the classification likelihood, whose penalty is the min-entropy H_∞ rather than H. The Rényi form of a label-only objective of this kind also appears in the deep clustering of Chen et al. (Chen et al. 2021).

The plug-in version. If q is not optimized but set to the posterior, one obtains

$$\mathbb{E}_{p_\theta(y|x)}\big[-\log p_\theta(x, y)\big] \;=\; -\log p_\theta(x) + H\big(p_\theta(\cdot \mid x)\big), \tag{7.7}$$

i.e., the marginal negative log-likelihood plus a Shannon entropy of weight one. This is the decomposition behind Hathaway's interpretation of EM for mixtures as coordinate descent on a single objective (Hathaway 1986): the mixture log-likelihood equals the fuzzy classification log-likelihood plus the entropy of the posterior partition. Classification EM instead maximizes the classification likelihood over hard partitions (Celeux and Govaert 1992). A general weight λ gives $-\log p_\theta(x) + \lambda H$, whose discriminative version is the entropy regularization of Grandvalet and Bengio (Grandvalet and Bengio 2004).

Minimax over the missing labels. Let the soft labels $q(\cdot \mid x_i)$ of the unlabelled inputs be outer variables as well, and let $p^\star_{\theta', q}$ be the maximum-entropy joint model obtained by fitting the head to the pseudo-labelled data $\hat{p}(x)\, q(y \mid x)$. Under the hypotheses of the joint bridge,

$$\min_q \min_{\theta'} H\big(p^\star_{\theta', q}\big) = \min_\theta \frac{1}{n} \sum_{i=1}^{n} \min_y \big[-\log p_\theta(x_i, y)\big], \tag{7.8}$$

which is the classification likelihood, the $T \to 0$ end of the dial. Indeed, by the joint bridge, $H(p^{\star}_{\theta',q}) = \min_{W,b} \mathbb{E}_{\hat{p}(x)\,q(y|x)}\big[-\log p_\theta(x,y)\big]$ for fixed (θ', q). After exchanging the minimizations, the objective is linear in q, so the optimal q sits at a vertex of the simplex, i.e., at the label that minimizes $-\log p_\theta(x_i, y)$ for each x_i. The minimax entropy principle with the missing labels among the outer variables is thus the hard end of the dial, and the label temperature interpolates between it and maximum-entropy learning at $T = 1$.

The structure has a direct precedent in crowdsourcing. Zhou et al. estimate the true labels from the noisy labels of many workers by a minimax entropy principle (Zhou et al. 2012). The inner problem maximizes the entropy of a distribution over workers, items, and labels, under per-item constraints suggested by majority voting and per-worker constraints suggested by the model of Dawid and Skene, and the outer problem minimizes the entropy of this maximum-entropy model over the unknown true labels. Their Theorem 2.1 shows that, when the true measurements are known, the Kullback–Leibler loss to the true distributions equals the entropy of the maximum-entropy model minus that of the truth, which is the view "closest to the truth" in the three views of Section 7.1.1. Since each item is usually labelled only a few times, they relax the constraints to prevent overfitting, i.e., they use the regularized maximum entropy of Section 5.6; a conditional version of the principle followed (Zhou et al. 2015).

7.3.2 What the density term buys, and what it does not

A joint model differs from a purely discriminative one by the term $-\log p_\theta(x)$. Let us look at what this term can and cannot do. In this subsection we write $p_x = p_\theta(\cdot \mid x)$ for short.

The energy–entropy identity. For every input x,

$$\begin{aligned} -E_\theta(x) &= \log \sum_y e^{f_\theta(x)[y]} \\ &= \max_{q \in \Delta_K} \Big\{ \mathbb{E}_q\big[f_\theta(x)[y]\big] + H(q) \Big\} \\ &= \big\langle f_\theta(x) \big\rangle_{p_x} + H(p_x), \end{aligned} \tag{7.9}$$

where $\langle f_\theta(x)\rangle_p$ denotes the expected logit under p. Hence, for uniform h, $-\log p_\theta(x) = -\langle f_\theta(x)\rangle_{p_x} - H(p_x) + \log Z(\theta) + \text{const}$. This is the Gibbs variational principle of the chapter on free energy, Chapter 3, applied to the label: the marginal energy of an input is the free energy of its label ensemble, as noted in Section 4.9. Park et al. state it as a Fenchel duality between the energy and the negative entropy (Park et al. 2025, Eq. 5). Three consequences are worth spelling out.

Firstly, the energy and the entropy of a prediction can move independently. Park et al. read Equation 7.9 in one direction: as the entropy goes to zero, the energy approaches minus the largest logit, so that, in their words, minimizing entropy "does not provide a clear momentum to reduce the overall energy". Read in the other direction, the identity says that lowering the energy need not sharpen the prediction. For the plug-in objective with a weight λ on the entropy, it gives

$$\begin{aligned} -\log p_\theta(x) + \lambda H(p_x) &= -\big\langle f_\theta(x) \big\rangle_{p_x} + (\lambda - 1)\, H(p_x) \\ &\quad + \log Z(\theta) + \text{const}, \end{aligned}$$

so at $\lambda = 1$ the explicit entropy cancels, and what remains is the soft self-labelled joint negative log-likelihood of Equation 7.7. This is an identity and not a statement that $\lambda = 1$ is a neutral point: the expected logit and the entropy are not independent coordinates, and, as the third consequence shows, the density term has no preference for any split of the mass among the labels.

Secondly, the density term makes the scale of the logits identifiable. For purely discriminative entropy minimization on unlabelled data, multiplying the logits by $c > 1$ changes no decision and lowers the en-

tropy, which tends to zero as $c \to \infty$; only other terms, e.g., weight decay, can stop the growth. With the density term this is no longer the case. Let $g_c(x) = \frac{1}{c} \log \sum_y e^{c f(x)[y]}$, which converges uniformly to $g(x) = \max_y f(x)[y]$ as $c \to \infty$, and let h have a positive continuous density on a compact $\mathcal{X}$, with g continuous. Write $A_c = \frac{1}{c} \log \int h(x') \, e^{c\, g_c(x')} \, dx'$, which tends to $\max_{x'} g(x')$ by a Laplace argument. Then

$$\begin{aligned} -\log p_{cf}(x) &= c\big(A_c - g_c(x)\big) - \log h(x) \\ &= c\Big(\max_{x'} g(x') - g(x)\Big) + o(c), \end{aligned} \tag{7.10}$$

so the density term grows linearly in c at every input that is not a maximizer of g. The scale can no longer be sent to infinity for free; it is set by the fit of $p_\theta(x)$. This does not contradict the remark in Section 5.1.4 that the temperature of a learned energy is not identifiable when the family is closed under rescaling. There the temperature is absorbed into the scale of the energy, and here that common scale, which governs both $p_\theta(x)$ and $p_\theta(y \mid x)$, is what the fit of $p_\theta(x)$ determines. The improved calibration of JEM (Grathwohl et al. 2020) is consistent with this mechanism, although it has not been shown to be caused by it.

Thirdly, the density term does not prevent collapse. By Equation 7.1, the term $-\log p_\theta(x)$ depends on the logits only through their log-sum-exp, at x and, through $Z(\theta)$, at every other input. Any change that keeps the log-sum-exp fixed at every input leaves the density term unchanged, including, in the limit, a change that moves all the posterior mass to one class. The density term shapes the landscape of the marginal energy and does not decide the split of the labels; it can influence the split only indirectly, through the shared network. Whether the labels collapse is therefore decided by the label-side terms, and a fairness term is needed, exactly as in Section 6.3.

In the language of joint maximum entropy, the fairness term has a clean place. The label statistic e_y of Equation 7.2 has the class frequencies as its moment, and its multiplier is the bias b_y. Without labels, a fairness term prescribes this moment, $\mathbb{E}_{\hat{p}(x)\, q(y|x)}[e_y] = \pi$, instead of estimating it. Adding this constraint to the sum of Equation 7.5 over the unlabelled inputs gives

$$q^\star(y \mid x) \;\propto\; \exp\Big(\frac{f_\theta(x)[y] - \gamma_y}{T}\Big), \tag{7.11}$$

where γ is the Lagrange multiplier of the constraint, chosen such that the constraint holds. This is the per-class offset that the history in Section 6.3 finds behind every fairness mechanism, in its hard-constraint form, i.e., Sinkhorn equipartition run to convergence. In practice the moment is computed with the batch-averaged posterior, as in distribution alignment (Berthelot et al. 2020) and in the diversity term of SHOT (Liang et al. 2020).

These observations place the three terms of Section 6.3 in one frame. *Firmness*, the conditional entropy down, is the outer minimization of $H(Y \mid X)$, i.e., a label temperature $T < 1$. *Fairness* is a prescribed moment of the inner model, Equation 7.11. *Capacity*, the term that makes the classes mean something, is where the maximum-entropy model of the inputs may contribute, since the shared features must then also explain $p_\theta(x)$; we state this as a hypothesis rather than a result. What the density term cannot do is replace fairness, since it is blind to collapse. Notably, Sustek et al. add the mutual information $I_\theta(X;Y) = H_\theta(Y) - H_\theta(Y \mid X)$ under the model distribution to the objective of JEM, which, in their words, encourages the classifier to make more certain decisions about output labels (Sustek et al. 2023). In the terms of this section, this puts a further weight on the Type II part $H_\theta(Y \mid X)$ together with a fairness term $H_\theta(Y)$, i.e., it places the mutual-information objective of Section 6.3 on top of a joint maximum-entropy learner.

Finally, it is worth writing the losses with and without labels in one form, because a sign is easily lost. With $\mathrm{CE}(a, b) = -\sum_y a_y \log b_y$, an entropy is a cross-entropy against the best target,

$$\begin{aligned} H(p) &= \min_{r \in \Delta_K} \mathrm{CE}(p, r), \\ H_\infty(p) &= -\log \max_y p_y = \min_{r \in \Delta_K} \mathrm{CE}(r, p), \end{aligned} \tag{7.12}$$

attained at $r = p$ and at the one-hot mode of p, respectively. With labels the target is given by the data, and without labels the model chooses it. Note that hard pseudo-labelling minimizes the min-entropy H_∞; this is one place where the min-entropy and minimum-entropy learning, which Section 2.1.4 keeps apart, coincide in form. The common forms are collected in Table 7.2, which shows in particular that the Kullback–Leibler divergence *to* the uniform distribution raises the entropy when it is minimized.

Table 7.2: Losses with and without labels. Here $p = p_\theta(\cdot \mid x)$, $\bar{p} = \mathbb{E}_x\, p_\theta(\cdot \mid x)$ is the batch-averaged prediction, u is the uniform distribution on the labels, and sg denotes a stop-gradient, i.e., the target is held fixed. In DINO the sharpened target comes from a teacher network.

case	loss	as a cross-entropy or KL	effect of minimizing
labelled	$-\log p_\theta(y \mid x)$	$\mathrm{CE}(\delta_y, p) = \mathrm{KL}(\delta_y \Vert p)$	fits the label
unlabelled, Shannon	$H(p)$	$\min_r \mathrm{CE}(p, r)$	sharpens (Grandvalet and Bengio 2004)
unlabelled, hard pseudo-label	$H_\infty(p)$	$\min_r \mathrm{CE}(r, p)$	hard self-training (Lee 2013)
unlabelled, sharpened target	$\mathrm{CE}\big(\mathrm{sg}[r_\tau], p\big)$	$r_\tau \propto p^{1/\tau}$, a target at temperature τ	soft self-training (Berthelot et al. 2020; Caron et al. 2021)
towards uniform	$\mathrm{KL}(p \Vert u)$	$\log K - H(p)$	**raises** the entropy: the confidence penalty (Pereyra et al. 2017)
fairness	$\mathrm{KL}(\bar{p} \Vert \pi)$	$\log K - H(\bar{p})$ for $\pi = u$	resists collapse (Bridle et al. 1992; Liang et al. 2020)

7.4 Boundaries, evidence, and a name clash

Why condition iii) is needed. Without it, every objective of the form "entropy minimization plus a Kullback–Leibler trust region to a reference" would be of Type II, since the optimum of such an objective is a Gibbs tilt of the reference (Equation 6.2). This would include EMPO, which minimizes the semantic entropy of the model's own answers (Zhang et al. 2025), and reinforcement learning with the negative entropy as the only reward (Agarwal et al. 2025). In these methods, however, the reference, if any, is frozen. It is a maximum-entropy model in the sense of i)–ii) of the definition in Section 5.1.4, not a learner on the current data. Moreover, the analysis in Section 6.6 shows that the released EMPO code switches the KL anchor off altogether, and that the resulting dynamics are a rich-get-richer process with collapse as its asymptotic behaviour. Condition iii) draws the line at whether the maximum-entropy side learns from the data on which the entropy is minimized, and these methods stay in Chapter 6.

The same condition sorts the semi-supervised and test-time methods. The entropy regularization of Grandvalet and Bengio is of Type II on the labelled inputs, through the conditional bridge, and belongs to the minimum-entropy learning of Chapter 6 on the unlabelled ones, where no model of the inputs is fitted; a joint model makes it Type II on both. Entropy minimization at test time (Wang et al. 2021) has no learner on the test inputs at all. TEA (Yuan et al. 2024) fits the marginal energy of the test inputs by contrastive divergence, which is maximum-entropy learning at $T = 1$. ReTTA (Park et al. 2025) pairs entropy minimization with a sliced-score-matching fit of the energy on the test inputs, and with a cross-entropy that takes the most probable class as its target. The latter equals $-\log \max_y p_\theta(y \mid x) = H_\infty$, so the objective of ReTTA contains both terms of the $T \to 0$ end of Equation 7.6, together with a Shannon entropy, and its density is fitted by score

matching: ReTTA is of Type II in the generalized sense. Its entropy term taken alone is the self-referential move of Section 9.4, and this is how Table 9.2 lists it.

Why $T = 1$ is not of Type II. At $T = 1$ the missing label keeps its full entropy, and nothing on the label side asks the model to commit; the objective is the marginal likelihood, and EM is coordinate descent on it (Hathaway 1986). This is the boundary between Chapter 5 and the present chapter. In the language of Chapter 3, the label temperature is the dial of that chapter applied to the missing label only, while the inputs stay at $T = 1$. The placement is summarized in Table 7.3.

Table 7.3: Methods placed by the definition of Section 7.2.1. The second column gives the data to which a maximum-entropy learner is fitted, and the third the entropy that the outer problem minimizes. The decisive column is the second, i.e., whether a maximum-entropy learner is fitted to the data on which the entropy is minimized.

method	maximum-entropy fit on	outer entropy	place
FRAME (Zhu et al. 1997)	images	the entropy of the model, over a filter bank	Type I
energy-based model by maximum likelihood (Section 5.6.2)	inputs	the entropy of the model, over a continuous family	Type I, implicit
softmax classifier (Equation 7.3)	labelled pairs, conditionally	$\hat{H}(Y \mid X)$, through the cross-entropy	Type II
JEM (Grathwohl et al. 2020) (Equation 7.4)	labelled pairs, jointly	$H_\theta(X)$ and $H_\theta(Y \mid X)$	Types I and II
semi-supervised JEM (Zhao et al. 2020); TEA (Yuan et al. 2024)	inputs	none on the missing label, at $T = 1$	maximum-entropy learning
joint model at $T < 1$; classification EM (Celeux and Govaert 1992)	inputs	$(1 - T)\, H_{1/T}$, or H_∞ as $T \to 0$	Type II
ReTTA (Park et al. 2025)	test inputs, by score matching	H and H_∞	Type II, generalized sense
entropy regularization (Grandvalet and Bengio 2004)	labelled pairs only	H on the unlabelled inputs	Type II on the labelled inputs only
Tent (Wang et al. 2021); EMPO (Zhang et al. 2025)	none on the current data	H, or a semantic entropy	minimum-entropy learning

Published evidence. The evidence on the two sides of the boundary is indirect, and it should be read with care. At $T = 1$, Zhao et al. report that semi-supervised JEM reaches 95.4% on MNIST with 100 labels, against 86.0% for the baseline classifier and 98.4% for virtual adversarial training, and 66.0% on SVHN with 1000 labels, against 62.7% and 62.8%, respectively; in their MNIST experiments, entropy regularization was not found to be helpful for either method (Zhao et al. 2020). Grathwohl et al. train JEM on CIFAR-10 with 4,000 labels and the rest of the training set as unlabelled data, and find a noticeable improvement in calibration but, surprisingly in their words, no improvement in generalization (Grathwohl et al. 2020, Appendix E.2). TEA reports an average gain of 4.7% over the best-performing test-time adaptation methods on corruption and domain-generalization benchmarks (Yuan et al. 2024). Of Type II, ReTTA reaches an average accuracy of 49.2% on ImageNet-C at severity 5 in its mild scenario, at least 0.6% above all compared methods. In its ablation under online label shift, adding the score-matching term to entropy minimization raises the accuracy from 43.9% to 45.6% for a ResNet-50 with group normalization, and from 62.3% to 64.7% for a ViT-Base (Park et al. 2025, Table 3). In summary, the published results are consistent with a density term that helps when it is fitted on the inputs where the prediction is made, but none of them tests one end of the dial against the other with everything else held fixed. We are not aware of a study that varies the label temperature of a joint energy-based model systematically; this is an open question rather than a result.

⚠ Terminology

Three neighbours share words with this chapter. i) The *minimax entropy* (MME) of Saito et al. is a method of semi-supervised domain adaptation, in which the conditional entropy of the unlabelled target data is maximized with respect to the classifier and minimized with respect to the feature extractor, through a gradient-reversal layer (Saito et al. 2019). It is an adversarial game on a single entropy rather than the nesting of a maximum-entropy model inside an entropy minimization, and it is unrelated to the principle of this chapter. ii) The *minimax approach* of Farnia and Tse (Farnia and Tse 2016), like the game against nature of Section 5.6, is a minimum over decision rules of a maximum over distributions, i.e., the inner problem of this chapter alone; in the same spirit, Section 5.6 calls the primal–dual form of maximum likelihood a saddle-point form and not minimax. iii) Maximum entropy discrimination (Jaakkola et al. 1999) and the principled hybrids of Lasserre et al. (Lasserre et al. 2006) also combine maximum entropy with discrimination, and generative with discriminative training, respectively, but along different axes: the former seeks a minimum-relative-entropy distribution over parameters under margin constraints, and the latter interpolates between a generative and a discriminative model through a prior that couples two sets of parameters.

What is known, and what may be new. The pieces of Section 7.2 and Section 7.3 are known. The conditional and joint bridges are instances of the duality between maximum likelihood and maximum entropy (Berger et al. 1996); JEM and its predecessors read a classifier as an energy-based model (Grathwohl et al. 2020; Xie et al. 2016); the label temperature is the deterministic annealing of Rose (Rose 1998), and its weights are those of Grandvalet and Bengio (Grandvalet and Bengio 2006); the plug-in identity is Hathaway's (Hathaway 1986); the energy–entropy identity is stated by Park et al. (Park et al. 2025); the minimax entropy principle over unknown labels appears in crowdsourcing (Zhou et al. 2012); and the Rényi form of a label-only objective appears in deep clustering (Chen et al. 2021). What we add is a reading that puts them together: a definition with the anchor condition iii), under which the softmax classifier, JEM, the label-temperature dial, and recent test-time adaptation sit in one nesting, and the observation that, written as Equation 7.6, the dial separates a maximum-entropy learner on the inputs from a Rényi penalty on the label. We make no claim of priority for any of the identities.

7.5 Takeaway

Minimax entropy couples the two preceding principles into a single criterion: **maximize entropy to model the world, minimize entropy to commit to what matters**. Maximum entropy supplies possibility, minimum entropy supplies certainty, and the minimax formulation decides which details deserve to be modelled at all.

On the joint space of inputs and labels, the slogan can be taken literally. A classifier with a linear head is a conditional maximum-entropy model, and the cross-entropy it minimizes is the largest conditional entropy that its features allow. Read as a joint energy-based model, its two training terms add up to the joint entropy of a maximum-entropy model, one part modelling the inputs and one part predicting the label. Without labels, a temperature on the label is the dial: at $T = 1$ the problem is maximum-entropy learning with missing data, below it the label side pays a Rényi entropy of order $1/T$, and as $T \to 0$ it becomes the classification likelihood. The density term makes the scale of the logits identifiable but is blind to collapse, so fairness remains a separate, prescribed moment.

The principle has recently yielded exactly solvable models of large neuronal populations (Lynn et al. 2025). Its implicit form runs inside every energy-based model trained by maximum likelihood (Section 5.6.2), and its second type is implicit in joint energy-based models and in test-time adaptation that pairs entropy minimization with a learned energy, as shown in Section 7.2 and Section 7.4. Its explicit form — the selection of a feature set under a description-length budget, with guarantees on sample complexity — remains largely

untapped in machine learning, and so does an explicit schedule for the label temperature. Open directions include scalability and sample complexity in high dimensions, realizations for LLMs, agents, and multimodal models, and the relationship to the free-energy principle (Friston 2010).

8 Energy-Based Guidance and Refinement

The previous chapters treated the energy as something to be learned from data. This chapter and the next treat a different situation, which is by now the most common one in practice: a large generative model is already trained, and we want to change what it generates without training it from scratch. We call the trained model the *reference*, written p_{ref}, and we call the change a *tilt*. Both chapters are organized around one object,

$$p^\star(x) \;=\; \frac{p_{\text{ref}}(x)\,\exp\big(-E(x)/T\big)}{Z}, \qquad Z \;=\; \sum_x p_{\text{ref}}(x)\,\exp\big(-E(x)/T\big), \tag{8.1}$$

which is the free-energy form of Section 5.1.1 with the reference as base measure, $h = p_{\text{ref}}$: $p^\star$ is the distribution closest to the reference, in Kullback–Leibler divergence, among those with a prescribed expected energy, and equivalently the minimizer of $\mathbb{E}_q[E] + T\,\mathrm{KL}(q \,\|\, p_{\text{ref}})$. In the vocabulary of Section 5.1, Equation 8.1 is the representation level. The two chapters differ in *where the tilt is executed*. In this chapter it is executed at inference time: the reference is frozen, and a sampler does the work — gradient guidance, reranking, importance sampling, sequential Monte Carlo, or iterative refinement. This is the inference level of Section 5.1.2 carried out at test time. In Chapter 9 the tilt is absorbed into the weights of the reference, or the energy is learned through the reference; that is the learning level. These are not rival approaches, but rather two ways to obtain the same target distribution: one pays the cost in computation at inference (test) time, while the other pays it upfront through additional training.

What is fixed and what is fitted. It is worth being precise about the phrase "the energy is given", since several methods in this chapter *train* an energy: a residual energy by noise-contrastive estimation, a verifier by pairwise ranking, a guidance network by regression. The distinction we emphasize is not whether the energy itself is trained or untrained, but rather which part in Equation 8.1 is being held fixed and which is being adjusted. In this chapter the *reference* is fixed; its parameters are never updated, and the energy, however it was obtained, enters only through the sampler. Training the energy is a learning problem in its own right — a maximum-entropy learner in the sense of Section 5.1.4 whose data are labels, preferences, or contrastive noise — but it is a separate learner, and its output is then handed to the sampler. So "energy given" should be read as "energy given to the sampler, reference frozen". Two methods at the end of the chapter blur even this line, because in them the energy *is* the generator: Energy Matching and energy-based transformers have no separate reference, and sampling is refinement on the energy itself. We treat them as the limit in which the tilt has nothing left to tilt.

8.1 The tilted reference

Three remarks on Equation 8.1 organize the chapter.

First, the temperature has the status of a genuine dial (Section 5.1.4): the energy is given, no constraint level is prescribed, and T is chosen. It appears in the literature under several names. In classifier guidance for diffusion models it is the reciprocal of the guidance scale w; in KL-regularized fine-tuning it is the KL coefficient β; in best-of-N selection, as we show in Section 8.4, the number of candidates N plays the role of an inverse temperature. Small T concentrates $p^\star$ on the low-energy region and moves it far from the reference; large T leaves the reference almost untouched. The zero-temperature limit is energy minimization on the support of the reference, and it is where guidance stops being a probabilistic method (Section 5.6.3).

Second, the energy is additive. If two energies E_1 and E_2 encode two requirements, the tilt by $E_1 + E_2$ is the product of the two tilts normalized once, i.e., a product of experts (Hinton 2002). This is why guidance composes: a classifier, a constraint, and a reward can be added without retraining anything. The price is a partition function that depends on all of them together, and most of the technical content of the chapter is about how each sampler avoids or estimates that partition function.

Third, the constrained reading is available. By the duality of Section 5.1.1, $p^\star$ is the I-projection of the reference onto the set $\{q : \mathbb{E}_q[E] = U\}$ for the level U implied by T. The reading matters for one practical question: how far from the reference a tilt is allowed to travel. The Kullback–Leibler divergence $\mathrm{KL}(p^\star \,\|\, p_{\mathrm{ref}})$ is a budget, and several results below are bounds on that budget.

8.2 Continuous guidance

For continuous data the reference is usually a diffusion or flow model, and the sampler has a gradient available. Classifier guidance (Dhariwal and Nichol 2021) is the original instance of Equation 8.1 in this setting. To generate from $p(x \mid y) \propto p(x)\, p(y \mid x)$ with a pretrained unconditional model $p(x)$ and a classifier $p(y \mid x)$, one adds the classifier gradient to the score at every noise level,

$$\nabla_{x_t} \log p_t(x_t \mid y) \;\approx\; \nabla_{x_t} \log p_t(x_t) \;+\; w\, \nabla_{x_t} \log p(y \mid x_t), \tag{8.2}$$

with a guidance scale $w \geq 1$ that sharpens the conditional. In our notation the energy is $E = -\log p(y \mid x)$ and $T = 1/w$; the target is a product of experts, and the classifier must be trained on noisy inputs x_t. Classifier-free guidance (Ho and Salimans 2022) removes the separate classifier by training the generator with and without the condition and extrapolating, $\hat{\epsilon} = (1+w)\,\epsilon_\theta(x_t, y) - w\,\epsilon_\theta(x_t)$, which amounts to using the implicit classifier $p(y \mid x_t) \propto p(x_t \mid y)/p(x_t)$ as the energy.

For a general energy J on clean data, i.e., a target $p'(x_0) \propto p(x_0)\, e^{-J(x_0)}$, the difficulty is that the sampler runs on noisy states x_t and needs the energy *there*. The exact intermediate energy is

$$E_t(x_t) \;=\; -\log \mathbb{E}_{p(x_0 \mid x_t)}\big[e^{-J(x_0)}\big], \tag{8.3}$$

a free energy of the posterior over clean samples given the noisy one, and its gradient is the exact guidance. Lu et al. (Lu et al. 2023) show that Equation 8.3 is the only guidance consistent with the marginals of the reverse process and propose contrastive energy prediction to learn it; the common shortcut evaluates J at the posterior mean $\hat{x}_0(x_t)$ instead, which is a first-order approximation. Feng et al. (Feng et al. 2025) carry the analysis to general flow matching, where the reference is a vector field v_t generating a probability path from noise to data. They ask for a guidance field g_t such that $v_t + g_t$ generates the path to $p'(x) \propto p(x)\, e^{-J(x)}$, and derive three families: an asymptotically exact Monte Carlo guidance that reweights samples of the coupling by e^{-J}, a local guidance obtained by Taylor expansion that is proportional to ∇J and recovers the classical gradient guidance of diffusion models under affine Gaussian paths, and training-based exact guidance with its own regression losses. The value of this framework for us is the hierarchy it makes explicit: exact guidance is an expectation under the posterior of the reference, and gradient guidance is its cheapest approximation.

When several energies are composed, the approximation error of gradient guidance compounds, since the reverse process only sees the sum of approximate scores. Du et al. (Du et al. 2023) show that running a few steps of Markov chain Monte Carlo (Langevin or Hamiltonian) on the composed energy inside each reverse step corrects the composition; the sampler then targets the product of experts rather than the sum of its approximations. This is refinement in the sense of Section 8.5, applied locally in noise level.

Energy Matching (Balcerak et al. 2025) takes the opposite route to the same end. Instead of tilting a flow with an energy, it learns a single time-independent scalar potential $V_\theta(x)$ whose gradient plays two roles: far from the data manifold it is a transport field along optimal transport paths, and near the manifold it is

the drift of Langevin dynamics, so that the sampler equilibrates in a Boltzmann well $\propto e^{-V_\theta}$ around the data. Additional requirements are added as energy terms — the authors use an interaction energy to spread samples over modes in protein design — which is the additivity of Section 8.1 with the potential itself as the reference. Here the energy and the generator coincide; there is no frozen reference to tilt, and we return to this case in Section 8.5.

8.3 Discrete guidance: residual energies over a proposal

For sequences the sampler has no gradient in the data space, and the tilt is implemented by *proposing* from the reference and *reweighting* by the energy. The construction is the residual energy-based model of Deng et al. (Deng et al. 2020),

$$p(x) \;=\; \frac{p_{\mathrm{AR}}(x)\,\exp\big(-E_\phi(x)\big)}{Z_\phi}, \tag{8.4}$$

in which a pretrained autoregressive language model p_{AR} is the proposal, E_ϕ is a bidirectional scorer of the whole sequence, and generation draws n candidates from p_{AR} and resamples them with weights $\propto e^{-E_\phi}$. The energy is trained by noise-contrastive estimation with the proposal as the noise distribution (Gutmann and Hyvärinen 2010), so that, in the classification of Section 5.6.3, the residual EBM is a maximum-entropy learner in the generalized sense whose inference is importance sampling from the tilt. The same construction at the token level is Electric (Clark et al. 2020b), an energy-based cloze model trained by conditional noise-contrastive estimation; its authors show that ELECTRA's replaced-token detection (Clark et al. 2020a) is a closely related negative-sampling objective, which gives a principled reading of what that discriminative pre-training learns: an energy over tokens in context.

Discrete diffusion language models sharpen the need for a residual energy. A masked diffusion model decodes several tokens in parallel at each step, but its denoiser factorizes over positions, $p_\theta(x_{t-1} \mid x_t) = \prod_i p_\theta(x^i_{t-1} \mid x_t)$, so the tokens decoded together are conditionally independent given the current state; with few steps the independence assumption is badly violated and quality degrades. The energy-based diffusion language model (EDLM) of Xu et al. (Xu et al. 2025) puts a residual energy on each denoising step,

$$p_{\mathrm{EDLM}}(x_{t-1} \mid x_t) \;\propto\; p_{\mathrm{diffusion}}(x_{t-1} \mid x_t)\,\exp\big(-E(x_{t-1}, x_t)\big), \tag{8.5}$$

and shows that the energy can be read off a pretrained autoregressive model or fitted by noise-contrastive estimation from a bidirectional transformer; the sampler is parallel importance sampling over candidates from the diffusion proposal. On language-modeling benchmarks EDLM approaches autoregressive perplexity and reaches the same generation quality as the diffusion baseline in fewer steps (a 1.3× sampling speedup is reported).

Our own contribution to this line is the *unified energy* (Uni-E) of Yan et al. (Yan et al. 2026), which starts from a decomposition of the gap between a diffusion language model and the data distribution. Writing the Kullback–Leibler divergence along a decoding order and expanding it, the gap splits into three factors: a *capacity* factor, i.e., how well the denoiser fits each position; a *dependency* factor, i.e., whether the tokens decoded in one step are independent given the context; and an *invariance* factor, i.e., whether a token decoded now would be decoded the same way once more context is revealed. The second factor is handled by the independent energy of EDLM, which compares the denoiser with a proxy autoregressive model on the fully decoded sequence,

$$E_{\mathrm{ind}}(x_0, x_t) \;=\; \log \frac{p_\theta(x_0 \mid x_t)}{p_{\mathrm{AR}}(x_0 \mid x_t)} - \log F_{\mathrm{ind}}(x_t). \tag{8.6}$$

For the third factor the paper introduces an *invariant energy*. Let x_s be a partially revealed sequence between x_t and x_0, obtained by re-masking; the decoding of x_s from x_t is a "future step", and a token is invariant if revealing it does not change what the proxy model believes about the rest. The invariant energy is

$$E_{\text{inv}}(x_0, x_s, x_t) \;=\; \log \frac{p_\theta(x_s \mid x_t)}{p_{\text{AR}}(x_s \mid x_t)} \;+\; \log \frac{p_{\text{AR}}(x_0 \mid x_t)}{p_{\text{AR}}(x_0 \mid x_s)} \;-\; \log F_{\text{inv}}(x_t), \tag{8.7}$$

whose second term is small exactly when the intermediate reveal x_s leaves the proxy's posterior over x_0 unchanged. By the additivity of energies the two are combined into the unified energy $E_{\text{uni}} = E_{\text{inv}} + E_{\text{ind}}$, parameterized without its normalizer as

$$E_\phi(x_0, x_s, x_t) \;=\; \log \frac{p_\theta(x_s \mid x_t)}{p_{\text{AR}}(x_s \mid x_t)} \;+\; \log \frac{p_\theta(x_0 \mid x_t)}{p_{\text{AR}}(x_0 \mid x_s)}. \tag{8.8}$$

The property that distinguishes Uni-E from other residual energies is that its partition function is available in closed form. Summing $p_\theta(x_0 \mid x_t)\, e^{-E_\phi}$ over x_0, the denoiser cancels against the second log-ratio and what remains is $p_{\text{AR}}(x_0 \mid x_s)$, which sums to one; hence

$$\log F(x_t) \;=\; \log \frac{p_{\text{AR}}(x_s \mid x_t)}{p_\theta(x_s \mid x_t)}, \tag{8.9}$$

and the tilted denoiser is the proxy model's posterior given the revealed sequence x_s. No Monte Carlo estimate of the normalizer is needed, which is the usual bottleneck of residual EBMs. The paper proves that the resulting decoder has a strictly smaller gap to the data distribution than the diffusion model whenever the dependency or invariance factor is non-zero, and reports that on diffusion language models Uni-E reaches quality comparable to autoregressive decoding with about a 22% speedup, while on diffusion large language models it mitigates the degradation that otherwise grows with the degree of parallelism. Two caveats should be stated. The invariant energy relies on a proxy autoregressive model and on a sampling estimate of the invariance term whose accuracy is characterized rather than exact (the paper's Lemma 4.1 gives the approximation factor). And the whole construction is a tilt of a *frozen* denoiser: the diffusion model is not retrained, and the energy, whether read off the proxy or fitted by noise-contrastive estimation, enters only through the reweighting of candidates. It is discrete guidance in the sense of this chapter, and its closed-form normalizer is what makes it exact rather than approximate.

8.4 Verifiers as energies, best-of-N, and sequential Monte Carlo

The simplest energy for a language model is a *verifier*: a scalar score of a complete output, low for good outputs and high for bad ones. Outcome reward models trained to predict correctness (Cobbe et al. 2021) and process reward models trained on step-level labels (Lightman et al. 2024) are energies of this kind. The learning signal is usually pairwise: given a preferred output y_w and a rejected one y_l, the Bradley–Terry model (Bradley and Terry 1952)

$$P(y_w \succ y_l) \;=\; \sigma\big(E(y_l) - E(y_w)\big) \tag{8.10}$$

is fitted by maximum likelihood, which is logistic regression on the difference of energies and hence the maximum-entropy classifier of Section 5.1.4 applied to pairs. The energy outcome reward model (EORM) of Jiang et al. (Jiang et al. 2025) shows how little is needed: a 55M-parameter encoder with a scalar energy head, trained with the pairwise loss over all correct–incorrect pairs of sampled chains of thought, using outcome labels only. At inference the verifier reranks a pool of N candidates from the language model and returns $\arg\min_y E_\theta(y)$; the partition function is never needed, since the candidates share it. In the terms of this chapter, reranking is the zero-temperature limit of the tilt on the finite support of the candidate pool, and the

reported gains (Llama 3 8B rises to 90.7% on GSM8K and 63.7% on MATH) are those of best-of-N with a learned energy in place of a large reward model.

Best-of-N itself deserves a precise statement, because it is the most widely used inference-time tilt and it is *not* the exponential tilt of Equation 8.1. Drawing N samples from the reference and keeping the one with the highest reward induces a distribution π_{BoN} that is a monotone reweighting of p_{ref} by the rank of the reward. Beirami et al. (Beirami et al. 2024) show that the expression $\log N - (N-1)/N$, often quoted as the KL divergence between π_{BoN} and the reference, is an *upper bound*,

$$\mathrm{KL}\big(\pi_{\mathrm{BoN}} \,\|\, p_{\mathrm{ref}}\big) \;\le\; \log N - \frac{N-1}{N}, \tag{8.11}$$

and that the win rate of π_{BoN} against the reference is bounded by $N/(N+1)$. Gui, Gârbacea, and Veitch (Gui et al. 2024) embed best-of-N and the distributions learned by alignment methods in a common class of tiltings of the reference and show that, within this class, best-of-N is essentially optimal for the trade-off between win rate and KL divergence. Read through Equation 8.1, N is an inverse temperature: the KL budget grows like $\log N$, exactly as the budget of an exponential tilt grows as T falls. The cost is N forward passes per output; distilling the best-of-N distribution into the weights is a post-training method, and we return to it in Section 9.2.

Between reranking whole outputs and gradient guidance sits sequential Monte Carlo. Zhao et al. (S. Zhao et al. 2024) observe that a large part of language model alignment, red-teaming, and constrained generation can be cast as sampling from a target $\sigma(x_{1:T}) \propto p_{\mathrm{ref}}(x_{1:T})\,\phi(x_{1:T})$ with a potential ϕ on complete sequences, i.e., Equation 8.1 with $\phi = e^{-E/T}$, and that the right inference-time tool is sequential Monte Carlo with *twist functions* $\psi_t(x_{1:t}) \approx \mathbb{E}[\phi(x_{1:T}) \mid x_{1:t}]$, the expected future potential given a prefix. The twists are soft value functions: their logarithms satisfy the soft Bellman recursion of Section 5.5.2, and the paper's contrastive method for learning them is a soft reinforcement learning algorithm. Resampling partial sequences by their twists focuses computation on promising prefixes, and the authors derive bidirectional bounds on $\log Z$ from the same machinery, which let one measure how far an inference method is from the target in KL divergence in both directions. Controlled decoding (Mudgal et al. 2024) is the same idea with a learned prefix scorer used either token by token or blockwise; in both cases the reference is frozen and the energy acts through the prefix values.

8.5 Refinement and reasoning as energy minimization

Rather than accepting a single forward pass, one can iteratively lower the energy of a candidate — an optimization view of "thinking longer". Pure descent,

$$x \;\leftarrow\; x - \eta\, \nabla_x E(x), \tag{8.12}$$

is the zero-temperature limit of Chapter 3: it consults the energy and ignores the entropy, and inherits the classical weakness of local minima. Adding noise and lowering it over the course of the process, i.e., annealing (Kirkpatrick et al. 1983), is the finite-temperature version, and it is the schedule that diffusion samplers implement under another name. For text, where the data space is discrete, COLD decoding (Qin et al. 2022) runs Langevin dynamics on a continuous relaxation of the sequence under a composed energy and rounds at the end; for images, the MCMC corrections of Section 8.2 are refinement inside each reverse step. In every case the number of steps is a compute knob that buys quality, and the target of the refinement is a tilt of a frozen reference.

Energy-based transformers (EBTs) (Gladstone et al. 2026) carry refinement to the point where it becomes the model. An EBT assigns an energy to a pair of input and candidate prediction and produces its prediction by gradient descent on the candidate until convergence; the number of descent steps is chosen at inference time,

which the authors frame as System-2 thinking learned from unsupervised pre-training alone, without verifiers or verifiable rewards. They report that EBTs scale faster than a strong autoregressive Transformer baseline during pre-training and improve with inference-time computation, with larger gains on data farther from the training distribution. Two points matter for this book. Learning here is the shaping of the landscape of Section 4.3 so that the correct continuation is the minimizer, and inference is the zero-temperature inference of Section 5.1.2 with a variable budget; and, as in Energy Matching, there is no separate reference: the energy is the generator, so the "tilt" of Equation 8.1 degenerates to the energy itself. This is the boundary case announced at the beginning of the chapter, and it is instructive because it shows the two roles of the energy — verifier and generator — collapsing into one function.

The unifying perspective, then, is the following. A reasoning problem defines an energy landscape over candidate solutions; reasoning is the search for low-energy candidates, by guidance, reranking, sampling, or refinement; and learning shapes the landscape so that correct or self-consistent solutions sit at its minima. Maximum-entropy learning (Chapter 5) defines the landscapes, minimum-entropy learning (Chapter 6) sharpens them from the model's own agreement, and the methods of this chapter navigate them at test time.

8.6 Guidance on the three levels

Table 8.1 reads the methods of this chapter through the three levels of Section 5.1. In every row the reference is fixed and the inference level is where the method lives; the energy column records where the energy came from, which is a separate learning problem when it was trained.

Table 8.1: Inference-time methods read through the three levels. The base measure is never updated; where an energy was trained, that training is a separate maximum-entropy learner whose output is handed to the sampler.

Method	Base measure (frozen)	Energy and its source	Sampler	Temperature
Classifier guidance (Dhariwal and Nichol 2021); classifier-free guidance (Ho and Salimans 2022)	diffusion prior	$-\log p(y \mid x_t)$: a classifier trained on noisy inputs, or the implicit classifier $p(x_t \mid y)/p(x_t)$	score plus gradient of the energy	guidance scale: $T = 1/w$ (classifier), $1/(1+w)$ (classifier-free)
Exact energy guidance, CEP (Lu et al. 2023); flow-matching guidance (Feng et al. 2025)	diffusion / flow	given J on clean data; intermediate energy Equation 8.3, exact (Monte Carlo, learned) or local (gradient)	guided score or vector field	T of the tilt
Compositional generation with MCMC (Du et al. 2023)	diffusion prior	sum of given energies (product of experts)	Langevin / HMC steps inside each reverse step	T; number of MCMC steps
Residual EBMs (Deng et al. 2020); Electric (Clark et al. 2020b)	autoregressive LM (Deng et al.); a cloze-model noise distribution (Electric)	residual energy, NCE-trained with the base measure as noise	sample-then-resample by e^{-E_ϕ}; reranking	absorbed in the scale of E_ϕ
EDLM (Xu et al. 2025); Uni-E (Yan et al. 2026)	diffusion LM denoiser	log-ratio to a proxy AR model, or NCE-trained	parallel importance sampling over candidates; closed-form Z for Uni-E	absorbed
COLD decoding (Qin et al. 2022)	LM, entering as a fluency energy	composed energies: fluency, constraints	Langevin on soft tokens, then rounding	T; step count

Method	Base measure (frozen)	Energy and its source	Sampler	Temperature
Verifier reranking: ORM (Cobbe et al. 2021), PRM (Lightman et al. 2024), EORM (Jiang et al. 2025)	LM candidate pool	verifier trained from labels (ORM, PRM) or Bradley–Terry pairs (EORM)	arg min of the energy over the pool	$T \to 0$
Best-of-N (Beirami et al. 2024; Gui et al. 2024)	LM	reward, given	keep the best of N	not a Gibbs tilt; KL budget $\leq \log N - (N-1)/N$
Twisted SMC (S. Zhao et al. 2024); controlled decoding (Mudgal et al. 2024)	LM	potential $\phi = e^{-E/T}$, given; twists / prefix scorer learned	resampling of prefixes by twists; token-wise or blockwise reweighting by the prefix scorer	T of the potential
Energy Matching (Balcerak et al. 2025); EBTs (Gladstone et al. 2026)	— (energy is the generator)	learned potential / energy	Langevin in the Boltzmann well; gradient descent on the candidate	T; number of steps

The table also makes the trade-off with the next chapter visible. Every row pays for the tilt at test time, in candidates, steps, or particles, and the payment is repeated for every query. Post-training pays once, by fitting the reference so that a single forward pass approximates the tilt, and gives up the ability to change the energy afterwards without retraining. The choice between them is a choice between computation and storage, or, in the terms of Gladstone et al. (2026), between System 2 and System 1.

8.7 Takeaway

Guidance, reranking, sequential Monte Carlo, and refinement are one operation performed by different samplers: the tilt of a frozen reference by an energy, Equation 8.1, which is the maximum-entropy model of Section 5.1 with the pretrained generator as base measure. The energy may be a classifier, a composed constraint, a residual log-ratio to a proxy model, a learned verifier, or a reward; when it was trained, that training is a separate learner, and the sampler treats its output as given. The temperature is a genuine dial that appears as a guidance scale, a KL coefficient, or the size N of a candidate pool, and the zero-temperature limit — descent and reranking — is where the method leaves the family. The next chapter asks what changes when the reference itself is allowed to move.

9 Maximum-Entropy Post-Training

Post-training is the name that has become standard for everything done to a pretrained model after pretraining: supervised fine-tuning on demonstrations, reinforcement learning from human or verifier feedback, preference optimization, and their many variants. The methods look different and are usually presented as a list. In this chapter we argue that most of them are one thing, and that the one thing is maximum-entropy learning in the sense of Section 5.1 with the pretrained model as base measure. Specifically, the object every method approximates is the tilt of the reference policy,

$$\begin{aligned} \pi^\star(y \mid x) &= \frac{\pi_{\mathrm{ref}}(y \mid x)\, \exp\big(r(x,y)/\beta\big)}{Z_\beta(x)} \\ &= \arg\min_{\pi}\ \mathbb{E}_\pi\big[-r(x,y)\big] + \beta\, \mathrm{KL}\big(\pi(\cdot \mid x) \,\|\, \pi_{\mathrm{ref}}(\cdot \mid x)\big), \end{aligned} \tag{9.1}$$

which is Equation 8.1 with the pretrained model as reference, the negative reward as energy, and the KL coefficient β as temperature; it was derived in Section 6.6 as Equation 6.2. Chapter 8 sampled from this tilt at inference time with the reference frozen. Post-training moves to the learning level: either the reference is fitted so that a single forward pass approximates $\pi^\star$ — we call this *amortizing* the tilt — or the energy r is itself learned from preferences, outcomes, or the model's own behaviour. We treat the two in turn, then the continuous case of diffusion and flow models, and finally the boundary of the family.

9.1 From the reference to its tilt

The KL-regularized objective on the right of Equation 9.1 has a longer history than its current use. In control it is the KL-control problem of Section 5.5.1, in which the base measure is the passive dynamics (Todorov 2006, 2009). For sequence models it was introduced by Jaques et al. (Jaques et al. 2017), who fine-tuned a recurrent network with a reward while penalizing the divergence from the maximum-likelihood model it started from, precisely to keep what the model had learned from data. Ziegler et al. (Ziegler et al. 2019) used the same penalty to fine-tune language models from human preferences, and the recipe of reward model plus KL-regularized policy optimization became reinforcement learning from human feedback (RLHF) (Christiano et al. 2017; Ouyang et al. 2022). It is worth noting why the penalty is not optional. Without it, maximizing $\mathbb{E}_\pi[r]$ over all policies places the entire mass on the arg-max of the reward, a degenerate distribution; Korbak, Perez, and Buckley (Korbak et al. 2022) make the point sharply and show that the KL-regularized problem is equivalent to variational inference: minimizing $\mathrm{KL}(\pi \,\|\, \pi^\star)$ for the target Equation 9.1, i.e., approximating a Bayesian posterior in which the reference is the prior and $e^{r/\beta}$ the likelihood. In the terms of this book, the entropy term is load-bearing (Section 5.1.4), and removing it is the zero-temperature boundary of Section 5.6.3.

The constrained form of Section 5.1.1 has an instance of its own. Generation with distributional control (GDC) (Khalifa et al. 2021) asks for the distribution p closest to a pretrained language model a that satisfies moment constraints on chosen features,

$$\begin{aligned} &\min_{p}\ \mathrm{KL}(p \,\|\, a) \quad \text{s.t.} \quad \mathbb{E}_p\big[\phi_i(x)\big] = \bar{\mu}_i,\ i = 1, \dots, K, \\ &\qquad\Longrightarrow \qquad p(x) = \frac{a(x)\, \exp\big(\boldsymbol{\lambda}^\top \boldsymbol{\phi}(x)\big)}{Z}. \end{aligned} \tag{9.2}$$

The constraints can be pointwise (every sample mentions sports) or distributional (half of the samples mention female characters), the solution is an energy-based model over the reference, and the authors describe the framework as a generalization of the maximum-entropy principle — which it is: Equation 9.2 is Equation 5.1 with $h = a$. The two forms sit side by side in Table 9.1. RLHF declares an energy and a temperature; GDC declares features and moments; the multipliers of one are the temperature of the other, and both are tilts of the same reference. Go et al. (Go et al. 2023) close the circle by showing that RLHF minimizes the reverse KL divergence to its implicit target while GDC minimizes the forward KL divergence to its explicit one, and by allowing any f-divergence in between; in their experiments the Jensen–Shannon divergence often gives the best balance between alignment and diversity. In the vocabulary of Section 5.6.3, the choice of divergence at the amortization step is the generalized sense of maximum-entropy learning, now applied to the fitting of the policy rather than of the energy.

Table 9.1: The two forms of the maximum-entropy model, instantiated in post-training.

	Free-energy form (RLHF, KL-control)	Constrained form (GDC)
Declared	reward r, temperature β	features $\boldsymbol{\phi}$, moments $\bar{\boldsymbol{\mu}}$
Target	$\pi_{\text{ref}}\, e^{r/\beta}/Z$	$a\, e^{\lambda^\top \boldsymbol{\phi}}/Z$
Temperature	a genuine dial	determined by the moments
Amortization	reverse KL, policy gradient	forward KL, distributional policy gradient
Generalization	any f-divergence (Go et al. 2023)	

The temperature of Equation 9.1 is a genuine dial in the sense of Section 5.1.4: the reward is given, no level is prescribed, and β is chosen. Small β moves the policy far from the reference and toward the arg-max of the reward; large β leaves it nearly unchanged. The same trade-off appears as a KL constraint rather than a penalty in trust-region methods (Schulman et al. 2015), which is the constrained form once more.

9.2 Amortizing the tilt

Amortization fits a parametric policy π_θ to $\pi^\star$ so that generation is one forward pass. The methods differ in the divergence they minimize and in how they estimate its gradient; none of them changes the target.

Reverse KL and policy gradients. Minimizing $\mathrm{KL}(\pi_\theta \,\|\, \pi^\star)$ is, up to a constant, maximizing the right-hand side of Equation 9.1 in θ, and its gradient is a policy gradient with a shaped reward,

$$\nabla_\theta \mathcal{J}(\theta) \;=\; \mathbb{E}_{y\sim\pi_\theta(\cdot|x)}\Big[\Big(r(x,y) - \beta \log \frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)} - b(x)\Big)\nabla_\theta \log \pi_\theta(y \mid x)\Big], \tag{9.3}$$

with any baseline $b(x)$ (Williams 1992). Proximal policy optimization (Schulman et al. 2017) estimates this with a learned value baseline and a clipped surrogate, and is the algorithm behind the RLHF of Ouyang et al. (2022). Two recent variants remove the value network. Group relative policy optimization (GRPO) (Shao et al. 2024) samples a group of G outputs $\{o_i\}$ for each prompt and uses the group as its own baseline,

$$\begin{aligned} \hat{A}_i \;&=\; \frac{r_i - \mathrm{mean}(r_1,\dots,r_G)}{\mathrm{std}(r_1,\dots,r_G)}, \\ \mathcal{L}_{\text{GRPO}} \;&=\; -\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\Big[\min\big(\rho_{i,t}\hat{A}_i,\; \mathrm{clip}(\rho_{i,t}, 1-\epsilon, 1+\epsilon)\hat{A}_i\big) - \beta\, \widehat{\mathrm{KL}}_{i,t}\Big], \end{aligned} \tag{9.4}$$

where $\rho_{i,t}$ is the token-level probability ratio to the sampling policy and the KL term is estimated per token by the unbiased, non-negative estimator

$$\widehat{\mathrm{KL}}_{i,t} \;=\; \frac{\pi_{\mathrm{ref}}(o_{i,t} \mid \cdot)}{\pi_\theta(o_{i,t} \mid \cdot)} - \log \frac{\pi_{\mathrm{ref}}(o_{i,t} \mid \cdot)}{\pi_\theta(o_{i,t} \mid \cdot)} - 1. \tag{9.5}$$

REINFORCE leave-one-out (RLOO) (Ahmadian et al. 2024) is the sequence-level version: the baseline for each sample is the mean reward of the other $G-1$ samples, there is no clipping, and the authors show that this simpler estimator matches or outperforms PPO in the RLHF setting. What we want the reader to see is that Equation 9.4 and RLOO are estimators of the same gradient Equation 9.3. The group normalization and the clipping are variance reduction; the target is unchanged, and the KL term is the entropy term of the free energy, estimated rather than computed. Every variant in this family is therefore an approximate maximum-entropy learner in the strict sense of Section 5.1.4, in which the inner problem is solved by amortization rather than by sampling.

Forward KL and distribution matching. The distributional policy gradient (DPG) of Parshakova, Andreoli, and Dymetman (Parshakova et al. 2019), which GDC (Khalifa et al. 2021) adopts, minimizes the forward divergence $\mathrm{KL}(p \,\|\, \pi_\theta)$ to the explicit target Equation 9.2 by importance sampling,

$$\nabla_\theta \,\mathrm{KL}(p \,\|\, \pi_\theta) \;=\; -\,\mathbb{E}_{x\sim q}\Big[\frac{p(x)}{q(x)}\,\nabla_\theta \log \pi_\theta(x)\Big], \tag{9.6}$$

with a proposal q that is periodically updated to the current policy and an estimate of Z. GFlowNet fine-tuning (Hu et al. 2024) trains the policy so that complete sequences are sampled with probability proportional to a reward, $\pi_\theta(x) \propto R(x)$, which is the tilt at $\beta = 1$ with $r = \log R$ and the reference absorbed into R; the training objective is a subtrajectory-balance loss rather than a divergence, and the authors call the result amortized inference, which is the exact name. Rejection-sampling fine-tuning — sample from the current model, keep the outputs the reward accepts, and fit by maximum likelihood on the survivors (Dong et al. 2023; Gulcehre et al. 2023) — is forward-KL amortization of a truncated tilt, and distilling the best-of-N distribution of Section 8.4 into the weights (Gui et al. 2024) is the same move with the best-of-N tilt as target, implemented as maximum likelihood on the best sample together with a pairwise term on the best and the worst of the N. The divergence matters: reverse KL is mode-seeking and drives the policy toward the reward's arg-max as β falls, whereas forward KL is mass-covering and needs samples from, or importance weights for, the target. Neither is right in general, which is the empirical finding of Go et al. (2023).

9.3 Learning the energy

The reward of Equation 9.1 is rarely given. It is learned, and the way it is learned is where post-training becomes a maximum-entropy learner in the sense of iii) in Section 5.1.4.

Reward models from pairs. The standard supervision is a comparison: for a prompt x, a preferred response y_w and a rejected one y_l. The Bradley–Terry model (Bradley and Terry 1952)

$$P(y_w \succ y_l \mid x) \;=\; \sigma\big(r_\theta(x, y_w) - r_\theta(x, y_l)\big) \tag{9.7}$$

is fitted by maximum likelihood, which is logistic regression on the difference of rewards and hence the maximum-entropy classifier of Section 5.1.4 applied to pairs; its likelihood equations match the empirical and predicted win frequencies. Outcome and process reward models (Cobbe et al. 2021; Lightman et al. 2024) and the small verifier of Section 8.4 are trained in this way, and the trained reward is then either handed to a sampler (Chapter 8) or amortized (Section 9.2).

Direct preference optimization: the energy through the closed form. Direct preference optimization (DPO) (Rafailov et al. 2023) removes the separate reward model by substituting the representation level into the learning level. Inverting Equation 9.1 gives

$$r(x, y) \;=\; \beta \log \frac{\pi^\star(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z_\beta(x), \tag{9.8}$$

and the partition function cancels in the difference that enters Equation 9.7. Replacing $\pi^\star$ by the policy being trained yields the loss

$$\mathcal{L}_{\text{DPO}}(\theta) \;=\; -\mathbb{E}_{(x,y_w,y_l)}\Big[\log \sigma\Big(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\Big)\Big], \tag{9.9}$$

whose gradient weights each pair by how wrongly the implicit reward $\hat{r}_\theta = \beta \log(\pi_\theta/\pi_{\text{ref}})$ currently orders it. Nothing is sampled during training, which is the practical appeal; the conceptual point is that the log-ratio of the policy to the reference *is* the energy, so that learning the policy and learning the energy are one step. Rafailov et al. (Rafailov et al. 2024) extend the reading to the token level: in the token Markov decision process, the token-wise log-ratio $\beta \log \pi^\star(a_t \mid s_t)/\pi_{\text{ref}}(a_t \mid s_t)$ is the soft advantage $Q^\star - V^\star$ of maximum-entropy reinforcement learning, so DPO fits soft Q-values of Section 5.5.2 with the reference as base measure. The limitations are those of any learned energy: the implicit reward is defined only relative to π_{ref}, the data are off-policy, and a small β lets the policy travel far from the reference on the strength of a few comparisons.

Energies from contrastive noise. The residual energies of Section 8.3 are the third route: noise-contrastive estimation with the reference as noise distribution (Deng et al. 2020; Xu et al. 2025; Yan et al. 2026). This is a maximum-entropy learner in the generalized sense of Section 5.6.3, and its output is used at inference time, so it belongs to both chapters — learned here, applied there.

9.4 Self-referential energies

A last class of methods takes the reward from the model itself. EMPO (Zhang et al. 2025) rewards a response by the frequency of its semantic cluster among the model's own samples; test-time reinforcement learning (TTRL) (Zuo et al. 2025) does the same with an exact-match majority vote as the pseudo-label; and in test-time adaptation the analogous move is to minimize the entropy of the model's predictions on the test stream, which Park et al. (Park et al. 2025) show should be paired with an explicit energy objective to be reliable. Formally these are instances of Equation 9.1 with r replaced by a functional of π itself, and the closed form becomes the self-consistency equation Equation 6.3. The consequence, worked out in Section 6.6, is a rich-get-richer dynamics with a collapse threshold, and the reason is that the outer criterion has been reversed: an external reward tilts the reference toward what the world values, whereas an internal one tilts it toward what the model already believes, and the entropy term now fights the objective rather than regularizing it. This is why we place these methods in Chapter 6 and not here. They share the inner Gibbs tilt with maximum-entropy post-training and differ in the sign of the outer problem; the two chapters are the two directions of the dial of Chapter 3.

9.5 The continuous case

The same structure recurs, almost verbatim, for diffusion and flow models, with the pretrained generator p_{base} as reference and a reward on clean samples as energy. Black et al. (Black et al. 2024) cast denoising as a multi-step decision process and optimize a reward by policy gradient (DDPO). It is worth noting that DDPO has no KL term: the authors control over-optimization by stopping early, so that on its own the method sits at the boundary of the family described in Section 9.6, and it is the concurrent DPOK of Fan et al. (Fan et al. 2023), which adds a per-step KL penalty to the pretrained model, that brings the construction back inside. Wallace et al. (Wallace et al. 2024) carry DPO over to diffusion models by writing the preference likelihood on denoising trajectories. Uehara et al. (Uehara et al. 2024) make the target explicit: fine-tuning

as entropy-regularized control against the pretrained model has the tilt $p_{\text{base}}(x)\, e^{r(x)/\alpha}$ as its solution, and the regularization is what prevents the "reward collapse" of over-optimizing an imperfect learned reward. Domingo-Enrich et al. (Domingo-Enrich et al. 2025) then show that the naive stochastic-optimal-control formulation carries an *initial value function bias*: the generated distribution picks up a term that depends on the initial noise and is not the tilt. The remedy is a particular memoryless noise schedule, under which the fine-tuned flow or diffusion model provably converges to $p_{\text{base}}\, e^{r}$, and their adjoint matching algorithm turns the control problem into a regression. Seen in light of Section 8.2, guidance and reward fine-tuning are two perspectives on the same underlying tilt: guidance injects the gradient correction from Equation 8.3 during sampling, while fine-tuning incorporates that same adjustment directly into the model's drift.

9.6 Post-training on the three levels, and where it stops

Table 9.2 places the methods of this chapter on the three levels of Section 5.1. The representation level is the same in every row; the rows differ in what is computed and in whether the energy is given, learned, or self-referential.

Table 9.2: Post-training methods on the three levels. Amortization fits the reference to a fixed tilt; energy learning fits the tilt itself; self-referential rewards reverse the outer criterion and belong to minimum-entropy learning. DDPO is listed for contrast: without a KL term it sits at the boundary of the family.

Method	Base measure π_{ref}	Energy	What is fitted	Level	Divergence or objective
RLHF with PPO (Ouyang et al. 2022); GRPO (Shao et al. 2024); RLOO (Ahmadian et al. 2024)	pretrained LM	reward, given: a learned reward model or a rule-based verifier	policy $\pi_\theta \approx \pi^\star$	amortized inference	reverse KL, by KL-penalized policy gradient
GDC, trained by DPG (Khalifa et al. 2021; Parshakova et al. 2019)	pretrained LM a	moments of declared features, i.e., $\lambda^\top \phi$	policy	amortized inference (constrained form)	forward KL, importance-weighted
f-DPG (Go et al. 2023)	pretrained LM	any unnormalized target P	policy	amortized inference	any f-divergence
GFlowNet fine-tuning (Hu et al. 2024)	pretrained LM (initialization; also evaluates R)	reward R, e.g., a posterior of the same LM	policy $\propto R$	amortized inference	subtrajectory-balance objective
RAFT (Dong et al. 2023), ReST (Gulcehre et al. 2023)	current / pretrained LM	reward, given	policy	amortized inference	forward KL to a truncated tilt: maximum likelihood on accepted samples
BoNBoN (Gui et al. 2024)	pretrained LM	reward, given	policy $\approx \pi_{\text{BoN}}$	amortized inference	maximum likelihood on the best of n plus an IPO-type term on best–worst pairs

Method	Base measure π_{ref}	Energy	What is fitted	Level	Divergence or objective
Reward models (Christiano et al. 2017; Ouyang et al. 2022); ORM (Cobbe et al. 2021); PRM (Lightman et al. 2024); EORM (Jiang et al. 2025)	—	learned from pairs, or from outcome / step labels	energy	learning	log score on pairs (Bradley–Terry) or on labels
DPO (Rafailov et al. 2023, 2024)	pretrained LM	implicit: $\beta \log \pi_\theta / \pi_{\text{ref}}$	policy = energy	learning through the closed form	log score on pairs
Residual energies (Deng et al. 2020); EDLM (Xu et al. 2025); Uni-E (Yan et al. 2026)	AR LM / DLM denoiser	residual, NCE-trained or read off a proxy AR model	energy	learning (generalized sense), or none when read off	NCE, i.e., the logistic score
DDPO (Black et al. 2024)	diffusion prior	reward, given	denoising policy	reward maximization with no KL term (boundary case; early stopping)	—
DPOK (Fan et al. 2023); entropy-regularized control (Uehara et al. 2024); adjoint matching (Domingo-Enrich et al. 2025)	diffusion / flow prior	reward, given	denoising policy / generator drift	amortized inference	reverse KL: per-step KL penalty, or stochastic optimal control
Diffusion-DPO (Wallace et al. 2024)	diffusion prior	implicit, from pairs	generator = energy	learning through the closed form	log score on pairs
EMPO (Zhang et al. 2025); TTRL (Zuo et al. 2025); ReTTA (Park et al. 2025)	pretrained model	the model's own agreement (EMPO, TTRL) or its own free energy (ReTTA)	policy	reversed outer criterion	— (Chapter 6)

The definition also says what is *not* maximum-entropy post-training. Reward maximization without a KL term is the zero-temperature boundary: its optimum is the degenerate distribution on the arg-max, which is the distribution collapse of Korbak et al. (2022). Best-of-N with $N \to \infty$ is the same limit reached from the sampling side. Supervised fine-tuning on demonstrations is maximum likelihood, i.e., forward KL from the data to the model, and involves no reference and no tilt; it is a maximum-entropy learner only in the trivial sense of Section 5.1.4, and it becomes part of the family here only as the fitting step of rejection-sampling fine-tuning, where the "data" are samples from a tilt. DPO with $\beta \to 0$ leaves the family in the same way as RLHF without a penalty.

What is open is, again, at least as interesting. Reward hacking is energy misspecification: the tilt faithfully concentrates on what the learned energy rewards, and the entropy term is the only thing that keeps an imperfect

energy from being exploited, which is why the KL budget, the choice of β, and the choice of divergence (Go et al. 2023) are not implementation details. The trade-off between test-time computation (Chapter 8) and train-time amortization has no theory yet beyond the KL accounting of best-of-N; a principled answer would say, for a given energy and budget, how much of the tilt to execute at inference and how much to store. And the self-referential methods of Section 9.4 show that the boundary between maximum- and minimum-entropy learning runs through post-training itself: the same tilt, with an external or an internal energy, gives a method that stays honest or one that collapses.

9.7 Takeaway

Post-training is maximum-entropy learning with the pretrained model as base measure. Its object is the tilt Equation 9.1, which has both forms of Section 5.1.1 as instances — RLHF and KL-control declare a reward and a temperature, GDC declares features and moments. Amortization (PPO, GRPO, RLOO, DPG, GFlowNet fine-tuning, rejection sampling) fits the reference to the tilt and differs only in the divergence and its estimator; energy learning (reward models, DPO, contrastive residual energies) fits the tilt itself, and in DPO the two coincide because the policy's log-ratio to the reference is the energy. The KL term is where the entropy is load-bearing, and removing it, or sending N or $1/\beta$ to infinity, is where the method leaves the family. The story so far is of a single distribution, accessible by paying in either of two currencies: computation expended at inference time, or training and storage invested in advance. The next part explores how and where these approaches enable progress in scientific discovery.

10 From Entropy-Based Learning to the Free-Energy Principle

The free-energy principle (FEP) is one of the most ambitious claims related to intelligence: that perception, learning, action, and — in its strongest form — the persistence of living things are all one optimization. It was developed in theoretical neuroscience, largely by Friston and collaborators (Friston 2005, 2010; Friston et al. 2006), and it is usually presented in a vocabulary of its own. This chapter examines the connection of the free-energy principle in the terms and concepts developed throughout this book. Some of the FEP is an identity with material of entroy-based learning: it is Equation 5.2 at $T = 1$, read in a particular way. Some of it is a correspondence, close enough to be illuminating but not a theorem. Other aspects — such as the physics of Markov blankets and non-equilibrium steady states — stand apart from the framework developed in this book, and we will make that distinction clear.

The reason to spend a chapter on this is not completeness. It is that the FEP supplies the one ingredient the preceding chapters leave implicit. Chapters 5 through 7 organize methods as an inner free-energy problem under an outer criterion, and that criterion either fits the data (Chapter 5), or sharpens the model's own output (Chapter 6), or chooses which features to measure and which assignments to commit to (Chapter 7). All three take the data as given. Active inference does not: its outer problem chooses *what to observe*. We propose that this constitutes a third distinct type of outer problem, as described in Section 7.2.1—precisely the kind that has been absent from the rest of the book. Making this explicit serves to clarify the structure and implications on both fronts.

Because the principle might be unfamiliar to many readers of this book, the chapter is arranged in four parts. The first, Section 10.1, presents the FEP on its own terms and in its own notation — where it came from, what it says about perception, learning, and action, and the physical claims that were later built on it — with worked examples, of perception and of planning. The second, Section 10.2, surveys the developments of the field since 2019, debates included. The third section, Section 10.3, acts as a bridge: it clarifies which aspects of the free-energy principle are mathematically identical to the free energy discussed in Chapter 3, which aspects are related but only by analogy, and which fall outside both categories. To illustrate these connections, it adds two additional examples—one focused on learning processes, and another on the case where preferences themselves are subject to learning. The fourth, Section 10.4, collects what the translation establishes and what it leaves open. The numbers in all four examples come from `code/fep-examples.py`.

10.1 Foundations of the free-energy principle

10.1.1 Origins

The idea behind the FEP is older than the name. Helmholtz described perception as *unconscious inference* about the causes of sensations, and the FEP begins by taking that description literally (Friston 2005). Its direct predecessors in machine learning are the Helmholtz machine (Dayan et al. 1995) and the free-energy view of EM (Neal and Hinton 1998), both of which read the evidence lower bound as a negative variational free energy, as the table of Chapter 3 records. What Friston, a neuroscientist at University College London, added was the claim that the brain — and, later, any system that persists — minimizes this quantity, and that perception, learning, and action are the three ways of doing so.

The principle took shape in roughly four steps. i) In 2005, Friston observed that perceptual inference and perceptual learning both rest on minimizing the brain's free energy, with learning arising from synaptic changes that minimize the free energy averaged over the stimuli encountered (Friston 2005); in 2006, with Kilner and Harrison, he termed this a free-energy principle for the brain (Friston et al. 2006). ii) In 2007, the variational Laplace method gave a practical algorithm for computing the free energy under a Gaussian approximation (Friston et al. 2007). iii) Two reviews of 2009–2010 promoted it to a unified brain theory (Friston 2009, 2010), and "Action and behavior" of 2010 brought in action, reading motor control as the fulfilment of prior expectations about proprioceptive sensations (Friston et al. 2010). iv) From 2012 on, *active inference* on discrete state spaces recast optimal control as a pure inference problem, with cost functions absorbed into prior beliefs about state transitions and terminal states (Friston, Samothrakis, et al. 2012); it then introduced the expected free energy and epistemic value (Friston et al. 2015), learning and habits (Friston et al. 2016), and a process theory for the brain (Friston et al. 2017). The textbook is *Active Inference* by Parr, Pezzulo, and Friston (Parr et al. 2022); a mathematical review is that of Buckley et al. (Buckley et al. 2017), and an introductory tutorial is that of Bogacz (Bogacz 2017), whose running example we adopt below.

In outline, the principle says the following. A system that persists must keep its sensory exchanges with the environment within bounds, i.e., it must avoid surprising sensations, where the surprise of an observation o under a model is $-\log p(o)$. Surprise cannot be evaluated directly, because the model has hidden causes that would have to be integrated out, so the system minimizes a tractable upper bound on it, the *variational free energy*. It has three ways of doing so: it can change its beliefs about the causes, which is perception; it can change the model, which is learning; or it can change the sensations themselves by acting on the world, which is action. The quantities behind these three sentences are developed in the next four subsections; the physical claim that makes "persists" into a theorem, by way of Markov blankets and steady states, is set out in Section 10.1.6.

10.1.2 Variational free energy and perception

The FEP starts from a *generative model*, namely a joint distribution $p(o, s) = p(s)\,p(o \mid s)$ over observations o and their hidden causes s, which the agent is assumed to embody. Given an observation, the agent cannot in general compute the posterior $p(s \mid o)$ or the evidence $p(o)$, so it maintains an approximate belief $q(s)$, termed the recognition density, and scores it by the variational free energy

$$\begin{aligned} F[q; o] &= \mathbb{E}_{q(s)}\big[-\log p(o, s)\big] - H[q] \\ &= \mathrm{KL}\big(q(s) \,\|\, p(s \mid o)\big) - \log p(o) \\ &= \underbrace{\mathrm{KL}\big(q(s) \,\|\, p(s)\big)}_{\text{complexity}} - \underbrace{\mathbb{E}_{q(s)}\big[\log p(o \mid s)\big]}_{\text{accuracy}} \;\geq\; -\log p(o). \end{aligned} \tag{10.1}$$

The three lines read "energy minus entropy", "divergence plus surprise", and "complexity minus accuracy". The second shows that F is an upper bound on the surprise $-\log p(o)$, with equality if and only if q is the exact posterior, so that minimizing F does two things at once: it brings q to the posterior, which is perception, and it brings F down to the surprise, which gives a computable bound on the model evidence — the negative of the evidence lower bound of Chapter 3. We note at once that what the FEP minimizes is the surprise of particular observations and not an entropy; the two are linked by a sentence of Friston's — under ergodic assumptions, the long-term average of surprise is entropy (Friston 2013) — to which we return in Section 10.1.6 and Section 10.3.9.

Example 1 (perception). We take the example of Bogacz's tutorial (Bogacz 2017): an animal infers the size v of a food item from an observed light intensity u, with the prior $v \sim \mathcal{N}(3, 1)$, the likelihood $u \mid v \sim \mathcal{N}(v^2, 1)$, and the observation $u = 2$. The model is nonlinear and the posterior has no closed form, but in one dimension it can be computed on a fine grid. The surprise is $-\log p(u) = 3.2368$ nats, and the posterior has mean 1.4797, standard deviation 0.3425, and mode 1.5675. Gradient descent on Equation 10.1 over the Gaussian family

$q = \mathcal{N}(m, s^2)$, started at the prior, converges to $m^\star = 1.4933$ and $s^\star = 0.3025$ with $F^\star = 3.2636$. The excess of $F^\star$ over the surprise, 0.026828 nats, agrees with $\mathrm{KL}(q^\star \| p(v \mid u))$ to within 10^{-13}, as the second line of Equation 10.1 says it must. At the starting point the free energy is 51.92 nats, since the prior predicts an intensity $v^2 \approx 9$, far from the observed value 2 (Figure 10.1 a, b).

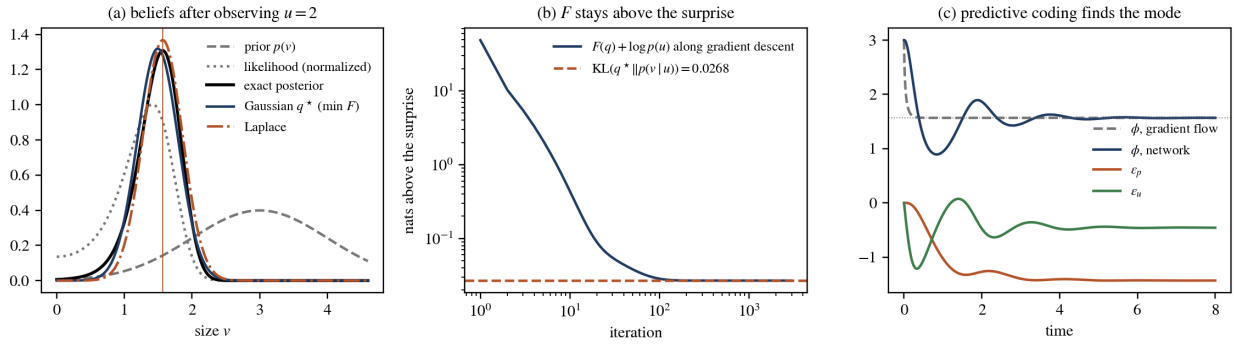


Figure 10.1: Perception as free-energy minimization (Example 1). (a) Prior, normalized likelihood, exact posterior, the best Gaussian $q^\star$, and the Laplace approximation after observing $u = 2$, with the posterior mode marked; (b) the excess of $F(q)$ over the surprise along gradient descent, converging to $\mathrm{KL}(q^\star \| p(v \mid u)) = 0.0268$; (c) the predictive-coding network, whose state ϕ and two error nodes converge to their values at the posterior mode. Produced by `code/fep-examples.py`.

10.1.3 Predictive coding, precision, and attention

Predictive coding is the process theory that implements the minimization in neural tissue, as message passing between units that carry states and units that carry prediction errors (Rao and Ballard 1999; Friston 2005). The name is older than the principle, and the idea is best seen where it started. In the engineering of signals, Elias proposed to transmit not a signal but the difference between the signal and a prediction of it formed from its own past: the difference is smaller and cheaper to send, and a receiver that forms the same prediction loses nothing (Elias 1955). The channel is then quiet as long as the world behaves as expected and speaks only when something unpredicted happens, as in Figure 10.2 (c). Srinivasan, Laughlin, and Dubs read the centre–surround organization of early visual neurons in the same way: the surround forms a prediction of the centre from its neighbours, the cell transmits the residual, and the best weighting of the prediction depends on the signal-to-noise ratio of the input (Srinivasan et al. 1982). Mumford proposed that the cortex as a whole is organized like this, with the feedback from a higher area carrying its reconstruction of the activity in the lower area and the feedforward path carrying only what remains unexplained (Mumford 1992), and Rao and Ballard built the hierarchical network and trained it on natural images (Rao and Ballard 1999). Their best-known result is the *end-stopping* of visual neurons, i.e., a unit that responds to a short bar responds progressively less when the bar is extended beyond its receptive field: the long bar is predicted from its surroundings, so there is little residual left to report.

Under the FEP this becomes a circuit with two kinds of units at every level of a hierarchy (Figure 10.2 a, b). State units hold the current estimate of the causes at their level and send a prediction to the level below; error units at that lower level compare the prediction with what is actually there and send back the difference, multiplied by a gain. Three rules govern the traffic: i) predictions descend and errors ascend; ii) an error that has been explained is silenced, so that at equilibrium only the unexplained part of the input travels upward, namely the news; and iii) a state unit moves in the direction that reduces the errors it receives, from below and from above. The picture to keep is of a hierarchy in which each level tries to quieten the level below. Perception in this scheme is not a reconstruction of the scene assembled upward from the retina but a guess issued downward and corrected by whatever it failed to predict; what climbs the hierarchy is a report of the misfit, not a copy of the world. The everyday analogy is proofreading one's own text: a fluent reader predicts the next word and looks only where the prediction fails, which is why a typo in a familiar word is so easily

missed, and why it is found when one reads slowly, i.e., when the gain on the sensory error is turned up. Anatomically, the FEP assigns the errors to the superficial pyramidal cells that send forward connections and the predictions to the deep pyramidal cells that send backward connections (Friston 2005; Bastos et al. 2012); the general scheme for hierarchies of dynamical models in continuous time is given in (Friston 2008).

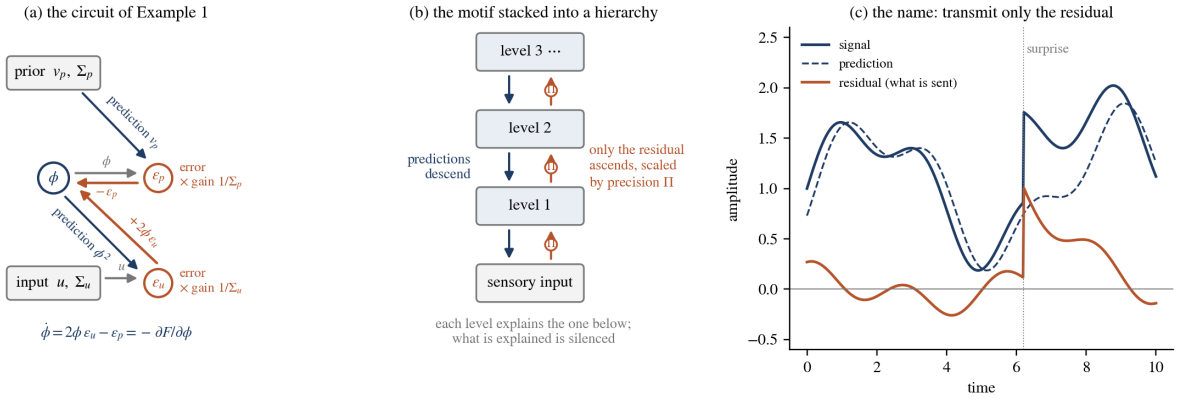


Figure 10.2: Predictive coding as a picture. (a) The circuit of Example 1: predictions (navy) descend from a state unit to the error unit of the level below, the state of each level enters its own error unit (grey), and the precision-weighted errors (rust) return to the state unit, which follows the negative gradient of the free energy. (b) The same motif stacked into a hierarchy: each level explains the one below, and only the residual, scaled by its precision Π, ascends. (c) The origin of the name: a channel that transmits the residual between a signal and its prediction carries little until something unpredicted happens. Drawn by `code/fep-predictive-coding.py`.

The rules are not an extra hypothesis; they are the gradient of the free energy read as a wiring diagram. For Example 1, with a point estimate ϕ in place of q, the free energy of Equation 10.1 is, up to a constant, a sum of two squared and precision-weighted errors, $F(\phi) = (\phi - v_p)^2/(2\Sigma_p) + (u - \phi^2)^2/(2\Sigma_u)$, and gradient descent on it is exactly Bogacz's network: three nodes, the state ϕ and the prediction errors $\varepsilon_p = (\phi - v_p)/\Sigma_p$ and $\varepsilon_u = (u - \phi^2)/\Sigma_u$, with the dynamics $\dot{\phi} = 2\phi\,\varepsilon_u - \varepsilon_p = -\partial F/\partial\phi$ (Bogacz 2017). Each error unit needs only the prediction from above and the signal from below, and the state unit needs only the two errors and the local slope 2ϕ of its own prediction. This locality is the point of the construction: no unit knows the posterior, yet the network finds it. Numerically ϕ converges to 1.5675 and the error nodes to $\varepsilon_p = -1.4325$ and $\varepsilon_u = -0.4570$, which are the analytic values at the mode (Figure 10.1 c). The network therefore computes the *mode* of the posterior, not its mean, and the uncertainty has to be supplied separately by the Laplace approximation (Friston et al. 2007): the curvature at the mode gives $s = 0.2918$, which understates both the posterior standard deviation 0.3425 and the 0.3025 of the best Gaussian.

The errors are weighted by their *precisions*, i.e., the inverse variances $1/\Sigma_p$ and $1/\Sigma_u$, and a precision is simply the gain of an error unit: it decides how loudly a given misfit is reported. Since the state unit balances the two reports, what matters for the estimate is their ratio. Specifically, in Example 1 the mode is 1.5675 at equal precisions; making the sensory channel four times more precise moves it to 1.4601, towards the $\sqrt{u} = 1.4142$ that the data alone would give, and making the prior four times more precise moves it to 1.8171, towards the prior mean 3, and the same 1.8171 is obtained by leaving the prior alone and making the sensory channel four times less precise. The FEP reads attention as inference about precision (Feldman and Friston 2010): a cue that a sensory channel is reliable raises the precision of its errors, and the state units then follow that channel more closely. In the circuit this is a change of gain, not a change of wiring, and it is, in effect, the knob that Srinivasan et al. had already found the retina turning with the signal-to-noise ratio. Notably, predictive coding is the process theory for continuous states; the discrete-state schemes of Section 10.1.5 minimize the same free energy with a different message passing (Friston et al. 2017). For the broader view of predictive processing in cognitive science, see the review of Clark and the introduction of Sprevak and Smith (Clark 2013; Sprevak and Smith 2023).

10.1.4 Learning and model selection

Learning uses the same quantity. Under the FEP, parameters are slowly varying hidden variables, and beliefs about them are updated by minimizing the free energy averaged over observations. In the standard discrete-state scheme the likelihood matrix **A** and the transition matrices **B** carry Dirichlet priors, and learning accumulates their concentration parameters after each observation (Friston et al. 2016; Da Costa et al. 2020); in the limit of point estimates this is EM, with the E-step minimizing over the hidden states and the M-step over the parameters (Neal and Hinton 1998). In the third line of Equation 10.1 the complexity term penalizes departures from the prior and the accuracy term rewards the fit, so the free energy is also a computable bound on the negative log model evidence, which serves for model comparison. Structure learning in active inference adds hidden-state slots for novel concepts and uses Bayesian model reduction to reset them in favour of a simpler model with higher evidence (Smith et al. 2020).

10.1.5 Action: active inference and the expected free energy

Action enters the FEP in two ways. i) In continuous-state formulations it minimizes F directly, by changing the observations so that they conform to predictions — motor control as the fulfilment of prior expectations about proprioception (Friston et al. 2010). ii) In discrete-state formulations the agent infers policies π, i.e., sequences of actions, scoring them by the *expected free energy* (Friston et al. 2015, 2017; Da Costa et al. 2020),

$$\begin{aligned} G(\pi) &= \underbrace{\mathrm{KL}\big(q(o \mid \pi) \,\|\, \tilde{p}(o)\big)}_{\text{risk}} + \underbrace{\mathbb{E}_{q(s|\pi)}\, H\big[p(o \mid s)\big]}_{\text{ambiguity}} \\ &= \underbrace{-\,\mathbb{E}_{q(o|\pi)}\big[\log \tilde{p}(o)\big]}_{-\text{ pragmatic value}} - \underbrace{I_\pi(S;O)}_{\text{epistemic value}} , \end{aligned} \tag{10.2}$$

where $q(o \mid \pi) = \mathbb{E}_{q(s|\pi)} p(o \mid s)$ is the predicted distribution of outcomes, $\tilde{p}(o)$ the *preferences*, termed prior preferences in the literature, and $I_\pi(S;O) = H[q(o \mid \pi)] - \mathbb{E}_{q(s|\pi)} H[p(o \mid s)]$ the mutual information between hidden states and outcomes, i.e., the expected information gain. The first line reads "risk plus ambiguity" and the second "minus pragmatic value minus epistemic value"; exploration and exploitation thus sit inside one quantity, and Friston et al. note the formal consistency with the InfoMax principle and with risk-sensitive KL control themselves (Friston et al. 2015). The posterior over policies is a softmax,

$$q(\pi) \;=\; \sigma\big(\log E(\pi) - \gamma\, G(\pi)\big), \tag{10.3}$$

with E a prior over policies and γ a precision. Da Costa et al. call γ an inverse temperature that denotes the confidence in selecting a policy (Da Costa et al. 2020); in the tutorial of Smith et al. the prior E reflects the number of times a policy has previously been chosen, which models habits, and γ has a prior of its own and is inferred (Smith et al. 2022). Experimentally, midbrain activity tracks trial-by-trial model-based estimates of precision (Schwartenbeck et al. 2015). It is worth being careful about what Equation 10.2 is and is not: it is not simply the free energy of the future (Millidge et al. 2021), and several non-equivalent objectives have been proposed for the same role (Parr and Friston 2019; Sajid, Ball, et al. 2021; Champion et al. 2024); we return to this in Section 10.2.

Example 2 (a T-maze with a cue, and the dark room). A hidden context $c \in \{L, R\}$ decides whether a reward lies in the left or the right arm, with prior $(1/2, 1/2)$. The agent may stay at the centre, visit a cue that reveals the context with validity v, enter an arm (reward or loss with probability 0.98, after which it cannot leave), or enter a dark room (after which it cannot leave). The preferences are $\tilde{p} \propto e^{C}$ with $C = +3$ for the reward, -3 for the loss, and 0 otherwise, so that $\tilde{p}$ is 0.8322 for the reward, 0.0021 for the loss, and 0.0414 for each neutral outcome.

Table 10.1: One-step expected free energy of each action in Example 2, in nats. Because the preferences are symmetric, the cross-entropy is the same for every action and the first choice is decided by the epistemic term alone; the rows for the centre, the arms, and the dark room do not depend on the cue validity v.

Action	Cross-entropy	$H[q(o)]$	Ambiguity	Risk	Information gain	G
centre	3.1837	0	0	3.1837	0	3.1837
cue, $v = 1$	3.1837	0.6931	0	2.4905	0.6931	**2.4905**
cue, $v = 0.9$	3.1837	0.6931	0.3251	2.4905	0.3681	2.8156
left (= right)	3.1837	0.6931	0.0980	2.4905	0.5951	2.5886
dark	3.1837	0	0	3.1837	0	3.1837

Three features of Table 10.1 are worth noting. First, the symmetry of the preferences makes the cross-entropy — minus the pragmatic value — equal to 3.1837 nats for every action, so the first move is decided by the epistemic term alone. Second, the two actions with the *lowest* outcome entropy, the centre and the dark room, have the *highest* expected free energy. This is the precise form of the answer to the dark-room objection. Critics asked why an agent that minimizes surprise does not retreat into a dark and unchanging room; Friston, Thornton, and Clark replied that average surprise is a function of the sensations *and of the agent predicting them*, whereas the entropy that a dark room minimizes is a function of the sensations alone (Friston, Thornton, et al. 2012), and Schwartenbeck et al. added that minimizing surprise leads naturally to exploration and to novelty bonuses (Schwartenbeck et al. 2013). In Table 10.1 the reply becomes an identity: the risk term $\mathrm{KL}(q(o \mid \pi)\|\tilde{p}) = \mathrm{CE} - H[q(o \mid \pi)]$ subtracts the outcome entropy from the expected surprise, so an agent minimizing Equation 10.2 does not minimize the entropy of its outcomes, and the dark room is chosen only if the preferences themselves favour it. Third, the epistemic term is sensitive to reliability: with a perfect cue the information gain is $\log 2 = 0.6931$ against 0.5951 for an arm, so the agent looks at the cue; at a validity of 0.9 the ambiguity rises to 0.3251 and the one-step choice flips to entering an arm. The precision γ sets how deterministic the choice is: with a reliable cue, the probability of visiting the cue under Equation 10.3 with a flat E is only 0.262 at $\gamma = 1$, 0.404 at $\gamma = 4$, and 0.706 at $\gamma = 16$ (Figure 10.3 b).

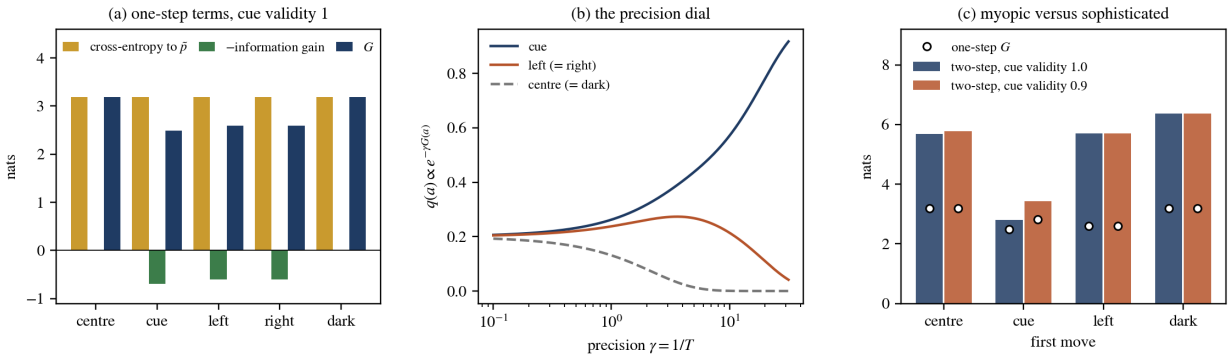


Figure 10.3: Planning with the expected free energy (Example 2). (a) Cross-entropy to the preferences, minus the information gain, and G for each action with a fully reliable cue; (b) the precision dial $q(a) \propto e^{-\gamma G(a)}$; (c) one-step G (dots) against the sophisticated two-step value (bars) at two cue validities. Produced by `code/fep-examples.py`.

That last flip is an artefact of myopia, and it is worth saying so, because the one-step expected free energy is often presented as the whole story. *Sophisticated inference* computes G recursively: the expected free energy of the next action is its own risk and ambiguity plus the average expected free energy of the beliefs that would follow under counterfactual outcomes and subsequent actions, the latter weighted by a softmax of their own G, so that it is a tree search over beliefs about beliefs (Friston, Da Costa, Hafner, et al. 2021). In the high-precision limit the two-step value is $G(a_1) + \mathbb{E}_{o_1} \min_{a_2} G(a_2 \mid o_1)$. Under that score, visiting the cue

first is best at both validities — 2.7942 at $v = 1$ and 3.4344 at $v = 0.9$, against 5.6742 for staying at the centre, 5.7049 for entering an arm, and 6.3674 for the dark room — which corrects the one-step choice (Figure 10.3 c). By contrast, a score that adds two one-step values of G under the same prior belief selects "cue, then cue again" (4.9811) at $v = 1$ and "left, then left" (5.1771) at $v = 0.9$, because it does not let the second step use the first observation. This matches the result of Da Costa et al.: the standard scheme is Bellman optimal for a planning horizon of one and not beyond, whereas sophisticated inference is Bellman optimal on any finite horizon (Da Costa et al. 2023).

10.1.6 Markov blankets, steady states, and Bayesian mechanics

The most ambitious part of the FEP is not an algorithm but a physical claim about existence. In "Life as we know it", Friston argues that if a set of dynamical systems is coupled only by short-range forces, then remote states are conditionally independent, which induces a *Markov blanket* separating internal from external states; the internal states then appear to minimize a free-energy functional of the blanket states, i.e., appear to infer the external world. He goes on to argue that, since the long-term average of surprise is an entropy, minimizing free energy by selectively sampling the world places an upper bound on the entropy of the sensory states, which in his words enables biological systems to resist the second law of thermodynamics (Friston 2013). Kirchhoff et al. nest the construction into blankets of blankets (Kirchhoff et al. 2018), and Ramstead et al. extend it to the capacity of living systems to avoid decay, in answer to Schrödinger's question (Ramstead et al. 2018).

After 2019 this part developed into *Bayesian mechanics*. Starting from a Langevin equation, sparse coupling yields a *particular partition*, and the paths of the internal states of a particle obey a principle of least action which, read teleologically, maximizes the marginal likelihood (Friston 2019; Friston, Da Costa, Sajid, et al. 2023); "strange" particles also infer their own actions (Friston, Da Costa, Sakthivadivel, et al. 2023). Da Costa et al. give a more careful treatment for stationary processes, in which the internal states appear to perform variational inference and the active states can be read as performing active inference, covering classical stochastic control such as PID control (Da Costa et al. 2021). Ramstead et al. distinguish path-tracking, mode-tracking, and mode-matching mechanics and examine a duality between the FEP and the constrained maximum entropy principle (Ramstead et al. 2023); Sakthivadivel argues that for a class of constrained dynamics the inference is equivalent to gradient ascent on a Shannon entropy functional (Sakthivadivel 2022); a quantum-information version is due to Fields et al. (Fields et al. 2022).

This is also where the controversy concentrates, and the critiques are taken up in Section 10.2.1. Two points matter for what follows. The mathematics of Section 10.1.2 to Section 10.1.5 does not depend on the physical assumptions about blankets and steady states; and, conversely, the physical claims do not follow from the entropy-based learning of this book, as we explain in Section 10.3.9.

10.1.7 A map of the principle

The FEP therefore operates at several levels at once, and much of the confusion around it comes from not saying which level is meant. Table 10.2 lists the levels, the object and the quantity minimized at each, and the counterpart in this book, which Section 10.3 establishes row by row.

Table 10.2: The levels of the free-energy principle and their counterparts in this book.

Level	Object in the FEP	Quantity minimized	Counterpart in this book
perception	belief $q(s)$	$F[q; o]$, Equation 10.1	Equation 5.2 at $T = 1$ with $h = p(s)$; Section 10.3.1
predictive coding	mode ϕ, precisions	$-\log p(o, \phi)$, plus curvature	the $T \to 0$ end of the dial with a Laplace correction; Section 10.3.2

Level	Object in the FEP	Quantity minimized	Counterpart in this book
learning	parameters θ	average F	maximum-entropy learning with missing data (Section 5.6.3); $T = 1$ on the dial of Section 7.3.1; Section 10.3.3
model selection	model m	F as a bound on $-\log p(o \mid m)$	Type I with a cost on the parameters; MDL (Section 7.1); Section 10.3.4
planning	policy π	$G(\pi)$, Equation 10.2	firmness plus fairness (Section 6.3); Type III, proposed in Section 10.3.5
policy selection	$q(\pi)$, Equation 10.3	$\mathbb{E}_q[G] + \gamma^{-1}\mathrm{KL}(q\|E)$	the Gibbs tilts Equation 6.2 and Equation 9.1; Section 10.3.6
existence	steady-state density, blankets	average surprise	not derivable; a duality with maximum entropy; Section 10.3.9

10.2 Recent developments, 2019–2026

10.2.1 Critiques and replies

The principle has been examined closely since 2020, and the examination has drawn detailed replies; in our reading the exchange is still open. We group the main lines under three headings, state each position as its authors state it, and do not attempt to settle the matter. Two cautions apply throughout. First, much of the technical discussion concerns the 2013 formulation of Section 10.1.6, and the critics themselves note that the later formulations parametrize the variational density by different variables and carry a different interpretation (Biehl et al. 2021), so that an objection to one formulation is not automatically an objection to another. Second, the variational free energy of Section 10.1.2 and its use in perception and learning, which is most of what this chapter draws on, is not what is in dispute; the disputes concern the physical claims about blankets and steady states, the objective for planning, and the epistemic status of the principle as a whole.

i) *Technical.* Biehl, Pollock, and Kanai examine the formulation of "Life as we know it" and report that the definitions of the Markov blanket used in different papers are not equivalent, that the steps which rewrite the equations of motion of a system with a blanket are not generally correct without additional assumptions that had not been stated, and that the free-energy lemma, taken at face value, admits counterexamples; where the lemma does hold, they show, the variational density coincides with the conditional density of the steady state, and since the interpretation in terms of Bayesian inference hinges on this point they conclude that it is not sufficiently justified. They add that the problems they raise do not rule out that the general ideas behind the principle are worth pursuing (Biehl et al. 2021). Friston, Da Costa, and Parr take the observations as bearing on an early formulation and use them to set out the technical arguments that underwrite the principle, with the discussion focusing on solenoidal coupling between subsets of states in sparsely coupled systems and on the distinction between exact and approximate Bayesian inference (Friston, Da Costa, and Parr 2021). Aguilera et al. examine the simplest setting, weakly coupled linear stochastic systems, and report that the blanket condition and the restrictions on solenoidal flow are met only in a narrow region of parameter space, requiring an absence of perception–action asymmetry that they consider unusual for living systems, and that the step from the dynamics to variational inference relies on an equivalence between the dynamics of the average states and the average of the dynamics which does not hold in general even for linear systems; they conclude that more development is needed before the theory can be applied to living and cognitive systems (Aguilera et al. 2022). Friston replied in a comment (Friston 2022b), and Heins and Da Costa worked out, for Gaussian steady states and for linear and nonlinear dynamics alike, the conditions under which sparse coupling does or does not give rise to a Markov blanket (Heins and Da Costa 2022); the authors' response moves the discussion toward what they term a confederation of Bayesian mechanics (Aguilera et al. 2023). The later formulations surveyed in Section 10.1.6 state their starting point as a partition defined by conditional independences that inherit from sparsely coupled dynamics (Friston, Da Costa, Sajid, et al. 2023; Da Costa

et al. 2021; Ramstead et al. 2023); how far this meets the objections is, in our reading, still being worked out on both sides.

ii) *Conceptual.* Bruineberg et al. trace the Markov blanket from its use in Bayesian networks to its use in active inference, distinguish *Pearl blankets*, an epistemic tool for Bayesian inference, from *Friston blankets*, a construct meant to demarcate the physical boundary between an agent and its environment, and argue that the literature has not kept the two apart; only the second, they hold, can do metaphysical work, and it requires philosophical premises that the success of the mathematics does not by itself supply (Bruineberg et al. 2022b). The paper appeared as a target article with thirty-five commentaries, which take many different stances, and the authors' reply (Bruineberg et al. 2022a). Raja et al. argue that the FEP generalizes Bayesian inference to all domains by way of the blanket formalism, which they term the Markov blanket trick, and that active inference presupposes successful perception and action instead of explaining them (Raja et al. 2021); Friston replied in a comment, and the authors responded (Friston 2022a; Raja et al. 2023). Colombo and Wright note that the FEP has been called a postulate, an unfalsifiable principle, a natural law, and an imperative, that its epistemic status is unclear, and that it might nevertheless afford a theoretical foundation for the relation between environment, life, and mind; they set out limits and prospects for it as a first principle in the life sciences (Colombo and Wright 2021). Andrews takes both the positive and the negative evaluations to date to have been largely in error: the FEP, on his account, is a model structure, i.e., a modelling language to which construals are added to make FEP models, so that demands that the principle itself be falsifiable or biologically realistic rest on a category error — the mathematics is not the territory (Andrews 2021).

iii) *On the expected free energy.* Millidge, Tschantz, and Buckley investigate where the expected free energy comes from and argue that it is not simply the free energy in the future: a functional that they take to be the natural extension of Equation 10.1 into the future actively discourages exploration, so that exploration does not directly follow from minimizing free energy ahead of time, and they develop the free energy of the expected future as an alternative objective (Millidge et al. 2021). Gottwald and Braun distinguish two notions of free energy in current models of intelligent agency, both of them applications of Bayesian inference to action selection, namely one that appears when trading off accuracy against uncertainty under a general maximum-entropy principle, and one that formulates action selection as minimizing the deviation of beliefs and policies from reference models, which is active inference (Gottwald and Braun 2020). The free energy of this book is of the first kind, as we make precise in Section 10.3.1; the subsequent theory of the expected free energy, which addresses part of the question raised by Millidge et al., is the subject of Section 10.2.2.

10.2.2 Planning and the theory of the expected free energy

The status of the expected free energy has nevertheless been clarified considerably. Parr and Friston compare two functionals: in the expected free energy, prior beliefs about outcomes are not part of the generative model but are absorbed into the prior over policies, whereas in the *generalized* free energy they become an explicit part of the model (Parr and Friston 2019). Da Costa et al. give a complete synthesis of active inference on discrete state spaces (Da Costa et al. 2020), and sophisticated inference and its Bellman optimality were discussed in Section 10.1.5. Sajid et al. compare active inference with reinforcement learning on the same environments and stress that reward can be treated as one more observation over which the agent has a preference (Sajid, Ball, et al. 2021). Champion et al. formalize the *unification problem* — deriving the various forms of the expected free energy from a single root definition — and find that in one setting all the forms are recovered, but that the root definition itself has no justification to date and admits only a limited class of preferences (Champion et al. 2024). Most recently, de Vries et al. and Nuijten et al. show that planning with the expected free energy can be written as variational free-energy minimization on a generative model augmented with *epistemic priors* (de Vries et al. 2025; Nuijten, van de Laar, et al. 2026); Nuijten et al. then rewrite the free energy of the augmented model as that of the predictive model plus explicit entropy corrections, and note that full planning also needs a correction that turns marginal inference into policy optimization (Nuijten, Lukashchuk, et al. 2026). Sweeney et al. develop the formal correspondences between expected-

free-energy minimization and Bayesian decision theory, resource rationality, optimal control, reinforcement learning, rate–distortion theory, and maximum entropy (Sweeney et al. 2025).

10.2.3 Deep and scalable active inference

Taking active inference from small discrete problems to high-dimensional tasks has been the main engineering direction. Ueltzhöffer combines it with variational inference in deep generative models and with evolution strategies, and obtains goal-directed behaviour on the mountain-car problem (Ueltzhöffer 2018); Millidge approximates the key densities by deep networks, obtains performance comparable with common reinforcement-learning baselines on OpenAI Gym benchmarks, and notes the similarities with maximum-entropy reinforcement learning and with policy gradients (Millidge 2020); Tschantz et al. take the free energy of the expected future as the objective and balance exploration and exploitation on benchmarks with sparse, well-shaped, and no rewards (Tschantz et al. 2020); and Fountas et al. build deep agents with Monte Carlo tree search, a habitual network, and MC dropout (Fountas et al. 2020). Surveys are those of Mazzaglia et al., from the side of deep learning (Mazzaglia et al. 2022), and of Lanillos et al., for robotics (Lanillos et al. 2021); software includes pymdp (Heins et al. 2022) and RxInfer (Bagaev et al. 2023). The latest step is the *renormalizing generative model*, which generalizes partially observed Markov decision processes to include paths as latent variables and uses the renormalization group to build deep, scale-invariant models, applied to image classification, to the compression and generation of movies and music, and to Atari-like games (Friston et al. 2025); AXIOM uses object-centric core priors to accelerate the learning of games in the low-data regime (Heins et al. 2025). Closest to machine learning, Hafner et al. introduce the *action–perception divergence*, a single principle that covers representation learning, information gain, empowerment, and skill discovery, with agents that use perception to align their beliefs with the world and action to align the world with their beliefs (Hafner et al. 2020).

10.2.4 Experimental tests

The evidence concentrates on in vitro neural networks and on human decision-making. Isomura and Friston stimulated in vitro cortical cultures with stimulus trains of known statistical structure and found the recorded responses consistent with variational message passing (Isomura and Friston 2018); Isomura et al. show that a class of canonical neural networks with delayed modulation of Hebbian plasticity implicitly performs active inference (Isomura et al. 2022); and in 2023 Isomura et al. used in vitro networks of rat cortical neurons that separate two mixed hidden sources to confirm quantitative predictions — pharmacological up- and down-regulation of excitability disrupted the inference in the way that a change of prior would, and changes in effective synaptic connectivity reduced the variational free energy (Isomura et al. 2023). The DishBrain of Kagan et al. embedded in vitro neurons in a simulated game of Pong and reports learning within minutes (Kagan et al. 2022); its use of the word "sentience" drew a public response (Balci et al. 2023). In humans, Schwartenbeck et al. found with fMRI that midbrain activity tracks model-based estimates of precision (Schwartenbeck et al. 2015). These results test particular process theories rather than the principle as such, and Isomura et al. say so explicitly: empirical applications entail a commitment to a particular process theory (Isomura et al. 2023).

10.2.5 Generative AI and agents

Since 2023 the FEP literature has addressed generative AI directly. Pezzulo et al. argue that the generative models of living organisms are anchored to the body and the world: unlike the passive models learned by generative AI, they must capture and control the sensory consequences of action, which in their view is what understanding rests on (Pezzulo et al. 2024). Friston et al. read intelligence as the capacity to accumulate evidence for a generative model of one's sensed world, i.e., self-evidencing in the sense of Hohwy (Hohwy

2016), and envisage an ecosystem of shared intelligence built on active inference (Friston et al. 2024). On the engineering side, Dutta et al. cast context acquisition by interactive AI agents — clarifying questions, retrieval, tool calls — as active inference: an inner step updates beliefs over a latent task state, and an outer decision selects the next action to minimize the expected free energy under a cost; in deterministic settings the epistemic term reduces to the expected information gain, and they benchmark frontier language models on an optimal-question-asking task (Dutta et al. 2026). We return to this structure, inference inside and the choice of data outside, in Section 10.3.7.

10.2.6 Timeline

The milestones of the principle, from its predecessors in machine learning to the developments of this section, are collected in Table 10.3.

Table 10.3: Milestones of the free-energy principle and active inference.

Year	Work	Point
1995–1998	Dayan et al.; Neal and Hinton	the evidence lower bound as a negative free energy (Dayan et al. 1995; Neal and Hinton 1998)
1999	Rao and Ballard	predictive coding in the visual cortex (Rao and Ballard 1999)
2005–2007	Friston; Friston, Kilner, and Harrison; Friston et al.	cortical responses; the FEP for the brain; variational Laplace (Friston 2005; Friston et al. 2006, 2007)
2010	Friston; Friston et al.	a unified brain theory; action and behaviour (Friston 2010; Friston et al. 2010)
2012–2013	Friston et al.; Friston	control without cost functions; the dark room; Markov blankets and life (Friston, Samothrakis, et al. 2012; Friston, Thornton, et al. 2012; Friston 2013)
2015–2017	Friston et al.	epistemic value; learning and habits; a process theory (Friston et al. 2015, 2016, 2017)
2019–2021	Parr and Friston; Da Costa et al.; Friston et al.	generalized free energy; the discrete synthesis; sophisticated inference (Parr and Friston 2019; Da Costa et al. 2020; Friston, Da Costa, Hafner, et al. 2021)
2020–2023	critiques and replies	two kinds of free energy; the technical critique and its reply; whence the EFE; linear systems and sparse coupling; Pearl and Friston blankets (Gottwald and Braun 2020; Biehl et al. 2021; Friston, Da Costa, and Parr 2021; Millidge et al. 2021; Aguilera et al. 2022; Heins and Da Costa 2022; Bruineberg et al. 2022b)
2021–2023	Bayesian mechanics	stationary processes; made simpler; the maximum-entropy duality (Da Costa et al. 2021; Friston, Da Costa, Sajid, et al. 2023; Ramstead et al. 2023)
2022–2023	in vitro tests	DishBrain; experimental validation (Kagan et al. 2022; Isomura et al. 2023)
2023	Da Costa et al.	Bellman optimality of sophisticated inference (Da Costa et al. 2023)
2024–2025	Pezzulo et al.; Friston et al.; Heins et al.	generative AI; renormalizing generative models; AXIOM (Pezzulo et al. 2024; Friston et al. 2025; Heins et al. 2025)
2024–2026	Champion et al.; de Vries et al.; Nuijten et al.; Sweeney et al.; Dutta et al.	the unification problem; planning as free-energy minimization with epistemic priors; formal equivalences; context acquisition by language-model agents (Champion et al. 2024; de Vries et al. 2025; Nuijten, van de Laar, et al. 2026; Nuijten, Lukashchuk, et al. 2026; Sweeney et al. 2025; Dutta et al. 2026)

10.3 Connections to entropy-based learning

This concludes our review of the principle in its own language. Now, let us reinterpret it in the framework developed throughout this book, moving step by step along the levels summarized in Table 10.2. The point to hold on to, at the outset, is that the central translations below are identities, not analogies; where a translation is only a correspondence, or a reading of ours, Table 10.5 says so.

10.3.1 One free energy

The first line of Equation 10.1 is the free energy of Section 5.1.1 with a particular assignment of base measure and energy.

> **Proposition 1 (one free energy)**
>
> For a generative model $p(o, s) = p(s)\, p(o \mid s)$, the variational free energy Equation 10.1 equals $\mathcal{F}_T(q; E)$ of Equation 5.2 with $h = p(s)$, $E(s) = -\log p(o \mid s)$, and $T = 1$. It also equals $\mathcal{F}_1$ with a flat base measure and the energy $E(s) = -\log p(o, s)$. In both readings the minimizer is the posterior $p(s \mid o)$ and the minimum is $-\log Z = -\log p(o)$. For a general temperature the two readings differ: the minimizers are $q_T \propto p(s)\, p(o \mid s)^{1/T}$ and $q_T \propto p(o, s)^{1/T}$, respectively.

The proof is a substitution: $\mathbb{E}_q[-\log p(o \mid s)] + \mathrm{KL}(q \| p(s))$ is the third line of Equation 10.1, and $Z = \int p(s)\, p(o \mid s)\, ds = p(o)$. Two remarks are worth making. First, in perception under the FEP, $T = 1$ is not a dial that someone chooses; it is the point at which Bayes' rule itself sits. Writing Bayes' rule as a variational problem is an old result in statistics, which goes back to the Donsker–Varadhan formula, to Williams' principle of minimum information, and to Zellner's optimal information processing (Donsker and Varadhan 1975; Williams 1980; Zellner 1988); Giffin and Caticha process data and moment constraints simultaneously by maximum relative entropy (Giffin and Caticha 2007), which is the constrained and the free-energy form of Section 5.1.1 unified in Bayesian updating. Second, replacing the negative log-likelihood by a general loss, or the KL divergence by another divergence, gives general Bayesian updating and generalized variational inference (Bissiri et al. 2016; Knoblauch et al. 2022), i.e., the generalized sense of Section 5.6.3.

Notably, the two readings of Proposition 1 are not a technicality. They are the two ways of splitting a model into a base measure and an energy, and the rest of this section shows that the FEP uses both, at different levels, and that several of its apparent puzzles dissolve once one asks which split is in force. Proposition 1 also concerns the variational free energy only, and it is worth placing it in the distinction of Gottwald and Braun (Section 10.2.1). Their first kind, the *free energy from constraints*, is an energy traded against an entropy with a multiplier that is determined by a constraint, which is Equation 5.2; their second kind, the *variational free energy*, is an error measure between a trial distribution q and an unnormalized reference function ϕ, namely $\mathbb{E}_q[-\log \phi] - H(q)$, introduced as a device for representing a normalization as an optimization problem, which is the first line of Equation 10.1 with $\phi = p(o, s)$. They note that the two expressions are transformed into each other by $\phi = h\, e^{-E/T}$, and insist that they nevertheless differ in method and motivation: the first presupposes a constrained quantity and a trade-off, the second presupposes neither (Gottwald and Braun 2020). Proposition 1 is that transformation at $T = 1$, and its second reading is exactly their second kind. The free energy of this book is of the first kind; the expected free energy Equation 10.2 is a further object of the second kind, built for action selection, which is not the same object as Equation 10.1, and we come to it in Section 10.3.5.

10.3.2 Perception: two readings of the dial

Example 1 showed that predictive coding computes the mode of the posterior and not its mean, with the uncertainty supplied afterwards by the Laplace approximation. This is the first place where the distinction of Proposition 1 earns its keep. Predictive coding is the $T \to 0$ end of the *second* reading, in which the energy is

$-\log p(o, s)$ and lowering the temperature concentrates the belief on the maximum a posteriori value without moving it.

Precision is a temperature. For a Gaussian likelihood, lowering the temperature of the likelihood term from 1 to T is the same as raising the sensory precision from $1/\Sigma_u$ to $1/(T\Sigma_u)$, since $p(u \mid v)^{1/T} \propto \mathcal{N}(u; v^2, T\Sigma_u)$. This is the *first* reading of Proposition 1, in which the prior is the base measure and the dial of Chapter 3 acts on the likelihood alone. In Example 1 the mode of $q_T \propto p(v)\, p(u \mid v)^{1/T}$ moves from 1.817 at $T = 4$, towards the prior mean 3, through 1.5675 at $T = 1$, to 1.418 at $T = 0.02$, approaching the maximum-likelihood value $\sqrt{2} = 1.4142$; under the second reading the mode stays at 1.5675 for every T, while the mean moves from 1.480 at $T = 1$ to 1.566 at $T = 0.02$, towards the mode. Attention as inference about precision (Section 10.1.3) is, in our terms, inference about a likelihood temperature. Perception itself stays at $T = 1$.

In the classification of Section 5.1, perception has only the levels i) and ii), representation and inference, with no learner: it is inference over a maximum-entropy model, and nothing is fitted.

10.3.3 Learning: minimax entropy at $T = 1$

We saw in Section 10.1.4 that learning under the FEP minimizes the same free energy, and that in the limit of point estimates it is EM. The question for this book is which outer problem that is. The answer is sharper than we expected.

> **Proposition 2 (learning as a difference of entropies)**
>
> Let the complete-data model $p_\theta(o, s) \propto h(o, s) \exp\big(\boldsymbol{\lambda}^\top \mathrm{T}(o, s)\big)$ be an exponential family in the sufficient statistics T with a uniform base measure h. Given data $o_1, \dots, o_n$ and any conditional distribution $q(s \mid o)$, let $\boldsymbol{\lambda}$ be the M-step fit to the completed data $\hat{p}(o)\, q(s \mid o)$, i.e., $\mathbb{E}_{\hat{p}q}[\mathrm{T}] = \mathbb{E}_{p_\theta}[\mathrm{T}]$. Then for every label temperature $T \geq 0$,
>
> $$\frac{1}{n} \sum_{i=1}^{n} \Big\{ \mathbb{E}_{q(\cdot \mid o_i)}\big[-\log p_\theta(o_i, s) \big] - T\, H\big(q(\cdot \mid o_i)\big) \Big\} \;=\; H\big(p_\theta(O, S)\big) - T\, \overline{H(q)}, \tag{10.4}$$
>
> where $\overline{H(q)} = \frac{1}{n} \sum_i H\big(q(\cdot \mid o_i)\big)$. At $T = 1$ the left-hand side is the average variational free energy Equation 10.1.

The proof is that of Equation 5.18. For a uniform h we have $-\log p_\theta(o, s) = -\boldsymbol{\lambda}^\top \mathrm{T}(o, s) + \log Z(\boldsymbol{\lambda})$ up to a constant, so the cross-entropy on the completed data equals $-\boldsymbol{\lambda}^\top \mathbb{E}_{\hat{p}q}[\mathrm{T}] + \log Z = -\boldsymbol{\lambda}^\top \mathbb{E}_{p_\theta}[\mathrm{T}] + \log Z = H(p_\theta)$; subtracting $T\, \overline{H(q)}$ gives Equation 10.4. If the statistics also depend on representation parameters θ', writing $p^\star_{\theta', q}$ for the maximum-entropy model of the completed data and ℓ_T for the label free energy Equation 7.5 gives

$$\min_{\theta} \; \frac{1}{n} \sum_{i=1}^{n} \ell_T(\theta; o_i) \;=\; \min_{\theta', q} \Big\{ H\big(p^\star_{\theta', q}\big) - T\, \overline{H(q)} \Big\}, \tag{10.5}$$

whose $T \to 0$ end is Equation 7.8, the classification likelihood, and whose $T = 1$ end is the marginal likelihood, i.e., EM.

Proposition 2 is a direct consequence of Equation 5.18, and Equation 7.7, the decomposition of Hathaway (Hathaway 1986), is its counterpart on the side of the E-step; we claim no priority for either. What we want the reader to take from it is the *reading*. The average free energy is the entropy of the world model minus the entropy of the inferred causes. Learning under the FEP is therefore minimax-entropy learning whose outer problem minimizes the entropy of the world model — which is Type I of Section 7.2.1 — while the inner problem keeps the entropy of the causes at full weight. The interpolation between the two ends that Section 7.3.1 describes is exactly the T of Equation 10.4, and the FEP sits at one particular setting of it.

Example 3 (learning). We draw $n = 2000$ samples from $0.4\,\mathcal{N}(-1, 0.7^2) + 0.6\,\mathcal{N}(1, 1)$ and run EM at a label temperature T, with the E-step $q(s \mid o) \propto p_\theta(o, s)^{1/T}$ — a hard assignment at $T = 0$ — and the M-step matching the moments of the completed data. At every temperature and every iteration the two sides of Equation 10.4 agree to within 3.8×10^{-15}, and the objective is non-increasing to within rounding (Figure 10.4 a). At $T = 1$, EM converges to $\hat{\pi} = (0.399, 0.601)$, $\hat{\mu} = (-1.034, 1.010)$, and $\hat{\sigma} = (0.703, 0.991)$; the average free energy 1.6791 equals the training negative log-likelihood to within 10^{-12}, and it splits into a world-model entropy of 1.9457 minus a cause entropy of 0.2666. On 200,000 held-out samples the negative log-likelihood is 0.4 millinats above that of the true distribution. At $T = 0$, which is classification EM (Celeux and Govaert 1992), the entropy of the world model falls to 1.7782 and that of the causes to zero, but the means are pushed apart and the variances shrink, $\hat{\mu} = (-0.963, 1.291)$ and $\hat{\sigma} = (0.657, 0.777)$, and the held-out negative log-likelihood is 17.0 millinats above the truth, against 12.3 at $T = 0.5$ (Figure 10.4 b, c).

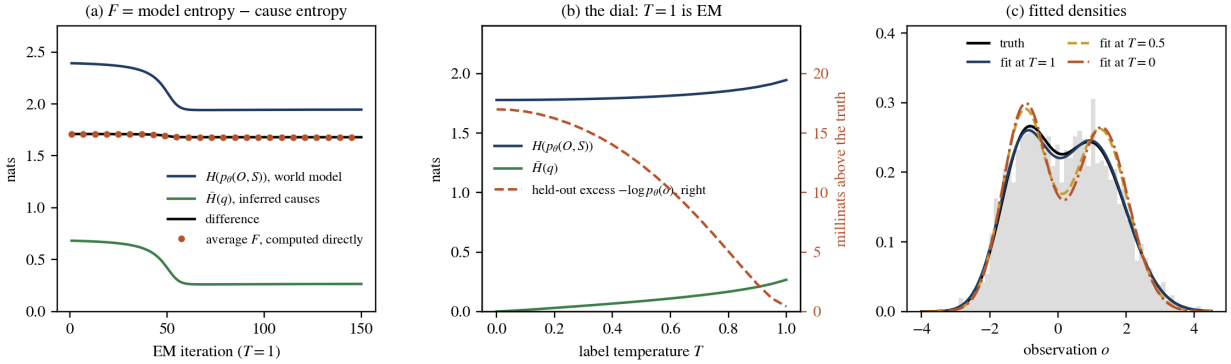


Figure 10.4: Learning as free-energy minimization (Example 3). (a) EM at $T = 1$: the entropy of the world model, the entropy of the inferred causes, and their difference, with the average free energy computed directly shown as dots; (b) the label-temperature dial, with the two entropies at convergence on the left axis and the excess of the held-out negative log-likelihood over the truth on the right, in millinats; (c) fitted densities at $T = 1$, 0.5, and 0 against the truth. Produced by `code/fep-examples.py`.

This is what Section 7.3.1 predicts: a label temperature below one makes the label side commit, which is Type II, and it pays for the commitment in the fit of the density to the inputs. Learning under the FEP declines to make that trade.

One detail deserves mention, since it looks like a contradiction and is not. We said in Section 5.6.3 that directed graphical models fitted by maximum likelihood or EM are maximum-entropy learners without being maximum-entropy models, which is true of the marginal $p_\theta(o)$. Proposition 2 says that at the level of the *completed* data, EM for an exponential-family mixture is a minimax-entropy learner. The two statements concern different sets of variables.

10.3.4 Model selection: Type I and description length

Model selection is the same story one level up. Recall that Section 7.1 starts from minimum description length: for a maximum-entropy model the description length equals the model's own entropy, $L(P_F) = S(P_F)$, so choosing features for the shortest description is Type I (Grünwald 2007; Feder 1986; Carcamo et al. 2025). Model selection under the FEP uses the free energy as a bound on the negative log model evidence (Section 10.1.4), and its complexity term for the parameters, $\mathrm{KL}(q(\theta) \| p(\theta))$, is their description length. This reading is not new to the FEP: Hinton and van Camp read variational learning as minimizing the description length of the weights (Hinton and van Camp 1993), Hinton and Zemel read the cost of an autoencoder in terms of MDL and the Helmholtz free energy (Hinton and Zemel 1994), and Honkela and Valpola explain variational Bayesian learning through bits-back coding (Honkela and Valpola 2004).

In the language of this book, then, model selection under the FEP is Type I with a cost on the parameters: the

budget of Section 7.1 is a number of features, whereas that of the FEP is counted in nats. The outer criteria differ slightly, in a way worth keeping straight. Type I minimizes $S(P_F)$, the cross-entropy at the maximum-likelihood fit; the FEP minimizes a free energy that also carries the uncertainty about the parameters. When the parameter posterior is concentrated, the free energy of n data points is approximately $n\,S(P_F)$ plus the cost of the parameters, and the two agree to leading order. The structure learning of Section 10.1.4, which adds hidden-state slots and reduces them by Bayesian model reduction, is the combinatorial search of Section 7.1 conducted in nats. Notably, the minimax entropy principle has recently yielded exactly solvable models of large neuronal populations (Lynn et al. 2025), a natural meeting point with the neuroscience from which the FEP arose.

10.3.5 Action: Type III, firm but fair

The first line of Equation 10.2 is an object this book has already met, with the direction of one arrow reversed.

Proposition 3 (the expected free energy is firm but fair)

Write $\bar{p}_\pi = q(o \mid \pi)$ for the predicted marginal of the outcomes. Then

$$G(\pi) \;=\; \underbrace{\mathrm{KL}\big(\bar{p}_\pi \,\|\, \tilde{p}\big)}_{\text{fairness, target } \tilde{p}} \;+\; \underbrace{\mathbb{E}_{q(s\mid\pi)} H\big[p(o \mid s)\big]}_{\text{firmness}}, \tag{10.6}$$

i.e., the fairness and firmness terms of Section 6.3, with the hidden state s in the place of the input, the outcome o in the place of the label, and the likelihood $p(o \mid s)$ in the place of the classifier. With flat preferences, $G(\pi) = \log|\mathcal{O}| - I_\pi(S;O)$, so that minimizing G maximizes the mutual information of Bridle et al. (Bridle et al. 1992). Unlike in Bridle's case, the variable being optimized is not the classifier but the distribution $q(s \mid \pi)$ of the inputs, or the channel $p(o \mid s, \pi)$.

The decomposition into risk and ambiguity is standard (Friston et al. 2015; Da Costa et al. 2020); what we add is the correspondence, which shows that both entropy terms of the expected free energy have been in this book all along. The classifier of Section 6.3 is optimized for a fixed distribution of data; active inference fixes the likelihood and optimizes *what to look at*. That direction is the one of channel capacity and of Bayesian experimental design: Lindley measures the information an experiment provides by the expected information between parameter and data (Lindley 1956), MacKay uses it for active data selection (MacKay 1992), and Linsker's InfoMax maximizes the mutual information between the input and the output of a perceptual network (Linsker 1988). Within the book, Chapter 7 already observes that Sustek et al., by adding $I_\theta(X;Y)$ to the objective of a joint energy-based model, place the mutual-information objective of Section 6.3 on top of a maximum-entropy learner (Sustek et al. 2023); the expected free energy places the same objective on top of a generative model and changes the optimized variable from the classifier to the action.

This is the gap in Section 7.2.1, and we propose to fill it.

Definition: minimax-entropy learning of Type III

In addition to Types I and II of Section 7.2.1, we term a method of **Type III** (*what to observe*) if its outer problem chooses an action, a policy, or an experiment ξ that decides the data to which the inner model will be fitted, so as to minimize the expected entropy of the inner model's posterior over its hidden variables,

$$\min_{\xi} \; \mathbb{E}_{o\sim q(o\mid\xi)}\, H\big(q(S \mid o, \xi)\big), \tag{10.7}$$

which, when the prior over the hidden variables does not depend on ξ, is the same as maximizing $I_\xi(S;O)$. The outer problem may add a prescribed marginal of the outcomes, the preferences, as the target of fairness. The posterior is computed at $T = 1$: the entropy is lowered by the choice of the data and not by sharpening the posterior. The anchor iii) is then met *through the world* — the outer entropy is an expectation under the inner model's own predictive distribution, and the inner model is afterwards fitted to the data that the chosen action actually produces.

Table 10.4: The three kinds of outer problem in minimax-entropy learning. Types I and II are those of Section 7.2.1; Type III is proposed here, and it is the one in which the data are not given.

Type	Outer variable	Entropy minimized	Temperature	Examples
I, what to measure	features, model, parameters	entropy of the maximum-entropy model, $S(P_F)$	inner at $T = 1$	FRAME; energy-based models by maximum likelihood; EM (Proposition 2); model selection under the FEP
II, what to commit to	assignments, classifier	predictive entropy of Y	label at $T < 1$	softmax classification; classification EM; joint energy-based models with entropy minimization
III, what to observe	action, policy, experiment	expected posterior entropy $\mathbb{E}_o H(S \mid o)$	posterior at $T = 1$; policies at $1/\gamma$	epistemic value of Equation 10.2; Bayesian experimental design; context acquisition by agents

Example 2 shows the type at work and separates it from Type II. The symmetry of the preferences made the first move a pure Type III decision, settled by the epistemic term alone. The agent's belief about the context is a Bayesian posterior at $T = 1$ throughout; it never commits by sharpening that posterior. It obtains certainty by *fetching the data* on which the posterior can be confident. Type II lowers the entropy of a belief on fixed data; Type III goes out and changes the data.

10.3.6 Policies: Gibbs tilts, and precision as a temperature

The posterior over policies, Equation 10.3, is an object from Chapter 9.

Proposition 4 (the posterior over policies is a Gibbs tilt)

$q(\pi) = \sigma\big(\log E(\pi) - \gamma G(\pi)\big)$ is the unique minimizer of $\mathbb{E}_q[G] + \gamma^{-1}\mathrm{KL}(q\|E)$, with minimum $-\gamma^{-1}\log\sum_\pi E(\pi)\,e^{-\gamma G(\pi)}$. It is therefore Equation 5.2 on the space of policies, with the energy G, the base measure E, and the temperature $T = 1/\gamma$; equivalently it is Equation 6.2 with reward $-G$ and reference policy E, and Equation 9.1 with the habits as reference.

Three comparisons follow, and each sharpens something stated loosely elsewhere. i) *Risk is maximum-entropy control on outcomes.* Since $\mathrm{KL}(\bar{p}_\pi\|\tilde{p}) = -H(\bar{p}_\pi) - \mathbb{E}_{\bar{p}_\pi}[C] + \log Z_C$ for $\tilde{p} \propto e^C$, minimizing the risk maximizes expected utility *plus the entropy of the outcomes*. The structure is that of Equation 5.14, but with the entropy bonus on the outcomes rather than on the actions; Millidge et al. likewise locate the difference between active inference and control as inference in how value enters the respective generative models (Millidge et al. 2020; Levine 2018). ii) *Precision is an inferred temperature.* We called the temperature of Equation 9.1 a genuine dial, chosen by the designer (Section 9.1). Active inference gives γ a prior and infers it, and the inferred value has a neural correlate (Section 10.1.5). The temperature of a learned energy is not identifiable when the family is closed under rescaling (Section 5.1.4); γ escapes this because G is measured in nats, on a fixed

scale. iii) *The case without preferences.* If $\tilde{p}$ is flat and E uniform, $q(\pi) \propto e^{\gamma I_\pi(S;O)}$ — a pure explorer that Gibbs-tilts by information gain.

10.3.7 The closed loop and the information ceiling

What Type III buys can be stated precisely with the ceiling of Section 6.7, which says that a self-referential training loop cannot create information about the truth: if every update uses only the model's own samples and a frozen equivalence relation, then $I(\pi_T; a^\star \mid \pi_0, q, \sim) = 0$, and supervision breaks the chain only because the verifier's reward makes the trained policy a function of the answer as well. Active inference breaks the chain in the same way, and this is, we think, the cleanest statement of what the closed loop is for. An observation $o \sim p(o \mid s^\star, \pi)$ depends on the *true* hidden state $s^\star$, so the agent's beliefs are not a function of themselves alone. Moreover, if the agent's model is correct, the epistemic term $I_\pi(S;O)$ is exactly the expected information, in nats, that the world deposits into the beliefs under the policy π. The epistemic value of Equation 10.2 therefore measures precisely the quantity that is missing from the ceiling of Section 6.7.

The dividing line is then clear. The self-referential methods of Chapter 6 are open-loop, and their rewards come from the model itself; perception and learning under the FEP are closed-loop and spend the incoming information honestly at $T = 1$; and action of Type III goes out to acquire it. The good regulator theorem of cybernetics says, as its title has it, that every good regulator of a system must be a model of that system (Conant and Ashby 1970); the closed loop is where that model is put to the test of the world. Type III is also where the FEP meets the agents of the present: the context acquisition of Dutta et al. (Section 10.2.5), with inference inside and the choice of data outside, is exactly the structure of Equation 10.7, and among the objectives that the action–perception divergence of Hafner et al. derives from one principle — representation learning, information gain, empowerment, and skill discovery (Hafner et al. 2020) — information gain is the one that Equation 10.7 names.

10.3.8 Self-referential tilts: learned preferences and habits

Two ingredients of the tilt Equation 10.3, the habits E and the preferences $\tilde{p}$, may themselves be learned from the agent's own behaviour, and this is where the one failure mode of the chapter appears. We observed in Section 9.4 that an external reward tilts the reference toward what the world values, whereas an internal one tilts it toward what the model already believes. Active inference has two places where an internal reward can appear: habits, where E reflects how often a policy has been chosen before (Smith et al. 2022; Friston et al. 2016), and learned preferences, where $\tilde{p}$ is estimated from experience (Sajid, Tigas, et al. 2021). The two behave differently, and the difference is instructive.

Proposition 5 (critical precision for learned preferences)

Let there be K actions, each deterministically yielding its own outcome, so that $G(a) = -\log \tilde{p}(o_a)$ and $q(a) \propto \tilde{p}(o_a)^\gamma$. Let the preferences have a fixed and a learned part, $\tilde{p} = (1-w)\,p + w\,u$ with u uniform, where p tracks the agent's own outcomes, $p_{t+1} = (1-\lambda)\,p_t + \lambda\,q_t$ for $0 < \lambda \le 1$. Then i) q minimizes $\mathrm{KL}(q\|\tilde{p}) + (1 - 1/\gamma)H(q)$; and ii) the balanced point $p = u$ is linearly stable if and only if $\gamma(1-w) < 1$, i.e.,

$$\gamma_c = \frac{1}{1-w}, \qquad T_c = 1 - w, \tag{10.8}$$

independently of K and λ.

Part i) follows because $q \propto \tilde{p}^\gamma$ minimizes $\mathrm{CE}(q, \tilde{p}) - \gamma^{-1}H(q)$ and $\mathrm{CE} = \mathrm{KL} + H$. For ii), put $p = u + \varepsilon$ with $\sum_a \varepsilon_a = 0$; then $\tilde{p} = u + (1-w)\varepsilon$, and to first order $q_a = 1/K + \gamma(1-w)\,\varepsilon_a$, so that $\varepsilon_{t+1} = \big[1 - \lambda\big(1 - \gamma(1-w)\big)\big]\varepsilon_t$. The reading of i) is the point. At $\gamma = 1$ the objective is pure fairness to the preferences and any $q = \tilde{p}$ is self-

consistent: this is the neutral point. For $\gamma > 1$ an entropy term of positive weight $1 - 1/\gamma$ appears, which is minimum-entropy learning at the level of policies, and if the preferences then follow the agent's own behaviour, this is exactly the self-referential tilt of Section 6.6. With $w = 0$ the critical precision is $\gamma_c = 1$, the Bayesian value.

Example 4 (learned preferences). We take K rooms, room 0 being dark, and iterate the dynamics of Proposition 5. The numerics agree with Equation 10.8: from a balanced start the symmetric state breaks at $\gamma = 1.005$, 1.255, and 2.005 for $w = 0$, 0.2, and 0.5, on a grid of step 0.005, against the predicted 1, 1.25, and 2. For $w > 0$ the picture is that of Equation 6.3. At $K = 2$ the transition is continuous; for $K \geq 3$ a collapsed state reached from a committed start remains stable below γ_c, giving a hysteresis window — $[1.225, 1.25]$ and $[1.87, 2.0]$ at $K = 4$ for $w = 0.2$ and 0.5 — across which the largest probability jumps by about 0.49 and 0.52 (Figure 10.5 b): a continuous transition at $K = 2$ and a first-order transition with hysteresis of mean-field Potts type for $K \geq 3$ (Wu 1982). The case $w = 0$ is degenerate. The fixed points of $p \mapsto p^\gamma / \sum_a p_a^\gamma$ are the distributions uniform on a subset of the actions, since $p_a = p_a^\gamma / Z$ forces $p_a^{\gamma-1} = Z$ on the support; the balanced state loses stability at $\gamma = 1$ exactly where the collapsed ones gain it, so the largest probability jumps from $1/K$ to 1 at $\gamma_c = 1$ for every K, with no window. The time courses make it concrete. With $K = 4$, $w = 0$, and the dark room ahead by 5 per cent, $q_t(\text{dark})$ returns from 0.257 to 0.250 at $\gamma = 0.8$, stays at 0.259 at $\gamma = 1$, and rises to 0.994 within 400 steps at $\gamma = 1.25$ (Figure 10.5 a).

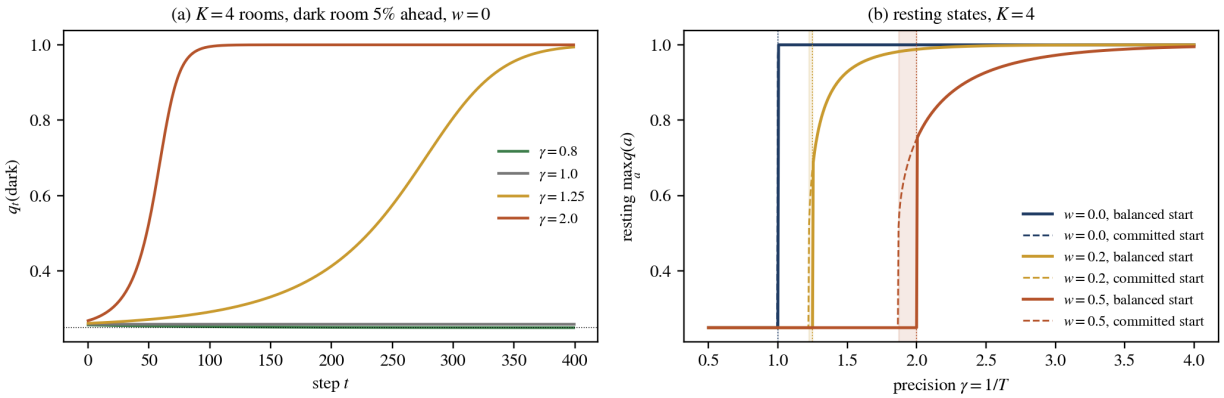


Figure 10.5: Learned preferences as a self-referential tilt (Example 4). (a) Time courses of $q_t(\text{dark})$ for $K = 4$ and $w = 0$, with the dark room ahead by 5 per cent; (b) the resting value of the largest probability against the precision at $K = 4$, from a balanced start (solid) and a committed start (dashed), with $\gamma_c = 1/(1-w)$ dotted and the hysteresis windows shaded. Produced by `code/fep-examples.py`.

So the dark room returns, but by a different door than the critics supposed. Fixed preferences rule it out, as Table 10.1 shows. Preferences learned from one's own behaviour bring it back above a critical precision: the agent learns to prefer what it happened to do more often, and then does it with more confidence. Habits, by contrast, do not collapse in this way. If $\tilde{p}$ is fixed and E tracks past choices, then $q \propto E$ whenever the policies have equal G, so E enters the softmax with unit weight, exactly at the neutral point — self-consistent but not amplifying. Collapse requires the precision to amplify an internal reward beyond one. In both cases it is the position of the precision relative to $T = 1$ that decides whether a self-referential tilt keeps its diversity, which is the counterpart of the critical value of Section 6.6.

We should be plain about the status of this example. It is a toy model constructed for this chapter, not a reproduction of any published agent; it exhibits a possible failure mode, not an observed one.

10.3.9 Schrödinger, steady states, and the maximum-entropy duality

The reading of Schrödinger in Section 6.8 stresses that he describes a stationary and fairly low entropy level — a balance, not a minimization — and warns against transplanting thermodynamic theorems onto the Shannon

entropy of a predictive distribution. The entropic claim of Section 10.1.6 is compatible with that warning, provided it is read with the same care.

The claim is that the long-term average of surprise is an entropy, so that minimizing free energy by selectively sampling the world bounds the entropy of the sensory states (Friston 2013). In the language of Chapter 7 the average surprise is a cross-entropy, $\mathbb{E}_p[-\log p_m] = H(p) + \mathrm{KL}(p\|p_m) \geq H(p)$, which is the identity behind the "closest to the truth" view, and which for a maximum-entropy model at its maximum-likelihood fit also equals the model's own entropy (Equation 5.18). Bounding the entropy therefore means that a cross-entropy bounds the true entropy from above. It does not mean that anything is pushed toward zero. And if the agent changes its sensory distribution by acting, the first term of $H(p^\pi) + \mathrm{KL}(p^\pi\|p_m)$ would reward the dark room — which is exactly the term that the risk of Equation 10.2 removes, as Example 2 showed.

The duality between the FEP and the constrained maximum entropy principle (Section 10.1.6) is where the physics of the FEP comes closest to this book, since Chapter 5 is the maximum-entropy side of it (Ramstead et al. 2023; Sakthivadivel 2022), and at the level of paths the counterpart is the principle of maximum caliber (Pressé et al. 2013). It is a conditional closeness, and the conditions are under discussion (Section 10.2.1): it has been argued that the premises of the blanket construction hold, even for linear stochastic systems, only in a narrow region of parameter space (Aguilera et al. 2022), that crucial steps of the original derivation needed assumptions that had not been stated (Biehl et al. 2021), and that the relation between the mathematical construct and the biological entities it is read onto needs premises which the mathematics does not by itself supply (Bruineberg et al. 2022b; Raja et al. 2021; Colombo and Wright 2021); the replies and the later formulations are cited there. Whatever the outcome, we record the Markov-blanket claims as *not derivable* from anything in this book: they need assumptions about dynamics — a Langevin equation, sparse coupling, a steady state — and about a partition, the Markov blanket, that the book does not contain, and we would discourage the reader from treating them as though they followed from it. The translations of Section 10.3.1 to Section 10.3.8 do not depend on them.

Two physical connections stand on firmer ground and deserve mention for that reason. Still et al. show that the non-predictive information retained by a stochastically driven system — the inefficiency of its implicit model — is equivalent to thermodynamic dissipation, so that a system which keeps memory about its environment and runs at maximal energetic efficiency must be predictive (Still et al. 2012). Ortega and Braun write bounded-rational decision-making as a free energy trading expected utility against information-processing cost measured in relative entropy (Ortega and Braun 2013). The first gives prediction a thermodynamic meaning that needs no Markov blanket; the second is the decision-theoretic reading of the Gibbs tilts of Chapter 5 and Chapter 9 — a free energy of the first kind, in the sense of Section 10.3.1.

10.3.10 The correspondence at a glance

The translations of this section are collected in Table 10.5, each with its status.

Table 10.5: The correspondence between the free-energy principle and this book.

In the FEP	In this book	Status
variational free energy $F[q; o]$	Equation 5.2 with $h = p(s)$, $T = 1$	identity (Proposition 1)
perception: minimize F over q	inference over a maximum-entropy model, levels i)–ii) of Section 5.1	identity
precision of a Gaussian likelihood	temperature of the likelihood term	identity
predictive coding	the $T \to 0$ mode with a Laplace correction	approximation
learning (EM)	maximum-entropy learning with missing data; $T = 1$ on the dial of Section 7.3.1; Type I on the completed data	identity (Proposition 2)
model evidence, complexity	Type I with a cost on the parameters; MDL, bits-back	correspondence

In the FEP	In this book	Status
risk plus ambiguity	fairness to $\tilde{p}$ plus firmness (Section 6.3)	identity (Proposition 3)
epistemic value	Type III (proposed); InfoMax; Lindley's information	new reading
$q(\pi) = \sigma(\log E - \gamma G)$	the Gibbs tilts Equation 6.2, Equation 9.1	identity (Proposition 4)
precision γ	an inverse temperature that is inferred	correspondence
closed loop	breaks the information ceiling of Section 6.7	new reading
learned preferences, habits	self-referential tilt, Equation 6.3	new (Proposition 5, Example 4)
average surprise bounds entropy	cross-entropy bounds entropy, the views of Chapter 7	identity, with a caveat
Markov blankets, steady states	none; the maximum-entropy duality is the nearest point	not derivable

10.4 Takeaway

The principle itself can be summarized in three sentences. The FEP says that a persisting system minimizes a variational bound on the surprise of its sensations, by perception, by learning, and by action; active inference scores actions by an expected free energy that combines a preference term with an information-gain term and selects among them by a precision-weighted softmax; and the physical claim that any system with a Markov blanket does this is contested in detail, whereas the mathematics of the first two sentences is not.

At its foundation, the translation is an exact identity. The free energy of the FEP is the free energy of this book at $T = 1$ (Proposition 1), and nearly everything else follows from asking which base measure, which temperature, and which outer problem is in force. Perception is inference over a maximum-entropy model with no learner, and predictive coding is its zero-temperature mode with a Laplace correction. Learning is maximum-entropy learning with missing data, and its average free energy is the entropy of the world model minus the entropy of the inferred causes (Proposition 2) — Type I on the completed data, sitting at the $T = 1$ end of the dial of Section 7.3.1 rather than committing like Type II. Model selection is Type I with a cost in nats. The posterior over policies is a Gibbs tilt (Proposition 4), the same object as Equation 9.1 with the precision as an inverse temperature that is, unusually, inferred rather than chosen. We note that the physics of Markov blankets and non-equilibrium steady states cannot be derived from the framework developed here.

Two things in the chapter are not translations. The first is Type III: the expected free energy is firm but fair (Proposition 3) with the action, not the classifier, as the optimized variable, and this is an outer problem the book did not have — the one in which the data are not given but fetched (Table 10.4). It is also what breaks the information ceiling of Section 6.7, since the epistemic value measures exactly the information the world deposits into the beliefs. The second concerns the failure mode: when preferences are learned from one's own predicted behaviour, setting the precision above its Bayesian value can re-open the dark room through a first-order transition (Proposition 5). This illustrates how the collapse mechanism described in Section 6.6 resurfaces within a theory that was originally intended to rule out such pathologies.

Significant open questions remain. The definition of expected free energy is not unique, and there is currently no consensus on a criterion for selecting among the various objectives proposed for this role (Millidge et al. 2021; Parr and Friston 2019; Champion et al. 2024). Whether recent approaches that recast planning as free-energy minimization within an expanded model framework (de Vries et al. 2025; Nuijten, van de Laar, et al. 2026) can provide such a criterion is, in our view, a key issue for future research. Proposition 5 is proved for a model without hidden states, and the collapse condition for learned preferences in a model that has them is unknown. Type III invites a concrete programme for agents that acquire their own context (Dutta et al. 2026), where the question is whether the expected posterior entropy of Equation 10.7 is a usable objective at scale or merely a correct one. And the label temperature of Equation 10.5 still has no schedule: this chapter shows

that the FEP pins it at one, which is a principled choice, but Chapter 7 leaves open whether anything better is available when the goal is to commit.

Part III — Applications

11 Entropy-Based Learning for Science

The ultimate aim of this research program is **scientific intelligence** — AI that is not only accurate and robust, but increasingly endowed with reasoning capabilities that enable new modes of scientific inquiry. Entropy- and energy-based learning offer principled tools for this goal.

11.1 Two synergistic directions

- **Advancing AI with Science.** Fundamental scientific and mathematical theories — statistical physics, information theory, the maximum-entropy principle — inspire principled AI methodologies. Energy-based games (Chapter 5), entropy-minimized reasoning (Chapter 6), and minimax-entropy model selection (Chapter 7) are three examples.
- **Advancing Science with AI.** The same methods accelerate discovery in the sciences, from molecular representation learning to chemical reasoning.

11.2 Molecular foundation models

A concrete thread runs from **self-supervised graph transformers** for molecular representation learning toward **molecule–language** models with reasoning capabilities. The trajectory mirrors the broader shift from *pretraining–finetuning* to *LLM-style reasoning*, and entropy-based objectives provide label-efficient signals where annotated data are scarce.

11.3 Outlook

A deeper unification may be within reach. The minimax view (Chapter 7) suggests that modelling the world and committing to a decision are two halves of one objective — an idea that echoes the **free-energy principle** proposed as a unified account of perception and action in the brain (Friston 2010). The quantity that principle asks the brain to minimize is the variational free energy of Chapter 3, which is what makes the correspondence more than a metaphor. Making it precise, and scaling it to LLMs, agents, and multimodal scientific models, is an open and appealing direction.

As the research matures, this book will grow to include worked examples, executable code, and reproducible experiments.

See the Blue Whale Lab and https://yataobian.com/ for the latest developments.

12 Summary

This book introduced **entropy-based learning** through three complementary principles and their shared energy-based substrate:

- **Maximum entropy** yields the least-biased model consistent with given constraints — grounding exponential families, Boltzmann machines, maximum-entropy reinforcement learning, and a principled family of game-theoretic valuations (Bian et al. 2022). When the energy is learned rather than given, the same principle trains neural set functions from optimal subsets (Ou et al. 2022; Xie, Bian, et al. 2024) and infers rewards from demonstrations (Ziebart et al. 2008). Applied to a pretrained model as base measure, it becomes energy-based guidance at inference time and maximum-entropy post-training — RLHF, GRPO, DPO — at training time, one Gibbs tilt paid for in two currencies. Read through the three levels of representation, inference, and learning, all of these methods belong to one family.
- **Minimum entropy** sharpens a capable model's own predictions, enabling *unsupervised* elicitation of reasoning (EMPO) in a latent semantic space (Zhang et al. 2025). It elicits but does not add capability, since the training loop never touches ground truth, and its collapse onto one confident answer is a first-order transition in a mean-field free energy, which is why the durable remedies hold the entropy above a floor rather than tune a coefficient.
- **Minimax entropy** couples the two: maximize entropy to model the world, minimize entropy to commit to what matters. Its first question is *what to measure*, and the answer is the features whose maximum-entropy model carries the least entropy, simultaneously the shortest, most accurate, and most informative description (Zhu et al. 1997; Carcamo et al. 2025). Its second question is *what to commit to*: a softmax classifier with a linear head is a conditional maximum-entropy model whose cross-entropy is the largest conditional entropy its features allow, and without labels a temperature on the label moves the problem from maximum-entropy learning with missing data to the classification likelihood.

Binding the three together is the **free energy** $F = U - TS$: temperature sets the exchange rate between energy and entropy, so the principles are three settings of one dial rather than three separate doctrines. Read at the level of algorithms, they share one inner problem, the minimization of a free energy over distributions, and differ only in the outer criterion. It is also the equation that carried statistical physics into machine learning, from the Hopfield network and the Boltzmann machine through to the evidence lower bound — a lineage recognized by the 2024 Nobel Prize in Physics (The Royal Swedish Academy of Sciences 2024).

This is an early scaffold; future revisions will add worked examples, code, and experiments. For updates, see https://yataobian.com/ and the Blue Whale Lab.

References

Abbeel, Pieter, and Andrew Y. Ng. 2004. "Apprenticeship Learning via Inverse Reinforcement Learning." *International Conference on Machine Learning (ICML)*.

Abe, Sumiyoshi. 2000. "Axioms and Uniqueness Theorem for Tsallis Entropy." *Physics Letters A* 271 (1–2): 74–79. https://doi.org/10.1016/S0375-9601(00)00337-6.

Ackley, David H., Geoffrey E. Hinton, and Terrence J. Sejnowski. 1985. "A Learning Algorithm for Boltzmann Machines." *Cognitive Science* 9 (1): 147–69. https://doi.org/10.1207/s15516709cog0901_7.

Aczél, János, and Zoltán Daróczy. 1963. "Über Verallgemeinerte Quasilineare Mittelwerte, Die Mit Gewichtsfunktionen Gebildet Sind." *Publicationes Mathematicae Debrecen* 10: 171–90. https://doi.org/10.5486/PMD.1963.10.1-4.24.

Agarwal, Shivam, Zimin Zhang, Lifan Yuan, Jiawei Han, and Hao Peng. 2025. "The Unreasonable Effectiveness of Entropy Minimization in LLM Reasoning." *Advances in Neural Information Processing Systems (NeurIPS)*.

Aguilera, Miguel, Beren Millidge, Alexander Tschantz, and Christopher L. Buckley. 2022. "How Particular Is the Physics of the Free Energy Principle?" *Physics of Life Reviews* 40: 24–50. https://doi.org/10.1016/j.plrev.2021.11.001.

Aguilera, Miguel, Beren Millidge, Alexander Tschantz, and Christopher L. Buckley. 2023. "From the Free Energy Principle to a Confederation of Bayesian Mechanics: Reply to Comments on 'How Particular Is the Physics of the Free Energy Principle?'" *Physics of Life Reviews* 44: 270–75. https://doi.org/10.1016/j.plrev.2023.01.018.

Ahmadian, Arash, Chris Cremer, Matthias Gallé, et al. 2024. "Back to Basics: Revisiting REINFORCE-Style Optimization for Learning from Human Feedback in LLMs." *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 12248–67.

Ahuja, Ravindra K., and James B. Orlin. 2001. "Inverse Optimization." *Operations Research* 49 (5): 771–83.

Altun, Yasemin, and Alexander J. Smola. 2006. "Unifying Divergence Minimization and Statistical Inference via Convex Duality." *Conference on Learning Theory (COLT)*.

Andrews, Mel. 2021. "The Math Is Not the Territory: Navigating the Free Energy Principle." *Biology & Philosophy* 36 (3): 30. https://doi.org/10.1007/s10539-021-09807-0.

Asano, Yuki M., Christian Rupprecht, and Andrea Vedaldi. 2020. "Self-Labelling via Simultaneous Clustering and Representation Learning." *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/1911.05371.

Baez, John C., and Tobias Fritz. 2014. "A Bayesian Characterization of Relative Entropy." *Theory and*

Applications of Categories 29 (16): 422–56. https://doi.org/10.70930/tac/myyolhqw.

Baez, John C., Tobias Fritz, and Tom Leinster. 2011. "A Characterization of Entropy in Terms of Information Loss." *Entropy* 13 (11): 1945–57. https://doi.org/10.3390/e13111945.

Bagaev, Dmitry, Albert Podusenko, and Bert de Vries. 2023. "RxInfer: A Julia Package for Reactive Real-Time Bayesian Inference." *Journal of Open Source Software* 8 (84): 5161. https://doi.org/10.21105/joss.05161.

Balcan, Maria-Florina, and Nicholas J. A. Harvey. 2018. "Submodular Functions: Learnability, Structure, and Optimization." *SIAM Journal on Computing* 47 (3): 703–54. https://doi.org/10.1137/120888909.

Balcerak, Michal, Tamaz Amiranashvili, Antonio Terpin, et al. 2025. "Energy Matching: Unifying Flow Matching and Energy-Based Models for Generative Modeling." *Advances in Neural Information Processing Systems 38 (NeurIPS)*.

Balci, Fuat, Suliann Ben Hamed, Thomas Boraud, et al. 2023. "A Response to Claims of Emergent Intelligence and Sentience in a Dish." *Neuron* 111 (5): 604–5. https://doi.org/10.1016/j.neuron.2023.02.009.

Barbera, Elvira. 1999. "On the Principle of Minimal Entropy Production for Navier–Stokes–Fourier Fluids." *Continuum Mechanics and Thermodynamics* 11 (5): 327–30. https://doi.org/10.1007/s001610050127.

Bastos, Andre M., W. Martin Usrey, Rick A. Adams, George R. Mangun, Pascal Fries, and Karl J. Friston. 2012. "Canonical Microcircuits for Predictive Coding." *Neuron* 76 (4): 695–711. https://doi.org/10.1016/j.neuron.2012.10.038.

Beirami, Ahmad, Alekh Agarwal, Jonathan Berant, et al. 2024. "Theoretical Guarantees on the Best-of-n Alignment Policy." *arXiv Preprint arXiv:2401.01879*.

Belanger, David, and Andrew McCallum. 2016. "Structured Prediction Energy Networks." *International Conference on Machine Learning (ICML)*.

Belanger, David, Bishan Yang, and Andrew McCallum. 2017. "End-to-End Learning for Structured Prediction Energy Networks." *International Conference on Machine Learning (ICML)*, 429–39.

Bengio, Yoshua, and Olivier Delalleau. 2009. "Justifying and Generalizing Contrastive Divergence." *Neural Computation* 21 (6): 1601–21.

Bennett, Charles H. 1976. "Efficient Estimation of Free Energy Differences from Monte Carlo Data." *Journal of Computational Physics* 22 (2): 245–68. https://doi.org/10.1016/0021-9991(76)90078-4.

Berger, Adam L., Stephen A. Della Pietra, and Vincent J. Della Pietra. 1996. "A Maximum Entropy Approach to Natural Language Processing." *Computational Linguistics* 22 (1): 39–71.

Berthelot, David, Nicholas Carlini, Ekin D. Cubuk, et al. 2020. "ReMixMatch: Semi-Supervised Learning with Distribution Alignment and Augmentation Anchoring." *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/1911.09785.

Besag, Julian. 1974. "Spatial Interaction and the Statistical Analysis of Lattice Systems." *Journal of the Royal Statistical Society: Series B (Methodological)* 36 (2): 192–236.

Bian, Andrew An, Joachim M. Buhmann, Andreas Krause, and Sebastian Tschiatschek. 2017. "Guarantees for Greedy Maximization of Non-Submodular Functions with Applications." *International Conference on Machine Learning (ICML)*, 498–507.

Bian, Yatao. 2020. *Awesome Energy-Based Models/Learning: A Comprehensive List of Energy-Based Learning Papers and Materials*. Https://github.com/yataobian/awesome-ebm.

Bian, Yatao, Joachim M. Buhmann, and Andreas Krause. 2019. "Optimal Continuous DR-Submodular Maximization and Applications to Provable Mean Field Inference." *International Conference on Machine Learning (ICML)*, 644–53.

Bian, Yatao, Yu Rong, Tingyang Xu, Jiaxiang Wu, Andreas Krause, and Junzhou Huang. 2022. "Energy-Based Learning for Cooperative Games, with Applications to Valuation Problems in Machine Learning." *International Conference on Learning Representations (ICLR)*. https://openreview.net/forum?id=xLfAgCroImw.

Biehl, Martin, Felix A. Pollock, and Ryota Kanai. 2021. "A Technical Critique of Some Parts of the Free Energy Principle." *Entropy* 23 (3): 293. https://doi.org/10.3390/e23030293.

Bilmes, Jeffrey A., and Wenruo Bai. 2017. "Deep Submodular Functions." *arXiv Preprint arXiv:1701.08939*.

Bissiri, Pier Giovanni, Chris C. Holmes, and Stephen G. Walker. 2016. "A General Framework for Updating Belief Distributions." *Journal of the Royal Statistical Society Series B: Statistical Methodology* 78 (5): 1103–30. https://doi.org/10.1111/rssb.12158.

Black, Kevin, Michael Janner, Yilun Du, Ilya Kostrikov, and Sergey Levine. 2024. "Training Diffusion Models with Reinforcement Learning." *International Conference on Learning Representations (ICLR)*.

Bloem-Reddy, Benjamin, and Yee Whye Teh. 2020. "Probabilistic Symmetries and Invariant Neural Networks." *Journal of Machine Learning Research* 21 (90): 1–61.

Bogacz, Rafal. 2017. "A Tutorial on the Free-Energy Framework for Modelling Perception and Learning." *Journal of Mathematical Psychology* 76: 198–211. https://doi.org/10.1016/j.jmp.2015.11.003.

Boltzmann, Ludwig. 1877. "Über Die Beziehung Zwischen Dem Zweiten Hauptsatze Der Mechanischen Wärmetheorie Und Der Wahrscheinlichkeitsrechnung Respektive Den Sätzen Über Das Wärmegleichgewicht." *Sitzungsberichte Der Kaiserlichen Akademie Der Wissenschaften in Wien* 76: 373–435.

Bradley, Ralph Allan, and Milton E. Terry. 1952. "Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons." *Biometrika* 39 (3/4): 324–45.

Bridle, John S., Anthony J. R. Heading, and David J. C. MacKay. 1992. "Unsupervised Classifiers, Mutual Information and 'Phantom Targets'." *Advances in Neural Information Processing Systems (NIPS) 4*, 1096–101.

Bruineberg, Jelle, Krzysztof Dołęga, Joe Dewhurst, and Manuel Baltieri. 2022a. "The Emperor Is Naked: Replies to Commentaries on the Target Article." *Behavioral and Brain Sciences* 45: e219. https://doi.org/10.1017/S0140525X22000656.

Bruineberg, Jelle, Krzysztof Dołęga, Joe Dewhurst, and Manuel Baltieri. 2022b. "The Emperor's New

Markov Blankets." *Behavioral and Brain Sciences* 45: e183. https://doi.org/10.1017/S0140525X21002351.

Buckley, Christopher L., Chang Sub Kim, Simon McGregor, and Anil K. Seth. 2017. "The Free Energy Principle for Action and Perception: A Mathematical Review." *Journal of Mathematical Psychology* 81: 55–79. https://doi.org/10.1016/j.jmp.2017.09.004.

Burton, Didier, and Philippe L. Toint. 1992. "On an Instance of the Inverse Shortest Paths Problem." *Mathematical Programming* 53: 45–61.

Calinescu, Gruia, Chandra Chekuri, Martin Pál, and Jan Vondrák. 2007. "Maximizing a Submodular Set Function Subject to a Matroid Constraint." *Integer Programming and Combinatorial Optimization (IPCO)*, 182–96.

Callen, Herbert B. 1985. *Thermodynamics and an Introduction to Thermostatistics*. 2nd ed. Wiley.

Carcamo, David P., Nicholas J. Weaver, Purushottam D. Dixit, and Christopher W. Lynn. 2025. "Minimax Entropy: The Statistical Physics of Optimal Models." *Physical Review E* 112 (6): 061001. https://doi.org/10.1103/kr9x-q59y.

Caron, Mathilde, Ishan Misra, Julien Mairal, Priya Goyal, Piotr Bojanowski, and Armand Joulin. 2020. "Unsupervised Learning of Visual Features by Contrasting Cluster Assignments." *Advances in Neural Information Processing Systems (NeurIPS)*. https://arxiv.org/abs/2006.09882.

Caron, Mathilde, Hugo Touvron, Ishan Misra, et al. 2021. "Emerging Properties in Self-Supervised Vision Transformers." *IEEE/CVF International Conference on Computer Vision (ICCV)*. https://arxiv.org/abs/2104.14294.

Carreira-Perpiñán, Miguel Á., and Geoffrey E. Hinton. 2005. "On Contrastive Divergence Learning." *International Workshop on Artificial Intelligence and Statistics (AISTATS)*, Proceedings of machine learning research, vol. R5: 33–40.

Celeux, Gilles, and Gérard Govaert. 1992. "A Classification EM Algorithm for Clustering and Two Stochastic Versions." *Computational Statistics & Data Analysis* 14 (3): 315–32.

Champion, Théophile, Howard Bowman, Dimitrije Marković, and Marek Grześ. 2024. "Reframing the Expected Free Energy: Four Formulations and a Unification." *arXiv Preprint arXiv:2402.14460*.

Chen, Minhua, Badrinath Jayakumar, Padmasundari Gopalakrishnan, Qiming Huang, Michael Johnston, and Patrick Haffner. 2021. "Deep Clustering with Measure Propagation." *arXiv Preprint arXiv:2104.08967*.

Christiano, Paul F., Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2017. "Deep Reinforcement Learning from Human Preferences." *Advances in Neural Information Processing Systems 30 (NeurIPS)*.

Clark, Andy. 2013. "Whatever Next? Predictive Brains, Situated Agents, and the Future of Cognitive Science." *Behavioral and Brain Sciences* 36 (3): 181–204. https://doi.org/10.1017/S0140525X12000477.

Clark, Kevin, Minh-Thang Luong, Quoc V. Le, and Christopher D. Manning. 2020a. "ELECTRA: Pre-Training Text Encoders as Discriminators Rather Than Generators." *International Conference on Learn-*

ing Representations (ICLR).

Clark, Kevin, Minh-Thang Luong, Quoc V. Le, and Christopher D. Manning. 2020b. "Pre-Training Transformers as Energy-Based Cloze Models." *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 285–94.

Clausius, Rudolf. 1865. "Über Verschiedene Für Die Anwendung Bequeme Formen Der Hauptgleichungen Der Mechanischen Wärmetheorie." *Annalen Der Physik Und Chemie* 125: 353–400.

Cobbe, Karl, Vineet Kosaraju, Mohammad Bavarian, et al. 2021. "Training Verifiers to Solve Math Word Problems." *arXiv Preprint arXiv:2110.14168*.

Cohen, Jacob. 1960. "A Coefficient of Agreement for Nominal Scales." *Educational and Psychological Measurement* 20 (1): 37–46. https://doi.org/10.1177/001316446002000104.

Collins, Michael. 2002. "Discriminative Training Methods for Hidden Markov Models: Theory and Experiments with Perceptron Algorithms." *Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Colombo, Matteo, and Cory Wright. 2021. "First Principles in the Life Sciences: The Free-Energy Principle, Organicism, and Mechanism." *Synthese* 198 (S14): 3463–88. https://doi.org/10.1007/s11229-018-01932-w.

Conant, Roger C., and W. Ross Ashby. 1970. "Every Good Regulator of a System Must Be a Model of That System." *International Journal of Systems Science* 1 (2): 89–97. https://doi.org/10.1080/00207727008920220.

Csiszár, Imre. 1975. "*I*-Divergence Geometry of Probability Distributions and Minimization Problems." *Annals of Probability* 3 (1): 146–58.

Csiszár, Imre. 2008. "Axiomatic Characterizations of Information Measures." *Entropy* 10 (3): 261–73. https://doi.org/10.3390/e10030261.

Cui, Ganqu, Yuchen Zhang, Jiacheng Chen, et al. 2025. "The Entropy Mechanism of Reinforcement Learning for Reasoning Language Models." *arXiv Preprint arXiv:2505.22617*. https://arxiv.org/abs/2505.22617.

Da Costa, Lancelot, Karl Friston, Conor Heins, and Grigorios A. Pavliotis. 2021. "Bayesian Mechanics for Stationary Processes." *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 477 (2256): 20210518. https://doi.org/10.1098/rspa.2021.0518.

Da Costa, Lancelot, Thomas Parr, Noor Sajid, Sebastijan Veselic, Victorita Neacsu, and Karl Friston. 2020. "Active Inference on Discrete State-Spaces: A Synthesis." *Journal of Mathematical Psychology* 99: 102447. https://doi.org/10.1016/j.jmp.2020.102447.

Da Costa, Lancelot, Noor Sajid, Thomas Parr, Karl Friston, and Ryan Smith. 2023. "Reward Maximization Through Discrete Active Inference." *Neural Computation* 35 (5): 807–52. https://doi.org/10.1162/neco_a_01574.

Dai, Bo, Zhen Liu, Hanjun Dai, et al. 2019. "Exponential Family Estimation via Adversarial Dynamics Embedding." *Advances in Neural Information Processing Systems (NeurIPS)* 32.

Dai, Jifeng, Yang Lu, and Ying Nian Wu. 2015. "Generative Modeling of Convolutional Neural Networks." *International Conference on Learning Representations (ICLR)*.

Daróczy, Zoltán. 1970. "Generalized Information Functions." *Information and Control* 16 (1): 36–51. https://doi.org/10.1016/S0019-9958(70)80040-7.

Dawid, A. Philip. 2007. "The Geometry of Proper Scoring Rules." *Annals of the Institute of Statistical Mathematics* 59 (1): 77–93. https://doi.org/10.1007/s10463-006-0099-8.

Dawid, Anna, and Yann LeCun. 2024. "Introduction to Latent Variable Energy-Based Models: A Path Towards Autonomous Machine Intelligence." *Journal of Statistical Mechanics: Theory and Experiment* 2024 (10): 104011.

Dayan, Peter, Geoffrey E. Hinton, Radford M. Neal, and Richard S. Zemel. 1995. "The Helmholtz Machine." *Neural Computation* 7 (5): 889–904. https://doi.org/10.1162/neco.1995.7.5.889.

de Vries, Bert, Wouter W. L. Nuijten, Thijs van de Laar, et al. 2025. "Expected Free Energy-Based Planning as Variational Inference." *arXiv Preprint arXiv:2504.14898*.

DeepSeek-AI. 2025. "DeepSeek-R1 Incentivizes Reasoning in LLMs Through Reinforcement Learning." *Nature* 645. https://doi.org/10.1038/s41586-025-09422-z.

Della Pietra, Stephen, Vincent Della Pietra, and John Lafferty. 1997. "Inducing Features of Random Fields." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 19 (4): 380–93. https://doi.org/10.1109/34.588021.

Deng, Yuntian, Anton Bakhtin, Myle Ott, Arthur Szlam, and Marc'Aurelio Ranzato. 2020. "Residual Energy-Based Models for Text Generation." *International Conference on Learning Representations (ICLR)*.

Dhariwal, Prafulla, and Alexander Nichol. 2021. "Diffusion Models Beat GANs on Image Synthesis." *Advances in Neural Information Processing Systems (NeurIPS)*.

Domingo-Enrich, Carles, Michal Drozdzal, Brian Karrer, and Ricky T. Q. Chen. 2025. "Adjoint Matching: Fine-Tuning Flow and Diffusion Generative Models with Memoryless Stochastic Optimal Control." *International Conference on Learning Representations (ICLR)*.

Domke, Justin. 2012. "Generic Methods for Optimization-Based Modeling." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 318–26.

Domke, Justin. 2013. "Learning Graphical Model Parameters with Approximate Marginal Inference." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 35 (10): 2454–67. https://doi.org/10.1109/TPAMI.2013.31.

Dong, Hanze, Wei Xiong, Deepanshu Goyal, et al. 2023. "RAFT: Reward rAnked FineTuning for Generative Foundation Model Alignment." *Transactions on Machine Learning Research*.

Donsker, Monroe D., and S. R. Srinivasa Varadhan. 1975. "Asymptotic Evaluation of Certain Markov Process Expectations for Large Time, I." *Communications on Pure and Applied Mathematics* 28 (1): 1–47. https://doi.org/10.1002/cpa.3160280102.

Du, Yilun, Conor Durkan, Robin Strudel, et al. 2023. "Reduce, Reuse, Recycle: Compositional Generation with Energy-Based Diffusion Models and MCMC." *Proceedings of the 40th International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 202.

Du, Yilun, Shuang Li, and Igor Mordatch. 2020. "Compositional Visual Generation with Energy Based Models." *Advances in Neural Information Processing Systems (NeurIPS)*.

Du, Yilun, and Igor Mordatch. 2019. "Implicit Generation and Modeling with Energy-Based Models." *Advances in Neural Information Processing Systems (NeurIPS)*.

Duďık, Miroslav, Steven J. Phillips, and Robert E. Schapire. 2007. "Maximum Entropy Density Estimation with Generalized Regularization and an Application to Species Distribution Modeling." *Journal of Machine Learning Research* 8: 1217–60.

Dutta, Sanchayan, Sai Niranjan Ramachandran, and Suvrit Sra. 2026. "Active Inference as Context Acquisition for AI Agents." *arXiv Preprint arXiv:2608.19202*.

Efron, Bradley. 1975. "Defining the Curvature of a Statistical Problem (with Applications to Second Order Efficiency)." *Annals of Statistics* 3 (6): 1189–242. https://doi.org/10.1214/aos/1176343282.

Elias, Peter. 1955. "Predictive Coding—I." *IRE Transactions on Information Theory* 1 (1): 16–24. https://doi.org/10.1109/TIT.1955.1055126.

Emmerich, Michael T. M., Ksenia Pereverdieva, and André H. Deutz. 2026a. "Maximum Solow–Polasky Diversity Subset Selection Is NP-Hard Even in the Euclidean Plane." *arXiv Preprint arXiv:2604.19484*. https://arxiv.org/abs/2604.19484.

Emmerich, Michael T. M., Ksenia Pereverdieva, and André H. Deutz. 2026b. "Selecting a Maximum Solow–Polasky Diversity Subset in General Metric Spaces Is NP-Hard." *arXiv Preprint arXiv:2604.05495*. https://arxiv.org/abs/2604.05495.

Faddeev, D. K. 1956. "On the Concept of Entropy of a Finite Probabilistic Scheme." *Uspekhi Matematicheskikh Nauk* 11 (1(67)): 227–31. https://www.mathnet.ru/eng/rm7756.

Fan, Ying, Olivia Watkins, Yuqing Du, et al. 2023. "DPOK: Reinforcement Learning for Fine-Tuning Text-to-Image Diffusion Models." *Advances in Neural Information Processing Systems 36 (NeurIPS)*.

Farnia, Farzan, and David Tse. 2016. "A Minimax Approach to Supervised Learning." *Advances in Neural Information Processing Systems (NeurIPS)* 29.

Farquhar, Sebastian, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. 2024. "Detecting Hallucinations in Large Language Models Using Semantic Entropy." *Nature* 630: 625–30. https://doi.org/10.1038/s41586-024-07421-0.

Feder, M. 1986. "Maximum Entropy as a Special Case of the Minimum Description Length Criterion." *IEEE Transactions on Information Theory* 32: 847–49.

Feldman, Harriet, and Karl Friston. 2010. "Attention, Uncertainty, and Free-Energy." *Frontiers in Human Neuroscience* 4: 215. https://doi.org/10.3389/fnhum.2010.00215.

Feng, Ruiqi, Chenglei Yu, Wenhao Deng, Peiyan Hu, and Tailin Wu. 2025. "On the Guidance of Flow Matching." *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 267: 16993–7029.

Fields, Chris, Karl Friston, James F. Glazebrook, and Michael Levin. 2022. "A Free Energy Principle for Generic Quantum Systems." *Progress in Biophysics and Molecular Biology* 173: 36–59. https://doi.org/10.1016/j.pbiomolbio.2022.05.006.

Fountas, Zafeirios, Noor Sajid, Pedro A. M. Mediano, and Karl Friston. 2020. "Deep Active Inference Agents Using Monte-Carlo Methods." *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 11662–75.

Fox, Roy, Ari Pakman, and Naftali Tishby. 2016. "Taming the Noise in Reinforcement Learning via Soft Updates." *Conference on Uncertainty in Artificial Intelligence (UAI)*.

Frieden, B. Roy. 1990. "Fisher Information, Disorder, and the Equilibrium Distributions of Physics." *Physical Review A* 41 (8): 4265–76. https://doi.org/10.1103/PhysRevA.41.4265.

Friedman, Dan, and Adji Bousso Dieng. 2023. "The Vendi Score: A Diversity Evaluation Metric for Machine Learning." *Transactions on Machine Learning Research*. https://arxiv.org/abs/2210.02410.

Friston, Karl. 2005. "A Theory of Cortical Responses." *Philosophical Transactions of the Royal Society B: Biological Sciences* 360 (1456): 815–36. https://doi.org/10.1098/rstb.2005.1622.

Friston, Karl. 2008. "Hierarchical Models in the Brain." *PLoS Computational Biology* 4 (11): e1000211. https://doi.org/10.1371/journal.pcbi.1000211.

Friston, Karl. 2009. "The Free-Energy Principle: A Rough Guide to the Brain?" *Trends in Cognitive Sciences* 13 (7): 293–301. https://doi.org/10.1016/j.tics.2009.04.005.

Friston, Karl. 2010. "The Free-Energy Principle: A Unified Brain Theory?" *Nature Reviews Neuroscience* 11 (2): 127–38.

Friston, Karl. 2013. "Life as We Know It." *Journal of the Royal Society Interface* 10 (86): 20130475. https://doi.org/10.1098/rsif.2013.0475.

Friston, Karl. 2019. "A Free Energy Principle for a Particular Physics." *arXiv Preprint arXiv:1906.10184*.

Friston, Karl. 2022a. "The Ultimate Trick? Comment on 'The Markov Blanket Trick: On the Scope of the Free Energy Principle and Active Inference' by Raja et al." *Physics of Life Reviews* 43: 10–16. https://doi.org/10.1016/j.plrev.2022.07.007.

Friston, Karl. 2022b. "Very Particular: Comment on 'How Particular Is the Physics of the Free Energy Principle?'" *Physics of Life Reviews* 41: 58–60. https://doi.org/10.1016/j.plrev.2022.05.002.

Friston, Karl, Lancelot Da Costa, Danijar Hafner, Casper Hesp, and Thomas Parr. 2021. "Sophisticated Inference." *Neural Computation* 33 (3): 713–63. https://doi.org/10.1162/neco_a_01351.

Friston, Karl, Lancelot Da Costa, and Thomas Parr. 2021. "Some Interesting Observations on the Free Energy Principle." *Entropy* 23 (8): 1076. https://doi.org/10.3390/e23081076.

Friston, Karl, Lancelot Da Costa, Noor Sajid, et al. 2023. "The Free Energy Principle Made Simpler but Not Too Simple." *Physics Reports* 1024: 1–29. https://doi.org/10.1016/j.physrep.2023.07.001.

Friston, Karl, Lancelot Da Costa, Dalton A. R. Sakthivadivel, et al. 2023. "Path Integrals, Particular Kinds, and Strange Things." *Physics of Life Reviews* 47: 35–62. https://doi.org/10.1016/j.plrev.2023.08.016.

Friston, Karl, Jean Daunizeau, James Kilner, and Stefan J. Kiebel. 2010. "Action and Behavior: A Free-Energy Formulation." *Biological Cybernetics* 102 (3): 227–60. https://doi.org/10.1007/s00422-010-0364-z.

Friston, Karl, Thomas H. B. FitzGerald, Francesco Rigoli, Philipp Schwartenbeck, John O'Doherty, and Giovanni Pezzulo. 2016. "Active Inference and Learning." *Neuroscience & Biobehavioral Reviews* 68: 862–79. https://doi.org/10.1016/j.neubiorev.2016.06.022.

Friston, Karl, Thomas H. B. FitzGerald, Francesco Rigoli, Philipp Schwartenbeck, and Giovanni Pezzulo. 2017. "Active Inference: A Process Theory." *Neural Computation* 29 (1): 1–49. https://doi.org/10.1162/NECO_a_00912.

Friston, Karl, Conor Heins, Tim Verbelen, et al. 2025. "From Pixels to Planning: Scale-Free Active Inference." *Frontiers in Network Physiology* 5: 1521963. https://doi.org/10.3389/fnetp.2025.1521963.

Friston, Karl, James Kilner, and Lee Harrison. 2006. "A Free Energy Principle for the Brain." *Journal of Physiology-Paris* 100 (1-3): 70–87. https://doi.org/10.1016/j.jphysparis.2006.10.001.

Friston, Karl, Jérémie Mattout, Nelson Trujillo-Barreto, John Ashburner, and Will Penny. 2007. "Variational Free Energy and the Laplace Approximation." *NeuroImage* 34 (1): 220–34. https://doi.org/10.1016/j.neuroimage.2006.08.035.

Friston, Karl, Maxwell J. D. Ramstead, Alex B. Kiefer, et al. 2024. "Designing Ecosystems of Intelligence from First Principles." *Collective Intelligence* 3 (1): 26339137231222481. https://doi.org/10.1177/26339137231222481.

Friston, Karl, Francesco Rigoli, Dimitri Ognibene, Christoph Mathys, Thomas Fitzgerald, and Giovanni Pezzulo. 2015. "Active Inference and Epistemic Value." *Cognitive Neuroscience* 6 (4): 187–214. https://doi.org/10.1080/17588928.2015.1020053.

Friston, Karl, Spyridon Samothrakis, and Read Montague. 2012. "Active Inference and Agency: Optimal Control Without Cost Functions." *Biological Cybernetics* 106 (8-9): 523–41. https://doi.org/10.1007/s00422-012-0512-8.

Friston, Karl, Christopher Thornton, and Andy Clark. 2012. "Free-Energy Minimization and the Dark-Room Problem." *Frontiers in Psychology* 3: 130. https://doi.org/10.3389/fpsyg.2012.00130.

Geiger, Dan, David Heckerman, Henry King, and Christopher Meek. 2001. "Stratified Exponential Families: Graphical Models and Model Selection." *The Annals of Statistics* 29 (2): 505–29. https://doi.org/10.1214/aos/1009210550.

Geist, Matthieu, Bruno Scherrer, and Olivier Pietquin. 2019. "A Theory of Regularized Markov Decision Processes." *International Conference on Machine Learning (ICML)*.

Gibbs, Josiah Willard. 1902. *Elementary Principles in Statistical Mechanics*. Charles Scribner's Sons.

Giffin, Adom, and Ariel Caticha. 2007. "Updating Probabilities with Data and Moments." *Bayesian Inference and Maximum Entropy Methods in Science and Engineering*, AIP conference proceedings, vol. 954: 74–84. https://doi.org/10.1063/1.2821302.

Gladstone, Alexi, Ganesh Nanduru, Md Mofijul Islam, et al. 2026. "Energy-Based Transformers Are Scalable Learners and Thinkers." *International Conference on Learning Representations (ICLR)*.

Gnaiger, Erich. 2009. "Open and Closed Systems: Styles of Thinking Explain Controversies on the 'Negative Entropy' Concept of Ludwig Boltzmann and Erwin Schrödinger." *Mitochondrial Physiology Network*. https://www.mitophysiology.org/images/2/23/Gnaiger_2009_OCESHS.pdf.

Go, Dongyoung, Tomasz Korbak, Germán Kruszewski, Jos Rozen, Nahyeon Ryu, and Marc Dymetman. 2023. "Aligning Language Models with Preferences Through f-Divergence Minimization." *Proceedings of the 40th International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 202.

Gomes, Ryan, Andreas Krause, and Pietro Perona. 2010. "Discriminative Clustering by Regularized Information Maximization." *Advances in Neural Information Processing Systems (NeurIPS)* 23.

Gottwald, Sebastian, and Daniel A. Braun. 2020. "The Two Kinds of Free Energy and the Bayesian Revolution." *PLOS Computational Biology* 16 (12): e1008420. https://doi.org/10.1371/journal.pcbi.1008420.

Grandvalet, Yves, and Yoshua Bengio. 2004. "Semi-Supervised Learning by Entropy Minimization." *Advances in Neural Information Processing Systems (NeurIPS)* 17.

Grandvalet, Yves, and Yoshua Bengio. 2006. "Entropy Regularization." In *Semi-Supervised Learning*, edited by Olivier Chapelle, Bernhard Schölkopf, and Alexander Zien. MIT Press.

Grathwohl, Will, Kevin Swersky, Milad Hashemi, David Duvenaud, and Chris Maddison. 2021. "Oops I Took a Gradient: Scalable Sampling for Discrete Distributions." *International Conference on Machine Learning (ICML)*, 3831–41.

Grathwohl, Will, Kuan-Chieh Wang, Jörn-Henrik Jacobsen, David Duvenaud, Mohammad Norouzi, and Kevin Swersky. 2020. "Your Classifier Is Secretly an Energy Based Model and You Should Treat It Like One." *International Conference on Learning Representations (ICLR)*.

Gregor, Karol, and Yann LeCun. 2010. "Learning Fast Approximations of Sparse Coding." *International Conference on Machine Learning (ICML)*, 399–406.

Grünwald, Peter D. 2007. *The Minimum Description Length Principle*. MIT Press.

Grünwald, Peter D., and A. Philip Dawid. 2004. "Game Theory, Maximum Entropy, Minimum Discrepancy and Robust Bayesian Decision Theory." *Annals of Statistics* 32 (4): 1367–433.

Gui, Lin, Cristina Gârbacea, and Victor Veitch. 2024. "BoNBoN Alignment for Large Language Models and the Sweetness of Best-of-n Sampling." *Advances in Neural Information Processing Systems 37 (NeurIPS)*.

Gulcehre, Caglar, Tom Le Paine, Srivatsan Srinivasan, et al. 2023. "Reinforced Self-Training (ReST) for Language Modeling." *arXiv Preprint arXiv:2308.08998*.

Gutmann, Michael U., and Aapo Hyvärinen. 2012. "Noise-Contrastive Estimation of Unnormalized Statistical Models, with Applications to Natural Image Statistics." *Journal of Machine Learning Research* 13: 307–61.

Gutmann, Michael, and Jun-ichiro Hirayama. 2011. "Bregman Divergence as General Framework to Estimate Unnormalized Statistical Models." *Conference on Uncertainty in Artificial Intelligence (UAI)*, 283–90.

Gutmann, Michael, and Aapo Hyvärinen. 2010. "Noise-Contrastive Estimation: A New Estimation Principle for Unnormalized Statistical Models." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 297–304.

GX-Chen, Anthony, Jatin Prakash, Jeff Guo, Rob Fergus, and Rajesh Ranganath. 2025. "KL-Regularized Reinforcement Learning Is Designed to Mode Collapse." *arXiv Preprint arXiv:2510.20817*. https://arxiv.org/abs/2510.20817.

Haarnoja, Tuomas, Haoran Tang, Pieter Abbeel, and Sergey Levine. 2017. "Reinforcement Learning with Deep Energy-Based Policies." *International Conference on Machine Learning (ICML)*, 1352–61.

Haarnoja, Tuomas, Aurick Zhou, Pieter Abbeel, and Sergey Levine. 2018. "Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor." *International Conference on Machine Learning (ICML)*, 1861–70.

Hafner, Danijar, Pedro A. Ortega, Jimmy Ba, Thomas Parr, Karl Friston, and Nicolas Heess. 2020. "Action and Perception as Divergence Minimization." *arXiv Preprint arXiv:2009.01791*.

Hanel, Rudolf, and Stefan Thurner. 2011. "A Comprehensive Classification of Complex Statistical Systems and an Axiomatic Derivation of Their Entropy and Distribution Functions." *EPL (Europhysics Letters)* 93 (2): 20006. https://doi.org/10.1209/0295-5075/93/20006.

Hassani, Hamed, Mahdi Soltanolkotabi, and Amin Karbasi. 2017. "Gradient Methods for Submodular Maximization." *Advances in Neural Information Processing Systems (NeurIPS)* 30.

Hathaway, Richard J. 1986. "Another Interpretation of the EM Algorithm for Mixture Distributions." *Statistics & Probability Letters* 4 (2): 53–56. https://doi.org/10.1016/0167-7152(86)90016-7.

Havrda, Jan, and František Charvát. 1967. "Quantification Method of Classification Processes. Concept of Structural a-Entropy." *Kybernetika* 3 (1): 30–35. https://www.kybernetika.cz/content/1967/1/30.

Heins, Conor, and Lancelot Da Costa. 2022. "Sparse Coupling and Markov Blankets: A Comment on 'How Particular Is the Physics of the Free Energy Principle?' By Aguilera, Millidge, Tschantz and Buckley." *Physics of Life Reviews* 42: 33–39. https://doi.org/10.1016/j.plrev.2022.06.001.

Heins, Conor, Beren Millidge, Daphne Demekas, et al. 2022. "pymdp: A Python Library for Active Inference in Discrete State Spaces." *Journal of Open Source Software* 7 (73): 4098. https://doi.org/10.21105/joss.04098.

Heins, Conor, Toon Van de Maele, Alexander Tschantz, et al. 2025. "AXIOM: Learning to Play Games in Minutes with Expanding Object-Centric Models." *arXiv Preprint arXiv:2505.24784*.

Hershey, John R., Jonathan Le Roux, and Felix Weninger. 2014. "Deep Unfolding: Model-Based Inspiration of Novel Deep Architectures." *arXiv Preprint arXiv:1409.2574*.

Heuberger, Clemens. 2004. "Inverse Combinatorial Optimization: A Survey on Problems, Methods, and Results." *Journal of Combinatorial Optimization* 8 (3): 329–61.

Hill, M. O. 1973. "Diversity and Evenness: A Unifying Notation and Its Consequences." *Ecology* 54 (2): 427–32. https://doi.org/10.2307/1934352.

Hinton, Geoffrey E. 2002. "Training Products of Experts by Minimizing Contrastive Divergence." *Neural Computation* 14 (8): 1771–800. https://doi.org/10.1162/089976602760128018.

Hinton, Geoffrey E., and Drew van Camp. 1993. "Keeping the Neural Networks Simple by Minimizing the Description Length of the Weights." *Proceedings of the Sixth Annual Conference on Computational Learning Theory (COLT)*, 5–13. https://doi.org/10.1145/168304.168306.

Hinton, Geoffrey E., and Richard S. Zemel. 1994. "Autoencoders, Minimum Description Length and Helmholtz Free Energy." *Advances in Neural Information Processing Systems 6*, 3–10.

Hirsh, Jacob B., Raymond A. Mar, and Jordan B. Peterson. 2012. "Psychological Entropy: A Framework for Understanding Uncertainty-Related Anxiety." *Psychological Review* 119 (2): 304–20. https://doi.org/10.1037/a0026767.

Ho, Jonathan, Ajay Jain, and Pieter Abbeel. 2020. "Denoising Diffusion Probabilistic Models." *Advances in Neural Information Processing Systems (NeurIPS)*.

Ho, Jonathan, and Tim Salimans. 2022. "Classifier-Free Diffusion Guidance." *arXiv Preprint arXiv:2207.12598*.

Hohwy, Jakob. 2016. "The Self-Evidencing Brain." *Noûs* 50 (2): 259–85. https://doi.org/10.1111/nous.12062.

Honkela, Antti, and Harri Valpola. 2004. "Variational Learning and Bits-Back Coding: An Information-Theoretic View to Bayesian Learning." *IEEE Transactions on Neural Networks* 15 (4): 800–810. https://doi.org/10.1109/TNN.2004.828762.

Hopfield, John J. 1982. "Neural Networks and Physical Systems with Emergent Collective Computational Abilities." *Proceedings of the National Academy of Sciences* 79 (8): 2554–58. https://doi.org/10.1073/pnas.79.8.2554.

Hu, Edward J., Moksh Jain, Eric Elmoznino, et al. 2024. "Amortizing Intractable Inference in Large Language Models." *International Conference on Learning Representations (ICLR)*.

Hu, Weihua, Takeru Miyato, Seiya Tokui, Eiichi Matsumoto, and Masashi Sugiyama. 2017. "Learning Discrete Representations via Information Maximizing Self-Augmented Training." *International Conference on Machine Learning (ICML)*. https://arxiv.org/abs/1702.08720.

Hyvärinen, Aapo. 2005. "Estimation of Non-Normalized Statistical Models by Score Matching." *Journal of Machine Learning Research* 6: 695–709.

Hyvärinen, Aapo. 2007. "Connections Between Score Matching, Contrastive Divergence, and Pseudolikelihood for Continuous-Valued Variables." *IEEE Transactions on Neural Networks* 18 (5): 1529–31.

Isomura, Takuya, and Karl Friston. 2018. "In Vitro Neural Networks Minimise Variational Free Energy." *Scientific Reports* 8: 16926. https://doi.org/10.1038/s41598-018-35221-w.

Isomura, Takuya, Kiyoshi Kotani, Yasuhiko Jimbo, and Karl Friston. 2023. "Experimental Validation of the Free-Energy Principle with in Vitro Neural Networks." *Nature Communications* 14: 4547. https://doi.org/10.1038/s41467-023-40141-z.

Isomura, Takuya, Hideaki Shimazaki, and Karl Friston. 2022. "Canonical Neural Networks Perform Active Inference." *Communications Biology* 5: 55. https://doi.org/10.1038/s42003-021-02994-2.

Jaakkola, Tommi, Marina Meila, and Tony Jebara. 1999. "Maximum Entropy Discrimination." *Advances in Neural Information Processing Systems (NeurIPS)* 12: 470–76.

Jalali, Mohammad, Cheuk Ting Li, and Farzan Farnia. 2023. "An Information-Theoretic Evaluation of Generative Models in Learning Multi-Modal Distributions." *Advances in Neural Information Processing Systems (NeurIPS)* 36: 9931–43. https://doi.org/10.52202/075280-0434.

Jaques, Natasha, Shixiang Gu, Dzmitry Bahdanau, José Miguel Hernández-Lobato, Richard E. Turner, and Douglas Eck. 2017. "Sequence Tutor: Conservative Fine-Tuning of Sequence Generation Models with KL-Control." *Proceedings of the 34th International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 70: 1645–54.

Jaynes, Edwin T. 1957. "Information Theory and Statistical Mechanics." *Physical Review* 106: 620.

Jaynes, Edwin T. 1968. "Prior Probabilities." *IEEE Transactions on Systems Science and Cybernetics* 4 (3): 227–41.

Jaynes, Edwin T. 1980. "The Minimum Entropy Production Principle." *Annual Review of Physical Chemistry* 31: 579–601. https://doi.org/10.1146/annurev.pc.31.100180.003051.

Jiang, Eric H., Haozheng Luo, Shengyuan Pang, et al. 2025. "Learning to Rank Chain-of-Thought: Using a Small Model." *arXiv Preprint arXiv:2505.14999*.

Jin, Long, Justin Lazarow, and Zhuowen Tu. 2017. "Introspective Classification with Convolutional Nets." *Advances in Neural Information Processing Systems (NeurIPS)* 30.

Jizba, Petr, and Toshihico Arimitsu. 2004. "The World According to Rényi: Thermodynamics of Multifractal Systems." *Annals of Physics* 312 (1): 17–59. https://doi.org/10.1016/j.aop.2004.01.002.

Jordan, Michael I., Zoubin Ghahramani, Tommi S. Jaakkola, and Lawrence K. Saul. 1999. "An Introduction to Variational Methods for Graphical Models." *Machine Learning* 37 (2): 183–233.

Jost, Lou. 2006. "Entropy and Diversity." *Oikos* 113 (2): 363–75. https://doi.org/10.1111/j.2006.0030-1299.14714.x.

Jost, Lou. 2009. "Mismeasuring Biological Diversity: Response to Hoffmann and Hoffmann (2008)." *Ecological Economics* 68 (4): 925–28. https://doi.org/10.1016/j.ecolecon.2008.10.015.

Jost, Lou, Philip DeVries, Thomas Walla, Harold Greeney, Anne Chao, and Carlo Ricotta. 2010. "Partitioning Diversity for Conservation Analyses." *Diversity and Distributions* 16 (1): 65–76. https://doi.org/10.1111/j.1472-4642.2009.00626.x.

Kagan, Brett J., Andy C. Kitchen, Nhi T. Tran, et al. 2022. "In Vitro Neurons Learn and Exhibit Sentience When Embodied in a Simulated Game-World." *Neuron* 110 (23): 3952–3969.e8. https://doi.org/10.1016/j.neuron.2022.09.001.

Kappen, Hilbert J. 2005. "Path Integrals and Symmetry Breaking for Optimal Control Theory." *Journal of Statistical Mechanics: Theory and Experiment* 2005 (11): P11011.

Kappen, Hilbert J., Vicenç Gómez, and Manfred Opper. 2012. "Optimal Control as a Graphical Model Inference Problem." *Machine Learning* 87 (2): 159–82.

Karalias, Nikolaos, Joshua Robinson, Andreas Loukas, and Stefanie Jegelka. 2022. "Neural Set Function Extensions: Learning with Discrete Functions in High Dimensions." *Advances in Neural Information Processing Systems (NeurIPS)* 35: 15338–52.

Khalifa, Muhammad, Hady Elsahar, and Marc Dymetman. 2021. "A Distributional Approach to Controlled Text Generation." *International Conference on Learning Representations (ICLR)*.

Khinchin, A. I. 1957. *Mathematical Foundations of Information Theory*. Dover Publications.

Kianercy, Ardeshir, and Aram Galstyan. 2012. "Dynamics of Boltzmann q Learning in Two-Player Two-Action Games." *Physical Review E* 85 (4): 041145. https://doi.org/10.1103/PhysRevE.85.041145.

Kim, Taesup, and Yoshua Bengio. 2016. "Deep Directed Generative Models with Energy-Based Probability Estimation." *arXiv Preprint arXiv:1606.03439*.

Kirchhoff, Michael, Thomas Parr, Ensor Palacios, Karl Friston, and Julian Kiverstein. 2018. "The Markov Blankets of Life: Autonomy, Active Inference and the Free Energy Principle." *Journal of the Royal Society Interface* 15 (138): 20170792. https://doi.org/10.1098/rsif.2017.0792.

Kirk, Robert, Ishita Mediratta, Christoforos Nalmpantis, et al. 2024. "Understanding the Effects of RLHF on LLM Generalisation and Diversity." *International Conference on Learning Representations (ICLR)*, 20620–53.

Kirkpatrick, Scott, C. Daniel Gelatt, and Mario P. Vecchi. 1983. "Optimization by Simulated Annealing." *Science* 220 (4598): 671–80. https://doi.org/10.1126/science.220.4598.671.

Knoblauch, Jeremias, Jack Jewson, and Theodoros Damoulas. 2022. "An Optimization-Centric View on Bayes' Rule: Reviewing and Generalizing Variational Inference." *Journal of Machine Learning Research* 23 (132): 1–109.

Koller, Daphne, and Nir Friedman. 2009. *Probabilistic Graphical Models: Principles and Techniques*. MIT Press.

Korbak, Tomasz, Ethan Perez, and Christopher L. Buckley. 2022. "RL with KL Penalties Is Better Viewed as Bayesian Inference." *Findings of the Association for Computational Linguistics: EMNLP 2022*.

Korbel, Jan. 2026. "Foundations of Entropy in Complex Systems." *arXiv Preprint arXiv:2606.16312*. https://arxiv.org/abs/2606.16312.

Krähenbühl, Philipp, and Vladlen Koltun. 2013. "Parameter Learning and Convergent Inference for Dense Random Fields." *International Conference on Machine Learning (ICML)*, 513–21.

Kuhn, Lorenz, Yarin Gal, and Sebastian Farquhar. 2023. "Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation." *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/2302.09664.

Kullback, Solomon. 1959. *Information Theory and Statistics*. Wiley.

Kumar, Rithesh, Sherjil Ozair, Anirudh Goyal, Aaron Courville, and Yoshua Bengio. 2019. "Maximum Entropy Generators for Energy-Based Models." *arXiv Preprint arXiv:1901.08508*.

Lafferty, John, Andrew McCallum, and Fernando Pereira. 2001. "Conditional Random Fields: Probabilistic Models for Segmenting and Labeling Sequence Data." *International Conference on Machine Learning (ICML)*, 282–89.

Landauer, Rolf. 1975. "Inadequacy of Entropy and Entropy Derivatives in Characterizing the Steady State." *Physical Review A* 12 (2): 636–38. https://doi.org/10.1103/PhysRevA.12.636.

Lanillos, Pablo, Cristian Meo, Corrado Pezzato, et al. 2021. "Active Inference in Robotics and Artificial Agents: Survey and Challenges." *arXiv Preprint arXiv:2112.01871*.

Lasserre, Julia A., Christopher M. Bishop, and Thomas P. Minka. 2006. "Principled Hybrids of Generative and Discriminative Models." *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 87–94.

Lazarev, Daniel. 2026. "A Structural Characterization of Entropy Functionals." *arXiv Preprint arXiv:2608.13917*. https://arxiv.org/abs/2608.13917.

LeCun, Yann. 2022. "A Path Towards Autonomous Machine Intelligence." *OpenReview Preprint*.

LeCun, Yann, Sumit Chopra, Raia Hadsell, Marc'Aurelio Ranzato, and Fu Jie Huang. 2006. "A Tutorial on Energy-Based Learning." In *Predicting Structured Data*, edited by Gökhan Bakir, Thomas Hofmann, Bernhard Schölkopf, Alexander J. Smola, and Ben Taskar. MIT Press.

Lee, Dong-Hyun. 2013. "Pseudo-Label: The Simple and Efficient Semi-Supervised Learning Method for Deep Neural Networks." *ICML Workshop on Challenges in Representation Learning*.

Lee, Jonghyun, Dahuin Jung, Saehyung Lee, et al. 2024. "Entropy Is Not Enough for Test-Time Adaptation: From the Perspective of Disentangled Factors." *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/2403.07366.

Leinster, Tom. 2009. "A Maximum Entropy Theorem with Applications to the Measurement of Biodiversity." *arXiv Preprint arXiv:0910.0906*. https://arxiv.org/abs/0910.0906.

Leinster, Tom. 2021. *Entropy and Diversity: The Axiomatic Approach*. Cambridge University Press. https://doi.org/10.1017/9781108963558.

Leinster, Tom, and Christina A. Cobbold. 2012. "Measuring Diversity: The Importance of Species Similarity." *Ecology* 93 (3): 477–89. https://doi.org/10.1890/10-2402.1.

Leinster, Tom, and Mark W. Meckes. 2016. "Maximizing Diversity in Biology and Beyond." *Entropy* 18 (3): 88. https://doi.org/10.3390/e18030088.

Levine, Sergey. 2018. "Reinforcement Learning and Control as Probabilistic Inference: Tutorial and Review." *arXiv Preprint arXiv:1805.00909*.

Liang, Jian, Dapeng Hu, and Jiashi Feng. 2020. "Do We Really Need to Access the Source Data? Source Hypothesis Transfer for Unsupervised Domain Adaptation." *International Conference on Machine Learning (ICML)*. https://arxiv.org/abs/2002.08546.

Lieb, Elliott H., and Jakob Yngvason. 1999. "The Physics and Mathematics of the Second Law of Thermodynamics." *Physics Reports* 310 (1): 1–96. https://doi.org/10.1016/S0370-1573(98)00082-9.

Lightman, Hunter, Vineet Kosaraju, Yuri Burda, et al. 2024. "Let's Verify Step by Step." *International Conference on Learning Representations (ICLR)*.

Lin, Hui, and Jeff A. Bilmes. 2012. "Learning Mixtures of Submodular Shells with Application to Document Summarization." *Conference on Uncertainty in Artificial Intelligence (UAI)*, 479–90.

Lindley, Dennis V. 1956. "On a Measure of the Information Provided by an Experiment." *The Annals of Mathematical Statistics* 27 (4): 986–1005. https://doi.org/10.1214/aoms/1177728069.

Linsker, Ralph. 1988. "Self-Organization in a Perceptual Network." *Computer* 21 (3): 105–17. https://doi.org/10.1109/2.36.

Liu, Weitang, Xiaoyun Wang, John D. Owens, and Yixuan Li. 2020. "Energy-Based Out-of-Distribution Detection." *Advances in Neural Information Processing Systems (NeurIPS)*.

Lu, Cheng, Huayu Chen, Jianfei Chen, Hang Su, Chongxuan Li, and Jun Zhu. 2023. "Contrastive Energy Prediction for Exact Energy-Guided Diffusion Sampling in Offline Reinforcement Learning." *Proceedings of the 40th International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 202.

Lynn, Christopher W., Qiwei Yu, Rich Pang, Stephanie E. Palmer, and William Bialek. 2025. "Exact Minimax Entropy Models of Large-Scale Neuronal Activity." *Physical Review E* 111 (5): 054411. https://doi.org/10.1103/PhysRevE.111.054411.

Lyu, Siwei. 2009. "Interpretation and Generalization of Score Matching." *Conference on Uncertainty in Artificial Intelligence (UAI)*, 359–66.

Ma, Xueguang, Qian Liu, Dongfu Jiang, Ge Zhang, Zejun Ma, and Wenhu Chen. 2025. "General-Reasoner: Advancing LLM Reasoning Across All Domains." *arXiv Preprint arXiv:2505.14652*. https://arxiv.org/abs/2505.14652.

MacArthur, Robert H. 1965. "Patterns of Species Diversity." *Biological Reviews* 40 (4): 510–33. https://doi.org/10.1111/j.1469-185X.1965.tb00815.x.

MacKay, David J. C. 1992. "Information-Based Objective Functions for Active Data Selection." *Neural Computation* 4 (4): 590–604. https://doi.org/10.1162/neco.1992.4.4.590.

Maes, Christian, and Karel Netočný. 2007. "Minimum Entropy Production Principle from a Dynamical Fluctuation Law." *Journal of Mathematical Physics* 48 (5): 053306. https://doi.org/10.1063/1.2738753.

Maes, Christian, and Karel Netočný. 2013. "Minimum Entropy Production Principle." *Scholarpedia* 8 (7): 9664. https://doi.org/10.4249/scholarpedia.9664.

Martyushev, Leonid M., A. S. Nazarova, and Vladimir D. Seleznev. 2007. "On the Problem of the Minimum Entropy Production in the Nonequilibrium Stationary State." *Journal of Physics A: Mathematical and Theoretical* 40 (3): 371–80. https://doi.org/10.1088/1751-8113/40/3/002.

Martyushev, Leonid M., and Vladimir D. Seleznev. 2006. "Maximum Entropy Production Principle in Physics, Chemistry and Biology." *Physics Reports* 426 (1): 1–45. https://doi.org/10.1016/j.physrep.2005.12.001.

Martyushev, Leonid M., and Vladimir D. Seleznev. 2013. "Entropy and Entropy Production: Old Misconceptions and New Breakthroughs." *Entropy* 15 (4): 1152–70. https://doi.org/10.3390/e15041152.

Mazzaglia, Pietro, Tim Verbelen, Ozan Çatal, and Bart Dhoedt. 2022. "The Free Energy Principle for Perception and Action: A Deep Learning Perspective." *Entropy* 24 (2): 301. https://doi.org/10.3390/e24020301.

Menon, Aditya Krishna, Sadeep Jayasumana, Ankit Singh Rawat, Himanshu Jain, Andreas Veit, and Sanjiv Kumar. 2021. "Long-Tail Learning via Logit Adjustment." *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/2007.07314.

Millidge, Beren. 2020. "Deep Active Inference as Variational Policy Gradients." *Journal of Mathematical Psychology* 96: 102348. https://doi.org/10.1016/j.jmp.2020.102348.

Millidge, Beren, Alexander Tschantz, and Christopher L. Buckley. 2021. "Whence the Expected Free Energy?" *Neural Computation* 33 (2): 447–82. https://doi.org/10.1162/neco_a_01354.

Millidge, Beren, Alexander Tschantz, Anil K. Seth, and Christopher L. Buckley. 2020. "On the Relationship Between Active Inference and Control as Inference." In *Active Inference: First International Workshop, IWAI 2020*, vol. 1326. Communications in Computer and Information Science. Springer International Publishing. https://doi.org/10.1007/978-3-030-64919-7_1.

Mironov, Mikhail, and Liudmila Prokhorenkova. 2025. "Measuring Diversity: Axioms and Challenges." *Proceedings of the 42nd International Conference on Machine Learning* 267: 44396–411. https://proceedings.mlr.press/v267/mironov25a.html.

Mnih, Andriy, and Yee Whye Teh. 2012. "A Fast and Simple Algorithm for Training Neural Probabilistic Language Models." *International Conference on Machine Learning (ICML)*, 419–26.

Moussouris, John. 1974. "Gibbs and Markov Random Systems with Constraints." *Journal of Statistical Physics* 10 (1): 11–33. https://doi.org/10.1007/BF01011714.

Mudgal, Sidharth, Jong Lee, Harish Ganapathy, et al. 2024. "Controlled Decoding from Language Models." *Proceedings of the 41st International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 235.

Mumford, David. 1992. "On the Computational Architecture of the Neocortex. II. The Role of Cortico-Cortical Loops." *Biological Cybernetics* 66 (3): 241–51. https://doi.org/10.1007/BF00198477.

Neal, Radford M., and Geoffrey E. Hinton. 1998. "A View of the EM Algorithm That Justifies Incremental, Sparse, and Other Variants." In *Learning in Graphical Models*, edited by Michael I. Jordan. Kluwer Academic Publishers.

Nehring, Klaus, and Clemens Puppe. 2002. "A Theory of Diversity." *Econometrica* 70 (3): 1155–98. https://doi.org/10.1111/1468-0262.00321.

Nemhauser, George L., Laurence A. Wolsey, and Marshall L. Fisher. 1978. "An Analysis of Approximations for Maximizing Submodular Set Functions—i." *Mathematical Programming* 14: 265–94.

Neu, Gergely, Anders Jonsson, and Vicenç Gómez. 2017. "A Unified View of Entropy-Regularized Markov Decision Processes." *arXiv Preprint arXiv:1705.07798*.

Neumann, John von. 1927. "Thermodynamik Quantenmechanischer Gesamtheiten." *Nachrichten von Der Gesellschaft Der Wissenschaften Zu Göttingen, Mathematisch-Physikalische Klasse* 1927: 273–91.

Ng, Andrew Y., and Stuart J. Russell. 2000. "Algorithms for Inverse Reinforcement Learning." *International Conference on Machine Learning (ICML)*.

Ngiam, Jiquan, Zhenghao Chen, Pang Wei Koh, and Andrew Y. Ng. 2011. "Learning Deep Energy Models." *International Conference on Machine Learning (ICML)*, 1105–12.

Nguyen, Quan, and Adji Bousso Dieng. 2024. "Quality-Weighted Vendi Scores and Their Application to Diverse Experimental Design." *Proceedings of the 41st International Conference on Machine Learning* 235: 37667–82. https://proceedings.mlr.press/v235/nguyen24d.html.

Nijkamp, Erik, Mitch Hill, Tian Han, Song-Chun Zhu, and Ying Nian Wu. 2020. "On the Anatomy of MCMC-Based Maximum Likelihood Learning of Energy-Based Models." *AAAI Conference on Artificial Intelligence* 34: 5272–80. https://doi.org/10.1609/aaai.v34i04.5973.

Nijkamp, Erik, Mitch Hill, Song-Chun Zhu, and Ying Nian Wu. 2019. "Learning Non-Convergent Non-Persistent Short-Run MCMC Toward Energy-Based Model." *Advances in Neural Information Processing Systems (NeurIPS)*.

Nikitin, Alexander, Jannik Kossen, Yarin Gal, and Pekka Marttinen. 2024. "Kernel Language Entropy: Fine-Grained Uncertainty Quantification for LLMs from Semantic Similarities." *Advances in Neural Information Processing Systems (NeurIPS)*. https://arxiv.org/abs/2405.20003.

Niu, Shuaicheng, Jiaxiang Wu, Yifan Zhang, et al. 2022. "Efficient Test-Time Model Adaptation Without Forgetting." *International Conference on Machine Learning (ICML)*. https://arxiv.org/abs/2204.02610.

Niu, Shuaicheng, Jiaxiang Wu, Yifan Zhang, et al. 2023. "Towards Stable Test-Time Adaptation in Dynamic Wild World." *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/2302

.12400.

Nuijten, Wouter W. L., Mykola Lukashchuk, Thijs van de Laar, and Bert de Vries. 2026. "What Type of Inference Is Active Inference?" *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI 2026)*, Proceedings of machine learning research, vol. 337: 5003–39.

Nuijten, Wouter W. L., Thijs van de Laar, and Bert de Vries. 2026. "Expected Free Energy-Based Planning as Variational Inference." *Transactions on Machine Learning Research*.

Ochs, W. 1975. "A New Axiomatic Characterization of the von Neumann Entropy." *Reports on Mathematical Physics* 8 (1): 109–20. https://doi.org/10.1016/0034-4877(75)90022-1.

Ortega, Pedro A., and Daniel A. Braun. 2013. "Thermodynamics as a Theory of Decision-Making with Information-Processing Costs." *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 469 (2153): 20120683. https://doi.org/10.1098/rspa.2012.0683.

Ospanov, Azim, Jingwei Zhang, Mohammad Jalali, Xuenan Cao, Andrej Bogdanov, and Farzan Farnia. 2024. "Towards a Scalable Reference-Free Evaluation of Generative Models." *Advances in Neural Information Processing Systems (NeurIPS)* 37: 120892–927. https://doi.org/10.52202/079017-3841.

Ou, Zijing, Tingyang Xu, Qinliang Su, Yingzhen Li, Peilin Zhao, and Yatao Bian. 2022. "Learning Neural Set Functions Under the Optimal Subset Oracle." *Advances in Neural Information Processing Systems (NeurIPS)* 35: 35021–34.

Ouyang, Long, Jeff Wu, Xu Jiang, et al. 2022. "Training Language Models to Follow Instructions with Human Feedback." *Advances in Neural Information Processing Systems 35 (NeurIPS)*.

Owen, Guillermo. 1972. "Multilinear Extensions of Games." *Management Science* 18 (5): P64–79. https://doi.org/10.1287/mnsc.18.5.P64.

Owen, Guillermo. 1975. "Multilinear Extensions and the Banzhaf Value." *Naval Research Logistics Quarterly* 22 (4): 741–50. https://doi.org/10.1002/nav.3800220409.

Özcan, Gözde, Chengzhi Shi, and Stratis Ioannidis. 2025. "Learning Set Functions with Implicit Differentiation." *AAAI Conference on Artificial Intelligence*, 19777–85.

Paltridge, Garth W. 1975. "Global Dynamics and Climate — a System of Minimum Entropy Exchange." *Quarterly Journal of the Royal Meteorological Society* 101 (429): 475–84. https://doi.org/10.1002/qj.49710142906.

Paninski, Liam. 2003. "Estimation of Entropy and Mutual Information." *Neural Computation* 15 (6): 1191–253. https://doi.org/10.1162/089976603321780272.

Park, Mincheol, Heeji Won, Won Woo Ro, and Suhyun Kim. 2025. "Rethinking Entropy in Test-Time Adaptation: The Missing Piece from Energy Duality." *Advances in Neural Information Processing Systems 38 (NeurIPS)*.

Parr, Thomas, and Karl Friston. 2019. "Generalised Free Energy and Active Inference." *Biological Cybernetics* 113 (5-6): 495–513. https://doi.org/10.1007/s00422-019-00805-w.

Parr, Thomas, Giovanni Pezzulo, and Karl Friston. 2022. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press. https://doi.org/10.7551/mitpress/12441.001.0001.

Parry, Matthew, A. Philip Dawid, and Steffen Lauritzen. 2012. "Proper Local Scoring Rules." *Annals of Statistics* 40 (1): 561–92. https://doi.org/10.1214/12-AOS971.

Parshakova, Tetiana, Jean-Marc Andreoli, and Marc Dymetman. 2019. "Distributional Reinforcement Learning for Energy-Based Sequential Models." *Optimization Foundations for Reinforcement Learning Workshop at NeurIPS 2019*.

Pasarkar, Amey P., and Adji Bousso Dieng. 2024. "Cousins of the Vendi Score: A Family of Similarity-Based Diversity Metrics for Science and Machine Learning." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, vol. 238. https://arxiv.org/abs/2310.12952.

Patil, G. P., and C. Taillie. 1982. "Diversity as a Concept and Its Measurement." *Journal of the American Statistical Association* 77 (379): 548–61. https://doi.org/10.1080/01621459.1982.10477845.

Pearl, Judea. 1988. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann.

Pereyra, Gabriel, George Tucker, Jan Chorowski, Łukasz Kaiser, and Geoffrey Hinton. 2017. "Regularizing Neural Networks by Penalizing Confident Output Distributions." *ICLR 2017 Workshop Track*.

Pezzulo, Giovanni, Thomas Parr, Paul Cisek, Andy Clark, and Karl Friston. 2024. "Generating Meaning: Active Inference and the Scope and Limits of Passive AI." *Trends in Cognitive Sciences* 28 (2): 97–112. https://doi.org/10.1016/j.tics.2023.10.002.

Pressé, Steve, Kingshuk Ghosh, Julian Lee, and Ken A. Dill. 2013. "Principles of Maximum Entropy and Maximum Caliber in Statistical Physics." *Reviews of Modern Physics* 85 (3): 1115–41. https://doi.org/10.1103/RevModPhys.85.1115.

Prigogine, Ilya. 1945. "Modération Et Transformations Irréversibles Des Systèmes Ouverts." *Bulletin de La Classe Des Sciences, Académie Royale de Belgique* 31: 600–606.

Prigogine, Ilya. 1947. *Étude Thermodynamique Des Phénomènes Irréversibles*. Dunod, Paris; Desoer, Liège.

Prigogine, Ilya. 1978. "Time, Structure, and Fluctuations." *Science* 201 (4358): 777–85. https://doi.org/10.1126/science.201.4358.777.

Prigogine, Ilya, and Jean-Marie Wiame. 1946. "Biologie Et Thermodynamique Des Phénomènes Irréversibles." *Experientia* 2 (11): 451–53. https://doi.org/10.1007/BF02153597.

Qin, Lianhui, Sean Welleck, Daniel Khashabi, and Yejin Choi. 2022. "COLD Decoding: Energy-Based Constrained Text Generation with Langevin Dynamics." *Advances in Neural Information Processing Systems (NeurIPS)*.

Rafailov, Rafael, Joey Hejna, Ryan Park, and Chelsea Finn. 2024. "From r to Q^*: Your Language Model Is Secretly a Q-Function." *arXiv Preprint arXiv:2404.12358*.

Rafailov, Rafael, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn.

2023. "Direct Preference Optimization: Your Language Model Is Secretly a Reward Model." *Advances in Neural Information Processing Systems 36 (NeurIPS)*.

Raja, Vicente, Edward Baggs, Anthony Chemero, and Michael L. Anderson. 2023. "Trick or Treat: A Response to Commentaries on 'The Markov Blanket Trick'." *Physics of Life Reviews* 44: 153–59. https://doi.org/10.1016/j.plrev.2023.01.001.

Raja, Vicente, Dinesh Valluri, Edward Baggs, Anthony Chemero, and Michael L. Anderson. 2021. "The Markov Blanket Trick: On the Scope of the Free Energy Principle and Active Inference." *Physics of Life Reviews* 39: 49–72. https://doi.org/10.1016/j.plrev.2021.09.001.

Ramstead, Maxwell J. D., Paul Benjamin Badcock, and Karl Friston. 2018. "Answering Schrödinger's Question: A Free-Energy Formulation." *Physics of Life Reviews* 24: 1–16. https://doi.org/10.1016/j.plrev.2017.09.001.

Ramstead, Maxwell J. D., Dalton A. R. Sakthivadivel, Conor Heins, et al. 2023. "On Bayesian Mechanics: A Physics of and by Beliefs." *Interface Focus* 13 (3): 20220029. https://doi.org/10.1098/rsfs.2022.0029.

Rao, C. Radhakrishna. 1982. "Diversity and Dissimilarity Coefficients: A Unified Approach." *Theoretical Population Biology* 21 (1): 24–43. https://doi.org/10.1016/0040-5809(82)90004-1.

Rao, Rajesh P. N., and Dana H. Ballard. 1999. "Predictive Coding in the Visual Cortex: A Functional Interpretation of Some Extra-Classical Receptive-Field Effects." *Nature Neuroscience* 2 (1): 79–87. https://doi.org/10.1038/4580.

Ratliff, Nathan D., J. Andrew Bagnell, and Martin A. Zinkevich. 2006. "Maximum Margin Planning." *International Conference on Machine Learning (ICML)*, 729–36.

Rényi, Alfréd. 1961. "On Measures of Entropy and Information." *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability, Volume 1: Contributions to the Theory of Statistics*, 547–61.

Rose, Kenneth. 1998. "Deterministic Annealing for Clustering, Compression, Classification, Regression, and Related Optimization Problems." *Proceedings of the IEEE* 86 (11): 2210–39.

Rose, Kenneth, Eitan Gurewitz, and Geoffrey C. Fox. 1990. "Statistical Mechanics and Phase Transitions in Clustering." *Physical Review Letters* 65 (8): 945–48. https://doi.org/10.1103/PhysRevLett.65.945.

Sahin, Aytunc, Yatao Bian, Joachim M. Buhmann, and Andreas Krause. 2020. "From Sets to Multisets: Provable Variational Inference for Probabilistic Integer Submodular Models." *International Conference on Machine Learning (ICML)*, 8388–97.

Saito, Kuniaki, Donghyun Kim, Stan Sclaroff, Trevor Darrell, and Kate Saenko. 2019. "Semi-Supervised Domain Adaptation via Minimax Entropy." *IEEE/CVF International Conference on Computer Vision (ICCV)*, 8050–58.

Sajid, Noor, Philip J. Ball, Thomas Parr, and Karl Friston. 2021. "Active Inference: Demystified and Compared." *Neural Computation* 33 (3): 674–712. https://doi.org/10.1162/neco_a_01357.

Sajid, Noor, Panagiotis Tigas, Alexey Zakharov, Zafeirios Fountas, and Karl Friston. 2021. "Exploration and

Preference Satisfaction Trade-Off in Reward-Free Learning." *arXiv Preprint arXiv:2106.04316*.

Sakthivadivel, Dalton A. R. 2022. "Towards a Geometry and Analysis for Bayesian Mechanics." *arXiv Preprint arXiv:2204.11900*.

Sánchez Giraldo, Luis Gonzalo, Murali Rao, and José C. Príncipe. 2015. "Measures of Entropy from Data Using Infinitely Divisible Kernels." *IEEE Transactions on Information Theory* 61 (1): 535–48. https://doi.org/10.1109/TIT.2014.2370058.

Santos, Roberto J. V. dos. 1997. "Generalization of Shannon's Theorem for Tsallis Entropy." *Journal of Mathematical Physics* 38 (8): 4104–7. https://doi.org/10.1063/1.532107.

Schrödinger, Erwin. 1944. *What Is Life? The Physical Aspect of the Living Cell*. Cambridge University Press.

Schulman, John, Sergey Levine, Pieter Abbeel, Michael I. Jordan, and Philipp Moritz. 2015. "Trust Region Policy Optimization." *International Conference on Machine Learning (ICML)*.

Schulman, John, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. "Proximal Policy Optimization Algorithms." *arXiv Preprint arXiv:1707.06347*.

Schwartenbeck, Philipp, Thomas H. B. FitzGerald, Raymond J. Dolan, and Karl Friston. 2013. "Exploration, Novelty, Surprise, and Free Energy Minimization." *Frontiers in Psychology* 4: 710. https://doi.org/10.3389/fpsyg.2013.00710.

Schwartenbeck, Philipp, Thomas H. B. FitzGerald, Christoph Mathys, Raymond J. Dolan, and Karl Friston. 2015. "The Dopaminergic Midbrain Encodes the Expected Certainty about Desired Outcomes." *Cerebral Cortex* 25 (10): 3434–45. https://doi.org/10.1093/cercor/bhu159.

Shannon, Claude E. 1948. "A Mathematical Theory of Communication." *Bell System Technical Journal* 27 (3): 379–423.

Shao, Zhihong, Peiyi Wang, Qihao Zhu, et al. 2024. "DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models." *arXiv Preprint arXiv:2402.03300*. https://arxiv.org/abs/2402.03300.

Shi, Chence, Shitong Luo, Minkai Xu, and Jian Tang. 2021. "Learning Gradient Fields for Molecular Conformation Generation." *International Conference on Machine Learning (ICML)*.

Shi, Yongquan, Zijing Ou, Shiping Wang, and Yatao Bian. 2026. "Advancing Optimal Subset Oracle via Learning Relaxation of Neural Set Functions." *arXiv Preprint arXiv:2607.11555*.

Shirts, Michael R., Eric Bair, Giles Hooker, and Vijay S. Pande. 2003. "Equilibrium Free Energies from Nonequilibrium Measurements Using Maximum-Likelihood Methods." *Physical Review Letters* 91: 140601. https://doi.org/10.1103/PhysRevLett.91.140601.

Shore, John E., and Rodney W. Johnson. 1980. "Axiomatic Derivation of the Principle of Maximum Entropy and the Principle of Minimum Cross-Entropy." *IEEE Transactions on Information Theory* 26 (1): 26–37.

Simpson, E. H. 1949. "Measurement of Diversity." *Nature* 163 (4148): 688. https://doi.org/10.1038/163688

a0.

Sipos, Ruben, Pannaga Shivaswamy, and Thorsten Joachims. 2012. "Large-Margin Learning of Submodular Summarization Models." *Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 224–33.

Smith, Ryan, Karl Friston, and Christopher J. Whyte. 2022. "A Step-by-Step Tutorial on Active Inference and Its Application to Empirical Data." *Journal of Mathematical Psychology* 107: 102632. https://doi.org/10.1016/j.jmp.2021.102632.

Smith, Ryan, Philipp Schwartenbeck, Thomas Parr, and Karl Friston. 2020. "An Active Inference Approach to Modeling Structure Learning: Concept Learning as an Example Case." *Frontiers in Computational Neuroscience* 14: 41. https://doi.org/10.3389/fncom.2020.00041.

Sohl-Dickstein, Jascha, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. 2015. "Deep Unsupervised Learning Using Nonequilibrium Thermodynamics." *International Conference on Machine Learning (ICML)*, 2256–65.

Solow, Andrew R., and Stephen Polasky. 1994. "Measuring Biological Diversity." *Environmental and Ecological Statistics* 1 (2): 95–103. https://doi.org/10.1007/BF02426650.

Song, Yang, Conor Durkan, Iain Murray, and Stefano Ermon. 2021. "Maximum Likelihood Training of Score-Based Diffusion Models." *Advances in Neural Information Processing Systems (NeurIPS)* 34.

Song, Yang, and Stefano Ermon. 2019. "Generative Modeling by Estimating Gradients of the Data Distribution." *Advances in Neural Information Processing Systems (NeurIPS)*.

Song, Yang, Sahaj Garg, Jiaxin Shi, and Stefano Ermon. 2019. "Sliced Score Matching: A Scalable Approach to Density and Score Estimation." *Uncertainty in Artificial Intelligence (UAI)*.

Song, Yang, and Diederik P. Kingma. 2021. "How to Train Your Energy-Based Models." *arXiv Preprint arXiv:2101.03288*.

Song, Yang, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. 2021. "Score-Based Generative Modeling Through Stochastic Differential Equations." *International Conference on Learning Representations (ICLR)*.

Sprevak, Mark, and Ryan Smith. 2023. "An Introduction to Predictive Processing Models of Perception and Decision-Making." *Topics in Cognitive Science*, ahead of print. https://doi.org/10.1111/tops.12704.

Srinivasan, Mandyam V., Simon B. Laughlin, and A. Dubs. 1982. "Predictive Coding: A Fresh View of Inhibition in the Retina." *Proceedings of the Royal Society of London. Series B. Biological Sciences* 216 (1205): 427–59. https://doi.org/10.1098/rspb.1982.0085.

Still, Susanne, David A. Sivak, Anthony J. Bell, and Gavin E. Crooks. 2012. "Thermodynamics of Prediction." *Physical Review Letters* 109 (12): 120604. https://doi.org/10.1103/PhysRevLett.109.120604.

Stirling, Andy. 2007. "A General Framework for Analysing Diversity in Science, Technology and Society." *Journal of the Royal Society Interface* 4 (15): 707–19. https://doi.org/10.1098/rsif.2007.0213.

Stoyanov, Veselin, Alexander Ropson, and Jason Eisner. 2011. "Empirical Risk Minimization of Graphical Model Parameters Given Approximate Inference, Decoding, and Model Structure." *International Conference on Artificial Intelligence and Statistics (AISTATS)*.

Sustek, Martin, Samik Sadhu, Lukáš Burget, et al. 2023. "Stabilized Training of Joint Energy-Based Models and Their Practical Applications." *arXiv Preprint arXiv:2303.04187*.

Sutskever, Ilya, and Tijmen Tieleman. 2010. "On the Convergence Properties of Contrastive Divergence." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, Proceedings of machine learning research, vol. 9: 789–95.

Sweeney, Patrick, Jaime Ruiz-Serra, and Michael S. Harré. 2025. "Decision, Inference, and Information: Formal Equivalences Under Active Inference." *Entropy* 28 (1): 1. https://doi.org/10.3390/e28010001.

Tanaka, Daiki, Daiki Ikami, Toshihiko Yamasaki, and Kiyoharu Aizawa. 2018. "Joint Optimization Framework for Learning with Noisy Labels." *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. https://arxiv.org/abs/1803.11364.

Taskar, Ben, Carlos Guestrin, and Daphne Koller. 2003. "Max-Margin Markov Networks." *Advances in Neural Information Processing Systems (NeurIPS)* 16.

Tempesta, Piergiulio. 2016. "Beyond the Shannon–Khinchin Formulation: The Composability Axiom and the Universal-Group Entropy." *Annals of Physics* 365: 180–97. https://doi.org/10.1016/j.aop.2015.08.013.

The Royal Swedish Academy of Sciences. 2024. *Scientific Background to the Nobel Prize in Physics 2024: For Foundational Discoveries and Inventions That Enable Machine Learning with Artificial Neural Networks*. https://www.nobelprize.org/prizes/physics/2024/advanced-information/.

Tieleman, Tijmen. 2008. "Training Restricted Boltzmann Machines Using Approximations to the Likelihood Gradient." *International Conference on Machine Learning (ICML)*, 1064–71.

Todorov, Emanuel. 2006. "Linearly-Solvable Markov Decision Problems." *Advances in Neural Information Processing Systems (NeurIPS)* 19: 1369–76.

Todorov, Emanuel. 2009. "Efficient Computation of Optimal Actions." *Proceedings of the National Academy of Sciences* 106 (28): 11478–83.

Topsøe, Flemming. 1979. "Information-Theoretical Optimization Techniques." *Kybernetika* 15 (1): 8–27.

Toussaint, Marc. 2009. "Robot Trajectory Optimization Using Approximate Inference." *International Conference on Machine Learning (ICML)*.

Tsallis, Constantino. 1988. "Possible Generalization of Boltzmann–Gibbs Statistics." *Journal of Statistical Physics* 52 (1–2): 479–87.

Tschantz, Alexander, Beren Millidge, Anil K. Seth, and Christopher L. Buckley. 2020. "Reinforcement Learning Through Active Inference." *arXiv Preprint arXiv:2002.12636*.

Tschiatschek, Sebastian, Aytunc Sahin, and Andreas Krause. 2018. "Differentiable Submodular Maximiza-

tion." *International Joint Conference on Artificial Intelligence (IJCAI)*, 2731–38.

Tsochantaridis, Ioannis, Thorsten Joachims, Thomas Hofmann, and Yasemin Altun. 2005. "Large Margin Methods for Structured and Interdependent Output Variables." *Journal of Machine Learning Research* 6: 1453–84.

Tu, Zhuowen. 2007. "Learning Generative Models via Discriminative Approaches." *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Uehara, Masatoshi, Yulai Zhao, Kevin Black, et al. 2024. "Fine-Tuning of Continuous-Time Diffusion Models as Entropy-Regularized Control." *arXiv Preprint arXiv:2402.15194*.

Ueltzhöffer, Kai. 2018. "Deep Active Inference." *Biological Cybernetics* 112 (6): 547–73. https://doi.org/10.1007/s00422-018-0785-7.

Velikonivtsev, Fedor, Mikhail Mironov, and Liudmila Prokhorenkova. 2024. "Challenges of Generating Structurally Diverse Graphs." *Advances in Neural Information Processing Systems (NeurIPS)* 37: 57993–8022. https://doi.org/10.52202/079017-1849.

Verschaffelt, Jules-Émile. 1954. "Sur Les Minima de Production d'entropie Et de Dissipation d'énergie." *Bulletin de La Classe Des Sciences, Académie Royale de Belgique* 40: 779–83.

Vincent, Pascal. 2011. "A Connection Between Score Matching and Denoising Autoencoders." *Neural Computation* 23 (7): 1661–74.

Wainwright, Martin J., and Michael I. Jordan. 2008. "Graphical Models, Exponential Families, and Variational Inference." *Foundations and Trends in Machine Learning* 1 (1–2): 1–305. https://doi.org/10.1561/2200000001.

Wallace, Bram, Meihua Dang, Rafael Rafailov, et al. 2024. "Diffusion Model Alignment Using Direct Preference Optimization." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 8228–38.

Wang, Dequan, Evan Shelhamer, Shaoteng Liu, Bruno Olshausen, and Trevor Darrell. 2021. "Tent: Fully Test-Time Adaptation by Entropy Minimization." *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/2006.10726.

Wang, Xudong, Zhirong Wu, Long Lian, and Stella X. Yu. 2022. "Debiased Learning from Naturally Imbalanced Pseudo-Labels." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 14647–57. https://doi.org/10.1109/CVPR52688.2022.01424.

Weilenmann, Mirjam, Lea Kraemer, Philippe Faist, and Renato Renner. 2016. "Axiomatic Relation Between Thermodynamic and Information-Theoretic Entropies." *Physical Review Letters* 117 (26): 260601. https://doi.org/10.1103/PhysRevLett.117.260601.

Weitzman, Martin L. 1992. "On Diversity." *The Quarterly Journal of Economics* 107 (2): 363–405. https://doi.org/10.2307/2118476.

Williams, P. M. 1980. "Bayesian Conditionalisation and the Principle of Minimum Information." *The British Journal for the Philosophy of Science* 31 (2): 131–44. https://doi.org/10.1093/bjps/31.2.131.

Williams, Ronald J. 1992. "Simple Statistical Gradient-Following Algorithms for Connectionist Reinforcement Learning." *Machine Learning* 8 (3–4): 229–56.

Wu, Fa-Yueh. 1982. "The Potts Model." *Reviews of Modern Physics* 54 (1): 235–68. https://doi.org/10.1103/RevModPhys.54.235.

Wu, Jiaxiang, Tao Shen, Haidong Lan, Yatao Bian, and Junzhou Huang. 2021. "SE(3)-Equivariant Energy-Based Models for End-to-End Protein Folding." *bioRxiv Preprint*.

Wu, Tailin, and Ian Fischer. 2020. "Phase Transitions for the Information Bottleneck in Representation Learning." *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/2001.01878.

Wu, Tailin, Ian Fischer, Isaac L. Chuang, and Max Tegmark. 2019. "Learnability for the Information Bottleneck." *Uncertainty in Artificial Intelligence (UAI)*. https://arxiv.org/abs/1907.07331.

Xie, Binghui, Yatao Bian, Kaiwen Zhou, et al. 2024. "Enhancing Neural Subset Selection: Integrating Background Information into Set Representations." *International Conference on Learning Representations (ICLR)*.

Xie, Binghui, Yixuan Wang, Yongqiang Chen, et al. 2024. "HORSE: Hierarchical Representation for Large-Scale Neural Subset Selection." *Advances in Neural Information Processing Systems (NeurIPS)*, 4852–77.

Xie, Jianwen, Yang Lu, Ruiqi Gao, and Ying Nian Wu. 2018. "Cooperative Learning of Energy-Based Model and Latent Variable Model via MCMC Teaching." *AAAI Conference on Artificial Intelligence*.

Xie, Jianwen, Yang Lu, Song-Chun Zhu, and Ying Nian Wu. 2016. "A Theory of Generative ConvNet." *International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 48: 2635–44.

Xie, Yutong, Ziqiao Xu, Jiaqi Ma, and Qiaozhu Mei. 2023. "How Much Space Has Been Explored? Measuring the Chemical Space Covered by Databases and Machine-Generated Molecules." *International Conference on Learning Representations (ICLR)*. https://openreview.net/forum?id=Yo06F8kfMa1.

Xu, Minkai, Tomas Geffner, Karsten Kreis, et al. 2025. "Energy-Based Diffusion Language Models for Text Generation." *International Conference on Learning Representations (ICLR)*.

Yan, Yuchen, Minkai Xu, Zaiquan Yang, and Yatao Bian. 2026. "Unified Energy for Invariant and Independent Decoding in Diffusion Language Models." *arXiv Preprint arXiv:2606.09159*.

Yang, Xiulong, and Shihao Ji. 2021. "JEM++: Improved Techniques for Training JEM." *IEEE/CVF International Conference on Computer Vision (ICCV)*, 6494–503.

Yang, Xiulong, Qing Su, and Shihao Ji. 2023. "Towards Bridging the Performance Gaps of Joint Energy-Based Models." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 15732–41.

Yuan, Suqin, Jinkun Chen, Jiyang Zheng, et al. 2026. "Understanding Diversity Collapse in RLVR via the Lens of Overtraining." *arXiv Preprint arXiv:2606.15455*. https://arxiv.org/abs/2606.15455.

Yuan, Yige, Bingbing Xu, Liang Hou, Fei Sun, Huawei Shen, and Xueqi Cheng. 2024. "TEA: Test-Time Energy Adaptation." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 23901–11.

Zaheer, Manzil, Satwik Kottur, Siamak Ravanbakhsh, Barnabás Póczos, Ruslan Salakhutdinov, and Alexander J. Smola. 2017. "Deep Sets." *Advances in Neural Information Processing Systems (NeurIPS)* 30: 3391–401.

Zellner, Arnold. 1988. "Optimal Information Processing and Bayes's Theorem." *The American Statistician* 42 (4): 278–80. https://doi.org/10.1080/00031305.1988.10475585.

Zhang, Qingyang, Haitao Wu, Changqing Zhang, Peilin Zhao, and Yatao Bian. 2025. "Right Question Is Already Half the Answer: Fully Unsupervised LLM Reasoning Incentivization." *Advances in Neural Information Processing Systems (NeurIPS)*. https://arxiv.org/abs/2504.05812.

Zhao, Dora, Jerone T. A. Andrews, Orestis Papakyriakopoulos, and Alice Xiang. 2024. "Position: Measure Dataset Diversity, Don't Just Claim It." *Proceedings of the 41st International Conference on Machine Learning* 235: 60644–73. https://proceedings.mlr.press/v235/zhao24a.html.

Zhao, Stephen, Rob Brekelmans, Alireza Makhzani, and Roger Grosse. 2024. "Probabilistic Inference in Language Models via Twisted Sequential Monte Carlo." *Proceedings of the 41st International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 235.

Zhao, Stephen, Jörn-Henrik Jacobsen, and Will Grathwohl. 2020. "Joint Energy-Based Models for Semi-Supervised Classification." *ICML 2020 Workshop on Uncertainty and Robustness in Deep Learning*.

Zhao, Xuandong, Zhewei Kang, Aosong Feng, Sergey Levine, and Dawn Song. 2025. "Learning to Reason Without External Rewards." *arXiv Preprint arXiv:2505.19590*. https://arxiv.org/abs/2505.19590.

Zheng, Shuai, Sadeep Jayasumana, Bernardino Romera-Paredes, et al. 2015. "Conditional Random Fields as Recurrent Neural Networks." *IEEE International Conference on Computer Vision (ICCV)*, 1529–37.

Zhou, Dengyong, Qiang Liu, John C. Platt, Christopher Meek, and Nihar B. Shah. 2015. "Regularized Minimax Conditional Entropy for Crowdsourcing." *arXiv Preprint arXiv:1503.07240*.

Zhou, Dengyong, John C. Platt, Sumit Basu, and Yi Mao. 2012. "Learning from the Wisdom of Crowds by Minimax Entropy." *Advances in Neural Information Processing Systems (NeurIPS)* 25: 2195–203.

Zhu, Song Chun, Ying Nian Wu, and David Mumford. 1997. "Minimax Entropy Principle and Its Application to Texture Modeling." *Neural Computation* 9 (8): 1627–60. https://doi.org/10.1162/neco.1997.9.8.1627.

Zhu, Song Chun, Ying Nian Wu, and David Mumford. 1998. "Filters, Random Fields and Maximum Entropy (FRAME): Towards a Unified Theory for Texture Modeling." *International Journal of Computer Vision* 27 (2): 107–26.

Zhu, Yuchang, Huizhe Zhang, Bingzhe Wu, et al. 2025. "Measuring Diversity in Synthetic Datasets." *Proceedings of the 42nd International Conference on Machine Learning* 267: 80373–97. https://proceedings.mlr.press/v267/zhu25ac.html.

Ziebart, Brian D., J. Andrew Bagnell, and Anind K. Dey. 2010. "Modeling Interaction via the Principle of

Maximum Causal Entropy." *International Conference on Machine Learning (ICML)*.

Ziebart, Brian D., Andrew Maas, J. Andrew Bagnell, and Anind K. Dey. 2008. "Maximum Entropy Inverse Reinforcement Learning." *AAAI Conference on Artificial Intelligence*, 1433–38.

Ziegler, Daniel M., Nisan Stiennon, Jeffrey Wu, et al. 2019. "Fine-Tuning Language Models from Human Preferences." *arXiv Preprint arXiv:1909.08593*.

Zuo, Yuxin, Kaiyan Zhang, Li Sheng, et al. 2025. "TTRL: Test-Time Reinforcement Learning." *Advances in Neural Information Processing Systems 38 (NeurIPS)*.